

Voltus IC Power Integrity Solution Menu Reference

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About This Manual

This manual describes the menus that you can use in Voltus IC Power Integrity Solution.

How this Manual Is Organized

The chapters in this manual are organized to correspond to the menus in the Voltus graphical user interface (GUI). Each chapter contains one or more of the following sections. A particular section is omitted from a chapter if there is no significant information related to the topic.

Form Overview	Introduces the features available through the various menu selections and provides an illustration of the forms and screens used to set options and run the features.
Forms and Options	Lists and describes the fields and options on each screen.
Related Topics	Provides links to related tasks and concepts in the <i>Voltus User Guide</i> .
Related Commands	Lists the text commands that correspond to the GUI forms for this feature.

Conventions Used in This Manual

This section describes the typographic and syntax conventions used in this manual.

text	Indicates text that you must type exactly as shown. For example: <code>report_annotated_check -missing_setup</code>
------	--

<i>text</i>	Indicates information for which you must substitute a name or value. In the following example, you must substitute the name of a specific file for <i>worst_entries</i>: <pre>report_clock_timing -type skew -nworst worst_entries</pre>
<i>text</i>	Indicates the following: • Text found in the graphical user interface (GUI), including form names, button labels, and field names • Terms that are new to the manual, are the subject of discussion, or need special emphasis • Titles of manuals
[]	Indicates optional arguments. In the following example, you can specify none, one, or both of the bracketed arguments: <pre>command [-arg1] [arg2 value]</pre>
[]	Indicates an optional choice from a mutually exclusive list. In the following example, you can specify any of the arguments or none of the arguments, but you cannot specify more than one: <pre>command [arg1 arg2 arg3 arg4]</pre>
{ }	Indicates a required choice from a mutually exclusive list. In the following example, you must specify one, and only one, of the arguments: <pre>command {arg1 arg2 arg3}</pre>
{ [] [] }	Indicates a required choice of one or more items in a list. In the following example, you must choose one argument from the list, but you can choose more than one: <pre>command {[arg1] [arg2] [arg3]}</pre>

{ }	Indicates curly braces that must be entered with the command syntax. In the following example, you must type the curly braces: <code>command arg1 {x y}</code>
...	Indicates that you can repeat the previous argument.
.	Indicates an omission in an example of computer output or input.
<i>Command - Subcommand</i>	Indicates a command sequence, which shows the order in which you choose commands and subcommands from the GUI menu. In the following example, you choose <i>Power & Rail</i> from the menu, then <i>Report</i> from the submenu, and then <i>Dynamic Movies</i> from the displayed list: <i>Power & Rail - Report - Dynamic Movies</i> This sequence opens the View Movies form.

Related Documents

For more information about the Voltus and Tempus family of products, see the following documents. You can access these and other Cadence documents with the Cadence Help online documentation system.

Voltus IC Power Integrity Solution Product Documentation

- [Voltus Known Problems and Solutions](#)
Describes important Cadence Change Requests (CCRs) for Voltus, including solutions for working around known problems.
- [Voltus Text Command Reference](#)
Describes the Voltus text commands, including syntax and examples.
- [Voltus User Guide](#)
Provides information on using various Voltus features.

Tempus Timing Signoff Solution Documentation

- *What's New in Tempus*
Provides information about new and changed features in this release of the Tempus Timing Signoff Solution.
- *Tempus Known Problems and Solutions*
Describes important Cadence Change Requests (CCRs) for Tempus, including solutions for working around known problems.
- *Tempus Text Command Reference*
Describes the Tempus text commands, including syntax and examples.
- *Tempus User Guide*
Provides information on using various Tempus features.
- *Tempus Menu Reference*
Describes Tempus graphical user interface.

The Main Window

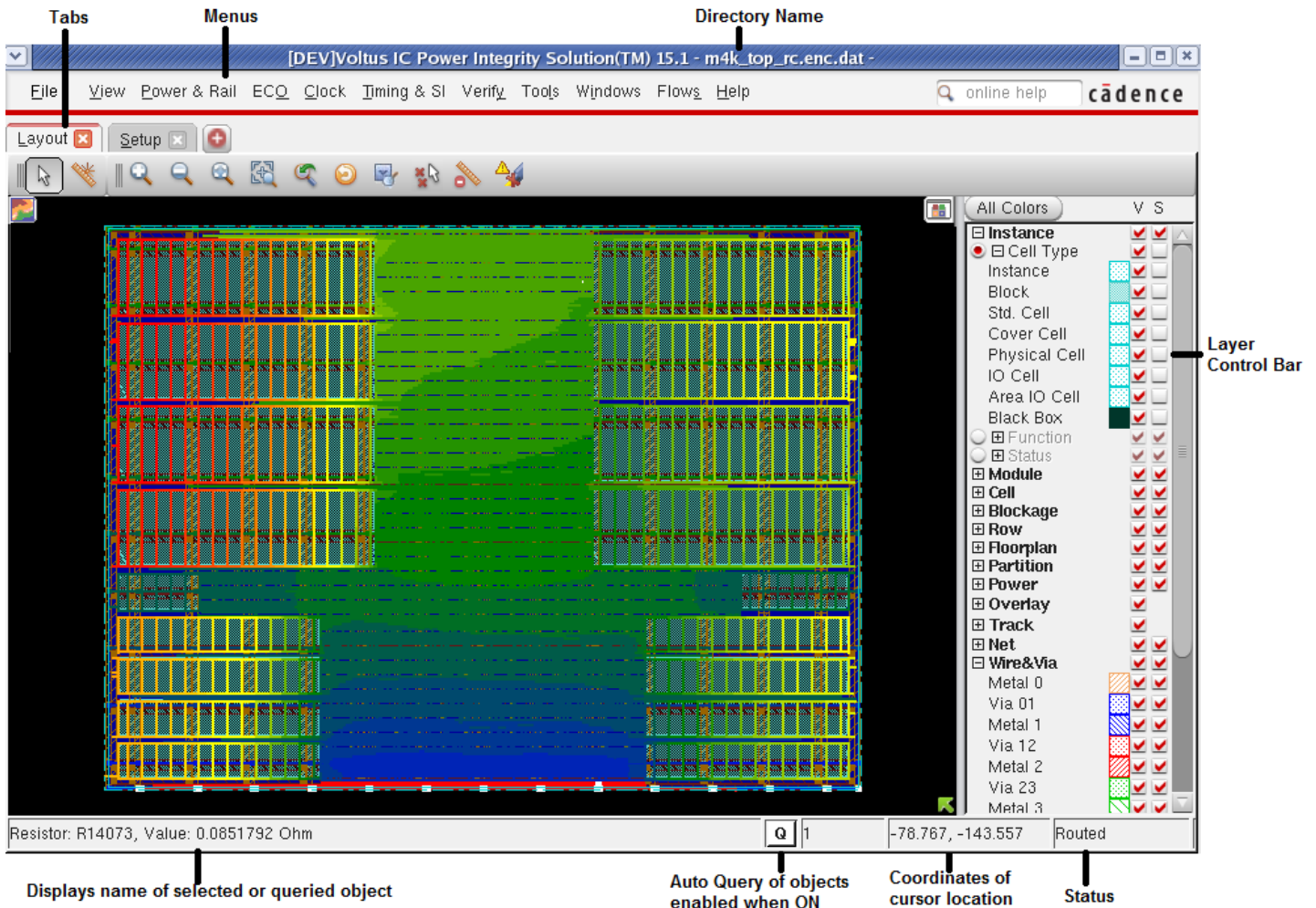
- [Main Window Components](#)
- [Menus](#)
- [Tabs](#)
- [Mouse Operation](#)

Main Window Components

The components in the main window are identified in the following figure:

Voltus IC Power Integrity Solution Menu Reference

The Main Window



Menus

There are eleven pull-down menus:

- File
- View
- Power and Rail
- ECO
- Clock
- Timing & SI
- Verify

- [Tools](#)
- [Windows](#)
- [Flows](#)
- [Help](#)

You can access the complete library of Voltus documentation through the *Help* menu. The *Help - Documentation Library* link opens the Cadence Help viewer with all the books in the Voltus documentation suite.

You can also view the individual books through their respective links from the *Help* menu. For example, to view the *Voltus Text Command Reference*, select *Help - Text Command Reference*.

Tabs

There are six tabs:

- [Layout](#)
- [Setup](#)
- [Schematics](#)
- [Design Browser](#)
- [Layout Viewer](#)
- [Violation Browser](#)

Mouse Operation

You can use the mouse buttons to perform various actions.

- Left-click the mouse to select a button or check box.
- To select multiple files in a directory, select the first item, then select the last item while holding the `shift` key.

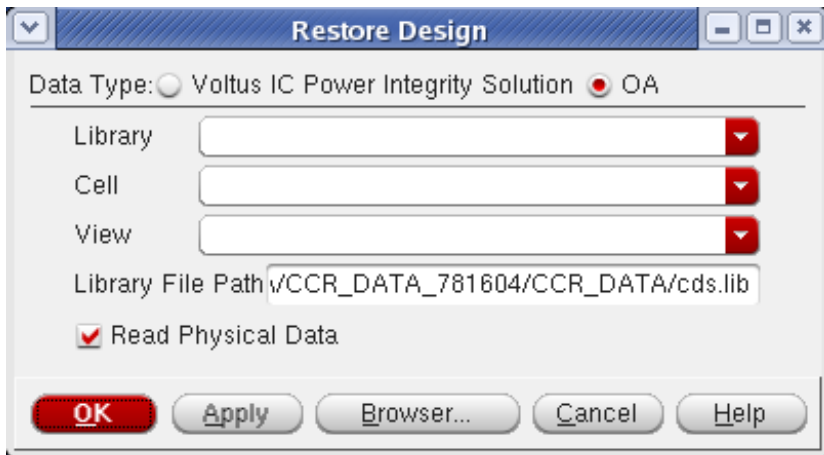
File Menu

- Read Design
- Save Design
- Read SDC
- Read SPEF
- Read SDF
- Read IR Drop
- Load/Commit CPF
- Specify Operating Condition/PVT
- Configure MMC
- Check Design
- Report Timing Library
- Check Timing Library
- Summary
- Netlist Statistics
- Hide GUI
- Exit

Read Design

Use the Restore Design form to read design data. This design data then can be used for calculating power and performing power analysis.

→ Choose *File - Read Design*.



The screenshot shows the 'Restore Design' dialog box. At the top, the title bar says 'Restore Design'. Below it, the 'Data Type' section has two radio buttons: 'Voltus IC Power Integrity Solution' (unselected) and 'OA' (selected). Underneath, there are three dropdown menus for 'Library', 'Cell', and 'View', all of which are currently empty. Below these is a text field for 'Library File Path' containing the path 'vCCR_DATA_781604/CCR_DATA/cds.lib'. A checkbox labeled 'Read Physical Data' is checked. At the bottom, there are five buttons: 'OK' (highlighted in red), 'Apply', 'Browser...', 'Cancel', and 'Help'.

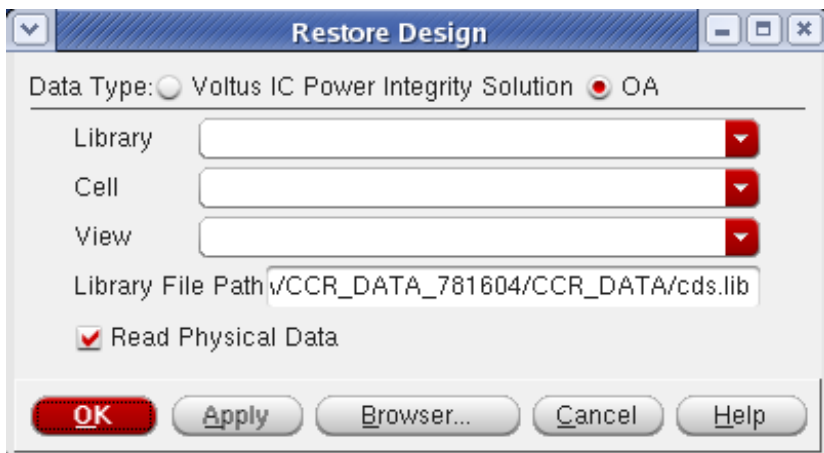
The Restore Design form has two pages:

- [Read Design - OA](#)
- [Read Design - Voltus IC Power Integrity Solution](#)

Read Design - OA

The OA page within the Restore Design form is used to restore all design information from the previous design session, using the OpenAccess database format.

→ Choose *File - Read Design*. The *OA* page opens by default.



This screenshot is identical to the one above, showing the 'Restore Design' dialog box with the 'OA' radio button selected. The 'Library', 'Cell', and 'View' dropdowns are empty, the 'Library File Path' is 'vCCR_DATA_781604/CCR_DATA/cds.lib', and the 'Read Physical Data' checkbox is checked. The 'OK' button is highlighted in red.

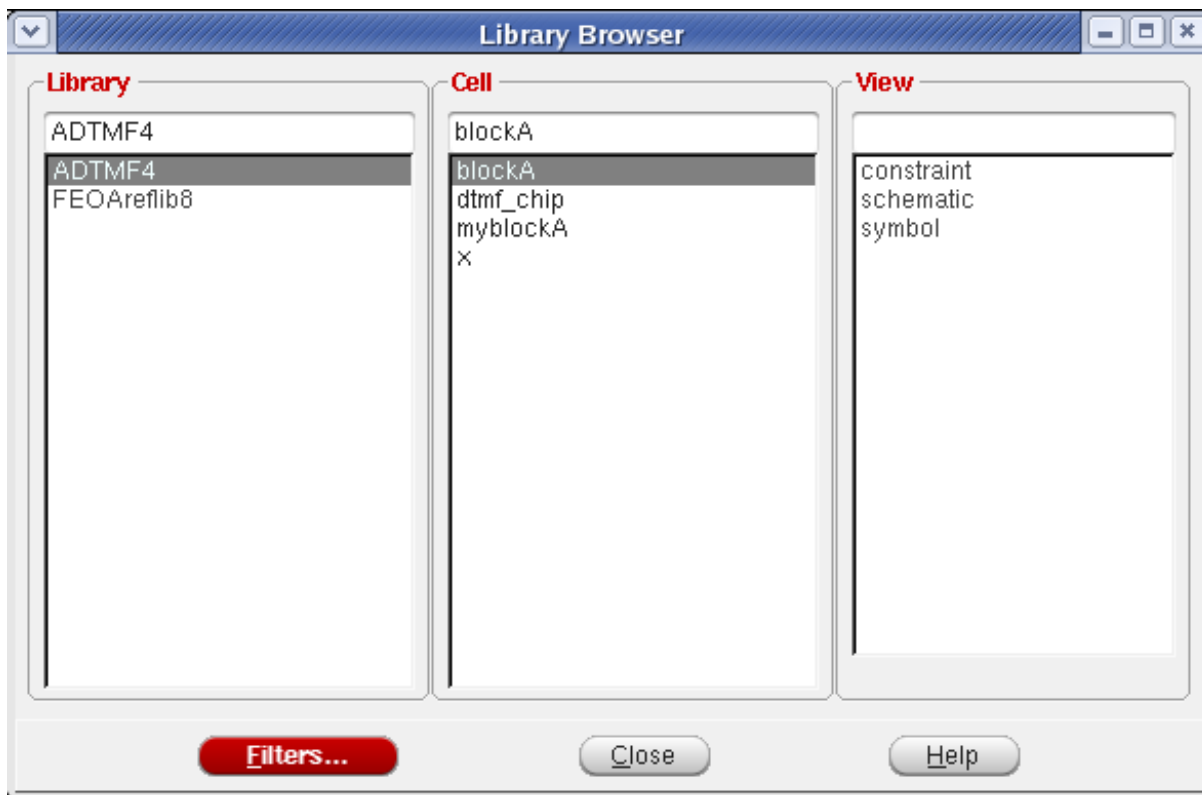
Fields and Options

<i>Library</i>	Specifies the location of the OpenAccess database to be restored. It contains the cell and the view to be restored. The OpenAccess library name should be specified in the <code>cds.lib</code> file. The <i>Setup</i> tab contains the <i>Verilog</i> and <i>OA</i> radio buttons that changes the form from the current LEF based style (default) to the OpenAccess-based style.
<i>Cell</i>	Specifies the design cell from the library that is to be restored.
<i>View</i>	Specifies the view name of the cell that is to be restored. <i>Default:</i> The layout view.
<i>Library File Path</i>	Specifies the full path to the library definitions file (<code>cds.lib</code>).
<i>Read Physical Data</i>	Reads physical data.
<i>Browser button</i>	Opens the Library Browser form, which enables you to browse available libraries, cells, views.

Library Browser

Use Library Browser to browse available library, cell, views, when restoring an OpenAccess design.

→ Choose *File - Read Design*, and click the *Browser* button.



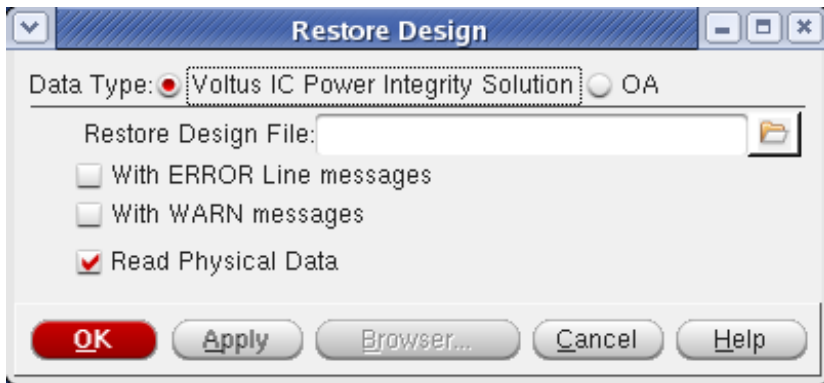
Fields and Options

<i>Library</i>	Lists the available libraries.
<i>Cell</i>	Lists the available cells for the selected library.
<i>View</i>	Lists the available views for the selected cell.

Read Design - Voltus IC Power Integrity Solution

Use the Restore Design form to load saved data from a previous design session.

→ Choose *File - Read Design* and click the *Voltus IC Power Integrity Solution* option button.



Fields and Options

<i>Restore Design File</i>	Specifies the name of the design file to load.
<i>With ERROR Line messages</i>	
	Reads the file, including error messages.
<i>With WARN messages</i>	
	Reads the file, including warnings.
<i>Read Physical Data</i>	Reads physical data.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `read_design`

Save Design

The Save Design form enables you to save design files from the Voltus session and to save all design information using the OpenAccess database format.

→ *Choose File - Save Design.*

The Save Design form contains the following two pages:

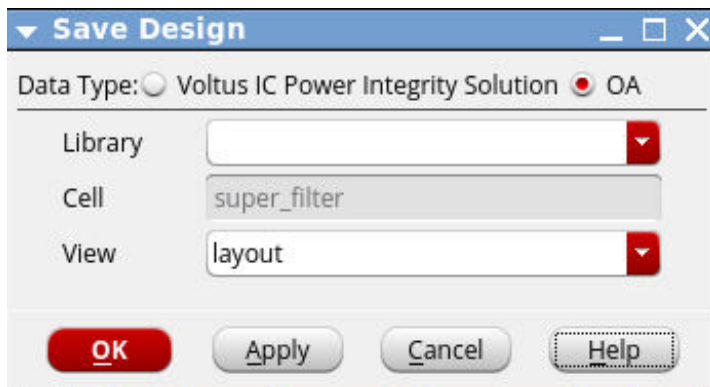
- [Save Design - OA](#)

- [Save Design - Voltus](#)

Save Design - OA

Use the OA page in the Save Design form to save all information from the current design session, using the OA database format.

→ Choose *File - Save Design*. The *OA* page opens by default.



The screenshot shows the 'Save Design' dialog box with the following fields and options:

- Data Type:** Two radio buttons are present. The first is labeled 'Voltus IC Power Integrity Solution' and is unselected. The second is labeled 'OA' and is selected.
- Library:** A text input field that is currently empty, with a red dropdown arrow on the right.
- Cell:** A text input field containing the text 'super_filter'.
- View:** A text input field containing the text 'layout', with a red dropdown arrow on the right.
- Buttons:** At the bottom, there are four buttons: 'OK' (red), 'Apply' (gray), 'Cancel' (gray), and 'Help' (dashed gray).

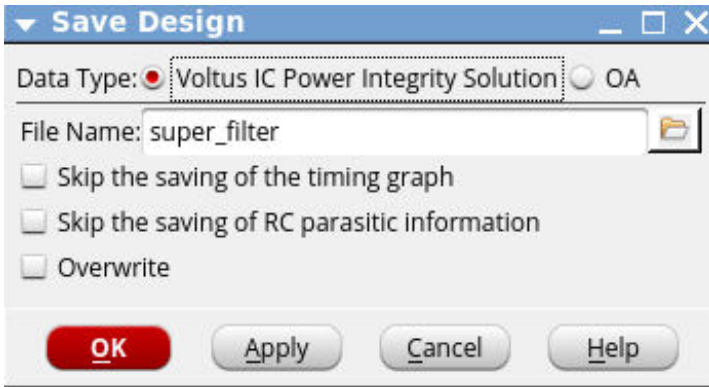
Fields and Options

<i>Library</i>	Specifies the target OpenAccess library name.
<i>Cell</i>	Specifies the top cell that is to be saved. Note: This field is read-only.
<i>View</i>	Specifies the view to save.

Save Design - Voltus

Use the Voltus IC Power Integrity Solution page of the Save Design form to save design files in a user-specified directory at anytime in a design session. Only files that exist during the save are copied to the directory.

→ Choose *File - Save Design* and click the *Voltus IC Power Integrity Solution* option button.



Fields and Options

<i>File Name</i>	Enter the name of the file to which you want to save the design. Click the button next to the field to choose the directory where you can save your file. You can select one of more of the sub-options to skip saving of timing graph, RC parasitic information, or to overwrite the previous file.
------------------	--

Related Text Commands

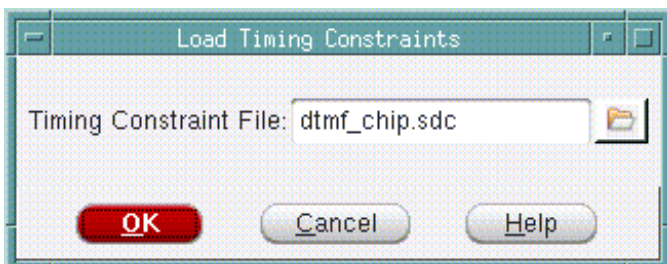
The following text command provides equivalent or additional functionality:

- `save_design`

Read SDC

Use the Load Timing Constraints form to read an SDC file.

→ Choose *File - Read SDC*.



Fields and Options

<i>Timing Constraint File</i>	Specifies the name of the timing constraints (SDC) file.
-------------------------------	--

Related Text Commands

The following text command provides equivalent or additional functionality:

- [read_sdc](#)

For more information, see "Setup Commands" in the *Voltus Text Command Reference*.

Read SPEF

Use the Load SPEF form to read a SPEF file.

→ Choose *File - Read SPEF*.



Fields and Options

<i>File name</i>	Specify the name of the SPEF file you want to load.
<i>RC Corner</i>	Specify the name of the corresponding RC Corner.

Related Text Commands

The following text command provides equivalent or additional functionality:

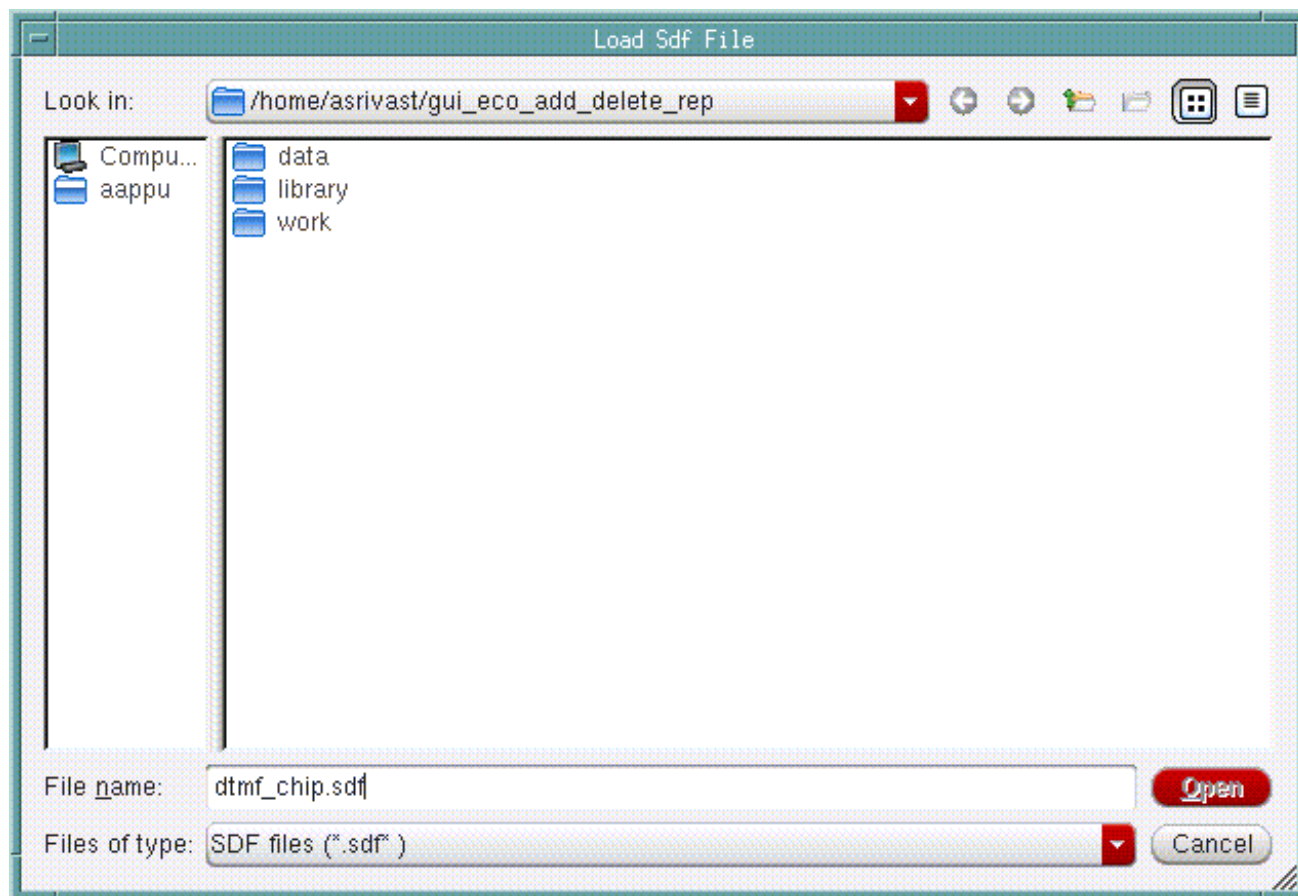
- [read_spef](#)

For more information, see "Setup Commands" in the *Voltus Text Command Reference*.

Read SDF

Use the Load Sdf File form to read a SDF file.

→ Choose *File - Read SDF*.



Fields and Options

<i>Directory</i>	Displays the current directory.
<i>File name</i>	Specifies the name of the SDF file.
<i>Files of type</i>	Specifies the type of files that are displayed. <i>Default:</i> SDF files

Read IR Drop

Use the Read IR Drop Files form to read an IR drop file.
→ Choose *File - Read IR Drop*.

Fields and Options

<i>Operating Condition</i>	Specify either Min or Max operating condition.
<i>Derating</i>	Applies the effects of IR-drop to the early path (data hold/clock setup) or late path (data setup/clock hold) in your design. Use the <i>Early</i> or <i>Late</i> parameter in conjunction with the <i>Min</i> or <i>Max</i> condition to account for the effects of early-max, early-min, late-max, or late-min IR-drop analysis.
<i>Reset</i>	Resets all IR-drop files, so a new one can be read.

Related Text Commands

The following text command provides equivalent or additional functionality:

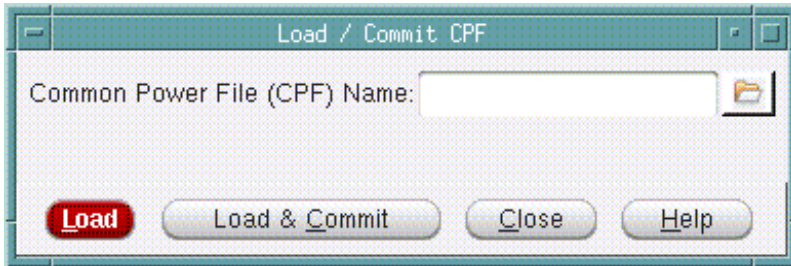
- `read_instance_voltage`

For more information, see "Setup Commands" in the *Voltus Text Command Reference*.

Load/Commit CPF

Use the Load/Commit CPF form to load a Common Power Format (CPF) form into the design, or commit a CPF file that you have already loaded.

→ To open the Load/Commit CPF form, choose *File - Load/Commit CPF*.



Fields and Options

<i>Commit</i>	Executes the commands provided in the loaded CPF file, such as <code>create_power_domain</code>, <code>create_power_ground_connection</code>, and <code>create_*_logic</code>. As a result, running this command does the following: • Creates power domains • Creates logical power/ground net connections • Specifies timing libraries for power domains
<i>Common Power File (CPF) Name</i>	
	Specifies the name of the CPF file to load or commit.
<i>Load</i>	Reads the CPF file and performs lint, parsing, and semantics checking. The <code>read_power_domain</code> command does not execute any of the commands within the CPF file. If the power/ground connections are incomplete, then the command automatically generates a power net specification template in CPF. You can then examine the template and update it to fully specify the power/ground connections.
<i>Load & Commit</i>	Reads the CPF file, as in <i>Load</i>, and runs the commands provided in the CPF file. Among many operations, this option does the following: • Creates power domains • Creates logical power/ground net connections • Specifies timing libraries for power domains

Related Text Commands

The following text command provides equivalent or additional functionality:

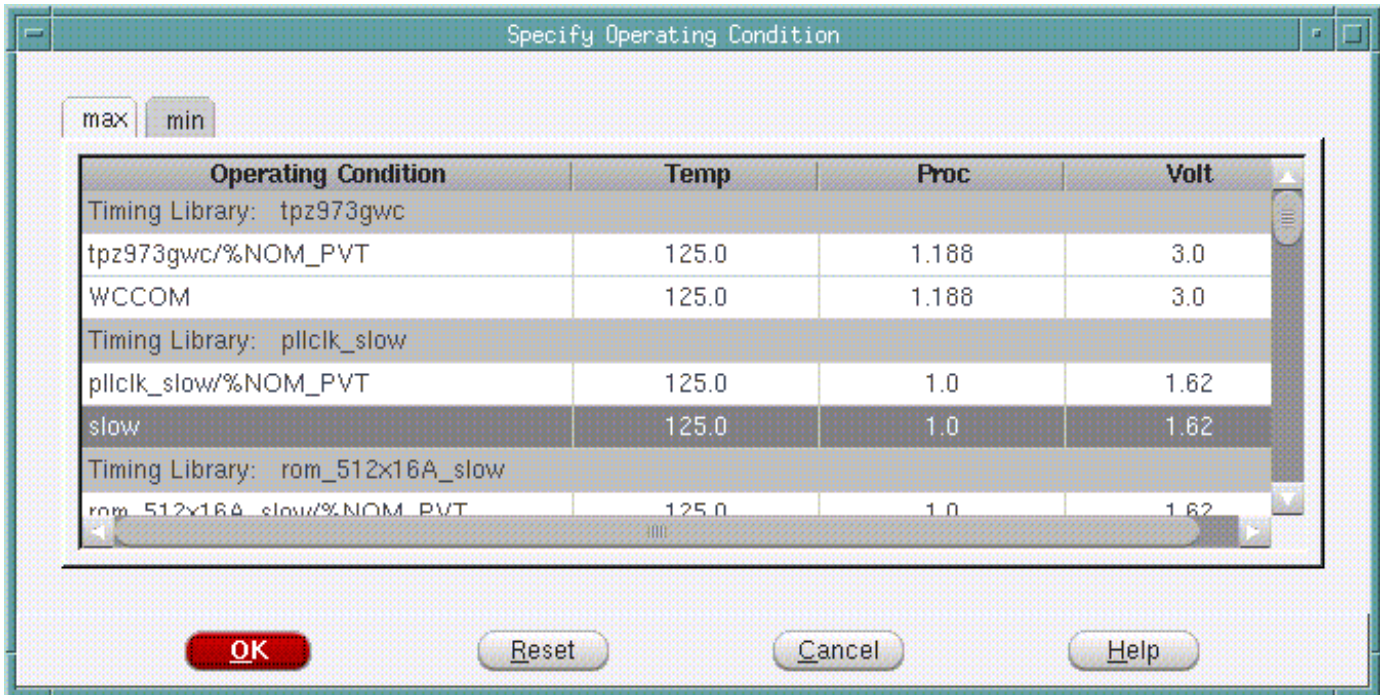
- [read power domain](#)

Specify Operating Condition/PVT

Use the Specify Operating Condition form to select the operating temperature, process, or voltage

conditions for the design. The operating conditions are contained in the timing library, which are normally read in during design import.

→ Choose *File - Specify Operating Conditions / PVT* and click on either the Min or Max tab to select the operating condition you want to specify.



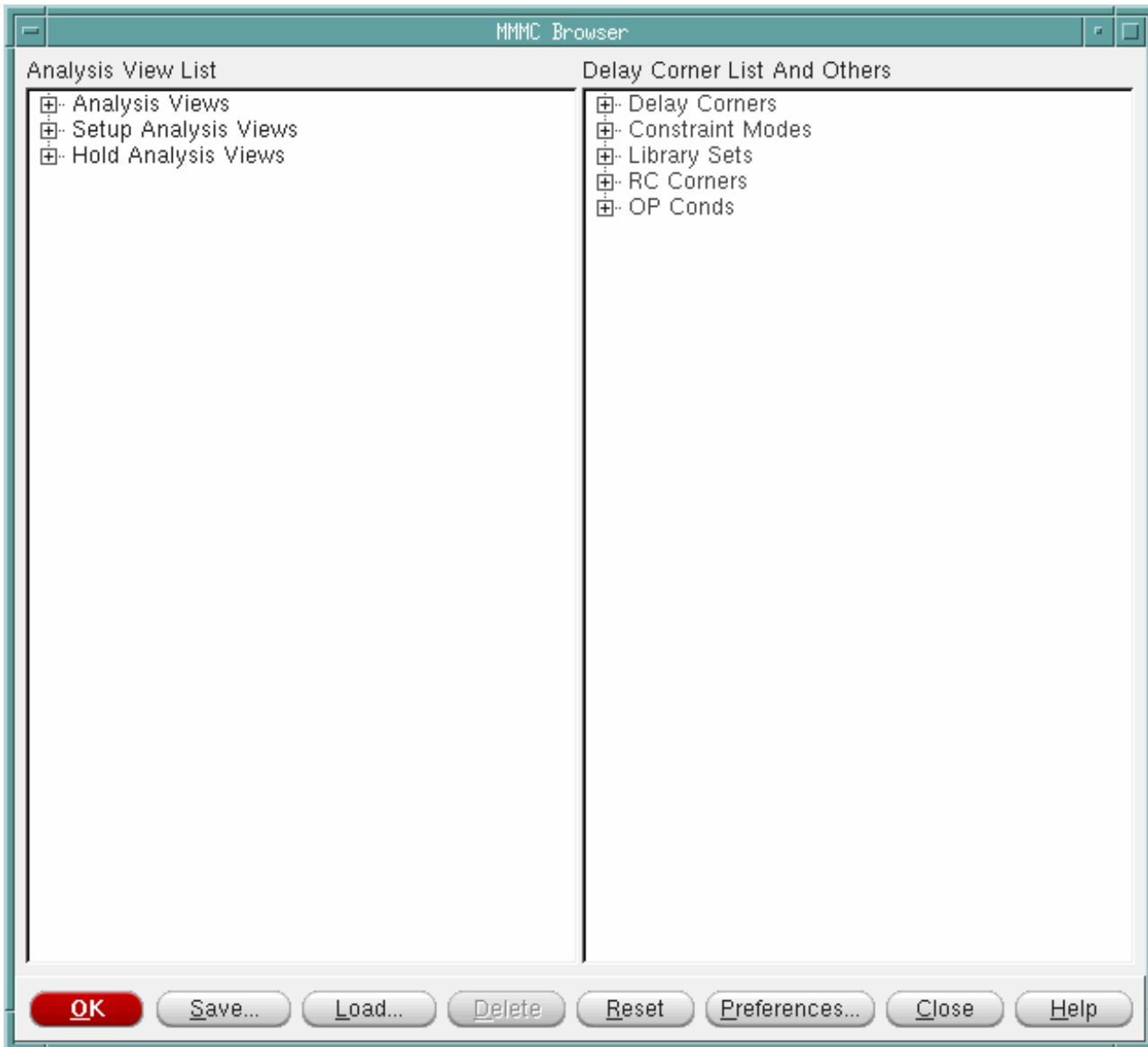
Fields and Options

<i>Operating Condition</i>	Displays the name of the operating condition. By default, the operating condition selected under the <i>Max</i> tab is used for setup analysis and the operating condition selected under the <i>Min</i> tab is used for hold analysis.
<i>Temp</i>	Displays either the minimum or maximum temperature for the operating condition.
<i>Proc</i>	Displays either the minimum or maximum process parameter for the operating condition.
<i>Volt</i>	Displays either the minimum or maximum voltage for the operating condition.

Configure MMMC

Use the MMMC Browser to view the configurations necessary for multi-mode multi-corner analysis. Each top-level set of information is called an analysis view, and is composed of a delay calculation corner and a constraint mode. The active analysis views defined in the software represent the different design variations that will be analyzed.

→ Choose *File - Configure MMMC*.



For more information on configuring the setup for multi-mode multi-corner analysis, see *Performing Multi-Mode Multi-Corner Timing Analysis*, in the *Tempus Timing Signoff Solution User Guide*.

<i>Name</i>	Specifies the name for the RC corner object being created. You can choose any name for an RC corner object, but it is recommended that you use a name that can be easily recognized as referring to a specific RC corner (for example, rc-min, rc-max, c-min, c-max, and so on).
<i>Temperature</i>	Specifies the Operating temperature, in units of Celsius, to be used to derate extracted resistance values. Note: The software does not require that the temperature specified by the PVT operating condition at the delay calculation corner level be the same as the RC nominal or user-specified temperature value. If you do not specify a temperature, the extracted resistance will not be derated.
<i>QRC Technology File</i>	Specifies the name of the QRC Technology file used by TQRC, IQRC, or sign-off QRC extraction. For most technologies, you can get the QRC Technology file from your foundry. For technologies that are not available through the technology vendors, you use the same ASCII-format interconnect technology (ICT) input file that is used to generate the captable. Techgen is the utility used to create the QRC Technology file. For more information on running TechGen, see <i>Quantus QRC Documentation</i>.

<i>Name</i>	Displays the name of the constraint mode object being edited.
<i>SDC Constraint Files</i>	Displays the SDC constraint files that are associated with the constraint mode object. Use the <i>Add</i> button to add SDC files to the list. Use the <i>Delete</i> button to delete SDC files from the list.
<i>ILM Constraint Files</i>	Displays the SDC constraint files for the ILM flow that are associated with the constraint mode object. Use the <i>Add</i> button to add SDC files to the list. Use the <i>Delete</i> button to delete SDC files from the list.

<i>Name</i>	Displays the name for the delay calculation corner object being edited.
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<i>Type</i>	Specifies the type of analysis for which you are configuring the multi-mode multi-corner environment. When running Single or Best-Case Worst-Case (BcWc) timing analysis, a delay calculation corner generally contains one library set and one RC corner object. When running on-chip-variation timing analysis, a delay calculation corner can contain early and late library sets and RC corners. Note: If you use a delay calculation corner containing early and late attributes when the software is in BC-WC mode, the software uses the early information for analysis of hold views, and the late information for the analysis of setup views.
<i>RC Corner</i>	Specifies an RC corner object to associate with this delay calculation corner object.
<i>Library Set</i>	Specifies a library set to associate with this delay calculation corner object.
<i>OpCond Lib</i>	Specifies the internal library name for the library in which the operating condition is defined. This is <i>not</i> the library file name.
<i>OpCond</i>	Specifies an operating condition to use for setup and hold analysis.
<i>IrDrop File</i>	Specifies the IR drop files to apply to both early and late delay calculation for this delay corner object.
<i>Early RC Corner</i>	Note: The 8.1 version of the software does not support the use of early and late RC corners; therefore, this field is disabled. Specifies an RC corner object to associate with this delay calculation corner object for calculating early arrival times at a single delay calculation corner.
<i>Early Library Set</i>	Specifies a library set to associate with this delay calculation corner object for calculating early arrival times at a single delay calculation corner.
<i>Early OpCond Lib</i>	Specifies the internal library name for the library in which the early operating condition is defined. This is <i>not</i> the library file name.
<i>Early OpCond</i>	Specifies an operating condition to use for calculating early arrival times at a single delay calculation corner.

<i>Early IrDrop File</i>	Specifies the IR drop files to apply when calculating early arrival times at a single delay corner.
<i>Late RC Corner</i>	Note: The 8.1 version of the software does not support the use of early and late RC corners; therefore, this field is disabled. Specifies an RC corner object to associate with this delay calculation corner object for calculating late arrival times at a single delay corner.
<i>Late Library Set</i>	Specifies a library set to associate with this delay calculation corner object for calculating late arrival times at a single delay calculation corner.
<i>Late OpCond Lib</i>	Specifies the internal library name for the library in which the late operating condition is defined. This is <i>not</i> the library file name.
<i>Late OpCond</i>	Specifies an operating condition to use for calculating late arrival times at a single delay calculation corner.
<i>Late IrDrop File</i>	Specifies the IR drop files to apply when calculating late arrival times at a single delay corner.
<i>Power Domain List</i>	Displays a list of delay calculation corner's power domain definitions. To add a power domain definition to the delay calculation corner, click the <i>Add</i> button to open the Add Power Domain form. To delete a power domain definition from the delay calculation corner, choose the power domain in the list and click the <i>Delete</i> button.
<i>Name</i>	Displays the name of the power domain being edited.
<i>Type</i>	Specifies the type of analysis for which you are configuring the multi-mode multi-corner environment. When running Single or Best-Case Worst-Case (BcWc) timing analysis, a power domain generally contains one library set and operating condition. When running on-chip-variation timing analysis, a power domain can contain early and late library sets and operating conditions.

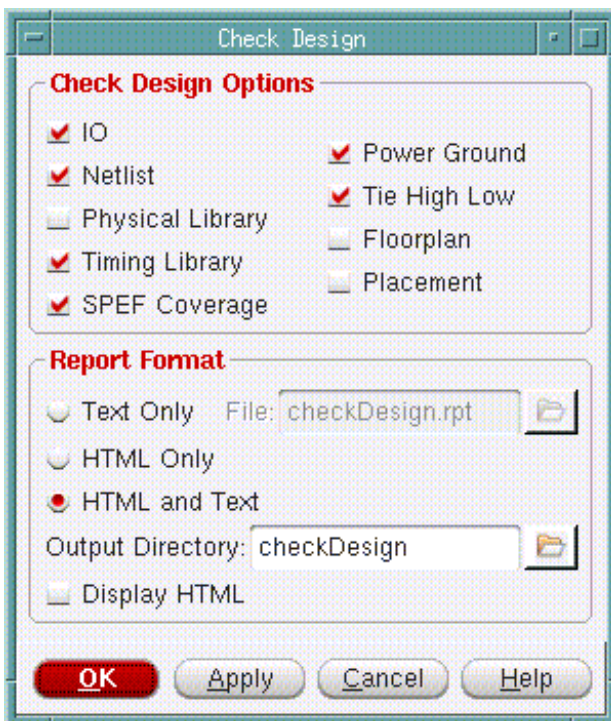
<i>Delay Corner Name</i>	Displays the delay calculation corner to which the power domain definition belongs.
<i>Library Set</i>	Specifies a library set to associate with the power domain. All of the cells and macros in the power domain get their timing and power information from these libraries.
<i>OpCond Lib</i>	Specifies the internal library name for the library in which the operating condition is defined. This is <i>not</i> the library file name.
<i>OpCond</i>	Specifies an operating condition to use for the power domain.
<i>IrDrop File</i>	Specifies the IR drop files to apply to both early and late delay calculation for this power domain object.
<i>Early Library Set</i>	Specifies a library set to associate with the power domain for calculating early arrival times at a single delay corner.
<i>Early OpCond Lib</i>	Specifies the internal library name for the library in which the early operating condition is defined. This is <i>not</i> the library file name.
<i>Early OpCond</i>	Specifies an operating condition to use for calculating early arrival times at a single delay corner.
<i>Early IrDrop File</i>	Specifies the IR drop files to apply when calculating early arrival times at a single power domain.
<i>Late Library Set</i>	Specifies a library set to associate with this delay corner object for calculating late arrival times at a single delay corner.
<i>Late OpCond Lib</i>	Specifies the internal library name for the library in which the late operating condition is defined. This is <i>not</i> the library file name.
<i>Late OpCond</i>	Specifies an operating condition to use for calculating late arrival times at a single delay corner.
<i>Late IrDrop File</i>	Specifies the IR drop files to apply when calculating late arrival times at a single power domain.

<i>Name</i>	Displays the name of the analysis view being edited.
<i>Constraint Mode</i>	Specifies the name of the constraint mode to associate with this analysis view.
<i>Delay Corner</i>	Specifies the name of the delay corner to associate with this analysis view.

Check Design

Use the Check Design form to check for missing or inconsistent library and design data at any stage of the design.

→ Choose *File - Check - Check Design*.



Fields and Options

<i>I/O</i>	Checks I/O pads, generates warning messages, and reports the following information: • Floating pins that belongs to I/O pads (warning) • I/O pins connected to non-I/O instances (directly connected to core cells) • Floating I/O pins
<i>Netlist</i>	Checks the netlist, generates error or warning messages, and reports the following information: • Output pins tied to power/ground nets (for example, <code>BUFX1 U1 (.A(net1), .Y1(1'b0))</code>) • Input pins floating (warning) • Multiple driver nets (warning)
<i>Physical Library</i>	Checks the physical libraries and reports whether all cells have LEF views. Issues error messages for the following conditions: • Cells not defined in LEF • Cells with missing power ground pins • Cells with missing dimensions • Pins with missing direction • Cells and pins with missing geometry • Cell dimensions are not an integer multiple of the core site dimensions • Cells and power ground pins with missing geometry
<i>Timing Library</i>	Checks whether the cells used in the design have been defined in the timing library. This option does not check for the presence of timing arcs. All physical cells are excluded from this check.
<i>SPEF Coverage</i>	Reports the following SPEF information: • SPEF coverage • Number of nets in the design that are not in SPEF • List of nets

<i>Power Ground</i>	Checks power and ground connections, generates warning messages, and reports the following information: • Power terminals connected to ground net • Ground terminals connected to power net • Floating power and ground terminals • Power and ground terminals connected to non-power and ground nets.
<i>Tie High Low</i>	Reports unconnected tie-high or tie-low terminals, and generates a warning message.
<i>Floorplan</i>	Checks the floorplan, generates error or warning messages, and reports the following information: • Off-grid horizontal and vertical tracks • Instances not snapped to row site • Unplaced I/O pins • Off Grid Power/Ground Pre-routes • Instances not on the manufacturing grid • Preroutes not on the manufacturing grid (error) • Regular preroutes not on tracks
<i>Placement</i>	Checks the floorplan, generates error or warning messages, and reports the following information: • Off-grid horizontal and vertical tracks • Instances not snapped to row site • Unplaced I/O pins • Off Grid Power/Ground Pre-routes • Instances not on the manufacturing grid • Preroutes not on the manufacturing grid (error) • Regular preroutes not on tracks

<i>Text Only</i>	Generates only a text version of the report. <i>Default:</i> Generates both an HTML and a text version of the report.
<i>File</i>	Writes the report information to the specified file. <i>Default:</i> <code>checkDesign.rpt</code>
<i>HTML Only</i>	Generates only an HTML version of the report. <i>Default:</i> Generates both an HTML and a text version of the report.
<i>HTML and Text</i>	Generates both an HTML version and a text version of the report.
<i>Output Directory</i>	Saves both the HTML version and the text version of the report to the specified directory. If you select the <i>HTML Only</i> option, the software saves the HTML version of the report to the specified directory. <i>Default:</i> <code>checkDesign</code>
<i>Display HTML</i>	Displays the HTML version of the report in the Report menu.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `check_design`

For more information, see "Setup Commands" in the *Voltus Text Command Reference*.

Report Timing Library

Use the Report Timing Library form to output the cell information to a report file after importing the design and loading the timing library. The report file is generated in ASCII format.

→ Choose *File - Check - Timing Library*



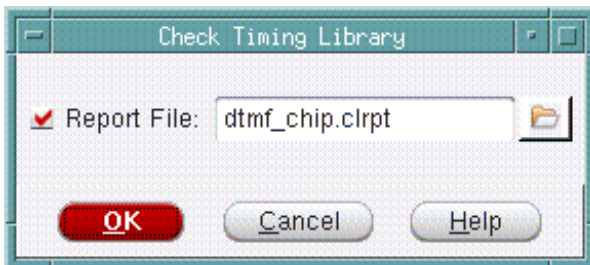
Fields and Options

<i>All Cell(s)</i>	Reports timing information for all cells. This is the default option.
<i>Cell Selection</i>	Reports timing information on the cells specified in the text box.
<i>Timing Library Report</i>	Specify a filename or click the file folder icon to choose a file. If you do not select this option, the report is displayed on the ETS console. The default filename is <i>TopCellName.talib</i> .

Check Timing Library

Use the Check Timing Library form to check the contents of the timing library and reports any inconsistencies in a report file. Use this form after importing the design and timing library.

→ Choose *File - Check - Library*



Fields and Options

<i>Report File</i>	Specifies the name of the report file. The default filename is <i>TopCellName.clrpt</i> .
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Related Text Commands

The following text command provides equivalent or additional functionality:

- `check_library`

For more information, see "Setup Commands" in the *Tempus Timing Signoff Solution Text Command Reference*.

Summary

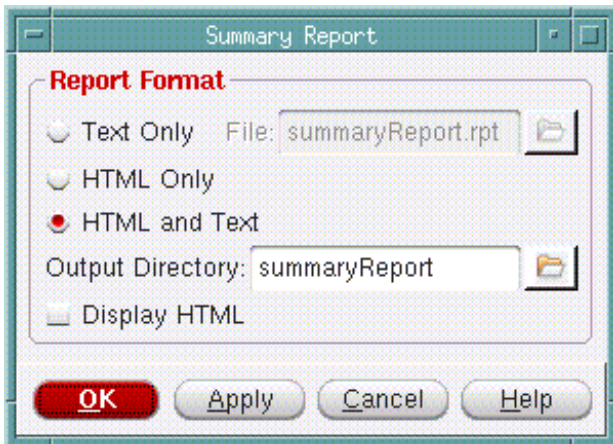
Use the Summary Report form to create a report that contains statistics for the entire design, or a selected object in the design.

A report on the entire design includes statistics for the following categories:

- General Design Information
- General Library Information
- Netlist Information
- Timing Information

The information contained in a report on a selected object varies depending on the object you select. You can select a module, an instance, or a net.

1. Select an object in the design display area. (If you want to display information about the entire design, skip Step 1.)
2. Choose *File - Report - Summary*.



Fields and Options

<i>Text Only</i>	Generates a text version of the summary report. <i>Default:</i> Off	
	<i>File</i>	Writes the report information to the specified file. <i>Default:</i> summaryReport.rpt
<i>HTML Only</i>	Generates an HTML version of the summary report. Some statistics in the sections include links to more detailed HTML reports. For example, if you click on the <i># Layers</i> link under <i>General Library Information</i> section of the HTML report, the software displays a detailed HTML report on the layers in the design. <i>Default:</i> Off	
<i>HTML and Text</i>	Generates both a text version and an HTML version of the summary report. <i>Default:</i> On	
<i>Output Directory</i>	Saves the HTML version of the report to the specified directory. If you select the <i>HTML and Text</i> option, the software saves both the HTML and text versions of the report to the specified directory. <i>Default:</i> summaryReport	
<i>Display HTML</i>	Opens a browser and displays the HTML version of the report. <i>Default:</i> Off	

Netlist Statistics

Use the *Netlist Statistics* menu command to create a list of design statistics, such as number of cells, nets, pins, and instances. A summary of the netlist is printed to the console and a log file, and saved in the run directory.

→ To print a list of design statistics, choose *File - Report - Netlist Statistics*.

Hide GUI

Use *Hide GUI* to hide the GUI. Type the `start_gui` command from command line to re-display the GUI.

Exit

To exit Voltus IC Power Integrity Solution.

Related Text Commands

See the `exit` command in the *Voltus Text Command Reference*.

View Menu

- [Zoom Menu](#)
- [Pan Menu](#)
- [Fit Menu](#)
- [Redraw](#)
- [Set Preferences](#)
- [Go To](#)
- [Find/Select Object](#)
- [Deselect All](#)
- [Highlight Selected](#)
- [Clear Highlight](#)
- [Edit Highlight Color](#)
- [Dim Background](#)

Zoom Menu

→ Choose *View - Zoom*.

The Zoom menu comprises the following items:

In

Use the Zoom-In menu command to zoom into a portion of the design and view it in greater detail. The equivalent bindkey is Z.

You can also use the scroll wheel of your mouse to zoom into a portion of the design. Each forward scroll of the scroll wheel zooms in one level.

Out

Use the Zoom-Out menu command to display a larger area of the design in less detail. The

equivalent bindkey is SHIFT+Z.

You can also use the scroll wheel of your mouse to zoom out of a detailed view of the design. Each backward scroll of the scroll wheel zooms in one level.

Selected

Select a design object and then use the Zoom-Selected menu command to zoom into the area that contains the selected object.

Previous

Use the Zoom-Previous menu command to toggle the display between the previous location/zoom level and the current location/zoom level. The equivalent bindkey is W.

Pan Menu

→ [Choose View - Pan.](#)

The Pan menu comprises the following items:

Up

Pans up the viewable window in pages. The size of each page is equal to the size of the current viewable window. The equivalent bindkey is the Up arrow key.

You can also use the scroll wheel of your mouse to pan up the viewable window. Press Shift and move the scroll wheel forward to pan upwards.

Down

Pans down the viewable window in pages. The size of each page is equal to the size of the current viewable window. The equivalent bindkey is the Down arrow key.

You can also use the scroll wheel of your mouse to pan down the viewable window. Press Shift and move the scroll wheel backward to pan downwards.

Left

Pans the viewable window to the left in pages. The size of each page is equal to the size of the current viewable window. The equivalent bindkey is the Left arrow key.

You can also use the scroll wheel of your mouse to pan the viewable window to the left. Press Ctrl and move the scroll wheel forward to pan to the left.

Right

Pans the viewable window to the right in pages. The size of each page is equal to the size of the current viewable window. The equivalent bindkey is the Right arrow key.

You can also use the scroll wheel of your mouse to pan the viewable window to the right. Press Ctrl and move the scroll wheel backward to pan to the right.

Fit Menu

Use the Fit menu command to fit the entire design within the design display area. The equivalent bindkey is F.

→ *Choose View - Fit.*

Redraw

Use the Zoom-Redraw menu command to refresh the display in the viewable window. The equivalent keyboard shortcut is Ctrl+R.

→ *Choose View - Redraw.*

Set Preferences

Use the Set Preference command in the View menu to access the Preferences form. This form allows you to set the design settings for the Voltus session.

→ *Choose View - Set Preferences.*

You can save custom preference settings to a file, load a previously saved file, or load default settings. The default settings are saved in the ssv.pref.tcl file; custom settings can be saved with a different filename.

Preferences - Design

Use the Design page of the Preferences form to set hierarchy levels in the netlist and DEF files, specify bus delimiters for PDEF files, and denote output of command logs. The default settings are saved in the ssv.pref.tcl file; custom settings can be saved with a different filename.

→ Choose *View - Set Preferences - Design* tab.

Fields and Options

Design Name	Specifies the name of the design for which to set the preferences.
Design Hierarchical Character	
	Denotes the character being used as the hierarchy delimiter in a netlist. If the netlist uses another character, enter it in the text box and import the design. Default: /

DEF Hierarchical Character	Denotes the character being used as the hierarchy delimiter in a DEF file. If the DEF file uses another character, enter it in the text box and load the DEF file. When writing a DEF file, the software uses the same character. Default: /	
PDEF Bus Delimiter	Denotes the character being used as the hierarchy delimiter in a PDEF file. If the PDEF file uses another character, enter it in the text box and load the PDEF file. When writing a PDEF file, the software uses the same character. Default: []	
Write Conformal/LEC dofile When Design is Saved		
	When saving a netlist, this option saves a dofile (a set of commands contained in a specified file) for the Conformal® Equivalence Checking capabilities. Default: Off	
Command Log Mode	Specifies where the command log is written. Default: Command and Log	
	Command Only	Writes to the command file only.
	Command and Log	Writes to the command file and log files.
	Command, Log, and Screen	Writes to the command file, log files, and screen.
	Note: When writing to log file and screen, the commands are preceded with <CMD>.	
Log Type-In Command	Logs whatever is manually entered (typed), to the command file. In the case of source commands, only the source command is logged, and not the commands in the sourced file. Default: Off	
Invoke viewLog at Start Up		
	Displays log viewer at startup. Default: Off	
Enable Rectilinear Design		

Enables you to set a rectilinear object or cut a rectilinear area in a design with IO cells.

Preferences - Display

Use the *Display* page of the *Preferences* form to set viewing preferences for standard cells, wires, route congestion ranges, modules, and objects. The default settings are saved in the `ssv.pref.tcl` file; custom settings can be saved with a different file name.

→ Choose *View - Set Preference - Display* tab.

The screenshot shows the 'Preferences (on sjfsb436)' dialog box with the 'Display' tab selected. The dialog has three tabs: 'Design', 'Display', and 'Selection'. The 'Display' tab contains various checkboxes and input fields for configuring the display of design elements.

Option	Value
Show All Constraint	☐
Show Attributes Popup	☒
Show Rectilinear Pad Cell	☐
Auto-redraw after Layer Change	☒
Always Show Via Metal	☐
Display Instance Pattern	☐
Show Crosshair When Select	☒
Use Digit Fill Pattern	☒
Draw Detail When Move Via	☒
Instance Display Threshold	0.0
Apply Threshold in 2D	☐
Scale of Zoom In/Out	3.0
Number of Previous Zoom Saved	6
Wire Display Threshold	2
Percentage of Window to Pan	30
Wheel Zoom with Modifiers: Shift+Ctrl	☐
Display Partition Pin	☒
Display Unplaced Instance(s)	☐
Show Lef Layer Name	☒
Enlarge Logical Pin In Fit View	☒
Ignore Viewable Amoeba Ratio	☐
Draw Instance By Placement Status	☒
Full View Outside Die Area	☐
Use X Pattern For Cut	☒
Show Child Partition With Level	
Automatic Detail Display For Large Design	☐
Special Wire	☐
Regular Wire	☐
Factor	16
Size of Resistor Via	2

Buttons at the bottom: OK, Apply, Use Defaults, Save, Load, Cancel, Help.

Fields and Options

<i>Show All Constraint</i>	Displays all constraints, such as fences, guides, regions, and softguides, regardless of the current hierarchy. When this option is selected, you can see the child guides of each fence as you shape the fence. <i>Default: Off</i>
<i>Display Partition Pin</i>	Displays partition pins. <i>Default: On</i>
<i>Show Attributes Popup</i>	Specifies that the Context Pop-up Attribute Viewer, which is displayed when you place the mouse cursor over an object, should be displayed.
<i>Display Unplaced Instance(s)</i>	Displays unplaced cells in the Physical view. <i>Default: Off</i>
<i>Show Rectilinear Pad Cell</i>	Displays bounding box if the pad cell has rectilinear shape. <i>Default: Off</i>
<i>Show Lef Layer Name</i>	Specifies that the layer name as specified in the LEF file should be displayed in the color bar. <i>Default: On</i>
<i>Auto-redraw after Layer Change</i>	
	Enables or disables auto-redraw when layers are changed. <i>Default: On</i>
<i>Enlarge Logical Pin In Fit View</i>	Displays larger symbols for logical pins in Fit view. Select this option to check the pin distribution in a block design. <i>Default: On</i>

<i>Always Show Via Metal</i>	Keeps the additional metal of a via visible even if you turn off the via cut layer. With the <i>Always Show Via Metal</i> option selected: • If you turn off the metal layer and keep via on, only the cut via is visible in the main window. • If you turn off the via and keep metal layer on, the top/bottom layer of the via is visible. This option makes it easy for you to view the top/bottom layer of the via by turning off the via. If this option is selected, you can select the via metal and view its attributes in Attribute Editor. <i>Default: Off</i>
<i>Ignore Viewable Amoeba Ratio</i>	Displays all modules in the <i>Amoeba View</i> when selected, irrespective of their display size. By default, if the display size of a module is smaller than 1/3rd the area of the current module or 1/20th of the fplan box, it is not displayed in the <i>Amoeba View</i>. <i>Default: Off</i>
<i>Draw Instance Pattern</i>	Marks overlapping instances with X in the display area. This makes it easier to identify overlapping instances as compared to the default fill pattern. <i>Default: Off</i>
<i>Draw Instance by Placement Status</i>	Controls thickness of instance borders on the basis of their placement status. Select the new <i>Draw Instance by Placement Status</i> option if you want fixed instances to be marked with a thick border. <i>Default: On</i>
<i>Instance Display Threshold</i>	Sets the draw size of the standard cell instance in the physical view. The default value of 0.0 draws all standard cells. A value of 2.0 draws fewer standard cells, thereby yielding faster draw speed. If you change this value, you must redraw the view by choosing <i>View - Redraw</i> from the menu.
<i>Wire Display Threshold</i>	Sets the minimum number of pixels used to display a wire or via. This value works in conjunction with the <i>Apply Threshold in 2D</i> option. At the chip level, set 1 for the <i>Wire Display Threshold</i> and turn off <i>Apply Threshold in 2D</i> to display all of the wires/vias in the design. If the display speed is too slow, then change either or both variables appropriately.

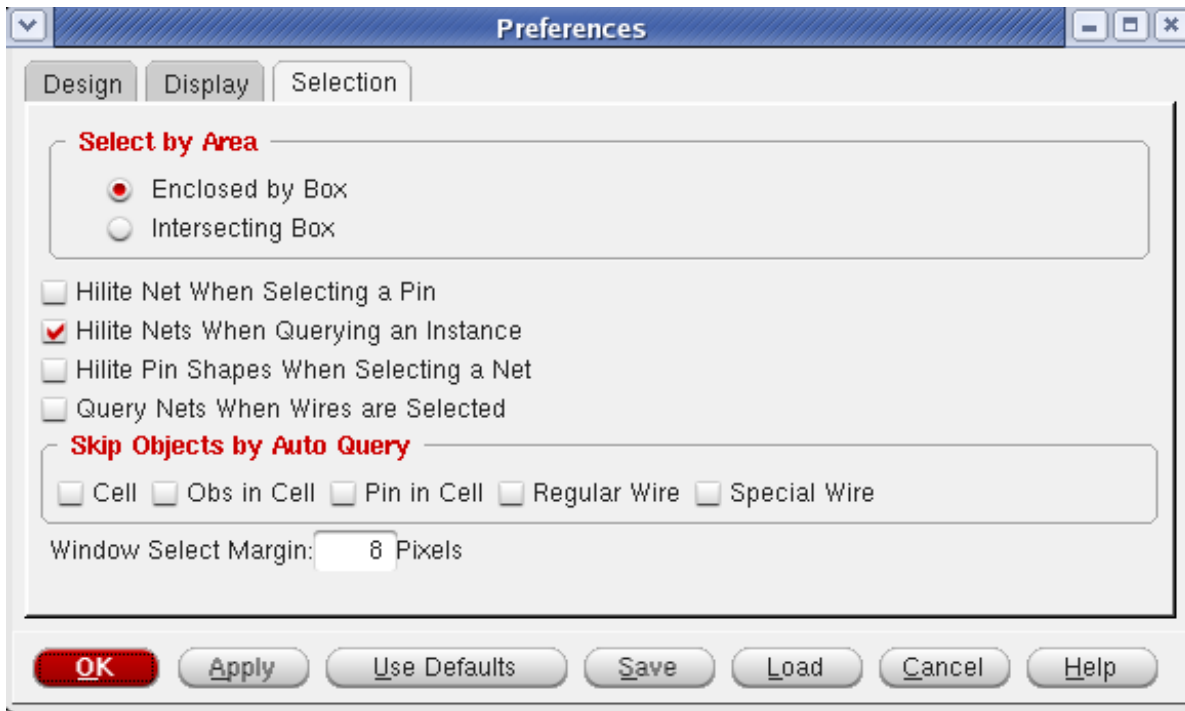
<i>Apply Threshold in 2D</i>	Checks both width and height of a wire or via when used with <i>Wire Display Threshold</i>. When off, the wire or via is displayed if either its width or height meets the threshold. When on, both width or height must meet the threshold for the wire or via to be displayed. <i>Default:</i> Off
<i>Percentage of Window to Pan</i>	Specifies the percentage of display area that will be panned when you pan the display using the right or the left arrow keys. <i>Default:</i> The default value is 95%.
<i>Scale of Zoom In/Out</i>	Specifies the scale for the zoom in and zoom out feature. You can use a decimal number for the scale.
Wheel Zoom with Modifiers: Shift+Ctrl	Enables use of scroll wheel to zoom in and out, when used along with the <code>Shift</code> and <code>Ctrl</code> keys.
<i>Number of Previous Zoom Saved</i>	Specifies the number of zoom views that are saved. You can then view the previous zoom views by clicking the Zoom Previous icon or the bindkey W. For example, if you set the number to 4, you can view four previous zoom views by clicking the Zoom Previous icon or the bindkey W. <i>Default:</i> 1
<i>Min. Floorplan Module Size</i>	Filters out modules from the display in the Floorplan view that have an area less than the specified value. For example, a value of 100 displays all the modules that contain 100 or more instances. Note: If modules or submodules contain at least one block, they are always displayed, regardless of the specified value. <i>Default:</i> 100
<i>Show Parent Module With Level</i>	Displays regions in hierarchy from the specified level. <i>Default:</i> 0

<i>Show Child Partition With Level</i>	Displays child partitions from the specified level. <i>Default: 0</i>	
<i>Automatic Detail Display for Large Design</i>	Controls the performance of redraw in large designs. If you select <i>Special Wire</i> and/or <i>Regular Wire</i> , the tool only draws these wires when you zoom in to the specified level (<i>Factor</i>). By default, the wires are drawn in full view (no zoom) itself.	
	<i>Special Wire</i>	Draws special wires only when you zoom in to the specified level. <i>Default: Off</i>
	<i>Regular Wire</i>	Draws regular wires only when you zoom in to the specified level. <i>Default: Off</i>
	<i>Factor</i>	Specifies the zoom level at which wires will be drawn in large designs. <i>Default: 16</i>
<i>Size of Resistor Via</i>	Controls the resistor via size displayed in GUI. The default resistor size is 1. This field is useful for debugging purpose when viewing and analyzing vias only.	

Preferences - Selection

Use the *Selection* page of the Preferences form to select and display objects and determine their behavior within a selected area. The default settings are saved in the `ssv.pref.tcl` file; custom settings can be saved with a different filename.

→ Choose *View - Set Preference - Selection* tab.



Fields and Options

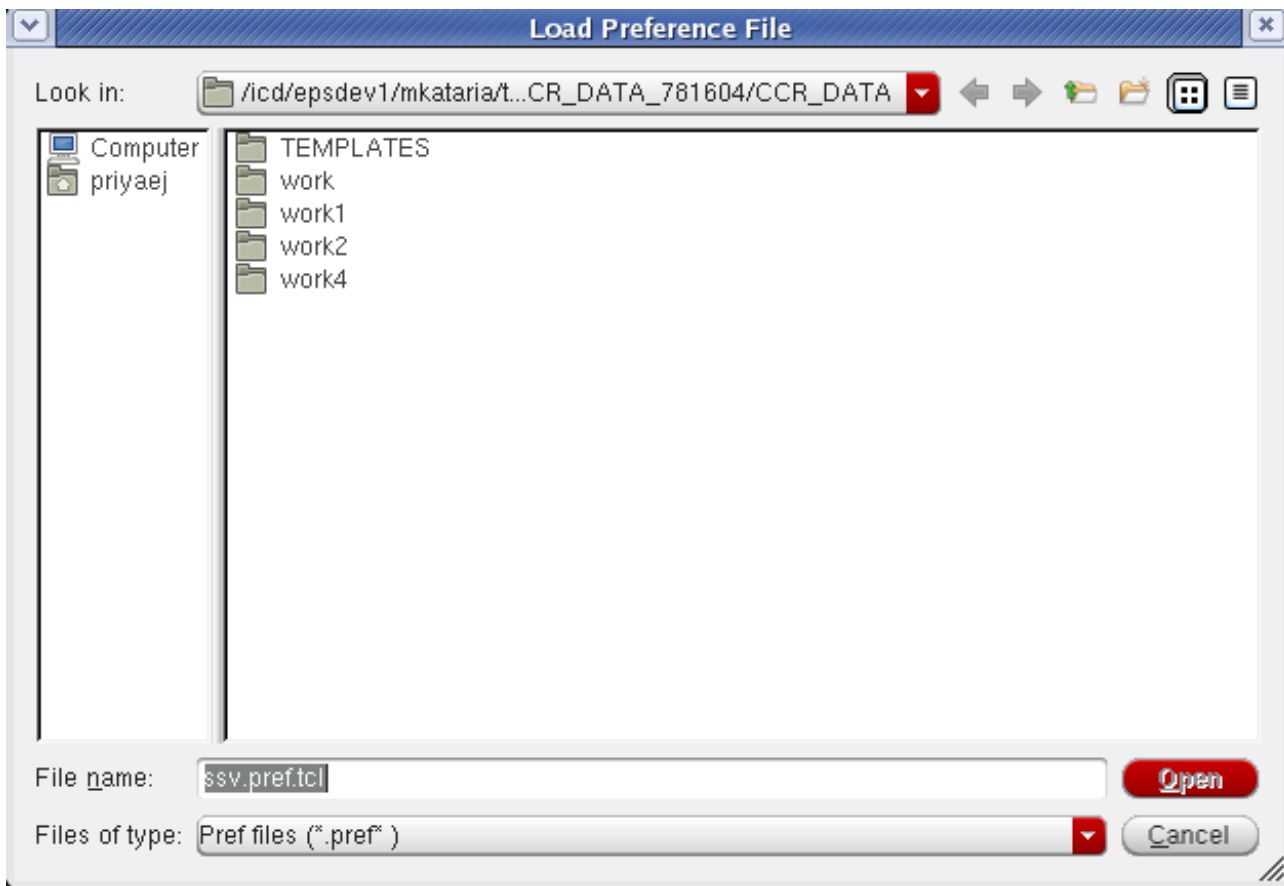
<i>Select by Area</i>		
	<i>Enclosed by Box</i>	Selects objects that are fully enclosed inside the specified area. Use the left mouse button to draw and define an area or bounding box. Any objects fully enclosed inside the drawing area are selected. <i>Default: On</i>
	<i>Intersecting Box</i>	Selects objects that are fully enclosed or intersected within the specified area. Use the left mouse button to draw and define an area or bounding box. Any objects enclosed inside or intersected with the area bounding box are selected. This mode should be used to select wire or row. <i>Default: Off</i>
<i>Hilite Net When Selecting a Pin</i>		
	Highlights a net through a pin selection. If this option is set when you select a pin, the corresponding net is highlighted. <i>Default: Off</i>	

<i>Hilite Nets When Querying an Instance</i>	
	Highlights nets that are connected to an instance when this instance is queried. For example, if you query a selected instance and this option is set, all nets that connect to this instance pin are also highlighted. <i>Default: On</i>
<i>Hilite Pin Shapes When Selecting a Net</i>	
	Highlights connected cell term shapes when you select a net. <i>Default: Off</i>
<i>Query Nets When Wires are Selected</i>	
	Queries all nets that connect to the selected wires. <i>Default: Off</i>
<i>Skip Objects by Auto Query</i>	
	Skips querying of specified objects. You can choose to skip Auto Query of: • <i>Cell</i> • <i>Obs in Cell</i> • <i>Pin in Cell</i> • <i>Regular Wire</i> • <i>Special Wire</i>
<i>Window Select Margin</i>	Specifies the number of pixels for the window margin. <i>Default: 8</i>

Load Preference File

Use the Load Preference File form to load a previously saved file. By default, settings are saved in the `ssv.pref.tcl` file; however, custom settings can be saved to a different file.

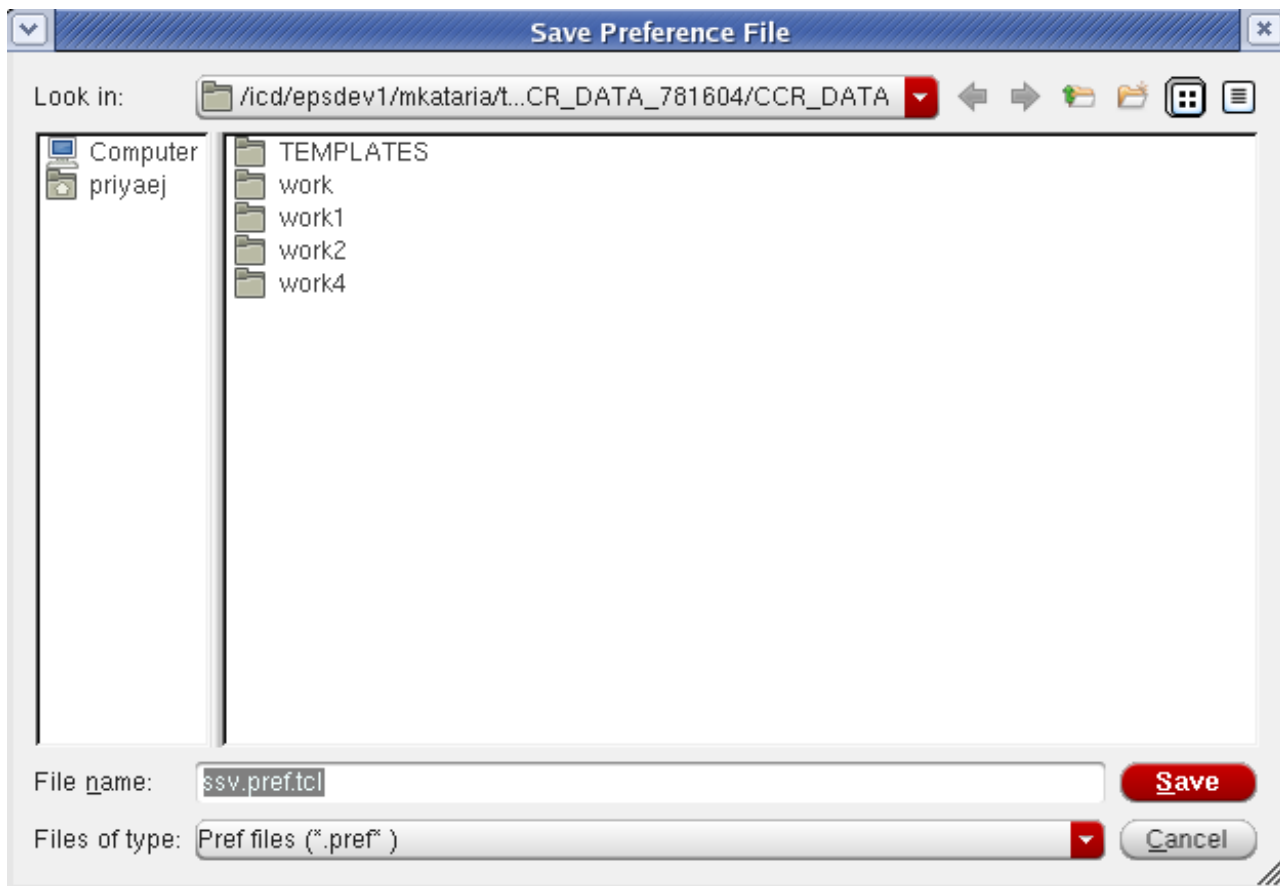
→ Choose *View - Set Preference* to open the Preferences form and click the *Load* button.



Save Preference

Use the Save Preference form to save custom preference design settings to a file. By default, settings are saved in the `ssv.pref.tcl` file; however, custom settings can be saved to a different file.

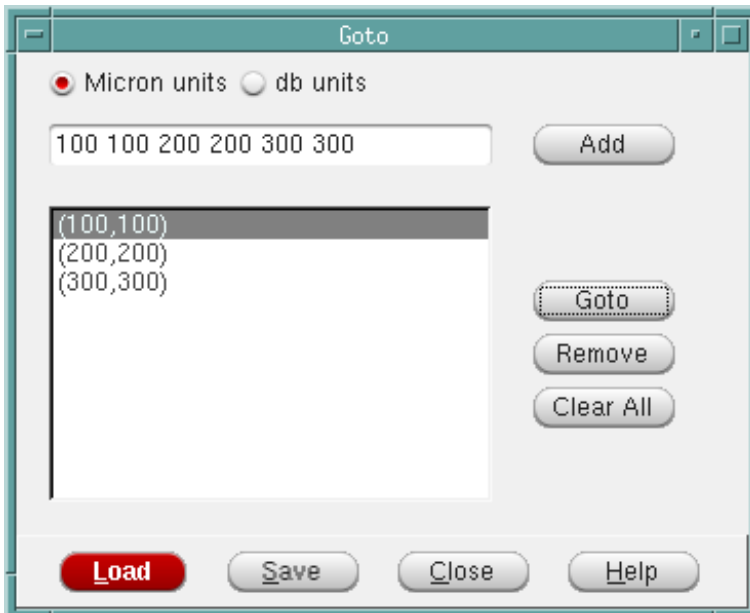
→ Choose *View - Set Preference* to open the Preferences form and click the *Save* button.



Go To

Use the Goto form to go to a specific point in the main window display. To open the Goto form:

→ Select *View - Go To*.



Fields and Options

<i>Micron units</i>	Specifies the coordinates in micron or db units.
<i>db units</i>	
<i>Add</i>	Adds the specified coordinates to the list box below.
<i>Goto</i>	Zooms in to the selected location in the main window display and places a marker on the location. Click this button after selecting a coordinate from the list.
<i>Remove</i>	Removes the selected coordinate from the list and the corresponding marker from the main window display.
<i>Clear All</i>	Clears the list and removes all <i>Goto</i> markers from the main window display.
<i>Load</i>	Loads the coordinates from a file.
<i>Save</i>	Saves the coordinates in the list to a file.
<i>Close</i>	Closes the Goto form.

Find/Select Object

Use the Find/Select Object form to find and select specific object types in the design display area.

Note: The Select operations are recorded in the command file and the log file.

→ Choose *View - Find/Select Object*.

Or

Press **Shift+F**.

Find/Select Object

Criteria

Object Type: **Instance** Mode: ☒ Single ☐ And List ☐ OR List ☐ Multiple

Property	Operator	Value
Name	Not Match	

Operation

☒ Find the object base on whole design ☐ Find the object base on selected objects

☒ Select Also

☐ Clear Result Window ☐ All ☐ Selected **Find**

Results

Object Type	Name
-------------	------

☐ Auto Zoom Last Found: 0

Save Result... **Close** **Help**

Fields and Options

<i>Object Type</i>	Specifies the type of object to find. Choose from: <i>Instance</i>, <i>Pin</i>, <i>Net</i>, <i>Module</i>, <i>Instance Group</i>, or <i>Bump</i>. The Pin object type pin can be used for I/O pins, partition pins, and instance pins. Names of instance pins and/or partition pins should be specified with the full path name. For example, pin A of instance DTMF_CHIP/alu should be specified as: DTMF_CHIP/alu/A You can use the <i>Net</i> option to find/display bus names.
<i>Mode</i>	Specifies the type of find/select operation. Select one of the following: • <i>Single</i>: Performs a simple search for a Property/Value combination. • <i>And List</i>: Performs a find/select operation that matches an AND list of two or more multiple Property/Value combinations. • <i>OR List</i>: Performs a find/select operation that matches an OR list of two or more multiple Property/Value combinations. • <i>Multiple</i>: Performs a find/select operation that matches multiple lists of Property/Value combinations; these lists can be a combination of AND lists and OR lists.
<i>Property</i>	Displays the available properties for the object that you selected in the Object Type field. You can use the pre-defined property Connection to find/select objects by connectivity. The Connection property supports connectivity search for the following objects: • Instance • Pin • Net • Module • Selected objects
<i>NOT</i>	Inverses the value of the operator in the Operator field. For example, If the value in the Operator field is <, clicking the NOT button effectively changes the value to NOT <

<i>Operator</i>	Specifies the relationship between the property and its value. This field is dynamically populated based on the specified object type.
<i>Value</i>	Specifies the value of the property. This field is dynamically populated based on the specified object type. In some cases, you may have to manually specify a value, for example, the name of a cell.
<i>Add</i>	Adds the search criteria based on the values of Property, Operator, and Value fields. The search criteria are displayed in the form. This button is not available if you select the <i>Single</i> mode.
<i>Delete</i>	Deletes the currently selected search criteria. This button is not available in the <i>Single</i> mode.
<i>Update</i>	Updates the currently selected search criteria based on the values of Property, Operator, and Value fields. This button is not available in the <i>Single</i> mode.
<i>Add List</i>	Adds a blank list to the search criteria. This button is available only in the <i>Multiple</i> mode. • If the And button is selected, an AND list is added • If the OR button is selected, an OR list is added. You can then add search criteria to the list by specifying values in the Property, Operator, and Value fields, selecting the list in the search criteria, and clicking the <i>Add</i> button. You can create a NOT AND list or a NOT OR list by selecting the <i>Not</i> button before you click the <i>Add List</i> button.
<i>And</i>	Specifies that the list added through the <i>Add List</i> button is an AND list. This button is available only in the <i>Multiple</i> mode.
<i>OR</i>	Specifies that the list added through the Add List button is an OR list. This button is available only in the <i>Multiple</i> mode.
<i>Not</i>	Creates a NOT AND list or a NOT OR List when the Add List button is clicked.

<i>Find the object base on whole design</i>	Searches for objects that match the specified criteria in the entire design and reports them in the <i>Results</i> section. This is the default operation if no objects are selected in the design display area when you open the Find/Select Object form.
<i>Select Also</i>	Selects the objects matching the criteria in the design display area. <i>Default:</i> This check box is turned off by default. Objects are reported in the results window but are not selected in the design display area.
<i>Find the object base on selected objects</i>	Searches for the required object within the selected object list, instead of the entire design. This is the default operation if you have selected objects in the design display area before opening the Find/Select Object form.
<i>Clear Result Window</i>	Clears objects previously listed in the <i>Results</i> list, then finds the objects that meet the specified criteria and reports them in the Results list. <i>Default:</i> The objects previously listed are retained in the results window along with the results of the current search. Additionally, you can choose one of the following clear options: • <i>All</i> - Clears all objects previously listed in the <i>Results</i> list. • <i>Selected</i> - Clears selected entries from the <i>Results</i> list.
<i>Find Only</i>	Finds all objects that meet the specified criteria and reports them in the <i>Results</i> window.
Zoom in Highlighted 	Zooms in the highlighted object(s). You can highlight object(s) by selecting them in the results window and clicking the <i>Highlight</i> button
<i>Select</i> 	When you click <i>Select</i>, the objects selected in the Results window are selected in the design display area.
<i>Select All</i> 	When you click <i>Select All</i>, all the objects in the Results window are selected in the design display area.

<i>Deselect</i> 	When you click <i>Deselect</i> , the objects selected in the Results window are deselected in the design display area.
<i>Deselect All</i> 	When you click <i>Deselect All</i> , all the objects in the Results window are deselected in the design display area.
<i>Highlight</i> 	When you click <i>Highlight</i> , the objects selected in the Results window are highlighted in the design display area.
<i>Dehighlight</i> 	When you click <i>Dehighlight</i> , the objects selected in the Results window are dehighlighted in the design display area.
<i>Highlight Color</i> 	Specifies color for highlighting objects in the design display area.
<i>Auto Zoom</i>	Automatically zooms in to the selected object in the design display area as you browse through the search results in the Find/Select Object form. <i>Auto Zoom</i> is off by default.

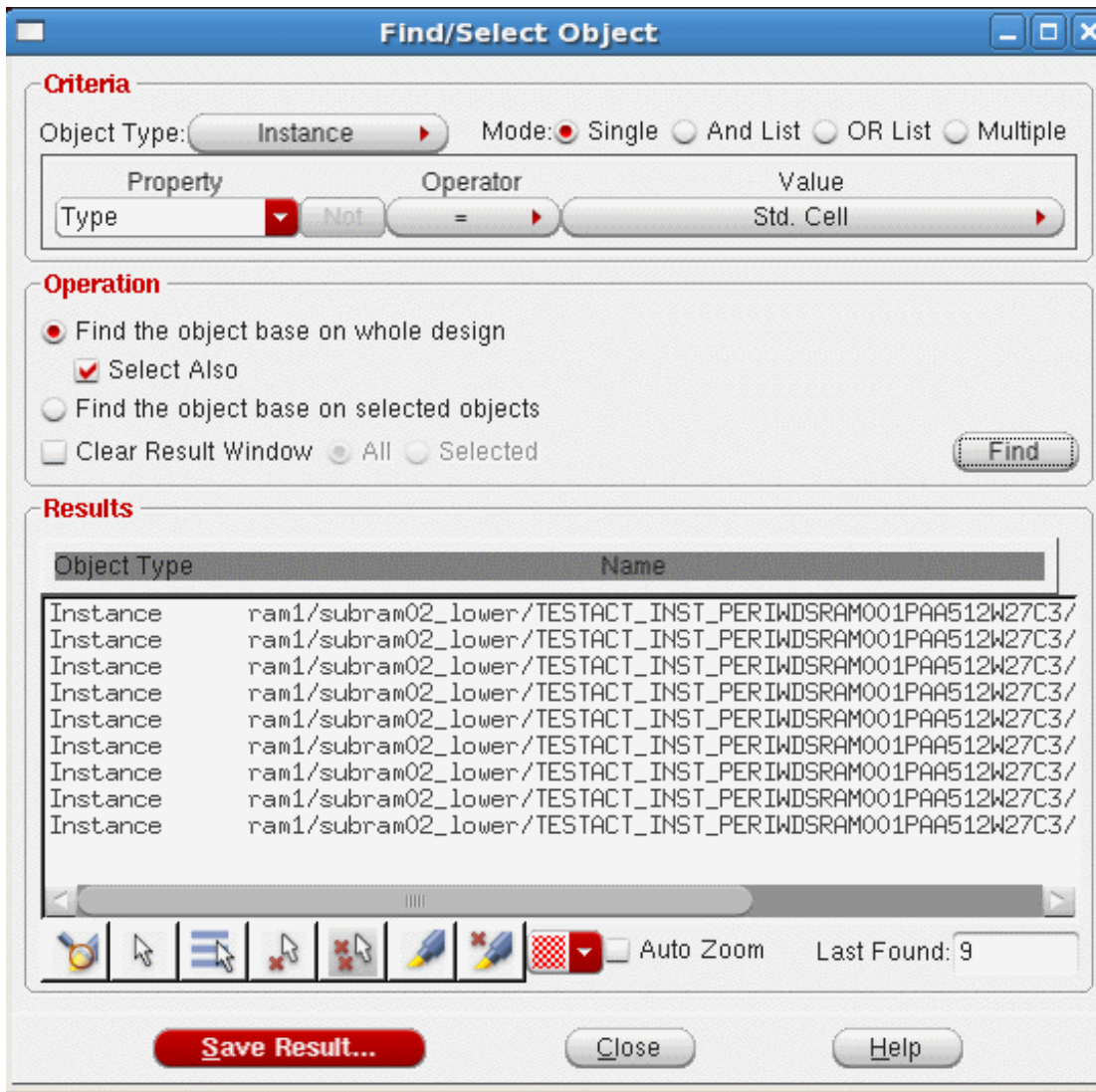
Mode Settings

→ Choose *View - Find/Select Object - Mode* field.

The Mode field provides the following options:

Single

Performs a simple search for a Property/Value combination. The following example illustrates the search for all instances of the type Standard Cell.



The select operation logged in this case by the `selectObjByProp` command would be:

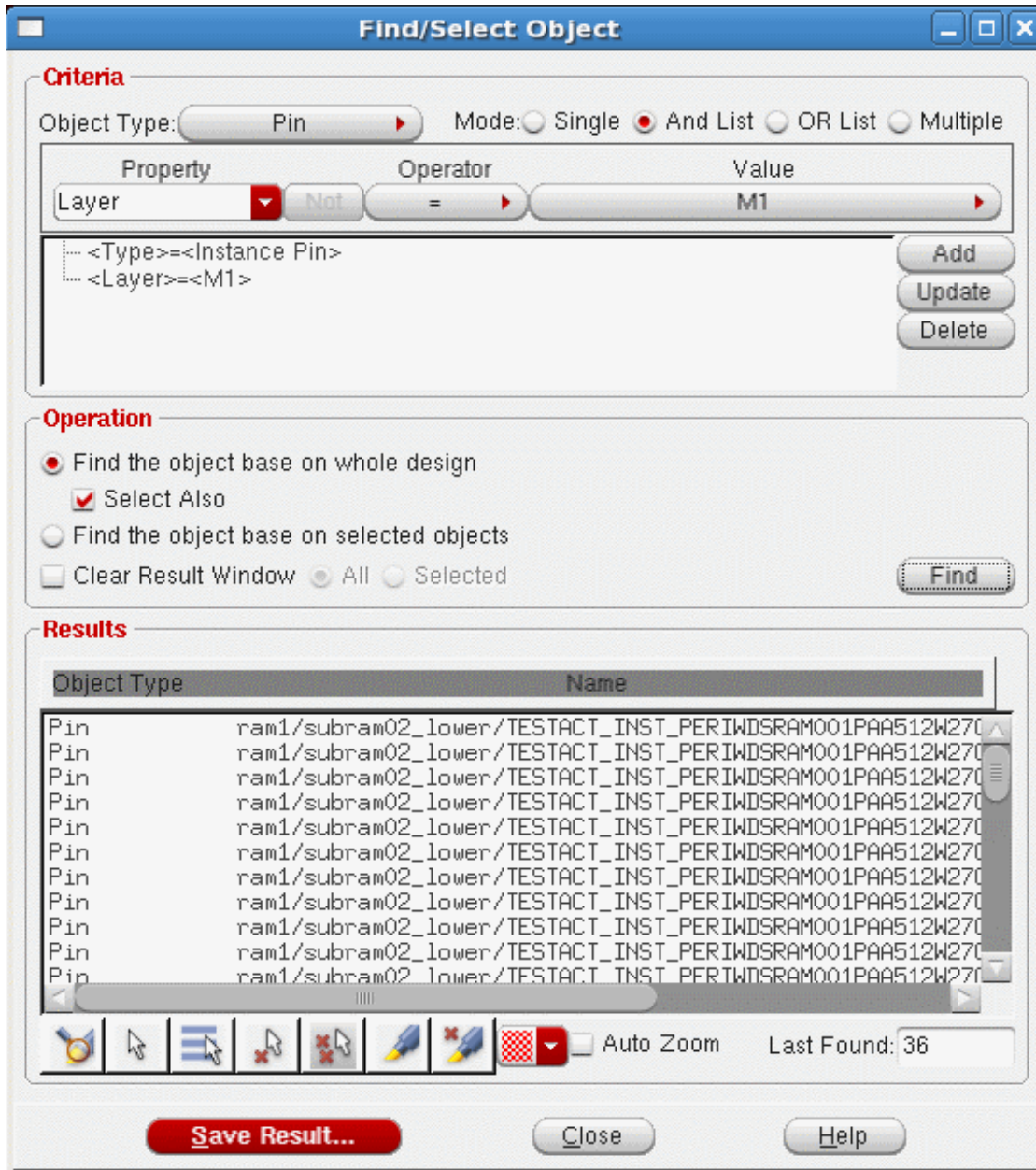
```
selectObjByProp Instance {<Type>=<Std. Cell>}
```

And List

Performs a find/select operation that matches an AND list of two or more Property/Value combinations. The following example illustrates the search for all pins that match *all* of the following criteria:

- Type is Instance Pin

- Layer is M1



The **Find/Select Object** dialog box is used to search for objects in the design. It is divided into three main sections: **Criteria**, **Operation**, and **Results**.

Criteria

Object Type: **Pin** Mode: ☐ Single ☒ And List ☐ OR List ☐ Multiple

Property	Operator	Value
Layer	Not =	M1

<Type>=<Instance Pin>
<Layer>=<M1>

Operation

☒ Find the object base on whole design
☒ Select Also
☐ Find the object base on selected objects
☐ Clear Result Window ☐ All ☐ Selected

Results

Object Type	Name
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270
Pin	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W270

Auto Zoom Last Found: 36

Save Result... **Close** **Help**

OR List

Performs a find/select operation that matches an OR list of two or more Property/Value combinations. The following example illustrates the search for all instances whose type is Block or Standard Cell.

Find/Select Object

Criteria

Object Type: Instance

Mode: ☐ Single ☐ And List ☒ OR List ☐ Multiple

Property	Operator	Value
Type	Not	=
<Type>=<Block> <Type>=<Std. Cell>		

Add
Update
Delete

Operation

☒ Find the object base on whole design

☒ Select Also

☐ Find the object base on selected objects

☐ Clear Result Window ☐ All ☐ Selected

Find

Results

Object Type	Name
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/
Instance	ram1/subram02_lower/TESTACT_INST_PERIWDSRAM001PAA512W27C3/

◀
|||
▶

Auto Zoom
Last Found: 9

Save Result...
Close
Help

Multiple

Performs a find/select operation that matches multiple lists of Property/Value combinations; these lists can be a combination of AND lists and OR lists. The following example illustrates the find/select operation for all pins of the modules DTMF_INST/TDSP_CORE_INST and DTMF_INST/ARB_INST that are on the top side and have a status of *Placed*.

The screenshot shows the 'Find/Select Object' dialog box with the following configuration:

- Criteria:**
 - Object Type: Pin
 - Mode: ☐ Single ☐ And List ☐ OR List ☒ Multiple
 - Property: Side, Operator: Not, Value: Top
 - Lists:
 - OR List
 - <Module>Match<DTMF_INST/TDSP_CORE_INST>
 - <Module>Match<DTMF_INST/ARB_INST>
 - And List
 - Buttons: Add, Add List, Update, Delete, And (selected), OR, Not
- Operation:**
 - ☒ Find the object base on whole design
 - ☒ Select Also
 - ☐ Find the object base on selected objects
 - ☐ Clear Result Window ☐ All ☐ Selected
 - Find button
- Results:**
 - Table with headers: Object Type, Name
 - Empty table body
 - Toolbar with icons for selection and zoom
 - Auto Zoom checkbox (checked)
 - Last Found: 0
- Buttons:** Save Result..., Close, Help

Deselect All

Use the *Deselect All* menu command to deselect all selected nets.

→ Choose *View - Deselect All*.

Highlight Selected

Use the *Highlight Selected* menu command to highlight the object or object set you have selected in the design display area. You can select object sets using the Find/Select Object form.

1. Select an object or object set.
2. Click *View - Highlight Selected* and select one of the following options:
 - Select *Original* to color the selected object in original color. This option is useful if you want to check a specific net or instance more closely with the background dimmed. An object highlighted in original color will stand out in the dimmed background.
 - Select the desired highlight set. The default number of highlight colors is 64.

Clear Highlight

Use the *Clear Highlight* submenu to clear highlights.

- Choose *View - Clear Highlight* and select one of the following options:
 - *Clear Original* - Clears all original highlights.
 - *Clear All*--Clears all highlights from the design display area.
 - *Clear Selected*--Clears highlight from the object selected in the design display area.
 - *Clear Set #<n>*--Clears the highlight set you choose. For example, if you select *Clear Set #2*, highlight is cleared from all objects that were highlighted with Highlight Set #2.

Note: The clear highlight options are also available from the context menu, which you can open by right-clicking an object in the design display area.

Edit Highlight Color

Use the Edit Highlight Color form to further customize the highlight color and pattern of selected object sets in the design display area.

→ Choose *View - Edit Highlight Color*.

Fields and Options

<i>Highlight Set</i>	Lists the available highlight sets. To customize a highlight set: • Select the set. • Choose the appropriate highlight color and pattern from the <i>Color</i> and <i>Stipple</i> area, respectively. • Click <i>Apply</i>.
<i>Color</i>	Specifies the highlight color for the selected highlight set.
<i>Stipple</i>	Specifies the highlight pattern for the selected set.
<i>Width of Highlight Boundary</i>	Specifies the width of highlight boundary. You can set the number as 1, 2, 3, 4, or 5. <i>Default: 1</i>
<i>Save</i>	Saves the highlight settings.
<i>Load</i>	Loads highlight settings previously saved in a file.
<i>Default</i>	Reverts to the default settings for all highlight sets.

Dim Background

Use the *View - Dim Background* menu command to dim the background of the Voltus display. This enhancement helps you view selected and highlighted objects more clearly. This is especially useful when you are debugging.

To enable the dim mode:

- Select the *View - Dim Background* menu command.

The equivalent bindkey is F12. The F12 key is used as a three-way toggle; so if you want to quit dim mode, press the F12 key again twice.

Power and Rail Menu

- [Set PG Library Mode](#)
- [Generate PG Library](#)
- [Set Power Analysis Mode](#)
- [Run Power Analysis](#)
- [Set Rail Analysis Mode](#)
- [Run Rail Analysis](#)
- [Run Resistance Analysis](#)
- [Cell Viewer](#)
- [Power and Rail Plots](#)
- [ESD Summary](#)
- [Power Density Map](#)
- [Power Library Report](#)
- [Power Report](#)
- [Power Histograms](#)
- [Dynamic Results](#)

Set PG Library Mode

→ Choose *Power Rail - Set PowerGrid Library Mode*.

<i>Basic</i>	Specifies what kind of power-grid library will be created. This form is used to define the type of input and configuration for generating power-grid libraries.
<i>Advanced</i>	Specifies the advanced power-grid library generation features.

Basic

The screenshot shows the 'Set PG Library Mode' dialog box with the 'Basic' tab selected. The dialog has a title bar and standard window controls. Inside, there are two tabs: 'Basic' and 'Advanced'. The 'Basic' tab contains the following fields and options:

- Cell Type:** Three radio buttons: 'Tech Only' (selected), 'Std Cells', and 'Macros/IO'.
- Filler Cell Names:** A text input field.
- Decap Cell Names:** A text input field.
- Capacitance:** A checked checkbox for 'Area Cap(fF/um^2)' with a value of '0.5'. There is also an unchecked checkbox for 'Simulation'.
- Spice subckt netlist:** A text input field.
- XYScaling:** A text input field with a folder icon to its left.
- Spice model:** A text input field.
- Corners:** A text input field with a folder icon to its left, and 'Add' and 'Remove' buttons to its right.
- Extraction:** A section containing:
 - Extraction tech file:** A text input field with a folder icon to its left.
 - GDS Files:** A text input field with a folder icon to its left.
 - GDS port generation:** A text input field with an unchecked checkbox to its left.
 - LEF Layermap:** A text input field with a folder icon to its left.
 - GDS Layermap:** A text input field with a folder icon to its left.
- Supply Pins:** A section containing:
 - Power Pins:** A text input field with the value 'VDD vdd!'.
 - Voltages:** A text input field.
 - Ground Pins:** A text input field with the value 'VSS gnd!'.

At the bottom of the dialog are six buttons: 'OK' (highlighted in red), 'Apply', 'Save...', 'Load...', 'Cancel', and 'Help'.

Fields and Options

<i>Cell Type</i>	Specifies the cell type used for library creation. The choices are <i>Tech Only</i>, <i>Std Cells</i>, <i>Macros/I/Os</i>. The default is <i>Tech Only</i>. • <i>Tech Only</i> Creates a technology Library that contains the tech file, area capacitance specification, decap/filler/powergate cells, and Tech view for all the cells. A technology library is a requirement for running rail analysis. • <i>Std Cells</i> Generates a power-grid library for all the standard cells using the Spice netlist. The stdcells power-grid library does not contain the tech file. Power-grid is connected at the LEF pin, and all the capacitance are derived from simulation. The power-grid library contains three kind of power-grid views (PGVs), Early/IR/EM, and the views are selected based on the accuracy mode of rail analysis. • <i>Macros/I/Os</i> Generates a power-grid library for memory, I/Os and macros using the Spice and GDS netlist. GDS is mandatory for the cell type macros. The <code>macros</code> power-grid library does not contain the tech file. Power-grid is connected down to contact/specified via, and all the capacitance are derived from simulation. The power-grid library contains three kind of PGVs, Early/IR/EM, and the views are selected based on the accuracy mode of rail analysis.
<i>Filler Cell Names</i>	Lists the names of the filler cells that have no GDS data. Wildcards (*) can be used. These cells can be swapped out for decaps during decap optimization.
<i>Decap Cell Names</i>	Specifies the names of cells that have explicit decoupling capacitance for use in decap optimization. These cells are nearly equivalent to filler cells, except that there is typically an active device between the power and ground nets to provide extra decoupling capacitance. Power-grid view library generation uses this list of cells to ensure that a capacitance value is associated with each cell. If no capacitance is associated with a cell, a warning message will be issued. Wildcards (*) can be used.
Capacitance	

<i>Area Cap</i> (fF/um^2)	Specifies the default amount of area based decoupling capacitance in a cell. This option is enabled for the cell type: <i>Tech Only</i> .
<i>Simulation</i>	Specifies that the capacitance is derived from netlist simulation. This option is enabled for the cell types: <i>Std Cells</i> and <i>Macros/IOs</i> .
<i>Spice subckt netlist</i>	Specifies the names of the SPICE netlist files containing the subcircuits corresponding to the cells being processed during power-grid library generation. These subcircuit netlists will be used to calculate pin capacitances using Voltus library simulator.
<i>XYScaling</i>	Specifies the current region scale factor for the Netlist-based flow. The current region scale factor is the factor by which the netlist coordinates are scaled before finding the closest node to attach the tap currents.
<i>Spice Model</i>	Specifies the names of the SPICE model files used by the SPICE netlist file.
<i>Corners</i>	Specifies a list of SPICE corners. This field is optional. If you specify SPICE corner(s), the number of corner items should be equal to the number of SPICE model files specified in the <i>Spice Model</i> field.
Extraction	
<i>Extraction Tech File</i>	Specifies the name of the technology file that will be used for extraction.
<i>LEF Layermap</i>	Specifies a file that provides the mapping of layers between the LEF and technology file.
<i>GDS Files</i>	Specifies the names of the GDSII files used during power-grid library generation.
<i>GDS Layermap</i>	Specifies a file that provides the mapping of layers between the LEF and GDS files.
<i>GDS Port Generation</i>	Identifies the layer name that library generation uses to compute the placement bounding box for GDSII cells that are placed by the DEF file.
Supply Pins	
<i>Power Pins</i>	Specifies the names of the nets considered power nets for all cells for which the power nets are not specified.
<i>Voltages</i>	Specifies the voltages for the power pins.

<i>Ground Pins</i>	Specifies the names of the nets considered power nets for all cells for which the power nets are not specified.
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Advanced

The screenshot shows the 'Set PG Library Mode' dialog box with the 'Advanced' tab selected. The dialog contains the following fields and options:

- Cell List File**: Text input field with a folder icon.
- Cell Decap File**: Text input field with a folder icon.
- Current Distribution**:
 - ☒ Activity Propagation
 - ☐ Current Region Files: Text input field with a folder icon.
 - ☐ Dynamic Simulation: Text input field with a folder icon.
- Bulk Power Pins**: Text input field.
- Voltages**: Text input field.
- Bulk Ground Pins**: Text input field.
- Stop@via**: Text input field.
- Temperature**: Text input field.
- ☐ Power gate Cell Characterization
 - Cell Name**: Text input field.
 - Supply Pin**: Text input field.
 - Switch Pin**: Text input field.
 - Ron(Ohm)**: Text input field.
 - Idsat(mA)**: Text input field.
 - Ileakage(mA)**: Text input field.
 - Add**: Button.
 - Remove**: Button.
- Table with 6 columns: CellName, SupplyPin, SwitchPin, Ron, Idsat, Ileakage.
- ☐ Power gate fine grain simulation
- Liberty Files:** Text input field with a folder icon.

At the bottom of the dialog are the following buttons: **OK** (highlighted in red), **Apply**, **Save...**, **Load...**, **Cancel**, and **Help**.

Fields and Options

<i>Cell List File</i>	Specifies a file containing a list of cells that will be processed. The <i>Cell List File</i> field is required when celltype <i>Macros/IOs</i> is selected. The program spawns the library generation job in serial mode for each macro in the cell list file. A separate library for each macro is generated. This facilitates debugging when library generation of any macro fails. It also helps in improving the overall performance.
<i>Cell Decap File</i>	Specifies the name of a file containing a list of cells and the decoupling capacitance value for each.
<i>Current Distribution</i>	Specifies the method of current distribution inside the cell. • Activity Propagation uses Voltus library simulator to determine the propagation activity to distribute the current. • Current Region Files specifies the amount of current to distribute within a cell region. • Dynamic simulation uses a trigger file to run detailed analysis to determine current distribution.
<i>Bulk Power Pins</i>	Specifies the names of nets that connect transistor bulk terminals to the power net. This information is used to prevent any current from being assigned to these nets. If a net is connected to any transistor terminals other than a bulk terminal, you must specify that net in the <i>Power Pins</i> field.
<i>Voltages</i>	Specifies the voltages for the bulk power pins.
<i>Bulk Power Pins</i>	Specifies the names of nets that connect transistor bulk terminals to the ground net. This information is used to prevent any current from being assigned to these nets. If a net is connected to any transistor terminals other than a bulk terminal, you must specify that net in the <i>Ground Pins</i> field.
<i>Stop@via</i>	Stops the extraction of the power-grid network inside a cell at the specified via layer.
<i>Temperature</i>	Specifies the temperature in degree celcius used in library characterization for thermal aware EM/IR analysis.

<i>Power Gate Cell Characterization</i>	Specifies the information that power-grid library generation needs to create a port power gate view. Specifies the cell name, the unswitched power pin, the switched power pin, the on resistance in ohms, the saturation current in milliamps, and the leakage current in milliamps. If <i>Ron</i> , <i>Idsat</i> , or <i>Ileakage</i> , values are specified, they will be used to override the data extracted by Thunder.
<i>Cell Name</i>	Specifies the power gate cell name.
<i>Supply Pin</i>	Specifies the power gate supply pin (unswitched power pin).
<i>Switch Pin</i>	Specifies the power gate switched power pin.
<i>Ron (ohms)</i>	Specifies the power gate on resistance.
<i>Idsat (mA)</i>	Specifies the power gate saturation current.
<i>Ileakage (mA)</i>	Specifies the power gate leakage current.
<i>Power gate fine grain simulation</i>	Specifies to enable finegrain powergate simulation.

Related Text Commands

See [set_pg_library_mode](#) and [set_advanced_pg_library_mode](#) command in the *Voltus Text Command Reference*.

Generate PG Library

Generates the output for power-grid library generation.

→ Choose *Power Rail - Generate PowerGrid Library*.



Fields and Options

<i>Result Directory</i>	Generates the output for power-grid library generation in the specified directory.
<i>Library Prefix</i>	Specifies a prefix for the power-grid library. When this option is specified, the default PGV for each cell type is as follows: • <code>techonly</code> - <code>prefix_techonly.cl</code> (default: <code>techonly.cl</code>) • <code>stdcells</code> - <code>prefix_stdcells.cl</code> (default: <code>stdcells.cl</code>) • <code>macros</code> - <code>prefix_cellname.cl</code> (default: <code>macros_cellname.cl</code>)

Related Text Commands

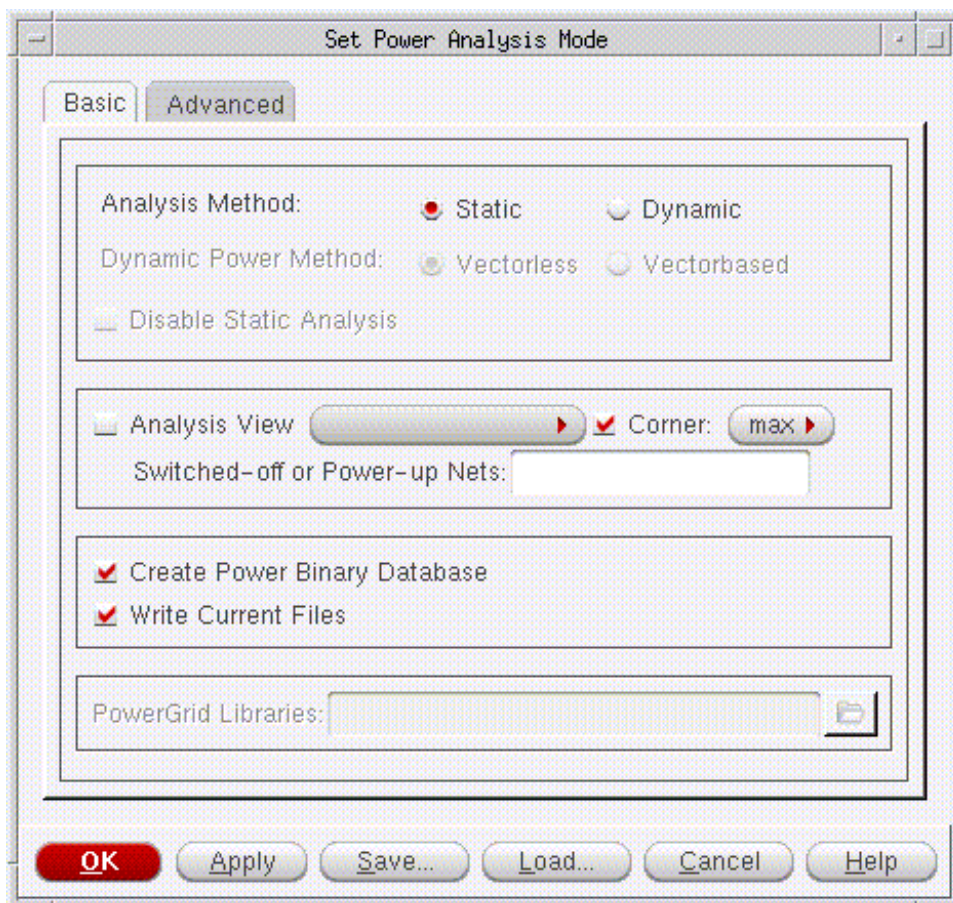
See [generate_pg_library](#) command in the *Voltus Text Command Reference*.

Set Power Analysis Mode

→ Choose *Power Rail - Set Power Analysis Mode*.

<i>Basic</i>	Specifies whether static or dynamic power analysis should be run and some additional settings.
<i>Advanced</i>	Specifies the advanced power analysis setup options.

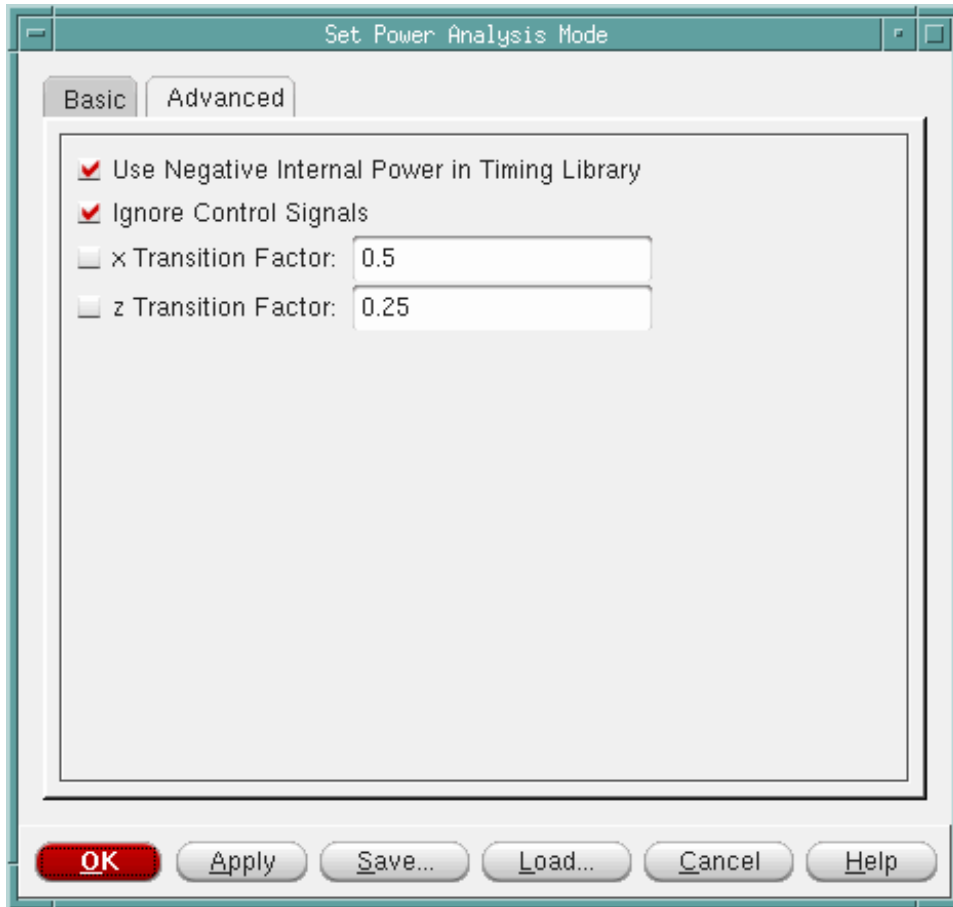
Basic



Fields and Options

<i>Analysis Method</i>	Specifies either static or dynamic power analysis
<i>Dynamic Power Method</i>	If dynamic power analysis is set this specifies if the method of analysis is vectorless or vector based.
<i>Disable Static Analysis</i>	When selected in the dynamic vector-based or dynamic vectorless flows, the command performs only dynamic analysis and turns-off static power calculation. <i>Default:</i> Not selected for <i>Static</i> analysis method, selected for <i>Dynamic</i> analysis method
<i>Analysis View</i>	Specifies the Analysis view.
<i>Corner</i>	Specifies the corner via a pulldown list. The choices are <code>min</code> , or <code>max</code> . <i>Default:</i> <code>max</code>
<i>Switch off Power Nets</i>	Specifies if power nets should be switched off
<i>Create Power Binary Database</i>	Specifies that a binary power database should be created. This can be used for viewing the power data graphically.
<i>Write Current Files</i>	Specifies that Voltus is to generate the current data files, one current data file per net. <i>Default:</i> <code>true</code> .
<i>PowerGrid Libraries</i>	Specifies the name of the Cadence power cell libraries (.cl)

Advanced



Fields and Options

<i>Use Negative Internal Power in Timing Library</i>	Specifies that Voltus will keep negative internal energy numbers from the <code>.lib</code> internal power table. When not set, negative power is treated as 0. <i>Default:</i> <code>true</code>
<i>Ignore Control Signals</i>	If set specifies that control signals will be ignored when propagating activity. <i>Default:</i> <code>true</code>
<i>X Transition Factor</i>	When using VCD one can specify how transitions to and from X are counted. These transitions are defined as shown in the Transitions from/to X and Z Values table. Transitions to and from the X state will be treated as full transitions multiplied by the specified factor. <i>Default:</i> <code>0.5</code>
<i>Z Transition Factor</i>	When using VCD one can specify how transitions to and from Z are counted. These transitions are defined as shown in the Transitions from/to X and Z Values table. Transitions to and from the Z state will be treated as full transitions multiplied by the specified factor. <i>Default:</i> <code>0.25</code>

Transitions from/to X and Z Values

Transition	Definition	Default
to/from X	[0 or 1 or Z] <-> X	0.5
to/from Z	[0 or 1] <-> Z	0.25

Related Text Commands

See `set_power_analysis_mode` command in the *Voltus Text Command Reference*.

Run Power Analysis

→ Choose *Power Rail - Run Power Analysis*.

<i>Basic</i>	Runs the power analysis with the provided parameters.
<i>Activity</i>	Specifies the global or custom switching activity for all primary nets, pins, and ports.
<i>Power</i>	Specifies the custom power information for a cell or instance.
<i>Advanced</i>	Specifies the options for additional power analysis setup.

Basic

The screenshot shows the 'Run Power Analysis' dialog box with the 'Basic' tab selected. The dialog has a title bar 'Run Power Analysis' and four tabs: 'Basic', 'Activity', 'Power', and 'Advanced'. The 'Basic' tab contains the following fields and controls:

- Input Activity:** A text field containing '0.2'.
- Dominant Frequency:** A text field containing '100' followed by '(MHz)'.
- Flop Activity:** An empty text field.
- Clock Gate Activity:** An empty text field.
- Activity File:** A section with a checkbox, two radio buttons ('VCD' and 'FSDB'), a file selection button (folder icon), and three buttons ('Add', 'Remove', 'Vector Profiler').
- Scope:** A text field.
- Start:** A text field followed by '(ns)'.
- Stop:** A text field followed by '(ns)'.
- Block:** A text field.
- Results Directory:** A text field containing './' with a folder selection button (folder icon).

At the bottom of the dialog are six buttons: 'OK' (highlighted in red), 'Apply', 'Save...', 'Load...', 'Cancel', and 'Help'.

Fields and Options

<i>Input Activity</i>	Specifies the average number of times as a (Specified as a fraction) that a primary input switches in a clock cycle. It can be any number between 0 and 1. <i>Default: 0.2</i>
<i>Dominant Frequency</i>	Specifies the dominant frequency of the design.
<i>Flop Activity</i>	Specifies the activity of outputs of sequential logic. There is no default.
<i>Clock Gate Activity</i>	Specifies the average number of times that a clock-gating cell switches in a clock cycle at the output of the pin. Enter any positive number. There is no default. Note: The value that you specify in GUI will overwrite the value in <code>set_default_switching_activity -clock_gates_output</code> parameter.
<i>Activity File</i>	If selected, activates the fields for entering VCD/FSDB information. A VCD/FSDB file must be specified.
<i>Vector Profiler</i>	When clicked, displays the Run Vector Profile window.
<i>Scope</i>	Specifies the scope of the VCD file.
<i>Start</i>	Specifies the start time in nanoseconds (ns) of the VCD file that will be used for power analysis.
<i>Stop</i>	Specifies the stop time in nanoseconds (ns) of the VCD file that will be used for power analysis.
<i>Block</i>	Specifies the block name.
<i>Results Directory</i>	Specifies the directory where the power analysis output will be placed.

Activity

The screenshot shows the 'Run Power Analysis' dialog box with the 'Activity' tab selected. The dialog has four tabs: 'Basic', 'Activity', 'Power', and 'Advanced'. The 'Activity' tab contains the following controls:

- ☐ Global Activity: [text field]
- ☐ Hierarchy(s): [text field] [Add] [Remove]
- Global Activity | Hierarchy(s)
- ☒ Activity ☐ Transition Density
- ☒ Net ☐ Pin ☐ port
- Name: [text field] Activity/Td: [text field]
- Duty: [text field] Period: [text field] [Add] [Remove]
- Table with columns: Type, Comp Type, Name, Value, Duty, Period
- ☐ TCF: [text field] [Folder icon]
- ☐ SAF: [text field] [Folder icon] SAF Scope: [text field]

At the bottom are buttons: OK, Apply, Save..., Load..., Cancel, and Help.

Fields and Options

<i>Global Activity</i>	Specifies the average number of times that all unset nodes switch in a clock cycle. If not specified the fastest domain frequency will be used unless one turns off clock domains. There is no default.
<i>Hierarchy(s)</i>	A hierarchical name that the <i>Global Activity</i> is being assigned to. If not provided the whole design is assumed. Selecting the <i>Add</i> button adds hierarchical activity information to a list. The <i>Remove</i> button removes a selected item in the list. To highlight multiple items, click them while pressing either the <i>Shift</i> or <i>Control</i> key.
<i>Activity Transition Density</i>	Specifies whether <i>Activity</i> or <i>Transition Density</i> numbers will be used when supplying <i>Net</i>, <i>Pin</i>, or <i>Port</i> activities.
<i>Net Port Pin</i>	Specifies that the activity numbers are for a <i>Net</i>, <i>Pin</i>, or <i>Port</i>.
<i>Name</i>	Name of the Net, Pin, or Port.
<i>Activity/Td</i>	The <i>Activity</i> or <i>Transition Density</i> value based on which of these two was selected.
<i>Duty Period</i>	Duty cycle and Period for the named Net, Pin, or Port. Duty is the fraction of time that a net, pin, or port has a value of 1.0 A duty cycle of 0.25 has a value of one 25% of the time. By selecting <i>Add</i> adds the supplied information to the list. Selecting <i>Remove</i> deletes selected items on the list. To highlight multiple items, click them while pressing either the <i>Shift</i> or <i>Control</i> key.
<i>Period</i>	Period for the named Net, Pin, or Port.
<i>TCF</i>	TCF file.
<i>SAF</i>	SAF file
<i>SAF Scope</i>	Scope of SAF file if <i>SAF</i> is selected.

Power

The screenshot shows the 'Run Power Analysis' dialog box with the 'Power' tab selected. The dialog has four tabs: 'Basic', 'Activity', 'Power', and 'Advanced'. The 'Power' tab contains the following controls:

- ☐ Custom Power
- ☒ Cell ☐ Instance Name: ☐ Sticky
- Power/PWL: Scale Factor: PG Net: ☐ Leakage ☐ Internal ☐ Switching
- A table with the following headers: Type, Name, PWL, Scale-Factor, PG-Net, Sticky, Component.
- ☒ Target Chip Power: (mw) ☐ Chip Scale Factor:
- ☐ Target Leakage (mw)

At the bottom of the dialog are the following buttons: (highlighted in red), , , , , and .

Fields and Options

<i>Custom Power</i>	When selected it activates the <i>Power/PWL</i> and <i>PG Net</i> fields for adding Cells
<i>Cell</i>	Specifies that the <i>Custom Power</i> information is for a cell.
<i>Instance</i>	Specifies that the <i>Custom Power</i> information is for an Instance.
<i>Name</i>	The name of the cell or instance. Must be provided.
<i>Sticky</i>	Applies the simulation-based PWL waveform as is to the specified cell/instance.
<i>Power/PWL</i>	After selecting <i>Custom Power</i> , <i>Cell</i> or <i>Instance</i> , the <i>Power/PWL</i> and <i>PG Net</i> fields can be entered. By selecting the <i>Add</i> button this information is added to the list. The <i>Remove</i> button deletes any selected items that were previously added to the list. To highlight multiple items, click them while pressing either the <i>Shift</i> or <i>Control</i> key.
<i>Scale Factor</i>	Specifies to scale the power value by the specified scale factor. You can use this option to specify the scale factor for specific cells and instances.
<i>PG Net</i>	Specifies the name of the power rail.
<i>Leakage</i>	Specifies the target leakage power of the design, cell, or instance. To specify leakage power for cells and instances, you can use the <i>Leakage</i> option in conjunction with the <i>Cell</i> or <i>Instance</i> option.
<i>Internal</i>	Specifies the target internal power of the design, cell, or instance. To specify internal power for cells and instances, you can use the <i>Internal</i> option in conjunction with the <i>Cell</i> or <i>Instance</i> option.
<i>Switching</i>	Specifies the target switching power of the design, cell, or instance. To specify switching power for cells and instances, you can use the <i>Switching</i> option in conjunction with the <i>Cell</i> or <i>Instance</i> option.
<i>Target Chip Power</i>	Specifies the power for the entire design or rail, in milliwatts (mW).
<i>Chip Scale Factor</i>	Scales the power value by the specified scale factor.
<i>Target Leakage</i>	Sets the total target leakage power of the entire design in milliwatts (mW).

Advanced

The screenshot shows the 'Run Power Analysis' dialog box with the 'Advanced' tab selected. The dialog has a title bar with a minus, maximize, and close button. Below the title bar are four tabs: 'Basic', 'Activity', 'Power', and 'Advanced'. The 'Advanced' tab contains the following options:

- ☐ Native Power Up
- ☐ Simulation Period: (ns) ☐ Step Size: (ps)
- Include File:
- ☐ Activity Mapping File: Golden: ☒ rtl ☐ gate

At the bottom of the dialog are five buttons: 'OK' (highlighted in red), 'Apply', 'Save...', 'Load...', and 'Help'.

Fields and Options

<i>Native Power Up</i>	Specifies to enable the native power up flow.
<i>Simulation Period</i>	The simulation period that dynamic rail analysis will be performed.
<i>Step Size</i>	Specifies the transient time step size or resolution.
<i>Include File</i>	Adds the Power Analysis options that do not have an equivalent option in Voltus. For more information, refer to Power Analysis Options in <i>Voltus Text Command Reference</i> .
<i>Activity Mapping File</i>	Specifies the rtl2gate mapping file which contains instance name mapping between RTL netlist and GATE level netlist. RTL level VCD or TCF are obtained from simulation on RTL Verilog netlist. Power analysis uses this instance name mapping to annotate the output nets of this instance from the pin based RTL, VCD, or TCF. By default, this option is always selected when either VCD/TCF is specified.
<i>Golden</i> <i>rtl</i> <i>gate</i>	Specifies whether <i>rtl</i> or <i>gate</i> is <i>Golden</i>. The one not specified as golden will be considered as Revised. Currently the Conformal MatchPoint file does not specify this. <i>Default: rtl</i>

Related Text Commands

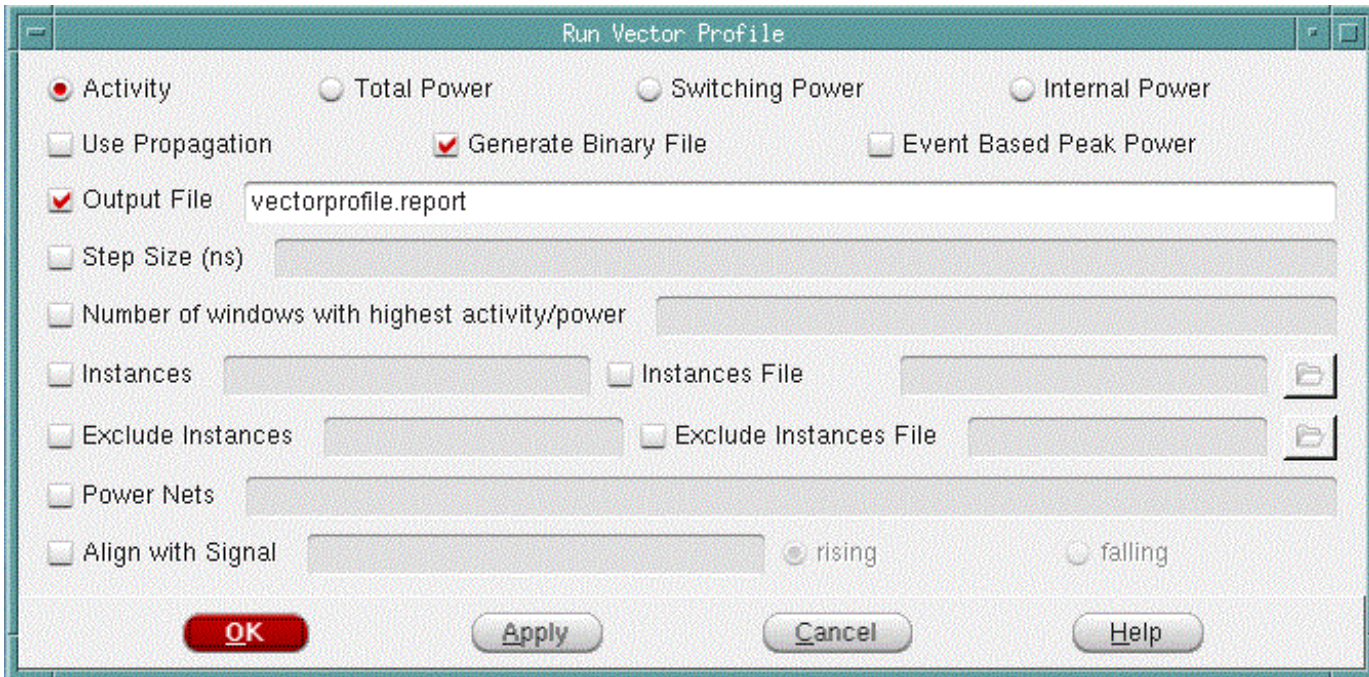
See [report_power](#), [set_default_switching_activity](#), and [set_power](#) command in the *Voltus Text Command Reference*.

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Run Vector Profile

Use this form to create Vector profile report for the specified VCD file.

→ Choose *Power & Rail - Run Power Analysis - Basic* tab, and then click the *Vector Profiler* button.



Fields and Options

<i>Activity</i>	Enables activity profiling. Reports the number of toggles for every window.
<i>Total Power</i>	Enables total power profiling. Reports total power in milliWatts (mW).
<i>Switching Power</i>	Enables switching power profiling. Reports switching power in milliWatts (mW).
<i>Internal Power</i>	Enables internal power profiling. Reports internal power in milliWatts (mW).
<i>Use Propagation</i>	Enables propagation when partial VCD is provided.
<i>Event Based Peak Power</i>	Specifies to enable event-based vector profiling. This method computes power profile of every event on each net. The event-based peak power profiling enables you to accurately capture vectors that could produce peak power using very small resolution (1ps) and with better performance than average power profiling which uses larger resolution to compute average toggle density.
<i>Output File</i>	Writes out the Vector profiling report into the file that you specify. By default, the report is written out into <code>vcdprofile.report</code> file.

<i>Step Size (ns)</i>	Specifies the step size, in nanoseconds (ns). Activity / Power profiling is carried out for every step size that you specify.
<i>Number of windows with highest activity/power</i>	Provides <i>n</i> top windows with maximum activity or power. <i>Default: 3</i>
<i>Instances</i>	Enables profiling of the instances that you specify.
<i>Instances File</i>	Enables profiling of the instances from the instance file that you specify. For example, instances in an instance file can be specified as follows: <pre> ----- inst_1 inst_2 inst_3 . . . inst_n ----- </pre>
<i>Exclude Instances</i>	Disables counting of toggles or calculating power for the specified instances while reporting activity or calculating power for the entire design.
<i>Exclude Instances File</i>	Disables counting of toggles or calculating power for the specified instances file while reporting activity or calculating power for the entire design.
<i>Power Nets</i>	Enables power profiling for the specified power net(s).
<i>Align with Signal</i>	Starts profiling at the first rise or fall of the signal. Select <i>rising</i> to start profiling at the first rising edge. Select <i>falling</i> to start profiling at the first falling edge.

Set Rail Analysis Mode

→ Choose *Power Rail - Set Rail Analysis Mode*.

<i>Basic</i>	Sets whether static or dynamic rail analysis should be performed along with some related parameters.
<i>Advanced</i>	Specifies the advanced rail analysis setup options.

Basic

The screenshot shows the 'Set Rail Analysis Mode' dialog box with the 'Basic' tab selected. The dialog has a title bar with a dropdown arrow, the text 'Set Rail Analysis Mode', and standard window controls. Below the title bar are two tabs: 'Basic' (selected) and 'Advanced'. The main area contains several groups of settings:

- Analysis Stage:** Radio buttons for 'Early' and 'Sign Off' (selected).
- Analysis Method:** Radio buttons for 'Static' (selected) and 'Dynamic'.
- Accuracy:** Radio buttons for 'XD' (selected) and 'HD'.
- PowerGrid Libraries:** A text field with a browse button.
- Analysis View:** A checkbox and a dropdown menu.
- Switched off Nets:** A checkbox and a text field.
- Powering up Nets:** A checkbox and a text field.
- Block Power Up Net:** A checkbox and a text field.
- Power Up Fast Mode:** A checked checkbox.
- Generate Power-Switched ECO:** An unchecked checkbox.
- Process EM Rule in Extraction Tech File:** An unchecked checkbox.
- Enable least resistance path analysis:** An unchecked checkbox.
- EM Analysis Models:** A checkbox and a text field with a browse button.
- ICT-EM Analysis Models:** A checkbox and a text field with a browse button.
- Default Package:** A checkbox and a text field with sub-fields for R= (Ohm), L= (H), and C=.
- Source Search Distance:** A checked checkbox, a text field with '50', and a unit '(um)'.
- Extraction Tech File:** A checkbox and a text field with a browse button.
- Thermal Map:** A checkbox and a text field with a browse button.
- Generate Movies:** An unchecked checkbox.
- Save Voltage Waveform Files:** An unchecked checkbox.

At the bottom are buttons for 'OK' (highlighted in red), 'Apply', 'Save...', 'Load...', 'Cancel', and 'Help'.

Fields and Options

Fields and Options

<i>Analysis Stage</i>	Specifies whether early rail analysis (<i>Early</i>) or signoff (<i>Sign Off</i>) rail analysis will be done. If you select <i>Early</i>, the <i>XD</i> mode is selected by default and <i>HD</i> mode is disabled. <i>Default:</i> <code>Sign Off</code>
<i>Analysis Method</i>	Specifies whether static or dynamic analysis will be done. <i>Default:</i> <code>static</code>
<i>Accuracy</i>	Specifies the accuracy mode of analysis. • <code>XD (accelerated definition)</code> – use for early implementation stage IR/EM analysis • <code>HD (high definition)</code> – use for final verification stage IR/EM analysis <i>Default:</i> <code>XD</code>
<i>PowerGrid Libraries</i>	Specifies the name of the power-grid view library directories. If you select the folder icon, the Set Rail Analysis form appears, which will be used to select and add the appropriate files.
<i>MMMC View</i>	Specifies the Mult-Mode Multi-Corner (MMMC) view.
<i>CPF mode</i>	Specifies Common Power Format (CPF) mode.
<i>Switched off Power Nets</i>	Specifies a list of nets to exclude from analysis.
<i>Powering up Nets</i>	Specifies a list of powering up nets.
<i>Block Power Up Net</i>	Specifies a net for block power-up analysis.
<i>Generate Power-Switched ECO</i>	Specifies that a power switch eco file should be created. <i>Default:</i> <code>false</code>

<i>Process EM Rule in Extraction Tech File</i>	Processes the EM rules defined in the extraction technology file. When <i>Process EM Rule in Extraction Tech File</i> is specified with <i>Extraction Tech File</i> , the extraction technology file will be processed. If <i>Process EM Rule in Extraction Tech File</i> is specified without <i>Extraction Tech File</i> , the .cl model will be processed.
<i>Enable Least Resistance Path Analysis</i>	Enables least resistance path analysis for all the instances in the design.
<i>EM Analysis Models</i>	Specifies the name of a file that includes the electromigration (em) models.
<i>ICT-EM Analysis Models</i>	Specifies the ICT-EM file which only contains the EM rules and excludes the RC extraction data in ICT. The software will use the EM rules in the specified ICT-EM file to calculate EM limits for power net EM analysis even if there are embedded EM rules in the Quantus QRC tech file.
<i>Default Package</i>	Specifies the default resistance, inductance and capacitance values of the package to be used for RLC model.
<i>Source Search Distance</i>	Specifies the search distance in microns(u) for adding voltage sources at power pin locations. <i>Default:</i> 50um
<i>Extraction Tech File</i>	Specifies the extraction technology file to be used for top-level power-grid extraction. If you do not specify this field, the software will use the extraction technology file stored inside the power-grid view library.
<i>Thermal Map</i>	Specifies to process the thermal map file generated for QRC to calculate resistance based on uniform temperature. <i>Default:</i> Not selected.
<i>Generate Movies</i>	Used to specify that dynamic movies will be created. For dynamic only. <i>Default:</i> true
<i>Save Voltage Waveforms Files</i>	

	Saves voltage waveform files in the state directory of analyzed net(s) to feed to SPICE critical path analysis and Substrate Noise Analysis. <i>Default:</i> Not selected. This option is disabled in the <i>Static</i> mode.
<i>Decap Optimization Method</i>	Specifies the decap optimization method to use. For dynamic only. Choices are <i>Placement-Aware</i> and <i>Timing-Aware</i>, <i>Area based</i>, and <i>Removal</i>. <i>Default:</i> <i>Area based</i>.
<i>Decap Removal Method</i>	Choices are <i>Conservative</i> and <i>Aggressive</i>. Used with <i>Removal</i> option of <i>Decap Optimization Method</i>. <i>Default:</i> <i>Aggressive</i>
<i>Maximum Leakage Current</i>	Specifies the maximum leakage current in A, for placement-aware and timing-aware decap optimization methods.
<i>Generate Decap ECO</i>	Specifies that a decap eco file should be created.
<i>Generate reduced die model:N-port</i>	
	Creates <code>n_port_freq_domain</code> as well as <code>n_port_time_domain</code> die-models simultaneously.

Advanced

Set Rail Analysis Mode

Basic | **Advanced**

☐ Generate Block Boundary Voltage:
Instance Name(s):

☐ Power Up Sequence File:

☐ Use Temp Directory: sv_tmpdir_18679_InhYFZ

☐ Report Via Current Direction

☐ Save Current Files for Hierarchical View

☐ Temperature: 25 Degree Celsius

☐ LifeTime: 87600 Hours

☐ GDS: ☒ Flip-Chip GDS ☐ Full-Chip GDS

GDS File:

GDS Map: GDS Cell:

☐ Don't Touch Decaps:

☐ Decap Cells:

☐ Filler Cells:

☐ Decap ECO:

☐ Specify Current Region Create

☐ Power Gate File

☐ Layer Mapping File

☐ Skin Layer(Pairs) for Virtual Via Insertion

OK Apply Save... Load... Cancel Help

Fields and Options

<i>Generate Block Boundary Voltage</i>	Generated the voltages on the block pins. This data can be used for doing rail analysis at the next level up in hierarchy.
<i>Instance Name(s)</i>	The name of the instance that the block boundary voltages will be generated for.
<i>Use Temp Directory</i>	Specifies a directory to use for temporary files.
<i>Report Via Current Direction</i>	Determines that the current direction of vias will be reported. <i>Default: false</i>
<i>Save Current Files for Hierarchical View</i>	Determines whether Voltus automatically saves current files. <i>Default: false</i>
<i>Temperature</i>	Specifies the temperature in degree celsius during rail analysis. The default temperature is 25 degree celsius.
<i>Lifetime</i>	Specifies the lifetime value in hours. <i>Default: 87,600 hours</i>
<i>GDS</i>	Specifies that a GDS file will be used.
<i>Flip-Chip GDS</i>	Specifies that the gds data is for flipchip.
<i>Full-Chip GDS</i>	Specifies that the gds data is for full chip.
<i>GDS File</i>	Specifies the name of the gds file.
<i>GDS Map</i>	Specifies the name of the gds map file.
<i>GDS Cell</i>	Specifies the name of the gds top cell.

<i>Don't Touch Decaps</i>	Specifies the decaps that will not be removed. For dynamic only and used if option: <code>-decap_opt_method removal</code> is set.
<i>Decap Cells</i>	Specifies the decap cells that will be used during decap optimization. This option is used when the decap cells are not tagged in the power-grid library. For dynamic only.
<i>Filler Cells</i>	Specifies the filler cells that will be used during <i>Placement-Aware</i> or <i>Timing-Aware</i> decap optimization. This option is used when the filler cells are not tagged in the power-grid library. For dynamic only.
<i>Decap ECO</i>	Specifies the name of the decap ECO file for domain based analysis that will be created after decap analysis is performed. This file is generated in <code>decap_opt/addcap.cmd</code> in the state directory. For dynamic only.
<i>Specify Current Region</i>	Specifies a file that includes a list of regions and the amount of current to be distributed within them for the power-grid. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab. If you select this checkbox, the <i>Create</i> button is enabled that lets you create current regions for Static and Dynamic analysis. Click <i>Create</i> to view the <i>Create Current Region</i> form. Default units in file are <code>mA</code>. You can also specify rectilinear current regions in the file.
<i>Power Gate File</i>	Specifies the name of the power gate (power switch) file that will be used to perform power gate steady-state analysis. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab.
<i>Layer Mapping File</i>	Specifies the layer map file to generate a port view. If a layer map file is not provided, it would be automatically inferred by the tool. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab.

<i>Skip Layer (Pair)s for Virtual Via Insertion</i>	Specifies to skip via insertion between stripes and non-stripes, on the specified LEF layer pairs. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab.
<i>Skip Virtual Via Insertion for Shape Type</i>	Specifies to skip a given via type. By default, ERA generates all virtual via layer types. whatif - vias are virtual vias that have connectivity to user-defined what if shapes. def - vias are virtual vias between two metal shapes defined in DEF. all - will skip all virtual via generation.
<i>Current Distribution Layer for Unplaced Instances</i>	Specifies the layer name for distributing unplaced current in the early rail analysis mode. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab.
<i>Enable Current Distribution for</i>	Controls the behavior of era current distribution. <code>set_power_data area</code> based power, or <code>current_region_file</code> need to be specified for ERA current distribution to work. Placed instanced without uti or ascii based power can also be considered for era current distribution. • Unplaced: Enable current distribution only for unplaced instances. If <code>set_power_data area</code> based power is specified, - <code>era_current_distribution_layer</code> will be required. • Placed: Enable current distribution for placed instances without any power specified. • All: Both unplaced and placed instances without power specified; will have ERA current. • None: Disable ERA current distribution. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab.
<i>Virtual Followpin Insertion</i>	Generates virtual followpins. The <code>extended</code> followpins will create followpins that extend from one stripe to another. The <code>standard</code> followpins may extend to previous stripe but does not reach the next stripe. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab.

<i>Current Distribution Factor for Placed Macros</i>	When you use <i>Distribute Current for Placed Instance</i> or <i>Generate Current Region File for Unplaced Area</i>, the ratio of power allocated to each macro/block is calculated based on its total area. This field controls the current distribution factors for the placed instances, hence power allocated for area-based power calculation. For example, if you specify 0.5, the software will assume all placed instances to be 50% of its actual size and distribute current accordingly. This field will be enabled only if you select <i>Early</i> analysis stage in the <i>Set Rail Analysis Mode - Basic</i> tab.
<i>Ignore Shorts</i>	Specifies to perform sign-off mode extraction while ignoring shorts, extract each net per geometry defined in LEF/DEF and honor manufacturing effects. This option allows you to ignore shorts and continue sign-off mode extraction.
<i>Enable Manufacturing Effects</i>	Specifies to honor manufacturing effects during static or dynamic analysis.

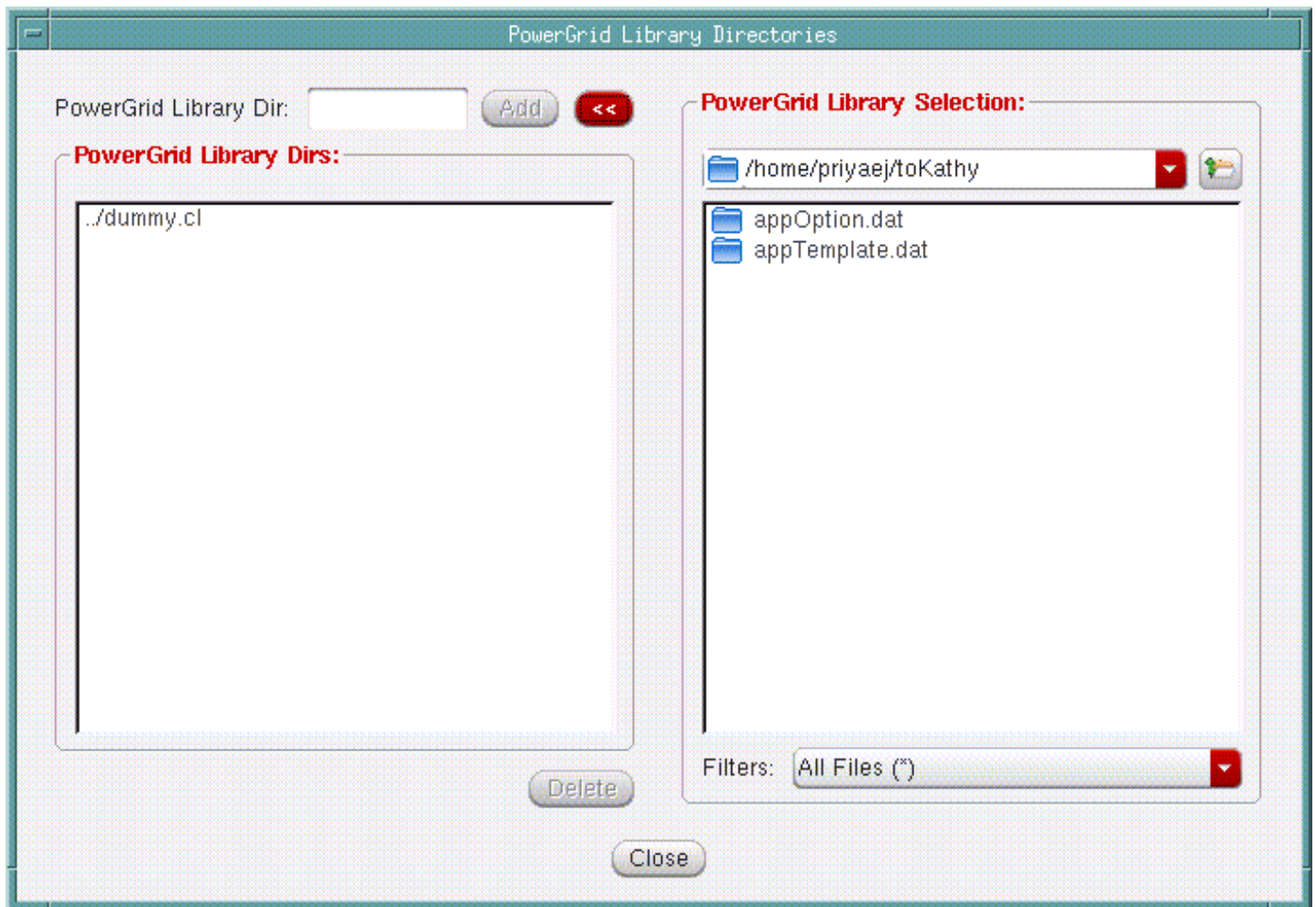
Related Text Commands

See [set_rail_analysis_mode](#) command in the *Voltus Text Command Reference*.

Power Grid Library Directories

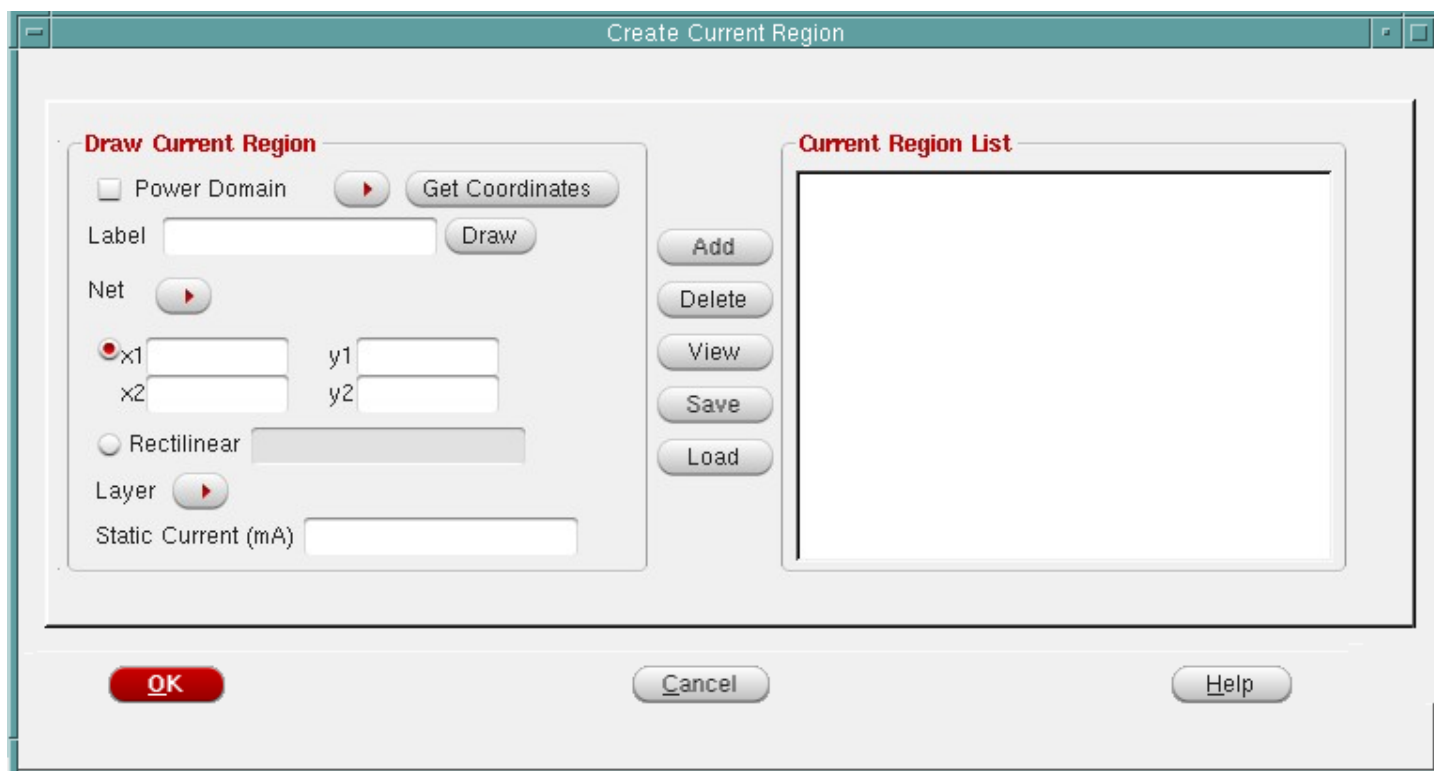
A form such as the one shown below appears when the folder icon is selected from a menu. Selecting the folder icon on this form will toggle the Selection part of the form. To add items, select a file, and then select the *Add* button. To add multiple items from the selection list, select the first item, then move the cursor to the last item and hold down the shift key while selecting the last item. The *Delete* button can be used for deleting items. Similarly the shift key can be used for deleting multiple items.

→ Choose *Power & Rail - Set Rail Analysis Mode - Basic* tab, and then click the *PowerGrid Libraries* button.



Create Current Region

→ Choose *Power Rail - Set Rail Analysis Mode - Advanced* tab, and then click the *Create* button.



Fields and Options

<i>Draw Current Region</i>	Specifies the coordinates of the region, the layer, and the static current. The <i>Add</i> button can be selected to add the region to the <i>Current Region List</i>. The <i>Delete</i> button will delete a selected item on the list. <i>Power Domain</i> lets you specify a power domain based current region. You can use the <i>Power Domain</i> field to select a power domain name and click <i>Get Coordinates</i> to automatically get the coordinates of the power domain. When the power domain is selected, the label name of the current region will be the power domain name and the boundary box coordinates will be the power domain boundary, and you cannot modify these fields. <i>Label</i> specifies a name for the region. If not specified, the tool will provide a name (region1, region2...) <i>Net</i> specifies the power/ground net name for domain-based analysis. This option allows you to specify current region values for a specific net. <i>x1, y1, x2, and y2</i> specifies a rectangular region that the current will be distributed within. <i>Rectilinear</i> specifies the rectilinear current region that the current will be distributed within. Note: In the static mode, ERA splits the rectilinear region into several rectangular regions and distributes current to the rectangular regions based on the area. In the dynamic mode, you can specify rectilinear regions interactively in the layout canvas by selecting the <i>Draw</i> button. You can select a rectilinear region by clicking each vertex of the rectilinear in sequence (clockwise or counter-clockwise), and then pressing the <i>ESC</i> key on the keyboard. This will quit the point selection mode, and the selected points appear in the <i>Rectilinear</i> field. Current in rectangular region = Area of the rectangle / Area of the rectilinear <i>Layer</i> specifies the metal layer that the current sink will be placed on. If you select <i>Draw</i>, then select a box in main window, the coordinate of this box will be automatically populated in the <i>x1 y1 x2 y2</i>. If you select <i>View</i> after selecting an item in the list, the selected region will be displayed in the main window.
<i>Static Current</i>	Specifies the current to be attached in the window.

<i>Dynamic PWL (0ns 0mA...)</i>	Specifies the dynamic PWL for regions without any placed instance, in time (in ns) and current (in mA) format. For example, 0ns 0mA 0.5ns 1mA 1ns 2mA 3ns 1mA 4ns 0. This option is displayed only if you select the analysis method <i>Dynamic</i> .
<i>Loading Cap (pf)</i>	Specifies loading capacitance in pf (picoFarads) waveform for the current region(s). This option is displayed only if you select the analysis method <i>Dynamic</i> .
<i>Estimate</i>	Estimates Gate Capacitance (C1) and Loading Capacitance (C2) values for the specified current regions. These estimated values of C1 and C2 are used along with the PWL current waveform for the regions to perform dynamic rail analysis. This option is displayed only if you select the analysis method <i>Dynamic</i> .
<i>Gate Cap (pf)</i>	Specifies gate capacitance in pf (picoFarads) waveform for the current region(s). This option is displayed only if you select the analysis method <i>Dynamic</i> .
<i>Current Region List</i>	Displays the current region list. This is created by specifying the <i>Draw Current Regions</i> information and selecting <i>Add</i> . Selecting an item on the list and then <i>Delete</i> will delete the selected item.

Related Text Commands

See [set_rail_analysis_mode](#) command in the *Voltus Text Command Reference*.

Run Rail Analysis

→ Choose *Power Rail - Run Rail Analysis*.

<i>Basic</i>	Runs rail analysis using the provided parameters.
<i>Advanced</i>	Specifies the advanced options for running rail analysis.

Basic

Run Rail Analysis

Basic | Advanced

☒ Net Based ☐ Domain Based Domain Name:

Power Net(s): VDD Voltage(s): (V) Threshold: (V) Add Remove

Ground Net(s): VSS Voltage(s): (V) Threshold: (V) Add Remove

Type	Net	Voltage	Threshold

Power Data: ☒ Current Files ☐ ASCII File ☐ Area Based

☐ Scale: ☐ Repeat: (ns) ☐ Total Power: (W) Bias (V)

Power/Current File(s): 📁

Power Pads: ☒ Def Pins ☐ Pad File ☐ XY File ☐ Boundary File Create

File: 📁 Net Name: Add Remove

Type	Net	File

☐ Package Model: 📁 Mapping File: 📁 Scale:

Offset: X: Y: Rotate: ☐ Flip

Results Directory: ./ 📁

OK
Apply
Save...
Load...
Cancel
Help

Fields and Options

<i>Net Based</i>	Set if net based analysis is being run.
<i>Domain Based</i>	Set if domain based analysis is being run.
<i>Domain Name</i>	Name of the power domain that rail analysis will be performed on. Only used for domain based analysis.
<i>Power Net(s)</i>	Name of the power nets that the rail analysis will be performed on.
<i>Voltages</i>	The voltage of the power net.

<i>V Threshold</i>	The voltage threshold of the power net
<i>Ground Net(s)</i>	Name of the ground nets that rail analysis will be performed on.
<i>Voltage(s)</i>	The voltage of the ground net.
<i>V Threshold</i>	The voltage threshold of the ground net.
<i>Add</i>	Click the <i>Add</i> button to add details of a specific net to the power/ground nets table for domain-based analysis.
<i>Remove</i>	Click the <i>Remove</i> button to remove details of a specific net from the power/ground nets table for domain-based analysis.
<i>Power Data</i>	The power data will be supplied by Current Files, an ASCII file or will be area based.
<i>Scale</i>	Specifies a scale factor to apply to the current data. You can use this parameter to scale currents to a more optimistic or pessimistic power-consumption estimate. This parameter is only for <code>-format current</code> and <code>-format ascii</code>.
<i>Repeat</i>	Repeats the dynamic current waveform for the specified time. You must ensure that the stop time for dynamic analysis is also specified and it matches the repeat duration. This option is only available for current format files during dynamic analysis. The repeat option can be used when multiple current files have different simulation periods or when IR drop analysis needs to be run for longer duration. <i>Default units:</i> ns.
<i>Total Power</i>	Specifies the power per area in watts. Only required for option <code>-format area</code> .
<i>Bias</i>	The value of the bias voltage. This parameter is only for <code>-format ascii</code> and <code>-format area</code> .
<i>Power File(s)</i>	Specifies the name of the power data files. The file is optional if <code>-format area</code> is specified.

<i>Power Pads</i>	<i>XYFile</i> specifies that voltage sources will be specified by x y coordinates. <i>Def Pins</i> specifies that voltage sources will be determined by power pins. <i>Pad File</i> specifies that voltage sources will be determined by pad cells given in the file. <i>Boundary Files</i> specifies that voltage source locations will be specified by information in the boundary file. <i>Default: Def Pins</i>
<i>File</i>	The name of the power file.
<i>Net Name</i>	The name of the power net. The <i>Add</i> button adds the net to the list. The <i>Remove</i> button removes the net from the list. To highlight multiple items, click them while pressing either the <i>Shift</i> or <i>Control</i> key.
<i>Package Model</i>	Specifies a file containing a SPICE subcircuit describing the package model. This file is generated by Allegro Package SI (also known as APE3D from Cadence) or other package-modeling products.
<i>Mapping File</i>	Specifies the name of the package terminal mapping file, which contains the mapping of the terminals of the SPICE subcircuit in the package model file (Specified by the <i>Package Model</i> field above) to the locations at the grid or top-level pins.
<i>Offset X</i> <i>Offset Y</i>	Specifies the lower left and upper right points coordinate of the die.
<i>Rotate</i>	Specifies the rotation angle of the die. The possible values are: 0/45/90/135/180/225/270/315. Rotation is always in the counter clockwise direction from the original point.
<i>Flip</i>	Specifies to flip the die along the Y axis.
<i>Results Directory</i>	The directory where the rail analysis output will be saved.

Advanced

The screenshot shows the 'Run Rail Analysis' dialog box with the 'Advanced' tab selected. The dialog has a title bar with a dropdown arrow, the text 'Run Rail Analysis', and standard window controls (minimize, maximize, close). Below the title bar are two tabs: 'Basic' and 'Advanced'. The 'Advanced' tab contains several input fields and buttons. The first section has two rows of labels and text boxes: 'Group Name:' followed by a text box, 'Power Nets:' followed by a text box, another 'Group Name:' followed by a text box, and 'Ground Nets:' followed by a text box. The second section has 'Dynamic Start Time:' followed by a text box and '(ns)', 'Stop Time:' followed by a text box and '(ns)', and 'Resolution:' followed by a text box and '(ps)'. The third section has 'Voltus Include File Begin:' followed by a text box and a folder icon, and 'Voltus Include File End:' followed by a text box and a folder icon. The fourth section has 'What-If Analysis:' followed by a 'Setup' button. At the bottom of the dialog are five buttons: 'OK' (highlighted in red), 'Apply', 'Save...', 'Load...', 'Cancel', and 'Help'.

Run Rail Analysis

Basic | **Advanced**

Group Name: Power Nets:
Group Name: Ground Nets:

Dynamic Start Time: (ns) Stop Time: (ns)
Resolution: (ps)

Voltus Include File Begin:  Voltus Include File End: 

What-If Analysis:

Fields and Options

<i>Group Name</i>	Specifies the name for a group of power or ground nets.
<i>Power Nets</i> <i>Ground Nets</i>	Replace power domain by group..
<i>Start Time</i>	Specifies the start time in nanoseconds (<i>ns</i>) for dynamic analysis.
<i>Stop Time</i>	Specifies the stop time in nanoseconds (<i>ns</i>) for dynamic analysis.
<i>Resolution</i>	Specifies the transient time step size or resolution in picoseconds (<i>ps</i>) .
<i>Voltus Include File Begin</i>	This field provides the ability to provide a file that includes <code>vstorm2</code> commands that will be run prior to the <code>analyze</code> command. This can be used for those VoltageStorm commands that do not have an Voltus GUI equivalent.
<i>Voltus Include File End</i>	This field provides the ability to provide a file that includes <code>vstorm2</code> commands that will be run after the <code>analyze</code> command. This can be used for those VoltageStorm commands that do not have an Voltus GUI equivalent.
<i>What-If Analysis</i> • <i>Setup</i>	Clicking the <i>Setup</i> button displays the GUI for • What-If Analysis Setup - Current • What-If Analysis Setup - Capacitance • What-If Analysis Setup - Resistance • What-If Analysis Setup - Virtual Shape and • What-If Analysis Setup - Regions.

Related Text Commands

See [set_pg_nets](#), [set_rail_analysis_domain](#), [set_power_data](#), [set_power_pads](#), [set_net_group](#), [set_package](#), [analyze_rail](#), [set_advanced_rail_options](#) commands in the *Voltus Text Command Reference*.

Edit Pad Location

Use the Edit Pad Location form to create or edit power pad location files (one for each power net) to mark points on the power structures as voltage source reference points. Power pad location files identify the voltage source reference points, such as VCC, VDD, and VSS. You can use these files during all stages of power analysis.

Note: Pad location displays are updated immediately after clicking *Auto Fetch*, *Delete*, or *Add*. The IR drops leading from the pad locations are displayed after running power analysis.

→ To open the Edit Pad Location form, click the *Create* button in the *Run Rail Analysis* form

Net Name:

Fetch Pad Location

☒ Auto Fetch

☐ Snap Distance: Lowest Snap Layer:

☐ Region On Layer:

x1: y1:

x2: y2:

xPitch: yPitch: ☐ Snap Distance:

☐ Specify:

☐ Instance Name: ☐ Instance Name and Pin:

☐ Cell Name: ☐ Cell Name and Pin:

☐ Honor Pin Connection

Pad Location List

Pad Location

X: Y: Layer:

Save File Format ☒ XY ☐ Padcell ☐ TSV ☐ Append to Existing File

Fields and Options

<i>Net Name</i>	The power net or ground net name of the pad cells. This field is required only when the file format is <code>Padcell</code>. Note: You can only edit the pad location file for one net at a time.
<i>Fetch Pad Location</i>	
<i>Auto Fetch</i>	Specifies to fetch all voltage sources. If you select this option, you must specify the net name.
<i>Snap Distance</i>	Specifies the distance of snapping distance constraint from IO/pad pins. The default value is 0. When selected, the software performs autofetch with snapping.
<i>Lowest Snap Layer</i>	Specifies the lowest layer used to fetch a DC source. Use a value such as M1 or M2. The default value is M5.
<i>Region</i>	
	Specifies a rectangular region that the current will be distributed within. A grid is created by using the <i>xPitch</i> and <i>yPitch</i> within the specified region. The current will be specified at each of the intersecting pitch lines. <i>Layers</i> specifies the metal layer that the current sink will be placed on. If you click <i>Draw</i>, and then select a box in the main window, the coordinate of this box will be automatically populated in the x1 y1 x2 y2. For region-based auto fetch, if the voltage source points that were fetched according to the start point and the specified region pitch (xPitch/yPitch) are not on the stripe, these points will be ignored during analysis. You can use the <i>Snap Distance</i> option to snap the voltage sources to a stripe within +/- snap distance/2. If no such stripe is found, this point will not be saved into the pad location file. As a result, the voltage sources are generated only on the stripes and these points will be saved to the pad location file.
<i>Specify</i>	Specifies an instance/cell and their pins to autofetch DC sources.
<i>Instance Name</i>	Specifies the instances from which voltage sources are determined.
<i>Instance Name and Pin</i>	Specifies the instances and their pins from which voltage sources are determined.
<i>Cell Name</i>	Specifies the cells from which voltage sources are determined.

<i>Cell Name and Pin</i>	Specifies the cells and their pins from which voltage sources are determined.
<i>Honor Pin Wire Connection</i>	Honors pin wire connection when fetching voltage source location from a power pad. When you select this option, only pins with wire connection are output as voltage sources.
<i>Fetch</i>	Automatically populates the <i>Pad Location List</i> with the pad locations based on properties defined in the LEF file. There are three ways to get the dc sources: • fetch all voltage sources - specifying <i>Net Name</i> and <i>Auto Fetch</i> • fetch the voltage sources within the specified region of the specified layer controlled by the specified xPitch/yPitch • fetch voltage source for the specified cell/cell pin/instance/instance pin Note: You may need to reissue the globalNetConnect command for <i>Auto Fetch</i> to function properly if the power pads were not defined in the database at the time that the globalNetConnect command was first issued.
<i>Pad Location List</i>	The list of coordinates for pad cells. To populate this field manually, click the <i>Add</i> button. To populate this field automatically, click the <i>Auto Fetch</i> button.
<i>Pad Location</i>	The X and Y coordinates of a pad cell.
<i>Layer</i>	The layer that contains the pad cell.
<i>Get coord</i>	Populates the X and Y coordinates in the <i>Pad Location</i> field. Click this button, then click a location in the design display area.
<i>Delete</i>	Removes one or more pad location items from the <i>Pad Location List</i> . Highlight the item be removed, then click this button. To highlight multiple items, click them while pressing either the Shift or Control key.
<i>Add</i>	Adds the X and Y coordinates of a pad cell to the <i>Pad Location List</i> .
<i>Save File Format</i>	The format of the saved Pad Location File. The choices are <i>XY file</i> (xy coordinates) , <i>Padcell file</i> (VoltageStorm), and <i>TSV</i> . You can create and save the voltage sources location file with a unique name for the voltage sources (both VDD and VSS).

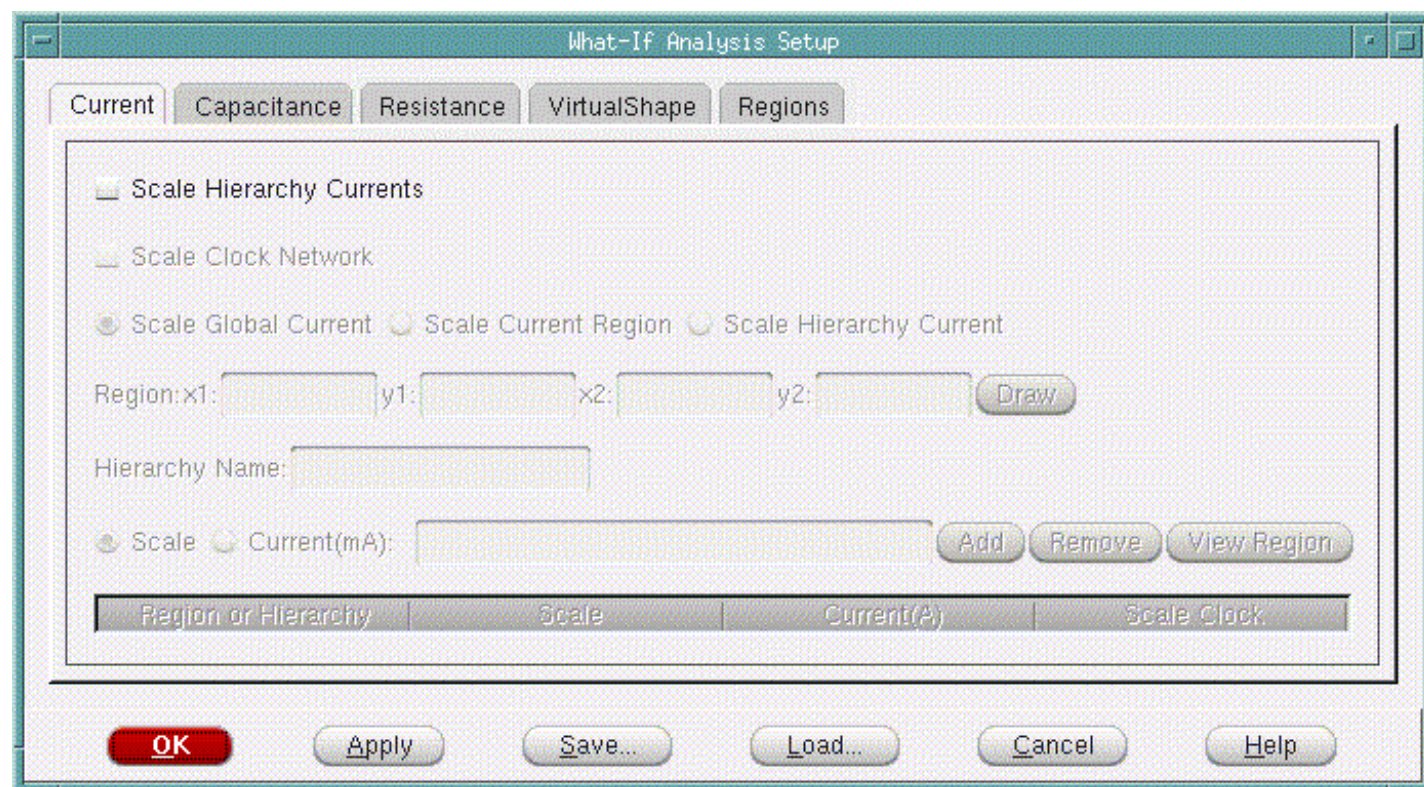
<i>Load</i>	Opens the Load Pad Location File form. Select a file from this list, then click <i>Open</i> to display the pad locations in the <i>Pad Location List</i> .
<i>Save</i>	Opens the Save Pad Location File form. Specify a location for the pad location file, then click <i>OK</i> . This saves the pad location file, clears the <i>Pad Location List</i> , and closes the Save Pad Location form.
<i>Clear</i>	Clears the <i>Pad Location List</i>. Once the <i>Pad Location List</i> is cleared, you can begin creating a pad location file for a new net. Note: Cadence recommends that you use the <i>Clear</i> button after you save the pad location file before you use <i>Auto Fetch</i> again.

What If Analysis Setup

→ Choose *Power & Rail > Run Rail Analysis > Advanced* form, click on the *What-If Analysis - Setup* button.

<i>Current</i>	Use this form to scale current to do what-if analysis for a hierarchical partition and assess its effect on IR drop.
<i>Capacitance</i>	Use this form to scale the intrinsic cell capacitance value to determine how much additional capacitance can be added in a region to reduce dynamic IR drop.
<i>Resistance</i>	Use this form to scale resistance globally or in a region to determine how much resistance can be modified in a region to reduce the entire IR drop.
<i>Virtual Shape</i>	Use this form to add, select, deselect, delete, list, and display the what if wires/vias to a design.
<i>Regions</i>	Use this form to specify static or dynamic current regions on global power-grids over the placed macro to accurately capture high and low current consuming regions during rail analysis.

Current



Fields and Options

<i>Scale Hierarchy Currents</i>	Specifies to scale hierarchical currents for all instances within the given hierarchy.
<i>Scale Clock Network</i>	Specifies to scale instances that belong to the clock network and scale currents for the rest of the instances within the given hierarchy to the specified value. You cannot specify this option with <code>-region x1 y1 x2 y2</code> or <code>-global</code> options. <i>Default:</i> Not selected
<i>Scale Global Current</i>	Specifies current scaling to be applied to the entire design.
<i>Scale Current Region</i>	Specifies current scaling to be applied to the specified current region in the design.
<i>Scale Hierarchy Current</i>	Specifies current scaling to be applied to the specified hierarchical block.
<i>Region</i>	Specifies a region where current scaling is to be applied. The default unit of <code>x</code> and <code>y</code> coordinates is in microns (um).
<i>Hierarchy Name</i>	Specifies the name of the hierarchical block. For example, <code>-hierarchy top/blockA</code> . To specify global scaling for the design use <code>" / "</code> . For example, <code>-hierarchy /</code> This is a required option.
<i>Scale</i>	Specifies any value greater than 0. For example, a value of 0.5 will scale the resistance in half. This is a required option.
<i>Current (mA)</i>	Specifies current value in milliAmperes. This is used as target peak current for the given hierarchy and is then scaled appropriately. You cannot specify this option with <code>-region x1 y1 x2 y2</code> or <code>-global</code> options.

Capacitance

The screenshot shows the 'What-If Analysis Setup' dialog box with the 'Capacitance' tab selected. The dialog has a title bar and standard window controls. Below the title bar are five tabs: 'Current', 'Capacitance' (selected), 'Resistance', 'VirtualShape', and 'Regions'. The main area contains several options and input fields:

- ☐ Scale Capacitance
- ☐ Intrinsic: [text field] — Loading: [text field] — Grid: [text field]
- ☒ Global
- ☐ Region: [Draw] x1 [text field] y1 [text field] x2 [text field] y2 [text field] [Add] [Remove]

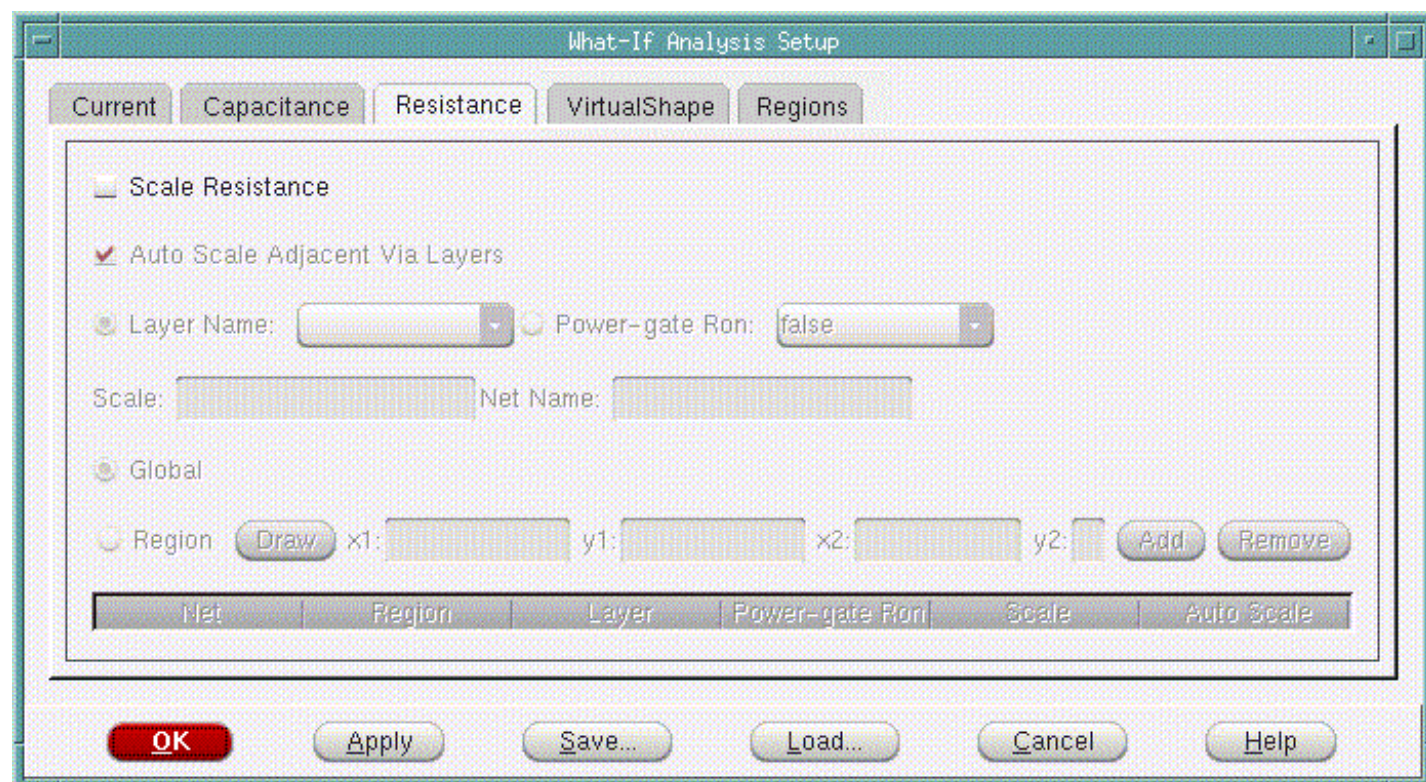
Below these options is a table with four columns: 'Region', 'Intrinsic Cap', 'Loading Cap', and 'Grid'. The table is currently empty.

At the bottom of the dialog are six buttons: 'OK' (highlighted in red), 'Apply', 'Save...', 'Load...', 'Cancel', and 'Help'.

Fields and Options

<i>Scale Capacitance</i>	Specifies to scale capacitance.
<i>Intrinsic</i>	Scales instance internal gate capacitance to the specified scale factor. The default scale value is 1.
<i>Loading</i>	Scales loading capacitance of the instance to the specified scale factor. The default scale value is 1.
<i>Grid</i>	Scales power-grid capacitance of the design to the specified scale factor. The default scale value is 1.
<i>Global</i>	Scales the specified capacitances for the entire design.
<i>Region</i> • <i>Draw</i>	Scales the specified capacitances in the region. The default unit of $x1\ y1\ x2\ y2$ is in database units.

Resistance



The image shows the 'What-If Analysis Setup' dialog box with the 'Resistance' tab selected. The dialog has a title bar and a close button. Below the title bar are five tabs: 'Current', 'Capacitance', 'Resistance' (selected), 'VirtualShape', and 'Regions'. The main area contains the following controls:

- ☐ Scale Resistance
- ☒ Auto Scale Adjacent Via Layers
- Layer Name: Power-gate Ron:
- Scale: Net Name:
- ☒ Global
- ☐ Region x1: y1: x2: y2:

At the bottom of the main area is a table with the following headers: Net, Region, Layer, Power-gate Ron, Scale, and Auto Scale. The table is currently empty.

At the bottom of the dialog are six buttons: **OK** (highlighted in red), Apply, Save..., Load..., Cancel, and Help.

Fields and Options

<i>Scale Resistance</i>	Specifies to scale resistance.
<i>Auto Scale Adjacent Via Layers</i>	Specifies to use auto via scaling for what-if resistance analysis.
<i>Scale</i>	Specifies any value greater than 0. For example, a value of 0.5 will scale the resistance in half. This is a required option.
<i>Layer Name</i>	Specifies the layer name (LEF layer or technology layer) on which the scaling is performed. Multiple commands can be issued to scale resistance on multiple layers. This is a required option.
<i>Net Name</i>	Scales resistance of the specified net in the domain which is being analyzed. This is a required option.
<i>Global</i>	Scales the resistance to the specified value for the entire design for a given layer.
<i>Region</i> • <i>Draw</i>	Scales the resistance in the specified region. The default unit of $x1\ y1\ x2\ y2$ is in database units.

Virtual Shape

What-If Analysis Setup

Current Capacitance Resistance **VirtualShape** Regions

☐ Create Virtual Shape

Operation

☒ Add ☐ Delete ☐ Select
☐ Deselect ☐ Display ☐ List

Setting

☒ Type : ☒ Wire ☐ Via
☒ Net :
☒ Layers: Top Bottom
☒ Area : x1 y1 x2 y2
☐ Pitch_Width Pitch Width ☐ Direction ☐ Method

Objects

☒ All ☐ Selected

☐ Save Virtual Shape File
☐ Append to Existing File

Fields and Options

Create Virtual Shape	Creates virtual shapes (wires/vias) to a design.
----------------------	--

<i>Operation</i>	• <i>Add</i> - Specifies to add what-if wires/vias. • <i>Delete</i> - Specifies to delete all or selected what-if shapes by constraint <i>Net/Layers/Type</i>. You can use this field in conjunction with the <i>Objects</i> field. • <i>Select</i> - Specifies to select all what-if shapes by constraint <i>Net/Layers/Type</i>. Use this field in conjunction with <i>Area</i> to select what-if shapes overlapping with the specified area. • <i>Deselect</i> - Specifies to deselect all what-if shapes by constraint <i>Net/Layers/Type</i>. Use this parameter in conjunction with <i>Area</i> to deselect what-if shapes overlapping with the specified area. • <i>Display</i> - Invokes the GUI to show what-if shapes and zooms to the selected shapes. • <i>List</i> - Specifies to list all or selected what-if shapes by constraint <i>Net/Layers/Type</i>. You can use this field in conjunction with the <i>Objects</i> field.
<i>Type</i>	Specifies what kind of shape is to be created.
<i>Net</i>	Specifies the name of the net for which what-if wires/vias is to be created.
<i>Layers</i>	Specifies the LEF layer name for wires, or layer pair for via to be added.
<i>Area</i>	Specifies the rectangular region in which the what if wires/vias are to be generated.
<i>Pitch Width</i>	Specifies the width of what-if wires and the distance between the centre line of two adjacent what-if wires (pitch) to be drawn in the specified <i>Area</i> in the preferred direction of the specified layer. The direction of created wires can also be controlled using the <i>Direction</i> field. If this field is not provided, a single rectangular shape of size {llx,lly,urx,ury} is created.
<i>Direction</i>	Controls the direction of the wires added with the <i>Pitch Width</i> field.
<i>Method</i>	Specifies the method used to add vias: • Auto: When area and layer pair are specified, the tool automatically adds vias on all intersecting points on the specified layer pairs in the given area. • Exact: When an area and layer pair are specified, the tool adds a via of exact size of the specified area between the specified layer pair.

<i>Objects</i>	Specify <i>Objects</i> in conjunction with the <i>Delete</i> or <i>List</i> fields to delete or list the selected what-if shapes. You can also delete or list all what-if shapes. The default value is <i>All</i> .
<i>Save Virtual Shape File</i>	Specifies to save all the in-memory what-if wires/vias to the specified file.
<i>Append to Existing File</i>	Specifies to append all what-if wires/vias to the existing file specified with the <i>Save Virtual Shape File</i> field.
<i>Load Virtual Shape Results</i>	Loads the saved files of all vias/wires from the <i>What-If Analysis Setup</i> form and results are displayed in the design layout.

Regions

The screenshot shows the 'What-If Analysis Setup' dialog box with the 'Regions' tab selected. The dialog has a title bar and a close button. Below the title bar are five tabs: 'Current', 'Capacitance', 'Resistance', 'VirtualShape', and 'Regions'. The 'Regions' tab is active and contains the following elements:

- A checkbox labeled 'Create Region'.
- A text field for 'Static Current (mA) / Dynamic PWL (ns,mA):'.
- A text field for 'Layer:'.
- Text fields for 'Intrinsic Cap (pF):' and 'Loading Cap (pF):'.
- A 'Region:' label followed by a 'Draw' button and four coordinate text fields: 'x1:', 'y1:', 'x2:', and 'y2:'.
- 'Add' and 'Remove' buttons.
- A table with five columns: 'Region', 'Layer', 'Current', 'Intrinsic Cap', and 'Loading Cap'.
- A row of buttons at the bottom: 'OK' (red), 'Apply', 'Save...', 'Load...', 'Cancel', and 'Help'.

Region	Layer	Current	Intrinsic Cap	Loading Cap
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Fields and Options

<i>Create Region</i>	Creates current regions on global power-grids.
<i>Static Current / Dynamic PWL</i>	<i>Static Current</i>: Specifies a single current value in mAmp. <i>Dynamic PWL</i>: Specifies a PWL current waveform in time (in ns) and current (in mA) format. For example, 0ns 0mA 0.5ns 1mA 1ns 2mA 3ns 1mA 4ns 0.
<i>Layer</i>	Specifies a layer name on which the scaling will be performed. This is a required option.
<i>Intrinsic Cap</i>	Specifies the cell intrinsic capacitance in pico farads (pf). This is required option when set in the dynamic mode (<code>set_rail_analysis_mode -method dynamic</code>). This option is disabled in static mode.
<i>Loading Cap</i>	Specifies the cell loading capacitance in pico farads (pf). This is required option when set in the dynamic mode (<code>set_rail_analysis_mode -method dynamic</code>). This option is disabled in static mode.
<i>Region</i> <i>Draw</i>	Specifies the region on the layer across which the specified current and capacitance (in dynamic mode) is distributed by the rail analysis engine. This is a required option.

Related topics

See [scale_what_if_current](#), [scale_what_if_capacitance](#), [scale_what_if_resistance](#), [create_what_if_shape](#), [create_current_region](#) commands in the *Rail Analysis Commands* chapter in the *Voltus Text Command Reference*.

Run Resistance Analysis

→ Choose *Power Rail - Run Resistance Analysis*.

Basic	Performs effective resistance calculation for all instances in a design. Effective resistance computes effective resistance between two nodes on the power-grid, or from a node to all the voltage sources in the design, or from an instance to all the voltage sources in the design.
--------------	---

Basic

Run Resistance Analysis

Basic

Analysis Method ☐ Effr

☒ Net Based ☐ Domain Based Domain Name:

Power Net(s): Voltage(s): (V) Threshold: (V)
 Ground Net(s): Voltage(s): (V) Threshold: (V)

☒ Power Pads: ☒ Def Pins ☐ Pad File ☐ XY File ☐ Boundary File

File Name: Net Name:

Type	Net	File
------	-----	------

Output Directory: Output Report Name:

☐ Effr To Power Pads ☒ Point-To-Point Effr

Node1: x1: y1: Layer1: Label:
 Node2: x2: y2: Layer2:

☐ Region: x1: y1: Nodes: 100
 x2: y2: Layer:

☐ Node File: ☐ Node-Pair File:

☐ AutoZoom

PASS/FAIL	Effr	Label	x1	y1	Layer1	x2	y2	Layer2
-----------	------	-------	----	----	--------	----	----	--------

Instance1: Pin1: Instance2: Pin2:

☐ Instance File: ☐ Instance-Pair File:

☐ AutoZoom

PASS/FAIL	Effr	Instance1	Instance2	Pin1	Pin2
-----------	------	-----------	-----------	------	------

Effr Threshold(Ohms): ☐ Force Reextraction ☐ Append

Fields and Options

<i>Analysis Method</i>	Specifies the resistance analysis method as <i>Efr</i> .
<i>Net Based</i>	Set if net-based analysis is being run. It is equivalent to the <code>set_pg_nets</code> command.
<i>Domain Based</i>	Set if domain-based analysis is being run. It is equivalent to the <code>set_rail_analysis_domain</code> command.
<i>Domain Name</i>	Specifies the domain name that is to be analyzed. Only used for domain based analysis.
<i>Power Net(s)</i>	Specifies names of the power nets for which resistance analysis will be performed.
<i>Voltage(s)</i>	The voltage of the power net.
<i>Threshold</i>	The voltage threshold of the power net
<i>Ground Net(s)</i>	Specifies names of the ground nets for which resistance analysis will be performed.
<i>Voltage(s)</i>	The voltage of the ground net.
<i>Threshold</i>	The voltage threshold of the ground net
<i>Power Pads</i>	• <i>Def Pins</i> - specifies that voltage sources will be determined by power pins. • <i>Pad File</i> - specifies that voltage sources will be determined by pad cells given in the file. • <i>XY File</i> - specifies that voltage sources will be specified by x y coordinates. • <i>Boundary File</i> - specifies that voltage source locations will be specified by information in the boundary file. <i>Default: Def Pins</i>
<i>File Name</i>	The name of the power file containing the locations of the power rail voltage sources (VDD, VSS). You can automatically generate this file by clicking the <i>Create</i> button. The Edit Pad Location form appears.

<i>Net Name</i>	The name of the power net. The <i>Add</i> button adds the net to the list. The <i>Remove</i> button removes the net from the list.
<i>Output Directory</i>	Specifies the name of the directory in which the output report file is saved.
<i>Output Report Name</i>	Specifies the name of the output report file that contains the report header and the sorted list for resistance values.
<i>Effr To Power Pads</i>	Specifies to check effective resistance of a node to the voltage sources or power pads. This option is enabled only when <i>Power Pads</i> is selected. When this option is selected, <i>Node2</i> and <i>Instance2</i> options are disabled.
<i>Point-To-Point Effr</i>	Specifies to check effective resistance between two nodes on the power-grid. When this option is selected, <i>Node2</i> and <i>Instance2</i> options are enabled.
<i>Node1</i>	Specifies the list of user-defined nodes for effective resistance calculation between the specified user-defined node and all voltage sources, or between two nodes on the power-grid. This option is enabled for both <i>Effr To Power Pads</i> and <i>Point-To-Point Effr</i>. You can add a list of nodes based on the region on a given layer. • <i>x1</i> and <i>y1</i> are the coordinates of the node. • <i>Layer1</i> is the generic metal routing layer name on which the node is located. • <i>Label</i> is any name chosen by you to identify node entries in the list. This is optional but helps during debugging. • Get
<i>Node2</i>	Specifies the second node for effective resistance calculation between two nodes on the power-grid. This option is enabled only for <i>Point-To-Point Effr</i>. • <i>x2</i> and <i>y2</i> are the coordinates of the second node. • <i>Layer2</i> is the generic metal routing layer name on which the second node is located. • Get

Region	Analyzes resistance in the specified region. You can draw a box on a specific layer, and specify the number of nodes to be selected (maximum limit of 100 nodes). This field allows you to compute effective resistance on a given layer across the chip and determine how the resistance is changing for the design from top to bottom layer. You can assign a name for the region. • <i>x1</i>, <i>y1</i>, <i>x2</i>, and <i>y2</i> are the coordinates of the box. • <i>Nodes</i> are the number of nodes to be selected • <i>Layer</i> is the generic metal routing layer name on which the nodes are located. • Get Region
Node File	Specifies the name of the node list file. This file contains the list of user-defined nodes in the same format as mentioned for the <code>-node_list</code> parameter of the <code>analyze_resistance</code> command. The file allows you to specify a large number of nodes.
Node-Pair File	Specifies the name of the node pair list file. This file contains the node pair list in the same format as mentioned for the <code>-node_pair_list</code> parameter of the <code>analyze_resistance</code> command. The file allows you to specify a large number of node pairs.
Instance1	Specifies the list of instances for which effective resistance is to be calculated. This option is enabled for both <i>Effr To Power Pads</i> and <i>Point-To-Point Effr</i>.
Pin1	Optionally, you can specify instance pin name. By default, the tool will use the instance pin associated with the net being analyzed. Pin name can be used in special cases when two different pins of an instance are connected to the same net.
Instance2	Specifies the second instance to enable effective resistance calculation between two instances in the design. This option is enabled only for <i>Point-To-Point Effr</i>.
Pin2	Specifies the pin name for the second instance.
Instance File	Specifies the name of the instance list file. This file contains the list of instances in the same format as mentioned for the <code>-instance_list</code> parameter of the <code>analyze_resistance</code> command. The file allows you to specify a large number of instances.

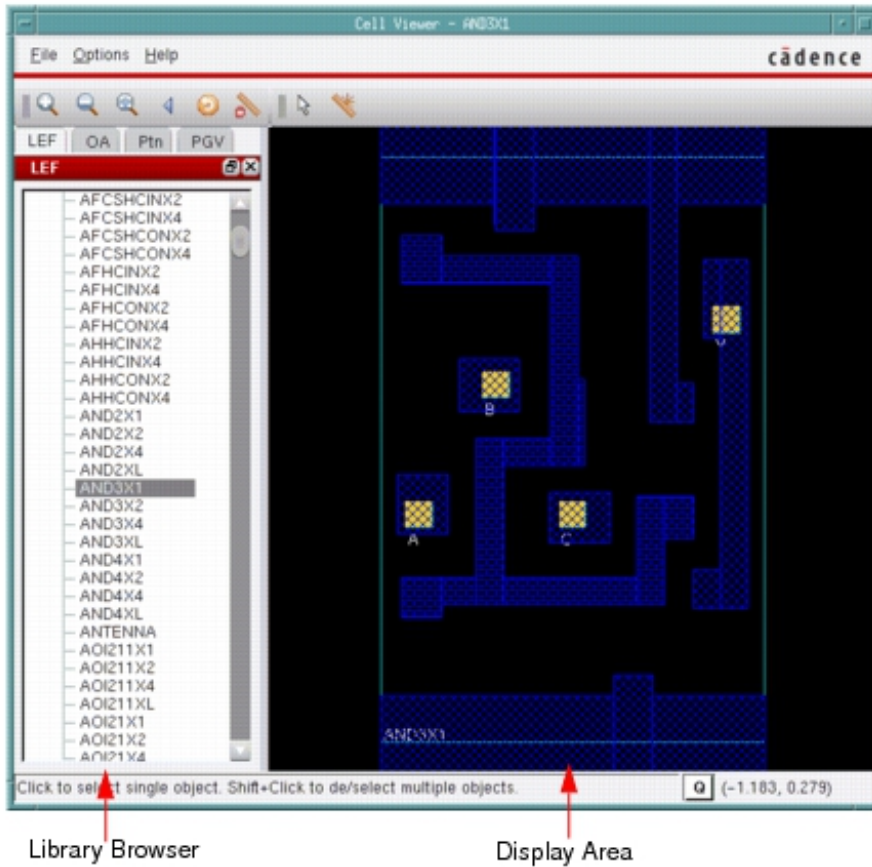
<i>Instance-Pair File</i>	Specifies the name of the instance pair list file. This file contains the instance pair list in the same format as mentioned for the <code>-instance_pair_list</code> parameter of the <code>analyze_resistance</code> command. The file allows you to specify a large number of instance pairs.
<i>Effr Threshold (Ohms)</i>	Specifies the minimum allowed effective resistance to display node/node-pair/instance/instance-pair that have a resistance value above this threshold value.
<i>Force Re-extraction</i>	Specifies to force a re-extract of resistances.
<i>Append</i>	When you select <i>Append</i> and click the <i>Load</i> button, the content of the effective resistance analysis result file is appended to the previous list in the node/instance list box. If <i>Append</i> is not selected, the previous list is cleared.
<i>Add</i>	Adds the specified node/node-pair/instance/instance-pair to the list box.
<i>Remove</i>	Removes the selected node/node-pair/instance/instance-pair to the list box.
<i>Auto Zoom</i>	Specifies to automatically zoom in to the selected node/instance in the list box.
<i>AutoFilter</i>	Specifies to show the nodes or instances with eight color filter. When you specify <i>Effr Threshold (Ohms)</i> and click <i>AutoFilter</i> , the layout displays the nodes or instances with eight color filter. When you change the threshold value, you will get different plots on the layout.

Cell Viewer

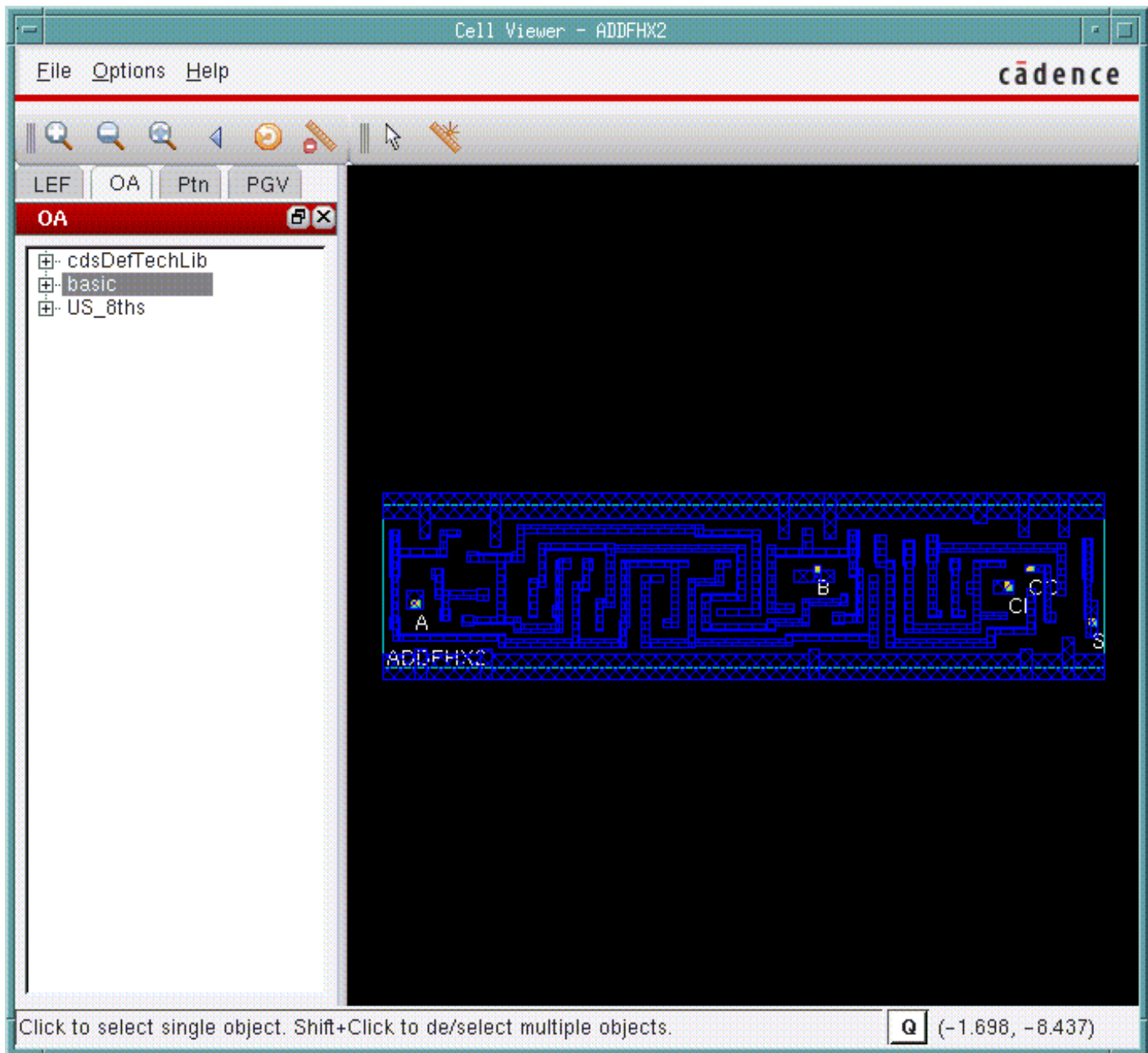
→ Choose *Tools - Library Cell Viewer*

<i>LEF</i>	Allows to view the details of library cells using the information in the LEF file.
<i>OA</i>	Allows to view the details of OpenAccess cells using the information in the OA library.
<i>Ptn</i>	Allows to display the abstract of any partition in the design.
<i>PGV</i>	Allows to view and debug power-grid libraries.

LEF



OA

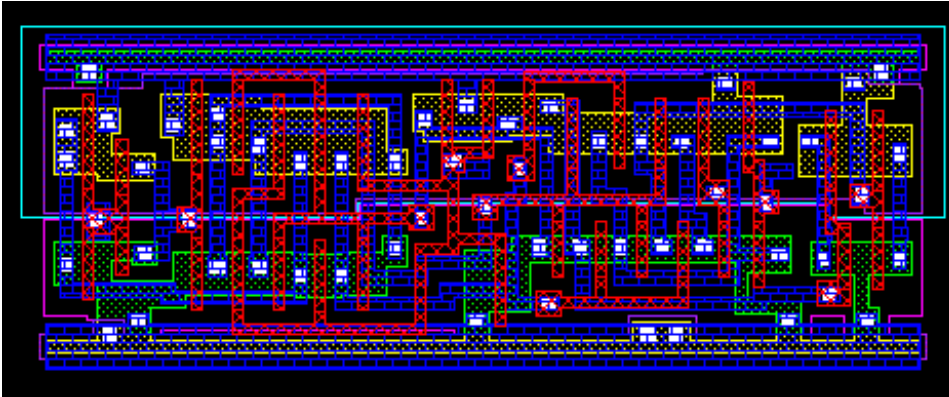


You can view an OA cell in two modes:

- Layout
- Abstract

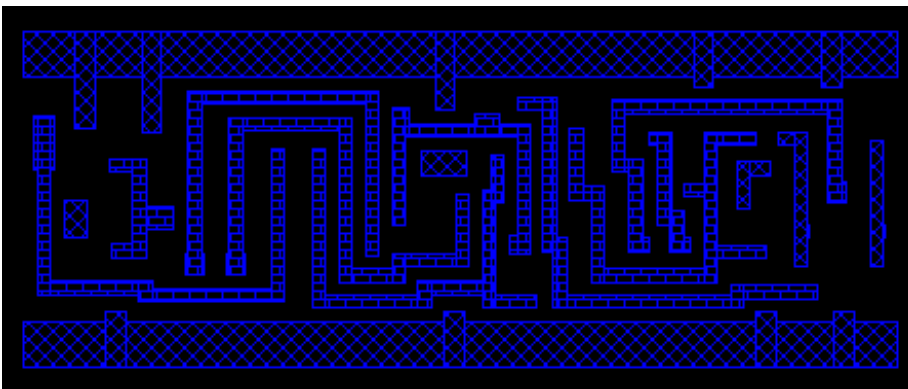
OA-Layout View

→ In the Library Browser, select the OA cell, and then click Layout.



OA-Abstract View

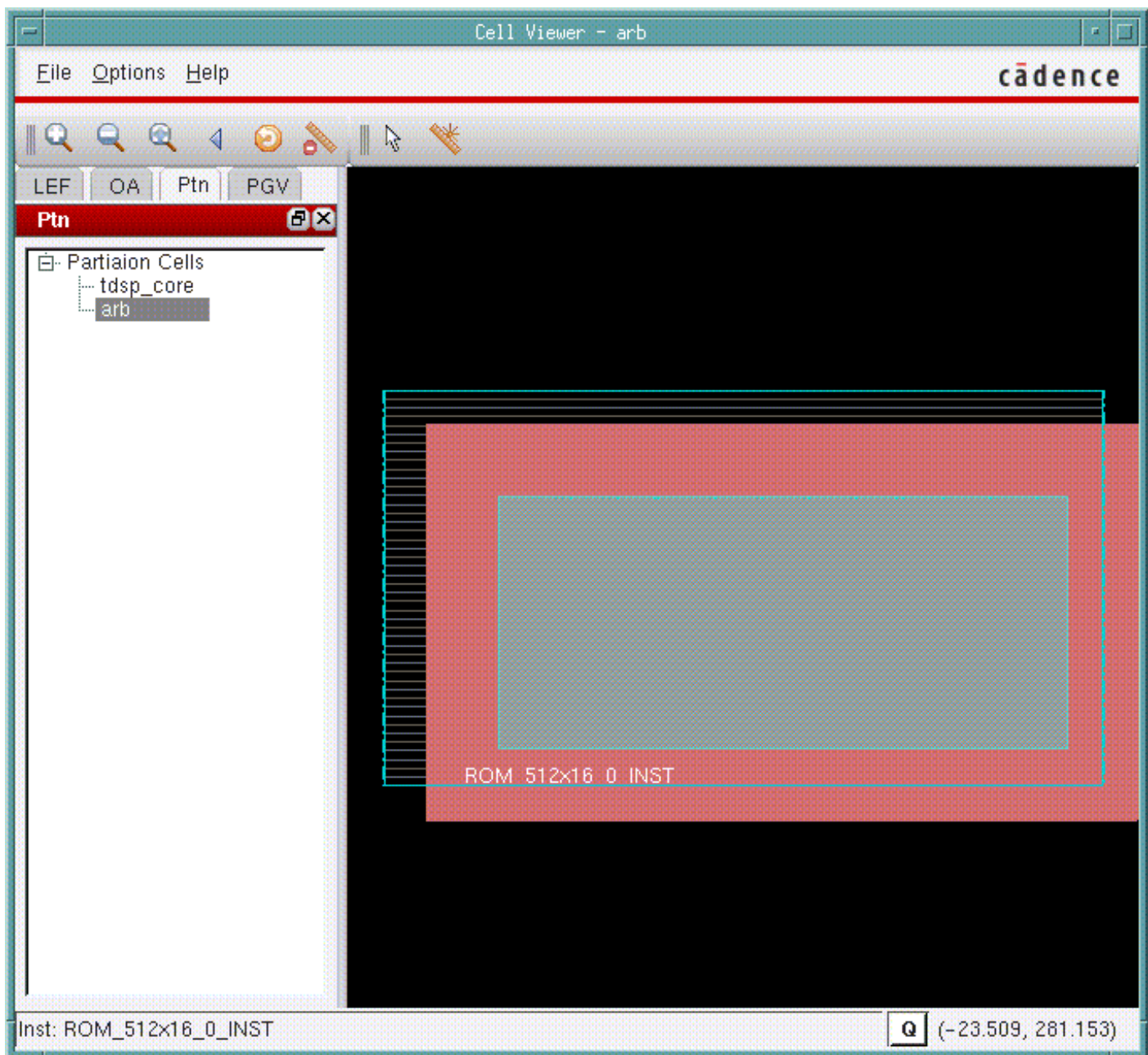
→ In the Library Browser, select the OA cell, and then click Abstract.



Ptn

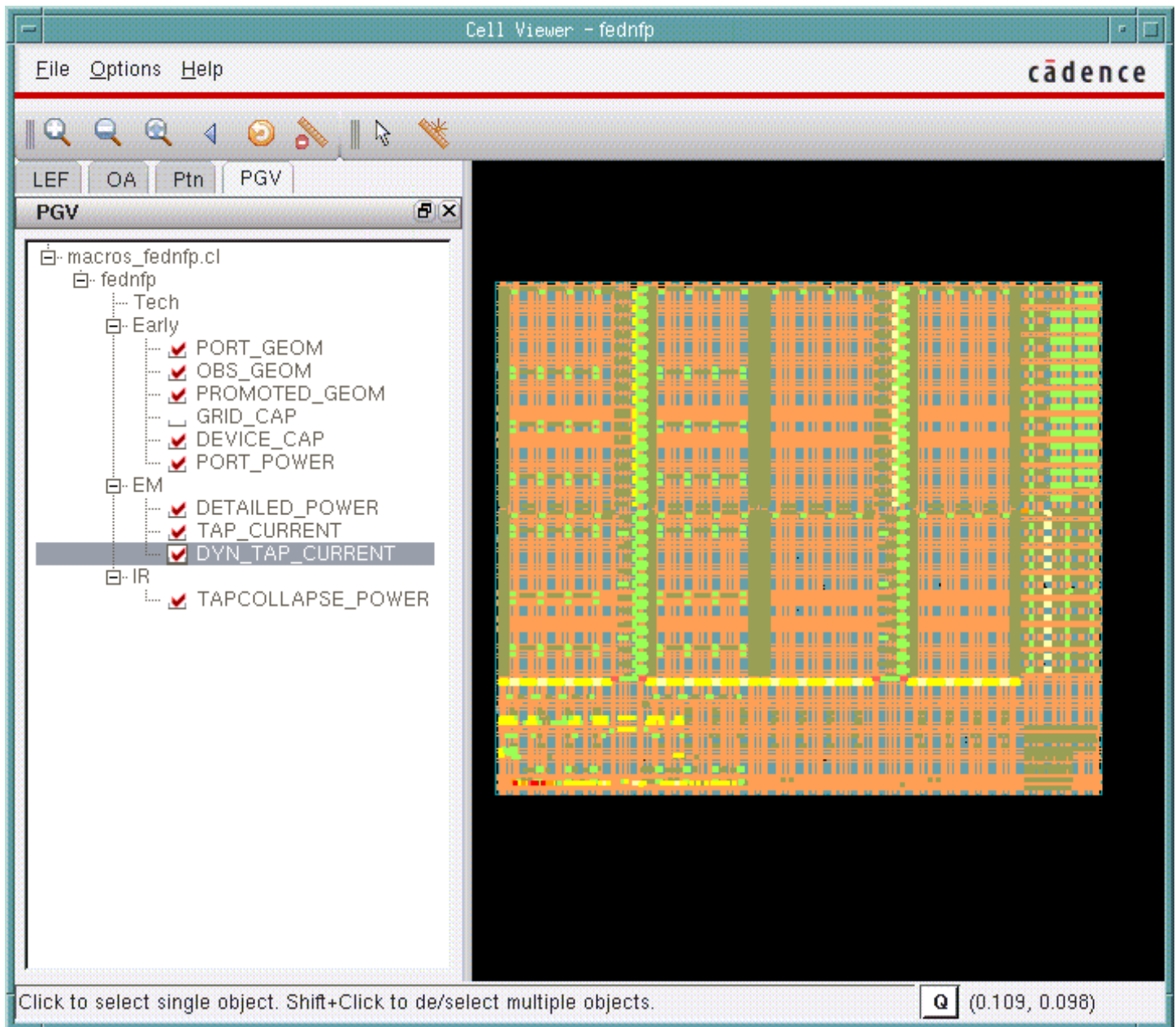
Use the Ptn page to display the abstract of any partition in the design. You can navigate to a partition through the Library Browser in the left side of the Cell Viewer.

→ Choose *Tools - Cell Viewer*, and click on the Ptn page.



PGV

You can navigate to a cell using the Library Browser, which is in the left side of the Cell Viewer. The details of the cell are displayed in the Display Area, which is in the right side of the Cell Viewer. You can expand a cell to view the default views corresponding to each PGV type (Tech, Early, EM, and IR).



This form allows you to take a physical layout and overlay it with a power-grid resistor network extracted by Power-Grid Library Generator.

Fields and Options

<i>File</i>	The <i>File</i> menu provides the following options:
<i>Open Power Grid Library</i>	Opens multiple power grid libraries.
Free PGV Database	Specifies to free all the loaded power-grid libraries.
<i>Quit</i>	Closes the Cell Viewer window.
<i>Options</i>	
Net Based View - Show Popup Automatically	
	If this menu item is selected, clicking on a view displays a dialog box automatically. This dialog box allows you to display the power-grid of a single net. By default, this menu item is not selected. The dialog box displays the power, ground, and signal nets. You can select a net and click <i>Plot</i> to view a single net. Alternatively, you can view this dialog box using the following method: 1. Right-click on a view to see the short-cut menu. 2. Click the view name option on the shortcut menu. Note: Clicking on the cell name returns to the whole view mode, and displays the selected views for the cell name.

PGV Cell Views

You can click the + button to expand a cell and view the default cell views. Each view has a check box for displaying the content specific to the selected view. You can view either one or multiple views overlapped over each other.

The default views for different power-grid view types are:

Tech

- Displays the basic port information of a cell.

Early

- Geometrical View - displays the LEF of a cell. The types of geometrical views are: PORT_GEOM, OBS_GEOM, PROMOTED_GEOM and OPTIMIZED_GEOM views.
- Port View - displays more accurate port and tap information of a cell when . This information is available in the PORT_POWER view of the library.
- Capacitance Views - displays the grid and device capacitance information of a cell. The types of capacitance views are: GRID_CAP and DEVICE_CAP views. These views are net specific.

EM

- Detailed View - displays the detailed power-grid information for all nets. This information is available in the DETAILED_POWER view of the library.
- Tap Current View - displays tap information of a cell. All the tap currents are shown in the viewer drawn in 8 colors for 8 linear filters according to the min max value. This information is available in the TAP_CURRENT view of the library.
- Dynamic Tap Current View - displays dynamic tap currents for each interval. The total current plot is net based, and supports multiple trigger view in a single PGV. This information is available in the DYN_TAP_CURRENT view of the library.
- Powergate View - displays the power switch information of a power-gate cell. This information is available in the POWERGATE view of the library.

IR

- Collapsed View - displays the collapsed current taps for the bit-cells inside the memory. This information is available in the TAPCOLLAPSE_POWER view of the library.

Cell Viewer Widgets

The Cell Viewer supports the following widgets: Zoom In, Zoom Out, Fit, View Last, Redraw, Clear All Rulers, Select and Ruler.



Auto Query

The Cell Viewer window includes an Auto Query (Q) button located at the bottom of the design display area that enables and disables the auto query. When you enable auto query, text information displays for any object that is under the cursor. You can use auto query for information on cell instances, instance pins, and resistors.

Power and Rail Plots

The Power & Rail Plots form is the main console window for displaying the power and rail analysis results. The data can be loaded into GUI using the *Power & Rail Setup* form accessed by clicking the *DB Setup* button. The displayed nets and layers selection can be changed using the *Power & Rail Layers/Nets* form accessed by clicking the *Layers/Nets* button. The rest of the display can be configured on the main window.

→ [Choose Power Rail - Power Rail Plots.](#)

Type: ☐ Power ☒ Rail ☐ Cap ☐ Sem

ivdd - Domain based Instance Voltage ▼

☐ Result Browser

☐ Voltage Source

☐ Percentage Range

☐ Legend ne ▼

RLRP Path Trace Clear

DB Setup Layers/Nets

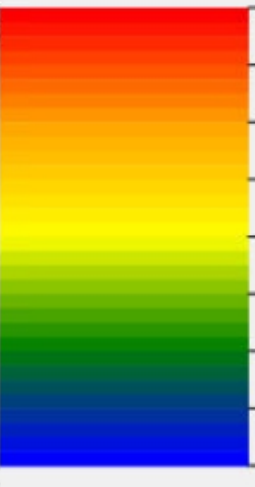
Measure

☒  186.26 mV 186.3 ▼

1.4297 ☒  1.43 ▼

☐ Auto Apply for Color Scale

Apply Basic Reset

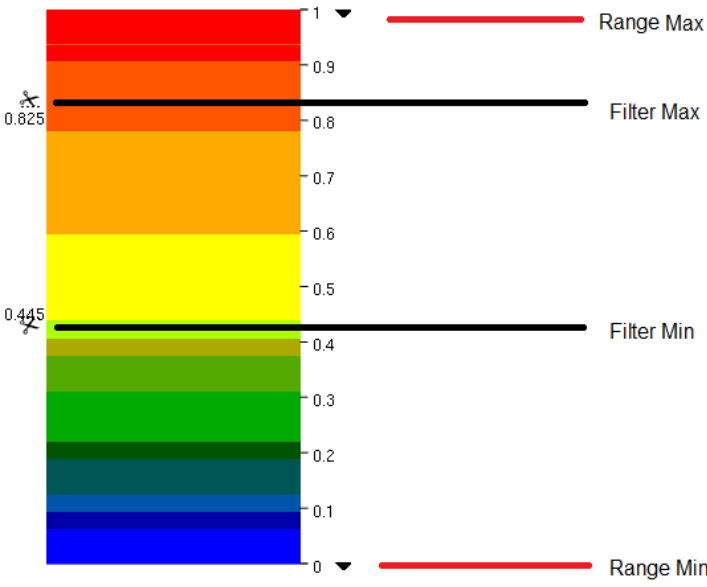


A vertical color scale bar representing voltage levels. The top is red (186.3 mV) and transitions through orange, yellow, green, and blue to the bottom (1.43 mV). Tick marks on the right side indicate values: 186.3, 163.2, 140, 116.9, 93.84, 70.74, 47.64, 24.53, and 1.43.

Fields and Options

<i>Type</i>	• <i>Power</i> - Plots the power analysis plot type. • <i>Rail</i> - Plots the rail analysis plot type. • <i>Cap</i> - Plots the capacitance plot type. • <i>Sem</i> - Plots the signal electromigration (EM) plot type. You can select a power, rail, capacitance, or signal EM plot type from the drop-down list. For a description of the various plot types, see the <code>set_power_rail_display</code> command.
<i>Result Browser</i>	Displays the Results Browser form.
<i>Voltage Source</i>	Displays voltage sources (power pads) using white pixels in the layout canvas.
<i>Percentage Range</i>	Displays color gradients for the following plots by percentage: • IR Drop • IVDD • IVDN By default, the Voltus GUI displays color gradients by the voltage value, and these gradients are applied to all rails/instances with varying voltage levels (rails/instances with max voltage value will be red and with min voltage value will be blue). When this option is selected, the GUI displays color gradients by percentage of voltages. The percentage value is based on the nominal voltage value of each rail/instance. When the <i>Percentage Range</i> option is selected, the GUI displays high percentage voltages in red and low percentage voltages in blue. This option allows you to be more intuitive when analyzing these plots.

<i>RLRP Path</i>	You can select an object (instance or resistor) and click the <i>Trace</i> button to highlight the least effective resistance path from the object to the nearest voltage source, and to open the <i>Resistance Path</i> form. The rail analysis must include RLRP analysis (<code>set_rail_analysis_mode -enable_rlrp_analysis true</code>) in order to have the required data for this functionality. You can click the <i>Clear</i> button to clear the path. The <i>Result Browser</i> will list each instance's total resistance, and the <i>Resistance Path</i> form will list the break down of each resistance in the rlrp path. You can select a resistor in this form, single click to select/zoom, and double click to query. Note: If the <i>Resistance Path</i> form is closed, you can open it again by first selecting the object that you want to trace, and then clicking the <i>Trace</i> button.
<i>Legend</i>	Displays the filter color and range details. When you select a location (<code>ne nw se sw</code>), this opens the legend at the specified location on the design layout. You can clear the checkbox to close the legend. Note: Legend is useful when dumping the screen capture to save the color range information.
<i>DB Setup</i>	Opens the Power & Rail Setup form that allows you to load power/rail/signal EM analysis results into GUI.
<i>Layer/Nets</i>	Opens the Power & Rail Layers/Nets form that allows you to control the visibility of the layers, the opacity of the rail analysis or design database, the visibility of power/ground nets, and visibility of the signal EM layers.
<i>Measure</i>	Provides the ability to measure tap current and capacitance of the specified region. The use model is as follows: 1. Click the <i>Measure</i> button. 2. Select a region in the main window. 3. Look for the results in the console window.

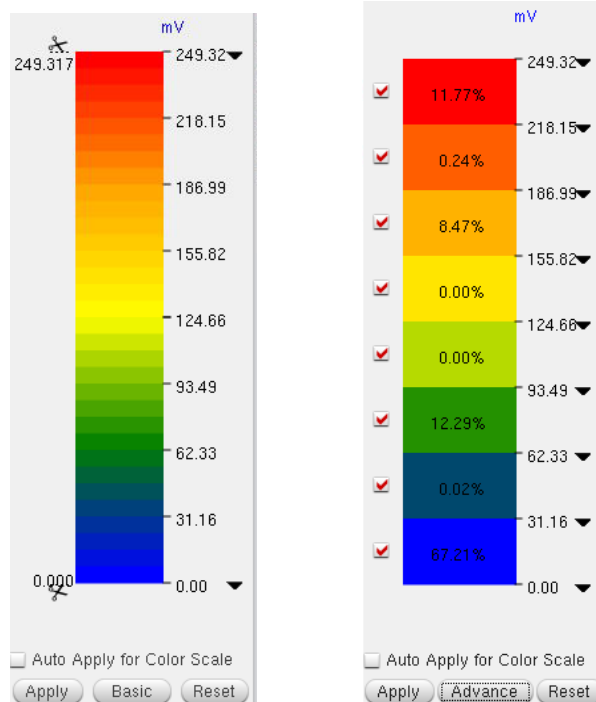
Range	When a plot type is selected, the filter values automatically get filled in.  • Range Min and Max - Click the ▼ icon to specify the minimum and maximum value to report. When the filter is at the default (max/min) position, any values greater than “Range Max” will take on the range max color, and vice-versa for “Range Min”. • Filter Min and Max - Click the ✂ icon to specify the minimum and maximum value to include in the report/display. Use the scissor icon to further filter out the data within the range specified. This is used to obtain statistic of filter range. “Filter Max” will hide any data point greater than the filtered value, and vice-versa for “Filter Min”.
<i>Auto Apply for Color Scale</i>	Allows you to to automatically redraw the layout after modifying the range values. If you do not select this checkbox, you can click the <i>Apply</i> button to manually redraw the layout.

Basic/Advanced

The *Basic/Advanced* toggle button allows you to alternate between the color gradient (*Basic*) and color range (*Advanced*) mode when it is clicked. By default, the *Basic* button and the color gradient based filter are displayed. When you click the *Basic* button, the button text changes to *Advanced*, and the form shows the color range based filter. The color gradient based filter allows you to modify the minimum and maximum values, and the color range based filter allows you to modify 8 equal filters between the minimum and maximum values.

In the color range mode, the filter ranges for the 8 buckets are evenly divided by default. You can customize these color scale range values for the 8 filters between the minimum and maximum values (specify 9 values).

Basic (Color Gradient) Advanced (Color Range)



Selecting *Apply* will accept the changes, and *Reset* will set it to default values.

Related Text Commands

See [report_power_rail_results](#), [read_power_rail_results](#), and [set_power_rail_display](#) commands in *Voltus Text Command Reference*.

Power and Rail Plots - DB Setup

Loads power/rail analysis results into GUI.

→ Choose *Power Rail - Power Rail Plots*, and then click the *DB Setup* button.

Power & Rail Setup

Database

Power Database: Browse...

Rail Database: Browse...

SEM Database: Browse...

Instance Files

Effective Res: Browse...

Inst Heat: Browse...

Tile Heat: Browse...

Detailed Heat: Browse...

Selective Loading

Nets: ☐ All ☐ Partial

IV Window: ☐ Timing ☐ Whole

IV Method Type: ☐ Worst ☐ WorstAvg ☐ Avg ☐ Best

Fields and Options

<i>Power Database</i>	Specifies the power database file. This file was created by using the <code>-create_binary_db</code> option of the <code>set_power_analysis_mode</code> command.
<i>Rail Database</i>	Specifies the rail analysis directory. You can load all the domain results by selecting the automatically generated analysis directory under the output directory (<code><domain>_<temp>_<dynamic/avg>_#</code>), or load individual nets by going one level further and selecting the net directory.
<i>SEM Database</i>	Specifies the database file generated by the <code>verify_AC_limit -report_db</code> command. This file contains the information required for loading and displaying signal EM data in GUI.

<i>Effective Res</i>	Specifies the result file from effective resistance analysis (only instance-based result is supported). When the rail analysis plot type <code>effr</code> is selected and a result file is specified, the layout shows effective resistance analysis result for each instance. Note: This field supports display of both net and domain based result file.
<i>Inst Heat</i>	Specifies the name of the instance-based delta temperature file from self-heating analysis. This is the device instance temperature.
<i>Tile Heat</i>	Specifies the name of the tile-based delta temperature file of self-heating analysis. This is the worst instance temperature among all wire segments in a tile. This is also layer-based. The FEOL or cell layer is used to show the device instance temperature.
<i>Detailed Heat</i>	Specifies the name of the detailed wire-based delta temperature file from self-heating analysis.
<i>Custom Value Instance Files</i>	Specifies the name of the custom value instance file (CVIF) that can be used to display a user-specified instance voltage file in GUI. After specifying the CVIF file, you can select the <i>cvil Rail</i> plot in the <i>Power & Rail Plots</i> form for displaying the EIV map.
<i>Selective Loading</i>	• <i>Nets</i> - Specifies whether all nets or selected nets are to be loaded into the GUI. ◦ <i>All</i> - loads all the nets in the domain-based state directory to the GUI. This option is selected by default. ◦ <i>Partial</i> - Loads only the specified nets in the domain-based state directory to the GUI. To specify nets: 1. Select <i>Partial</i> and click the <i>Select</i> button. The <i>Selective Nets</i> form appears. 2. Select the required power and ground nets, and click <i>OK</i>. • <i>IV Window</i> - Loads the instance voltage data for the selected EIV window. By default, the instance voltage results for both the timing and switching windows, and the entire rail simulation are loaded. • <i>IV Method Type</i> - Loads the instance voltage data for the specified EIV computing method. By default, the instance voltage results for all the four methods are loaded.

Related Text Commands

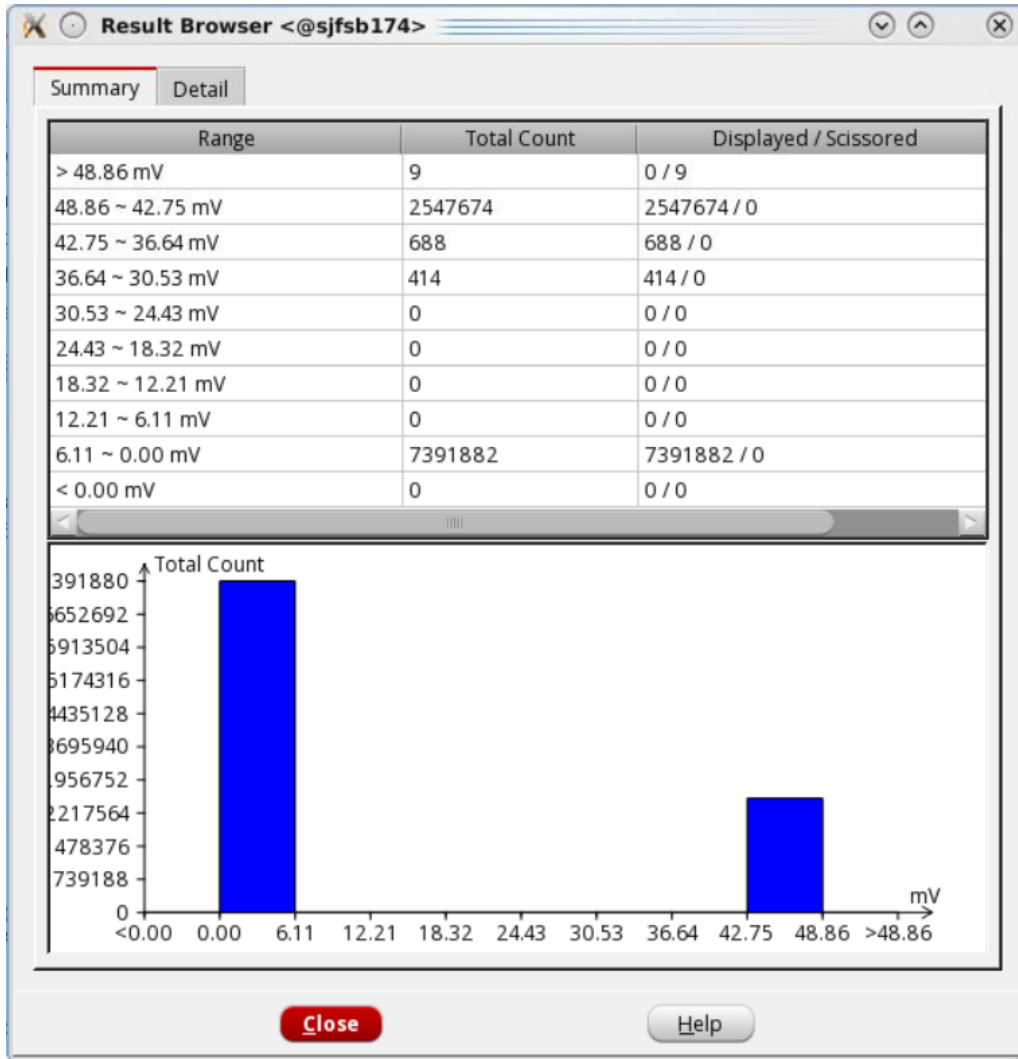
See `read_power_rail_results` command in *Voltus Text Command Reference*.

Result Browser

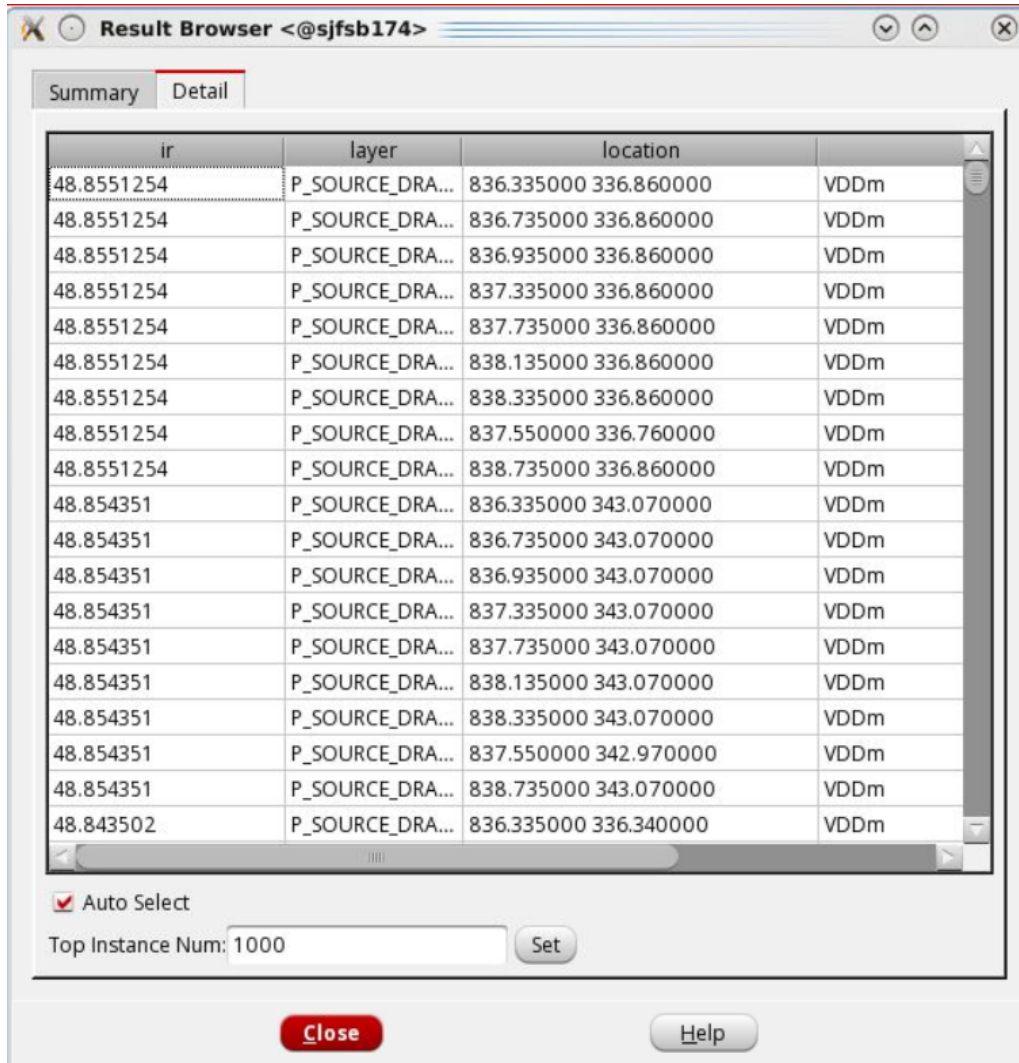
→ Choose *Power Rail - Power Rail Plots*, and then click the *Result Browser* button.

<i>Summary</i>	Provides summary information about the plot displayed. The data points are divided into buckets based on the range specified, and the summary information provides the statistic of each bucket. The tab also displays the rail analysis results in the histogram form.
<i>Detail</i>	Provides details of the individual data points of the plot displayed. By default, the first 1000 data points are displayed.

Summary



Detail



Power and Rail Plots - Layers/Nets/Sem Layers

→ Choose *Power Rail - Power Rail Plots*, and then click the *Layers/Nets* button.

<i>Layers</i>	Provides the ability to set the visibility of the layers and the opacity of the rail analysis or design database. This makes it easier to view when overlaying the rail analysis and the design database.
<i>Nets</i>	Controls the visibility of power/ground nets.

<i>Sem Layers</i>	Controls the visibility of the signal EM layers.
<i>Esd EM</i>	Provides the ability to selectively load the required ESD EM result directories if there are multiple result directories under the <code>ESD</code> directory.

Layers

Power & Rail Layers/Nets — sjfcl014 x

Layers Nets SemLayers EsdEM

Rail Analysis Opacity 0 — 100

Design DB Opacity 0 — 100

Via Size 0 — 6 1.00

Node Size 1 — 6

Vsrc Size 1 — 6

Resistor Size 1 — 6

IV Window whole

IV Method Type Worst

Enable Empty Layer ☐

Thermal Layers METAL_9

Layer Name	Report	visible
All	☒	☒

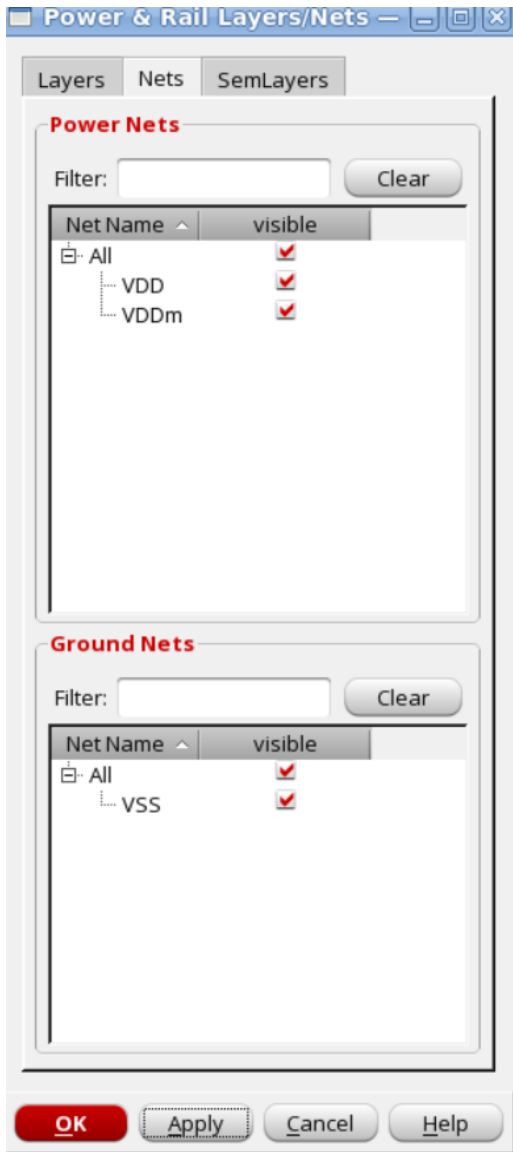
OK Apply Cancel Help

Fields and Options

<i>Rail Analysis Opacity</i>	Sets the opacity of the rail analysis output. These will be the resistor/node database load from rail analysis.
<i>Design DB Opacity</i>	Sets the opacity of the design database. These will be the shapes coming from DEF (main control panel).
<i>Via Size</i>	Specifies to modify the size of vias in the GUI. The default via size is 1. The possible via size values are 1, 2, 3, 4, 5, and 6.
<i>Node Size</i>	Specifies to modify the node size in the GUI. The default node size is 1. The possible node size values are 1, 2, 3, 4, 5, and 6.
<i>Vsrc Size</i>	Specifies to modify the size of voltage source (Vsrc) in the GUI. The default Vsrc size is 1. The possible Vsrc size values are 1, 2, 3, 4, 5, and 6.
<i>Resistor Size</i>	Specifies to modify the size of resistors in the GUI. The default resistor size is 1. The possible resistor size values are 1, 2, 3, 4, 5, and 6.
<i>IV Window</i>	Controls the window where effective instance voltage (EIV) is evaluated. EIV is obtained by processing the voltage waveforms and finding the worse-case effective voltage between the power and ground pins during a specific window. There are two possible options to evaluate EIV: • <i>timing/switching</i> – both switching activity (current is non-leakage) and timing window • <i>whole</i> – entire rail simulation not filtered to a specific window
<i>IV Method Type</i>	Allows different methods of computing effective instance voltage during rail analysis. Using this option, you can write out the <i>Best</i>, <i>Worst</i>, or <i>Average</i> effective instance voltage between power and ground nets. The <i>WorstAvg</i> method computes the average instance voltage during each evaluation window defined by the <i>IV Window</i> field, and reports the worst average instance voltage out of all the windows. If <i>IV Method</i> is <i>WorstAvg</i> and <i>IV Window</i> is <i>whole</i>, then this method will report average voltage across all time steps in simulation. <i>Default:</i> Worst
<i>Thermal Layers</i>	Displays the specified layer of the temperature map.

<i>Enable Empty Layer</i>	Specifies to enable or disable the empty layers.
<i>Layer Name</i>	You can enable or disable the reporting and visibility of the displayed layers.
<i>Report</i>	Specifies to select the user-defined layers to be used for processing.
<i>Visible</i>	Specifies which layers should be made visible when plotting.

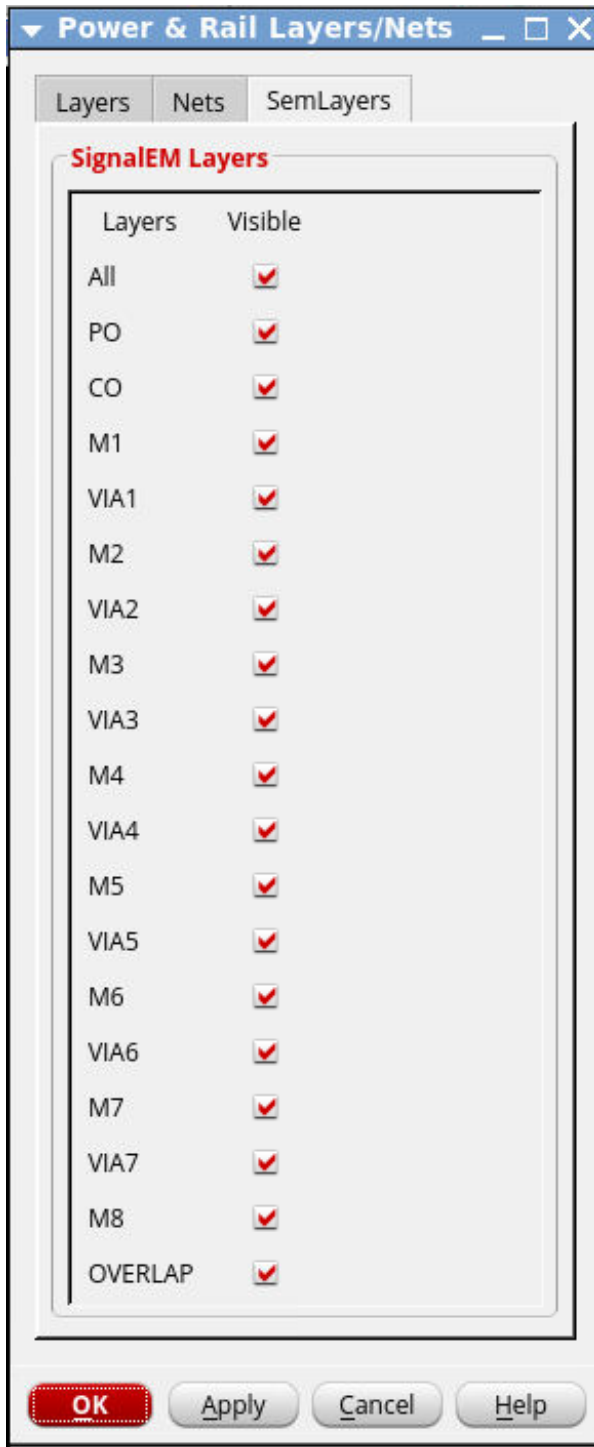
Nets



Fields and Options

<i>Filter</i>	Type a value in the <i>Filter</i> text field to locate a net in the list that matches the name you typed.
<i>Clear</i>	Click the <i>Clear</i> button to clear the filtered list.
<i>Power Nets</i>	Displays the list of power nets. - Click the <i>Net Name</i> column header to sort nets by names. You can switch between the ascending and descending order by clicking the column header. The directional arrow on the column points down to indicate that the net names are sorted in descending order, and points up to indicate that the net names are sorted in ascending order. - Select the <i>visible</i> checkbox next to the power nets to be viewed.
<i>Ground Nets</i>	Displays the list of ground nets. - Click the <i>Net Name</i> column header to sort nets by names. You can switch between the ascending and descending order by clicking the column header. The directional arrow on the column points down to indicate that the net names are sorted in descending order, and points up to indicate that the net names are sorted in ascending order. - Select the <i>visible</i> checkbox next to the ground nets to be viewed.

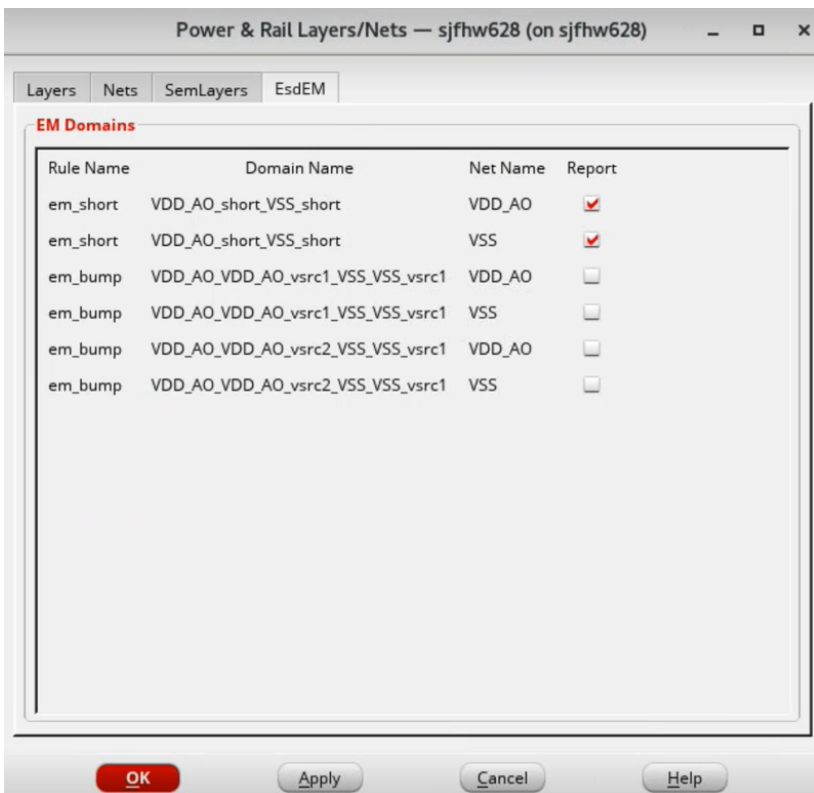
Sem Layers



Fields and Options

<i>Layers</i>	You can enable or disable the visibility of the displayed signal EM layers.
<i>Visible</i>	Specifies which layers should be made visible when plotting.

EsdEM



Field and Options

<i>Rule Name</i>	Displays the list of EM rule name.
<i>Domain Name</i>	Displays the list of domain name associated with the EM rule.
<i>Net Name</i>	Displays the list of nets associated with the EM rule. Note: You cannot select the same net from different rule or domain.
<i>Report</i>	Specifies which ESD EM result directory should be loaded.

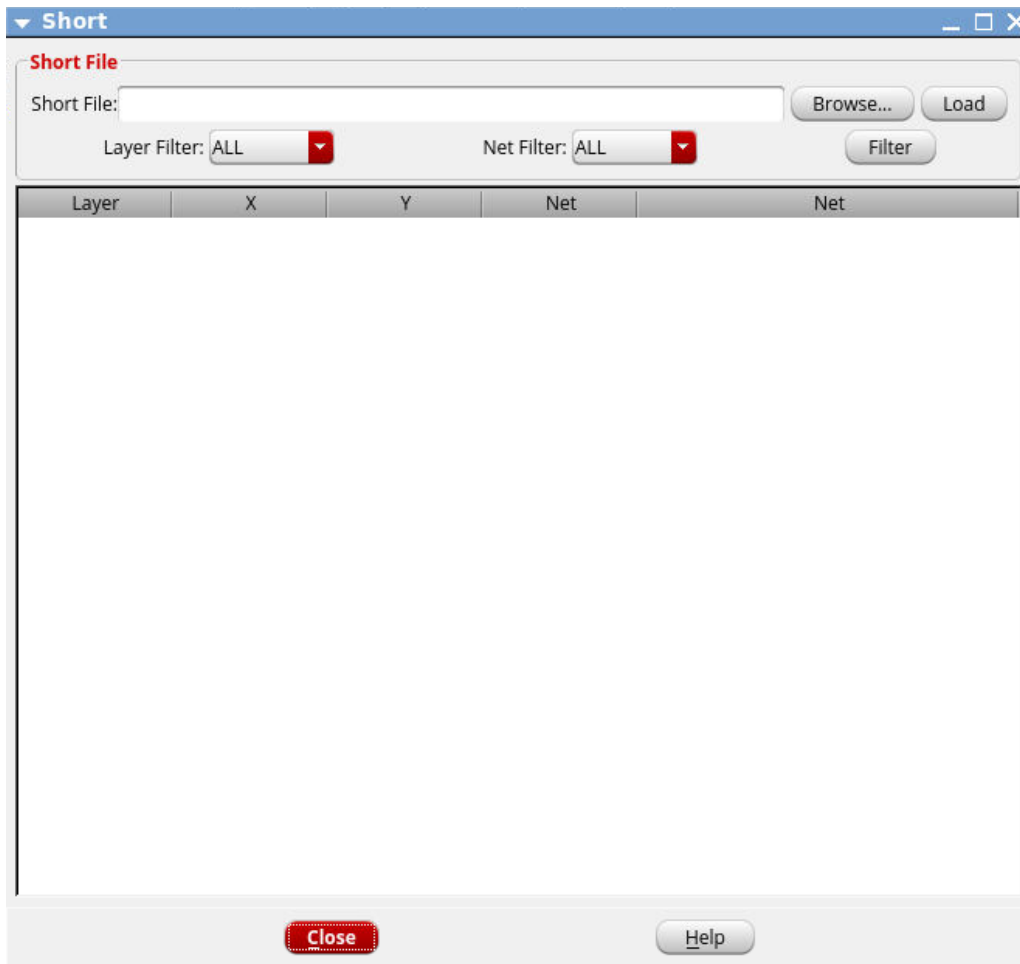
Related Text Commands

See [set_power_rail_layers_nets](#) command in *Voltus Text Command Reference*.

Short

Loads the short violations report (`voltus_short.report`) in the GUI. This file is generated using the `set_rail_analysis_mode -report_shorts true` command parameter, and is located in the work directory.

→ [Choose Power Rail - Short](#).



Fields and Options

<i>Short File</i>	Specifies the path to the short violations report.
<i>Browse button</i>	Opens the Short File browser form, which enables you to browse to the short violations report.
<i>Load button</i>	Loads the report file and displays the list of short violations on the power/ground nets.
<i>Layer Filter</i>	Allows you to filter the records based on the selected layer.
<i>Net Filter</i>	Allows you to filter the records based on the selected net.
<i>Filter button</i>	Displays the list of short violations based on the selected options from the <i>Layer Filter</i> and <i>Net Filter</i> drop-down lists.
<i>Layer</i>	The layer on which the short is detected.
<i>X</i>	The x coordinate where the short is detected.
<i>Y</i>	The y coordinate where the short is detected.
<i>Net</i>	The short violations report lists shorts between two geometries belonging to different nets. This field specifies the name of the first power or ground net.
<i>Net</i>	The short violations report lists shorts between two geometries belonging to different nets. This field specifies the name of the second power or ground net.

Related Text Commands

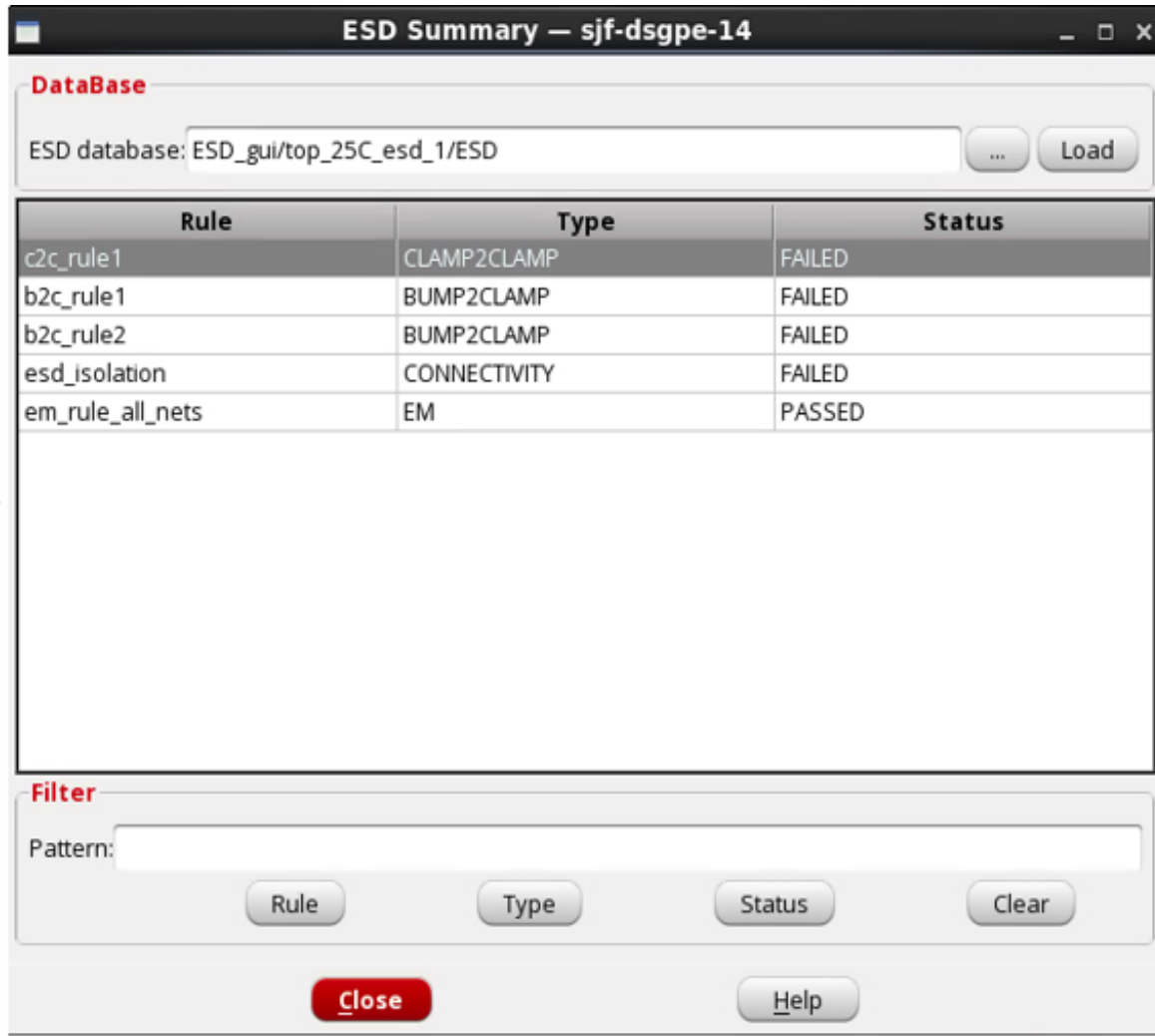
See [set_rail_analysis_mode](#) and [read_power_rail_results](#) commands in the *Voltus Text Command Reference*.

ESD Summary

Loads the ESD analysis results in the GUI. The ESD directory is generated using the

`analyze_esd_network` command, and is located in the state directory.

→ Choose *Power Rail - ESD Results*.



Fields and Options

<i>ESD Database</i>	Specifies the path to the ESD directory created using the <code>analyze_esd_network</code> command.
<i>Load button</i>	Loads the specified ESD directory. It displays the list of rules that were analyzed, and the rule type and rule status for each rule. You can select a rule to open the Rule Detail form.
<i>Pattern</i>	You can specify a pattern and click the <i>Rule/Type/Status</i> button to filter the rule records according to the specified pattern. For example, if you specify <i>FAIL</i> in the <i>Pattern</i> field, and click <i>Status</i> , all the rules with the <i>FAILED</i> status will be displayed.

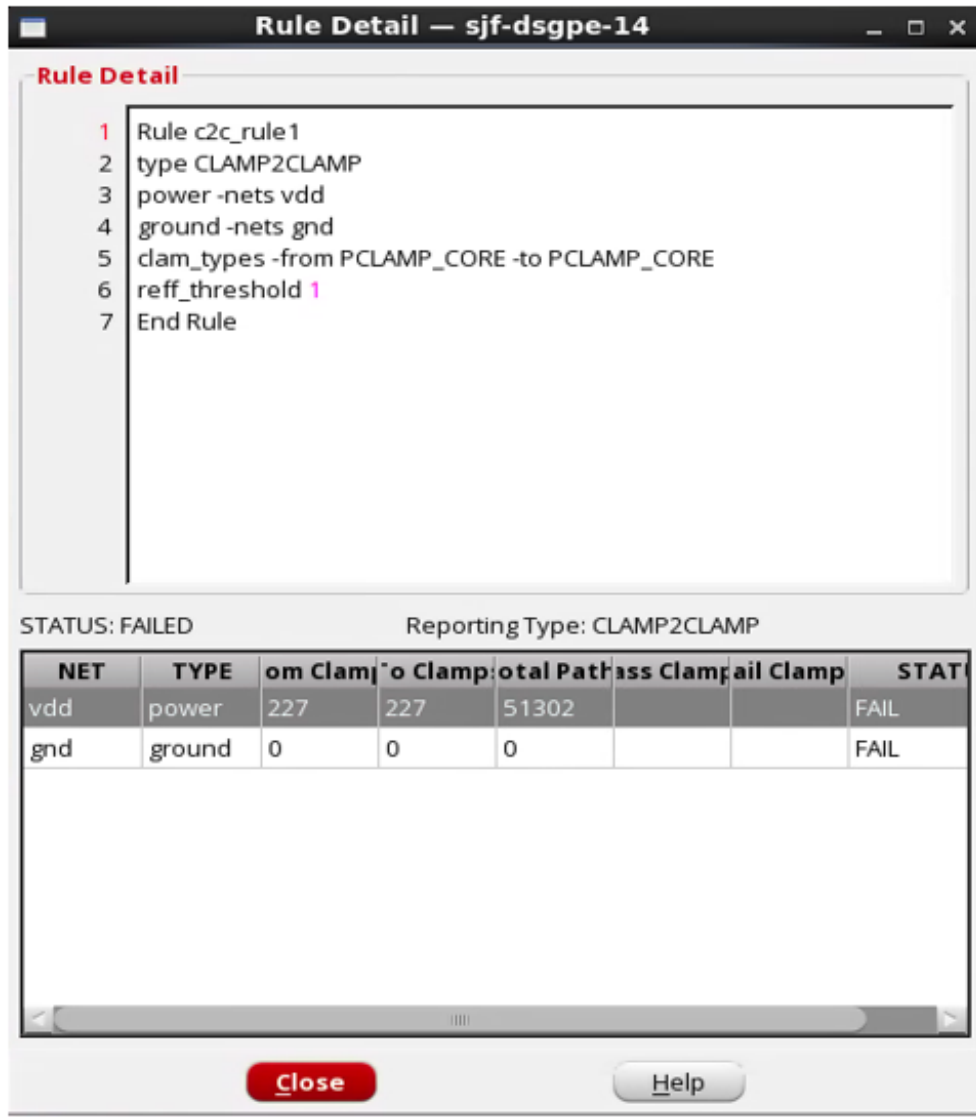
Related Text Commands

See `analyze_esd_network` commands in the *Voltus Text Command Reference*.

Rule Detail

Displays a summary of the rule selected in the *ESD Summary* form.

→ To open the *Rule Detail* Form, select a rule in the *ESD Summary* form.



This form has two parts:

- The top section displays the rule details as specified in the ESD rule file.
- The bottom section gives the rule status summary. It contains the rule details from the summary report.

For information on the ESD rule file and the summary report columns, refer to the "[ESD Analysis](#)" chapter in the *Voltus User Guide*.

Related Text Commands

See [analyze_esd_network](#) commands in the *Voltus Text Command Reference*.

ESD Net Window

Displays the From and To sections for the net selected in the *Rule Detail* form.

→ To open the *ESD Net Window* form, select a net in the bottom section of the *Rule Detail* form.

Net — sjf-dsgpe-14

Net: vdd Rule: c2c_rule1 Type: CLAMP2CLAMP

Clamp	PIN	Layer	X(um)	Y(um)	Instance
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	3277.889893	6321.200195	ESDCLAMP_1V2_INH_2M4PINS_inst_106
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	3546.576904	731.770020	ESDCLAMP_1V2_INH_2M4PINS_inst_108
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	3546.576904	1045.405029	ESDCLAMP_1V2_INH_2M4PINS_inst_109
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	1382.977051	2927.215088	ESDCLAMP_1V2_INH_2M4PINS_inst_11
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	3546.576904	1359.040039	ESDCLAMP_1V2_INH_2M4PINS_inst_110
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	3546.576904	1672.675049	ESDCLAMP_1V2_INH_2M4PINS_inst_111
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	3549.221924	1986.310059	ESDCLAMP_1V2_INH_2M4PINS_inst_112
ESDCLAMP_1V2_INH_2M4P...	VDDCORE...	M4	3548.340088	2299.945068	ESDCLAMP_1V2_INH_2M4PINS_inst_113

To Clamps

From Clamp	To Clamp	To Clamp Cell Type	X(um)	Y(um)	PIN	Layer
ESDCLAMP_1V2_INH_2M4PINS_inst_10	ESDCLAMP_1V2_INH_2M4PINS_inst_16	ESDCLAMP_1V2_...	1656.953003	2598.580078	VDDCORE...	M4
ESDCLAMP_1V2_INH_2M4PINS_inst_10	ESDCLAMP_1V2_INH_2M4PINS_inst_22	ESDCLAMP_1V2_...	1927.402954	2509.580078	VDDCORE...	M4
ESDCLAMP_1V2_INH_2M4PINS_inst_10	ESDCLAMP_1V2_INH_2M4PINS_inst_11	ESDCLAMP_1V2_...	1382.977051	2927.215088	VDDCORE...	M4
ESDCLAMP_1V2_INH_2M4PINS_inst_10	ESDCLAMP_1V2_INH_2M4PINS_inst_17	ESDCLAMP_1V2_...	1653.427002	2927.215088	VDDCORE...	M4
ESDCLAMP_1V2_INH_2M4PINS_inst_10	ESDCLAMP_1V2_INH_2M4PINS_inst_15	ESDCLAMP_1V2_...	1653.427002	2299.945068	VDDCORE...	M4
ESDCLAMP_1V2_INH_2M4PINS_inst_10	ESDCLAMP_1V2_INH_2M4PINS_inst_9	ESDCLAMP_1V2_...	1385.621948	2298.945068	VDDCORE...	M4
ESDCLAMP_1V2_INH_2M4PINS_inst_10	ESDCLAMP_1V2_INH_2M4PINS_inst_23	ESDCLAMP_1V2_...	1923.876953	2927.215088	VDDCORE...	M4

Filter

Pattern:

From Clamps/Bumps To Clamps/Bumps Clear

Close Help

This form has two parts:

- The top section displays the details of the From Clamps/Bumps.
- The bottom section displays the details of the To Clamps/Bumps. You can select a row from the top section to display the corresponding To Clamps/Bumps in the bottom section. When you select a “To Clamp/Bump”, the GUI will highlight the fly lines between the From Clamp/Bump and the To Clamp/Bump.

You can specify a pattern in the *Pattern* field, and click the *From Clamps/Bumps* or *To Clamps/Bumps* button to filter the rule records according to the specified pattern. For example, if

you specify *ESDCLAMP* in the *Pattern* field, and click *To Clamps/Bumps*, all the bumps/clamps with *ESDCLAMP* text in their names will be displayed in the bottom section.

For information on the From and To sections of the net, refer to the "*ESD Analysis*" chapter in the *Voltus User Guide*.

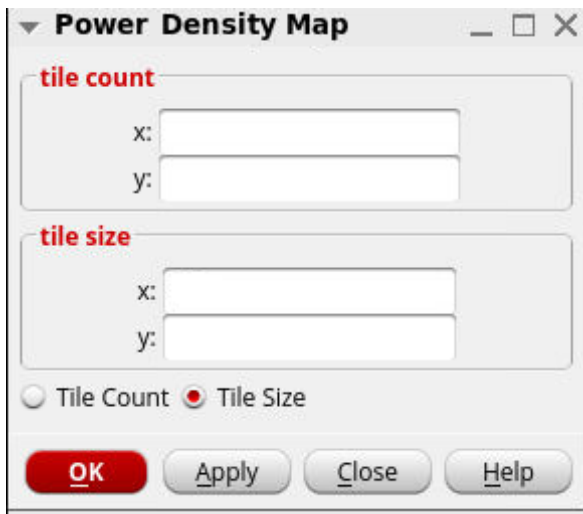
Related Text Commands

See [analyze_esd_network](#) commands in the *Voltus Text Command Reference*.

Power Density Map

Allows to resize the tile power density plot (*tpd* power plot). This form allows you to resize the *tpd* plot without having to restore the power database (*power.db*) again.

→ Choose *Power Rail - Tile Power Density*.



The screenshot shows a dialog box titled "Power Density Map". It contains two sections: "tile count" and "tile size". Each section has "x:" and "y:" input fields. At the bottom, there are two radio buttons: "Tile Count" (unselected) and "Tile Size" (selected). Below the radio buttons are four buttons: "OK" (highlighted in red), "Apply", "Close", and "Help".

Fields and Options

<i>Tile Count X</i>	Specifies the number of tiles in the x direction.
<i>Tile Count Y</i>	Specifies the number of tiles in the y direction.
<i>Tile Size X</i>	Specifies the horizontal dimension of each tile polygon in μm .
<i>Tile Size Y</i>	Specifies the vertical dimension of each tile polygon in μm .
<i>Tile Count/Tille Size</i>	You can either specify the tiles by count or specify the tiles by size. If you select <i>Tile Count</i> , specify the number of tiles in the <i>Tile Count X</i> and <i>Y</i> fields. Alternatively, if you select <i>Tile Size</i> , specify the tile size in the <i>Tile Size X</i> and <i>Y</i> fields,

Related Text Commands

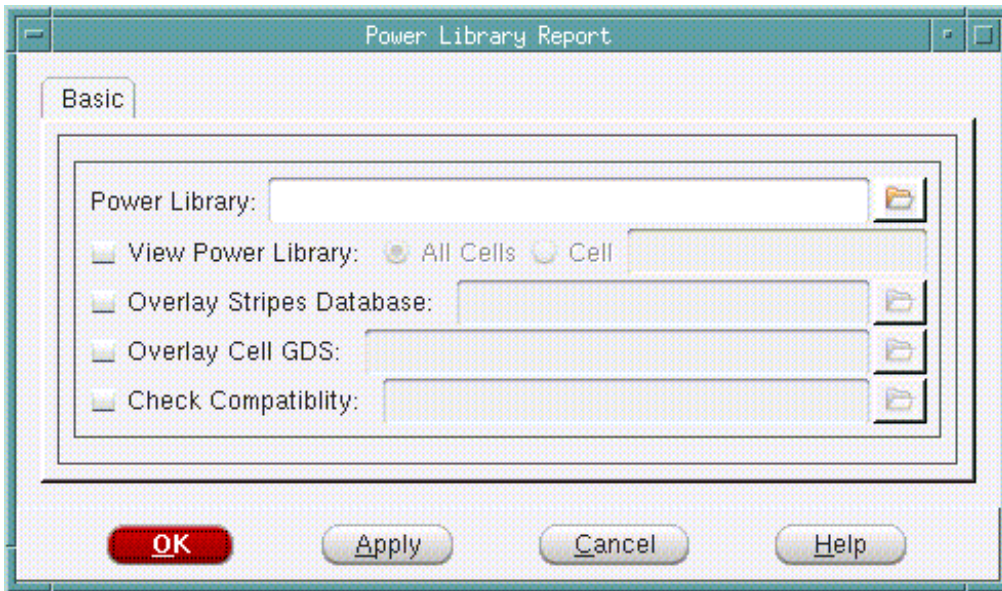
See [set_power_rail_display](#) command in the *Voltus Text Command Reference*.

Power Library Report

→ Choose *Power Rail - Textual Reports - PowerGrid Library*.

<i>Basic</i>	Reports contents of the already generated power library and physical view of the library power-grid network.
--------------	--

Basic



Fields and Options

<i>Power Library</i>	Specifies the power-grid library to generate the power library report.
<i>View Power Library</i>	Specifies the following options to view the power library: • <i>All Cells</i> • <i>Cells</i>
<i>Overlay Stripes Database</i>	Specifies the stripes database to be overlaid during the psviewer session.
<i>Overlay Cell GDS</i>	Specifies the cell gds design file to be overlaid, during the psviewer session.
<i>Check Compatibility</i>	Checks the compatibility between the specified cell libraries.

Related Text Commands

See [check_pg_library](#) command in the *Voltus Text Command Reference*.

Power Report

Provides the ability to provide reports of power information in various ways.

→ Choose *Power Rail - Textual Reports - Power Analysis*.

<i>Report Power</i>	Generates a power report for the specified options.
<i>Format</i>	Species the format for the power report.

Report Power

The screenshot shows the 'Report Power' dialog box with the following fields and options:

- Report Power** (selected tab) and **Format** (disabled tab).
- Output File**: A text field with a folder icon on the right.
- Clock Network**: A dropdown menu set to 'all'.
- ☐ **Count Sequential Elements in clock Network**.
- Instances**: A text field.
- Hierarchy Level**: A dropdown menu set to 'all'.
- Cell**: A text field.
- Cell Type**: A group box containing five radio buttons:
 - ☒ **all**
 - ☐ **macro**
 - ☐ **io**
 - ☐ **combinational**
 - ☐ **sequential**
- Power Domain**: A dropdown menu set to 'all'.
- PG Net**: A dropdown menu set to 'all'.
- Net**: A text field.
- Number of Nets Consuming Highest Power**: A text field.
- MMMC View**: A dropdown menu.
- Buttons**: **OK** (highlighted in red), **Apply**, **Cancel**, and **Help**.

Fields and Options

<i>Output File</i>	Specifies the name of the report file in which power analysis output information is written. You can include the directory name. Note: If you do not specify this parameter, the software directs the power report to the command line, as well as the log file, and no power report is written to a separate file.
<i>Clock Network</i>	Reports the power consumption of the clock network, including the power for generated clocks. Specify a list to report the power consumed by specific clocks. Specify <code>all</code> to report all clock networks. <i>Default:</i> <code>all</code>
<i>Count Sequential Elements in clock Network</i>	If selected, report will include the leaf flip-flop power in the clock network power report. <i>Default:</i> The leaf flip-flop power will not be included in the clock network power report.
<i>Instances</i>	Specifies a list of instances to include in the power report. The power report provides the amount of power consumed by a specified list of hierarchical or leaf-level instances. This field accepts wildcards (*) for leaf-level instances only.
<i>Hierarchy Level</i>	Specifies the hierarchy level that is included in the power report. You can specify the <code>Hierarchy_level</code> as a number that corresponds to a specific hierarchy level in the design. A <code>Hierarchy_level</code> of 0 specifies the top level. <i>Default:</i> <code>all</code>
<i>Cell</i>	Specifies the cells to include in the power report. Accepts wildcards (*)
<i>Cell Type</i>	Specifies the cell type to included in the power report. The choices are <code>all</code> or any combination of macro, io, combinational, or sequential. <i>Default:</i> <code>all</code>
<i>Power Domain</i>	Specifies all power domains or a list of power domains to be included in report.
<i>PG Net</i>	Specifies <code>all</code> power nets or those of a particular hierarchy level to be included in power report.

<i>Net</i>	Reports the net switching power, as well as the load capacitance, toggle rate, and switching values for each net in the design.
<i>Number of Nets Consuming Highest Power</i>	Reports the number of nets with the highest net switching power. Note: You must use this parameter in conjunction with the <i>Net</i> parameter.
<i>MMMC View</i>	Specifies the view that was created using the multi-mode multi-corner (MMMC) <code>create_analysis_view</code> command. Power analysis is performed for the specified view. Default: Uses current view in Voltus.

Format

The screenshot shows the 'Report Power' dialog box with the 'Format' tab selected. The dialog has a title bar 'Report Power' and two tabs: 'Report Power' and 'Format'. The 'Format' tab contains the following elements:

- Sort By:** A group box containing four radio buttons: 'internal', 'leakage', 'switching', and 'total'. The 'total' radio button is selected.
- Threshold for Power Reporting:** A checkbox labeled 'Threshold for Power Reporting' followed by a text input field containing the value '0'.
- For Rail Analysis:** A group box containing a checkbox labeled 'Set Rail Analysis'. Below this checkbox are two radio buttons: 'FE' and 'VS'. Below these radio buttons is the text: 'All the other options are ignored when rail analysis format is selected.'
- Buttons:** At the bottom of the dialog are four buttons: 'OK' (highlighted in red), 'Apply', 'Cancel', and 'Help'.

Fields and Options

Sort By: • <i>internal</i> • <i>leakage</i> • <i>switching</i> • <i>total</i>	Sorts power report based on the power (internal / leakage / switching / total) option that you select.
Threshold for Power Reporting	
	Specifies the threshold value for power reporting.
For Rail Analysis	• <i>Set Rail Analysis: Specifies to report power in rail analysis format.</i> • <i>FE:</i> • <i>VS:</i>

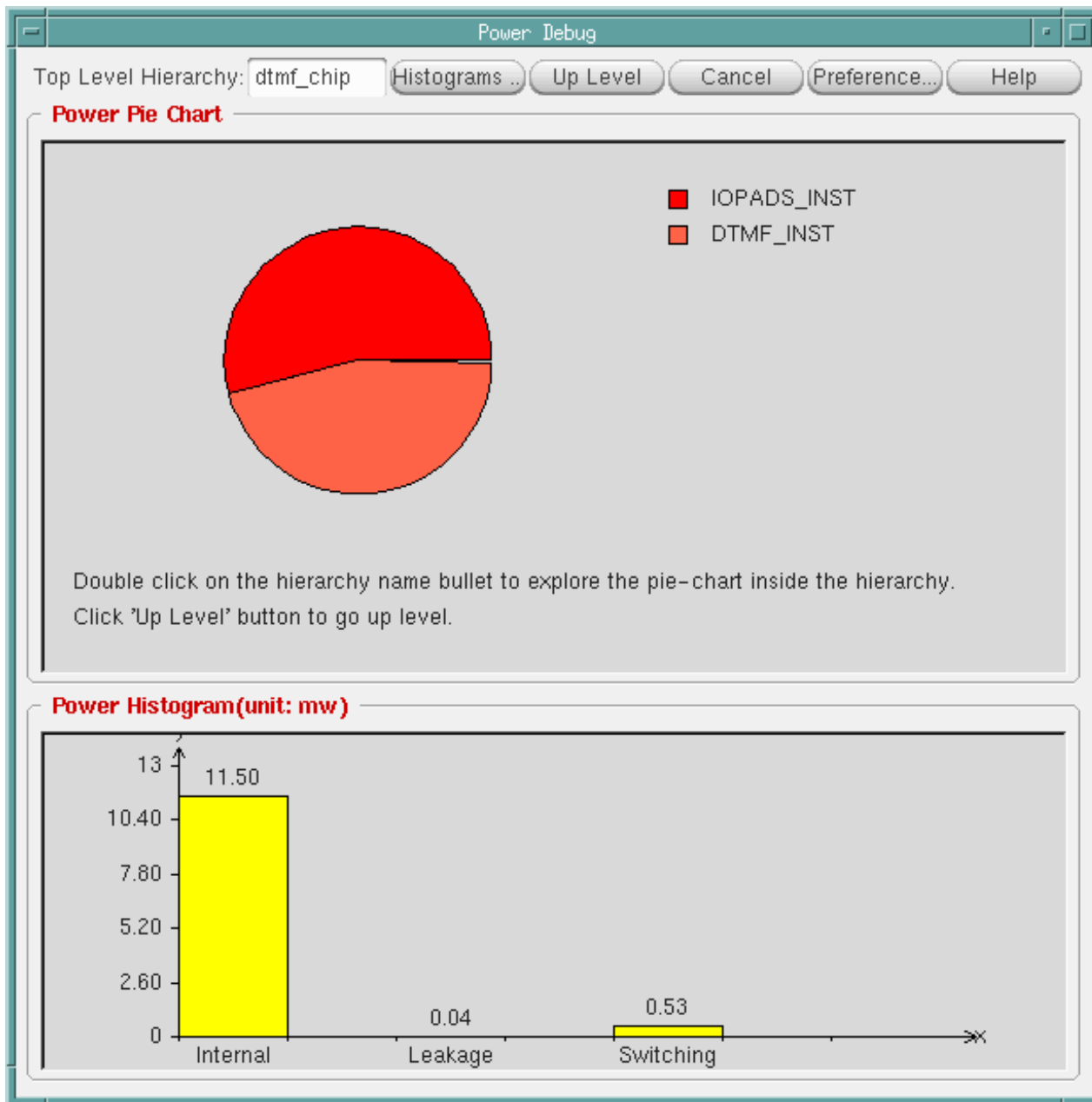
Power Histograms

The Power Debug form displays power histograms in various ways.

→ Choose *Power & Rail - Report - Power Histograms*.

→ To display information from one level down in the hierarchy, double-click on a listed instance.

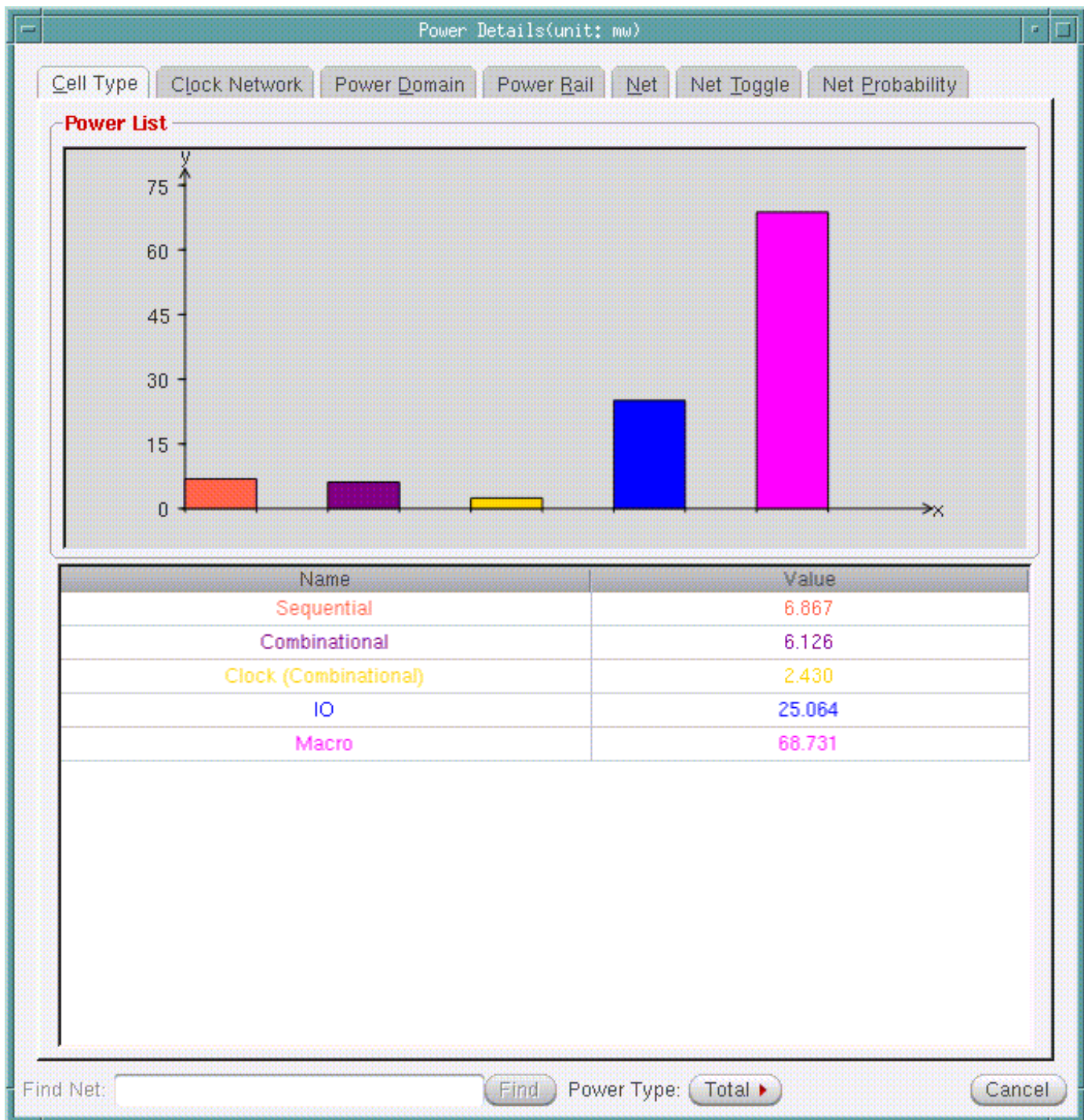
→ To display information from one level higher in the hierarchy, select *Up Level*.



Power Details

The Power Details form displays histograms by cell type, clock domain, power domain, power rail, net, net toggle, or net probability.

→ To open the Power Details form, click on the *Histograms* button on the Power Debug form.



Power Debug Preference

The Power Debug Preference form is used to specify Y-max/Y-min/Y-step for each of the power

debug histogram, such as *Cell Type*, *Clock Network*, and so on. You can also save and load these setting for all the histograms as a `.tcl` file. The default file name is `pdg.pref.tcl`.

→ To open the Power Debug Preference form, click on the *Preference* button on the Power Debug form or right-click on any of the power debug histograms in the Power Details form.

Power Debug Preference

Power Histogram
Ymax: 13 Ymin: 0 Ystep: 2.6

Cell Type
Ymax: 7 Ymin: 0 Ystep: 1.4

Clock Network
Ymax: 5 Ymin: 0 Ystep: 1.0

Power Domain
Ymax: 12 Ymin: 0 Ystep: 2.4

Power Rail
Ymax: 5 Ymin: 0 Ystep: 1.0

Net Power
Ymax: 0.0639837 Ymin: 0 Ystep: 0.01279674

Net Toggle
Ymax: 137500000 Ymin: 0 Ystep: 27500000.0

Net Probability
Ymax: 1.1 Ymin: 0 Ystep: 0.22

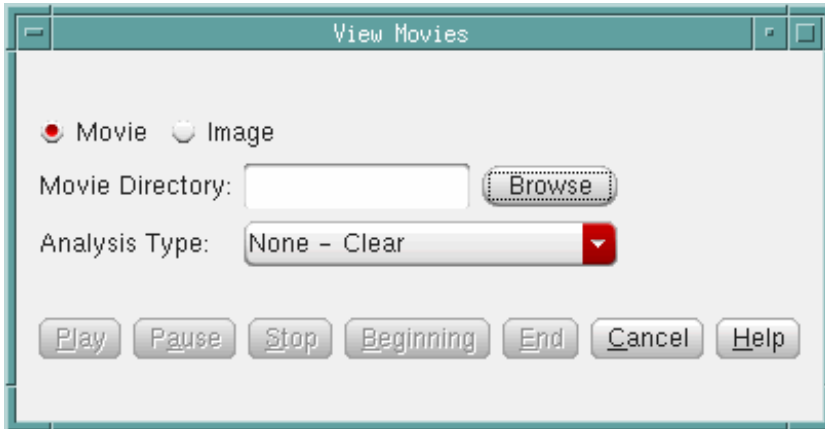
OK Apply Save... Load... Cancel Help

Dynamic Results

→ Choose *Power Rail - Dynamic Results*.

<i>Movies</i>	Provides the ability to view dynamic movies..
<i>Waveforms</i>	Provides the ability to display dynamic waveforms. This functionality requires a Simvision license.

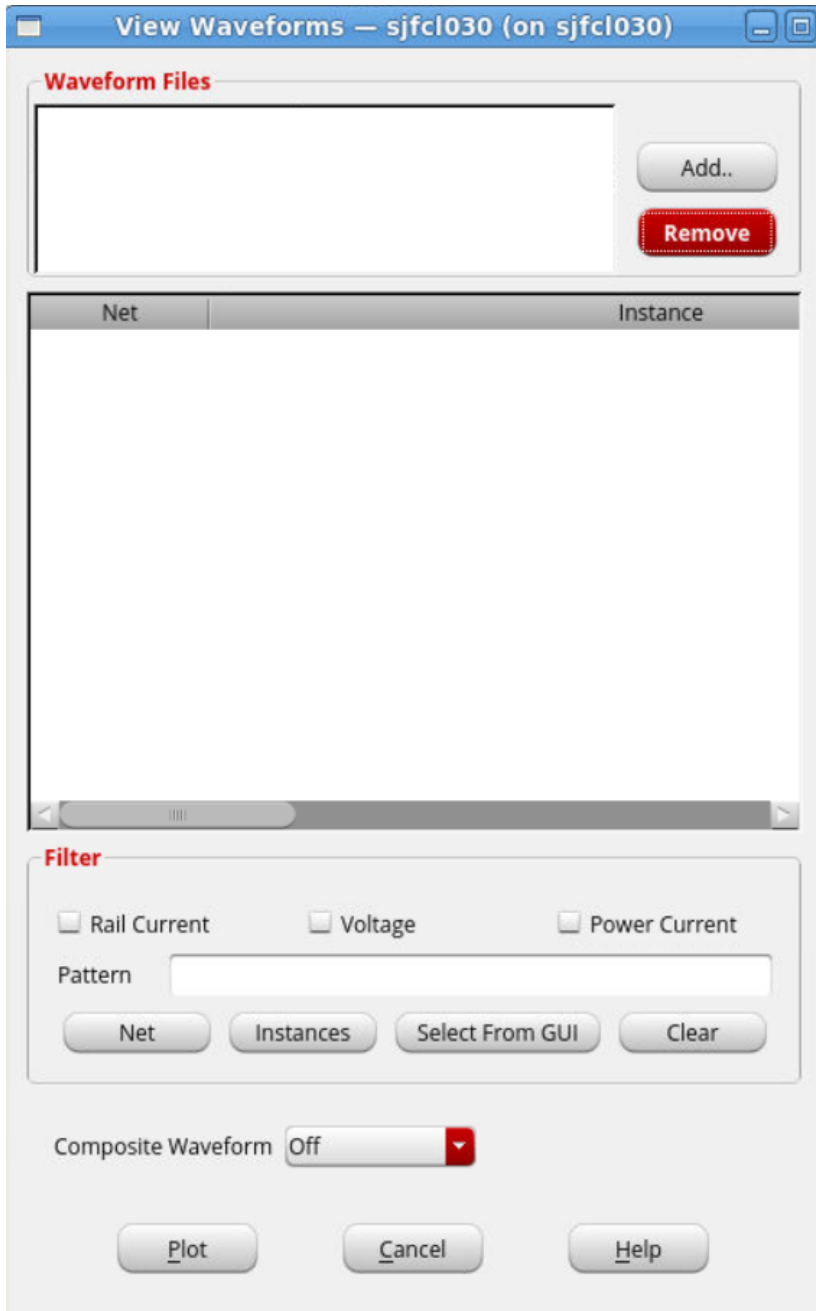
Movies



Fields and Options

<i>Movie Directory</i>	Specifies the directory that includes the movie files.
<i>Analysis Type</i>	Specifies the analysis type that will be viewed.
<i>Movie</i>	Specifies that a movie will be viewed.
<i>Image</i>	Specifies that an image or plot will be viewed.
<i>Play</i>	Plays the movie.
<i>Pause</i>	Pauses the movie.
<i>Stop</i>	Stops the movie.
<i>Beginning</i>	Goes to the beginning of the movie.
<i>End</i>	Goes to the end of the movie.
<i>Cancel</i>	Cancels the movie.

Waveforms



Fields and Options

<i>Waveform Files</i>	Specifies the waveforms that will be viewed. To view multiple .ptiavg waveform files (current and voltage waveform) and .trn profiling databases, click <i>Add</i> to browse to the required directory and select the corresponding .ptiavg/.trn file. These files can be loaded or removed using the <i>Add</i> and <i>Remove</i> buttons respectively.
<i>Filter - Pattern</i>	The list of nets/instances can be filtered basis the following: • Select <i>Rail Current</i>, <i>Voltage</i>, and/or <i>Power Current</i> to filter instances based on the waveform type. • Enter a value in the <i>Pattern</i> field and select either <i>Net</i> or <i>Instance</i> to filter instances based on the specified pattern.
<i>Net</i>	Specifies that the nets will be filtered by the text in the <i>Pattern</i> field.
<i>Instances</i>	Specifies that the instances will be filtered by the text in the <i>Pattern</i> field.  You can use multiple CPUs to enable the search of instances in parallel, and thereby improve the search time in large designs. Multi-threading is supported for instance search using both the <i>Instances</i> and <i>Select From GUI</i> buttons. You must set up multi-CPU usage before accessing the <i>View Waveforms</i> form, as shown below: <pre>set_multi_cpu_usage -localCpu number_of_cpus</pre>
<i>Select From GUI</i>	This button can be used to plot a waveform for an instance that is selected in the window.
<i>Clear</i>	The Pattern field can be emptied by selecting the <code>Clear</code> button.
<i>Plot</i>	To view a current or voltage waveform, select an item on the list, and select the <i>Plot</i> button. This will result in Simvision being run and a Simvision window appearing with the selected current waveform displayed.
<i>Cancel</i>	Closes the View Waveforms form.

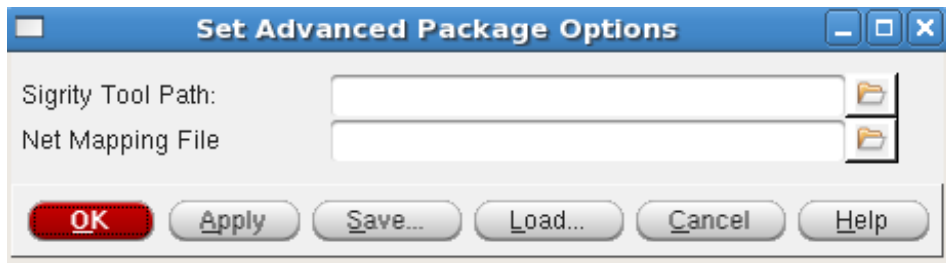
Package Menu

- [Set Advanced Package Options](#)
- [Extract Package](#)
- [Analyze Package](#)
- [Package Plots](#)
- [Generate Mapping File](#)
- [Merge Circuit Models](#)
- [Sigridity](#)

Set Advanced Package Options

Use the *Set Advanced Package Options* form to specify the path to the bin folder of package extraction and analysis tools.

→ Select *Package - Set Advanced Package Options*.



Fields and Options

<i>Sigrity Tool Path</i>	Specifies the full path to the bin folder of the package extraction and analysis tools (XtractIM and PowerDC).
<i>Net Mapping File</i>	Specifies the name of the two column mapping file that contains the mapping between the net names of package with the net names of die.

Related Text Commands

The following text command provides equivalent or additional functionality:

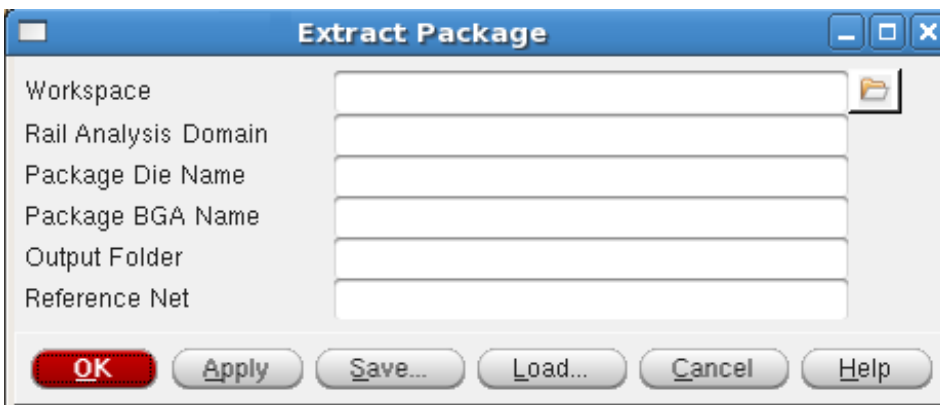
- [set_advanced_package_options](#)

For more information, see "[Package Analysis](#)" in the *Voltus User Guide*.

Extract Package

Specifies to start package model extraction.

→ *Select Package - Extract Package*.



Fields and Options

<i>Workspace</i>	Specify the workspace name (package model database) of the package extraction tool (XtractIM). The file name extension is <code>.ximx</code> .
<i>Rail Analysis Domain</i>	Specifies the domain name.
<i>Package Die Name</i>	Defines the die circuit name. When specified, the software automatically creates a mapping between the die name in the package extraction tool and the die name in the software.
<i>Package BGA Name</i>	Defines the ball grid array (BGA) circuit name. It is the PCB component in the package model, which eventually connects to the PCB model.
<i>Output Folder</i>	Specifies the output/result directory. By default, the package model will be created in the workspace. However, if you specify an output directory, then a copy of the result will be made in the new directory
<i>Reference Net</i>	Specifies the ground net that is selected as reference net for pin-based model generation.

Related Text Commands

The following text command provides equivalent or additional functionality:

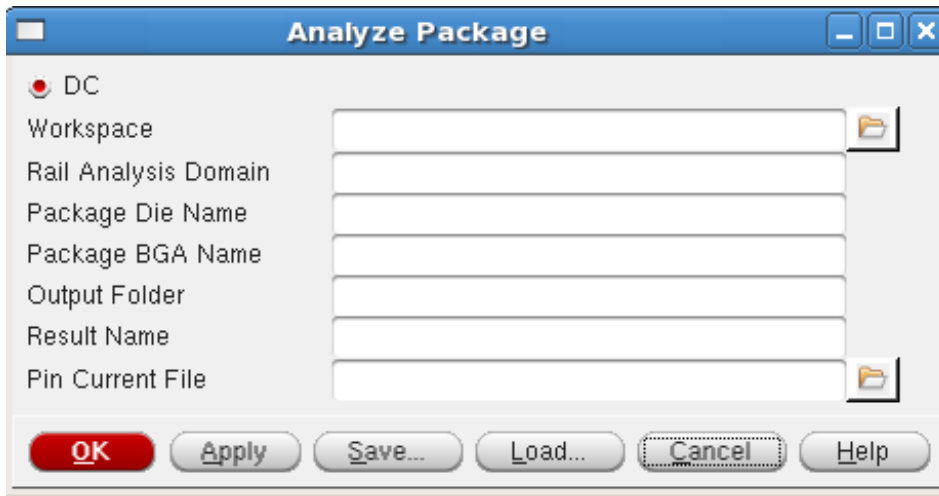
- [extract_package](#)

For more information, see "[Package Analysis](#)" in the *Voltus User Guide*.

Analyze Package

Specifies how the package analysis will be performed, and to run package analysis.

→ *Select Package - Analyze Package*.



Fields and Options

<i>Workspace</i>	Specify the workspace name for package analysis (powerDC). The file name extension is .pdcx.
<i>Rail Analysis Domain</i>	Specifies the domain name.
<i>Package Die Name</i>	Defines the die circuit name. When specified, the software automatically creates a mapping between the die name in the package analysis tool and the die name in the software.
<i>Package BGA Name</i>	Defines the ball grid array (BGA) circuit name.
<i>Output Folder</i>	Specifies the package analysis results directory. By default, the package model results will be created in the workspace.
<i>Result Name</i>	Specifies the name of the .xml result file.
<i>Pin Current File</i>	Specifies the name of the generated pin current file from rail analysis that contains the package pin names and the current values. It is automatically generated after the package model is created, and IR drop analysis is performed using the package model.

Related Text Commands

The following text command provides equivalent or additional functionality:

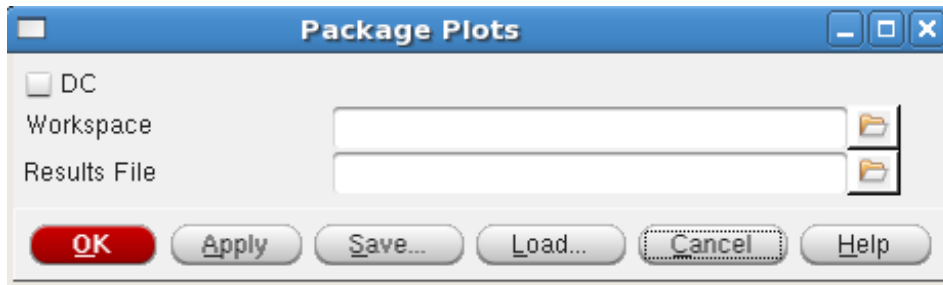
- `analyze_package`

For more information, see "[Package Analysis](#)" in the *Voltus User Guide*.

Package Plots

Allows you to view the package IR drop analysis results in the PowerDC GUI which will be invoked from within Voltus.

→ *Select Package - Package Plots.*



Fields and Descriptions

<i>Workspace</i>	Specifies the workspace name for package analysis (powerDC).
<i>Results File</i>	Specifies the name of the .xml result file.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `view_package_results`

For more information, see "[Package Analysis](#)" in the *Voltus User Guide*.

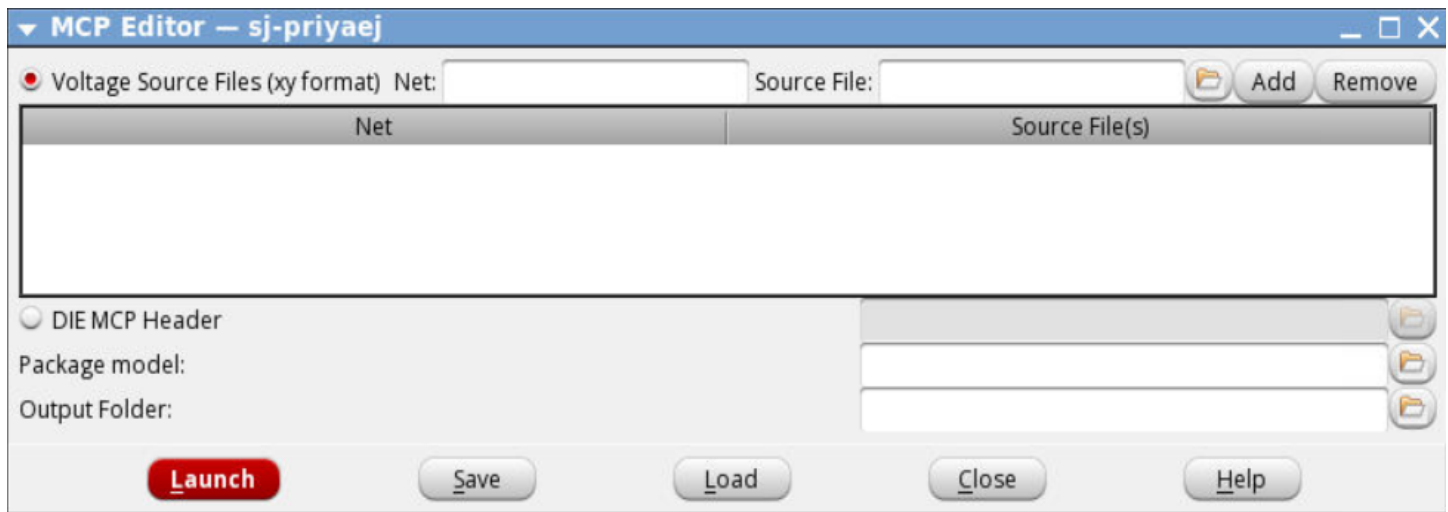
Generate Mapping File

Allows you to:

- launch the *MCP Editor* GUI to make manual connections between package and die pins.

- map the voltage source files with the package model to check and correct mapping information before rail analysis.
- generate a mapping file that can be used in previous versions of the software in case there is no mapping between die and package pins.

→ *Select Package - Sigrity - MCP Editor. - Generate Mapping File*



Fields and Descriptions

<i>Voltage Source Files (xy format)</i>	
<i>Net</i>	Specifies the net name of the package.
<i>Source File</i>	Specifies the voltage source file mapping to each net.
<i>Add</i>	Click the <i>Add</i> button to add details of the specific net and the corresponding source file to the power/ground nets table.
<i>Remove</i>	Click the <i>Remove</i> button to remove details of the specific net from the power/ground nets table.
<i>DIE MCP Header</i>	Specifies a circuit file with the MCP header containing the die abstract.
Package Model	Specifies the package model file.
Output Folder	Specifies the output directory containing the mapping results.

Related Text Commands

The following text command provides equivalent or additional functionality:

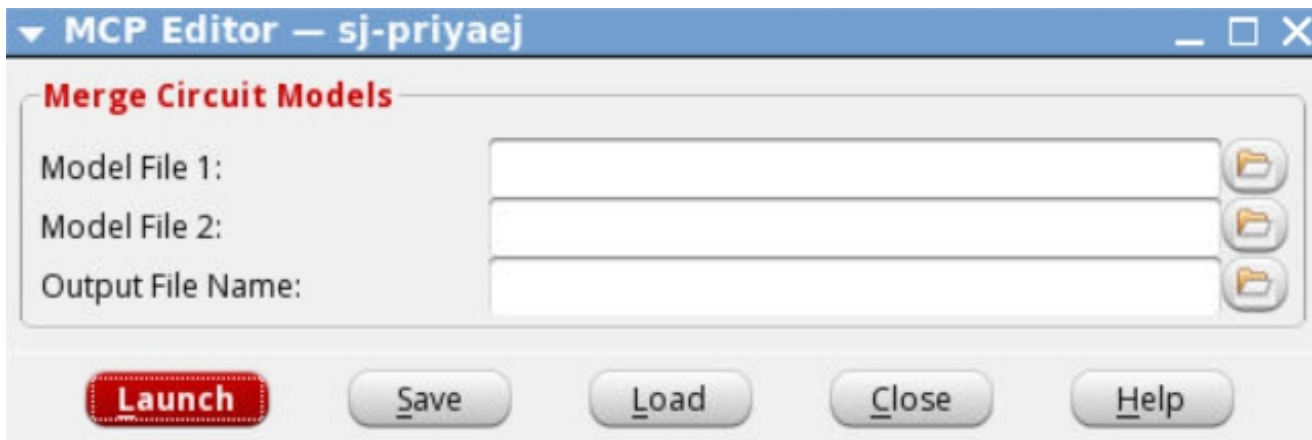
- `map_die_package`

For more information, see "[Package Analysis](#)" in the *Voltus User Guide*.

Merge Circuit Models

Allows you to merge two circuit model files. For example, you can use this form to merge the PCB and package model files. The merged package model file can be saved and specified with the `set_package -spice` parameter.

→ *Select Package - Sigrity - MCP Editor - Merge Circuit Models*



Fields and Options

<i>Model File 1</i>	Specifies the name of the first input circuit model file.
<i>Model File 2</i>	Specifies the name of the second input circuit model file.
<i>Output File Name</i>	Specifies the name of the output file containing the merged results. You can either type in the new filename or browse for an existing file with the Browse button.
<i>Launch</i>	Click this button to open the <i>MCP Editor</i> form. This form allows you map the two input model files to check and correct mapping information before rail analysis. Once the mapping is complete, click <i>OK</i> to close the <i>MCP Editor</i> form and save the merged file.

Related Text Commands

The following text command provides equivalent or additional functionality:

- `map_die_package`

For more information, see "[Package Analysis](#)" in the *Voltus User Guide*.

Sigrity

→ *Select Package - Sigrity.*

- *XtractIM*

- *PowerDC*

These are additional options that allow you to access the package extraction (*XtractIM*) and analysis (*PowerDC*) tools to have more control over package modeling and to easily make modifications (if needed), and then proceed with extraction or package analysis.

For details on Sigrity tools, refer to product documentation for Sigrity.

ECO Menu

The items in this menu require Cadence® Tempus™ Timing Signoff Solution license. See the *Tempus Timing Signoff Solution Menu Reference* for details.

Clock Menu

The items in this menu require Cadence® Tempus™ Timing Signoff Solution license. See the *Tempus Timing Signoff Solution Menu Reference* for details.

Timing and SI Menu

The items in this menu require Cadence® Tempus™ Timing Signoff Solution license. See the *Tempus Timing Signoff Solution Menu Reference* for details.

Verify Menu

- [Verify Connectivity](#)
- [Verify Power Via](#)

Verify Connectivity

Use the Verify Connectivity form to verify the connectivity of the nets in the design.
→ Select *Verify - Connectivity*.

Verify Connectivity

Net Type

☒ All
 ☐ Regular Only
 ☐ Special Only

Nets

☒ All
 ☐ Selected
 ☐ Named:

Check

☒ Open
 ☒ UnConnected Pin
 ☒ Unrouted Net

☐ Connectivity Loop
 ☒ DanglingWire (Antenna)
 ☒ Weakly Connected Pin

☐ Geometry Loop
 ☐ Geometry Connectivity
 ☐ Keep Previous Results

☐ Divide Power Net
 ☒ Soft PG Connect
 ☐ Raw Violations Mark

☐ Use new open Vio
 ☐ Remove old open Vio
 ☐ Use Virtual Connection

☐ TSV Die Abstract File

Verify Connectivity Report: dtmf_chip.conn.rpt

Report Limits

Error: 1000
 Warning: 50

Set Multiple CPU...

OK

Apply

Cancel

Help

Fields and Options

Net Type	Select one of the following options. Default: All	
	All	Checks all nets, including those that have been previously verified.
	Regular Only	Checks only regular nets.

	<i>Special Only</i>	Checks only special nets.
<i>Nets</i>	Select one of the following options. <i>Default: All</i>	
	<i>All</i>	Checks all nets of the type specified in the <i>Net Type</i> panel.
	<i>Selected</i>	Checks only nets in the selected set that are of the type specified in the <i>Net Type</i> panel.
	<i>Named</i>	Checks only nets of the name specified that are of the type specified in the <i>Net Type</i> panel. You can use wildcards (* and ?) when you specify the net names.
<i>Check</i>	Select one or more of the following options:	
	<i>Open</i>	Checks for open violations. Opens are parts of nets, such as wires or pins, that are connected to each other but are missing a connection to the net as a whole.
	<i>UnConnected Pin</i>	Checks for pins that are not connected to any objects.
	<i>Unrouted Net</i>	Checks for nets that are not routed.
	<i>Connectivity Loop</i>	Checks for connectivity loops in regular wires. The software detects connectivity loops based on the end points of the center line of a regular wire segment or the center of a via.  <i>Connectivity Loop, Geometry Loop, and Geometry Connectivity are mutually exclusive.</i>
	<i>DanglingWire Antenna</i>	Checks for violations due to unconnected wires (also called geometrical antennas or dangling wires).
	<i>Weakly Connected Pin</i>	Checks for routing to more than one port of the weakly connected pin ports. When this option is selected, Verify Connectivity marks a Weakly Connected Pin violation in one of the ports of the pin.

	<i>Geometry Loop</i>	Checks for loop violations of regular nets using a geometrical model. The nets do not have to overlap on the center line. When you specify this option, the software does not perform any other connectivity checks. Select this option if you use a third-party router that does not route using the centerline connection routing technique. In this case, <i>Connectivity Loop</i> might not detect connectivity loop violations.  <i>Connectivity Loop, Geometry Loop, and Geometry Connectivity</i> are mutually exclusive.
	<i>Geometry Connectivity</i>	Checks for connectivity violations of regular wires. Uses a geometrical model instead of the center-line model. In other words, if the wires overlap at any point, they are considered to be connected--they do not have to connect at the center line. Use this parameter if you manually change the routing or use a third-party router that does not route using the center-line connection routing technique.  <i>Connectivity Loop, Geometry Loop, and Geometry Connectivity</i> are mutually exclusive.
	<i>Keep Previous Results</i>	Displays incremental results in the Violation Browser. When this option is selected, Verify Connectivity appends violation markers to the previous results instead of overwriting them.
	<i>Divide Power Net</i>	Divides power nets into four subareas for connectivity verification. Use this option to decrease memory usage in 32-bit machines with limited memory. This option might increase or decrease the run time, depending on the design.
	<i>Soft PG Connect</i>	Checks soft Power/Ground connects on masterslice layers. This option is enabled by default. Deselect this option if you want connectivity on masterslice layers to be ignored.

	<i>Raw Violations Mark</i>	Displays violation markers for opens as the bounding box of the island. By default, violation markers for opens are displayed as polygons that include all wires, pins, and vias that connect to the island.
	<i>Use new open Vio</i>	Checks using the new open violations.
	<i>Remove old open Vio</i>	Checks by removing the old open violations.
	<i>Use Virtual Connection</i>	Implies a virtual connection for all bumps and external I/O pins of the same net. Select this option to override default behavior of Verify Connectivity in which bumps and external I/O pins of the same net are not considered to be interconnected. It is useful in flip chip designs, where the power bumps are connected outside the chip.
	<i>TSV Die Abstract File</i>	Specifies the input file for TSV die.
Verify Connectivity Report		
	Specifies the report file for connectivity violation data.	
Report Limits	Specifies error and warning limits	
	<i>Error</i>	Specifies the maximum number of errors to report. The software generates a warning message if this number is exceeded. <i>Default:</i> 1000
	<i>Warning</i>	Specifies the maximum number of warnings to report. <i>Default:</i> 50
Set Multiple CPU	Displays the Multiple CPU Processing form in which you can specify multi-threading settings for verifying connectivity.	

Related Text Commands

The following text command provides equivalent or additional functionality:

- `verify_connectivity`

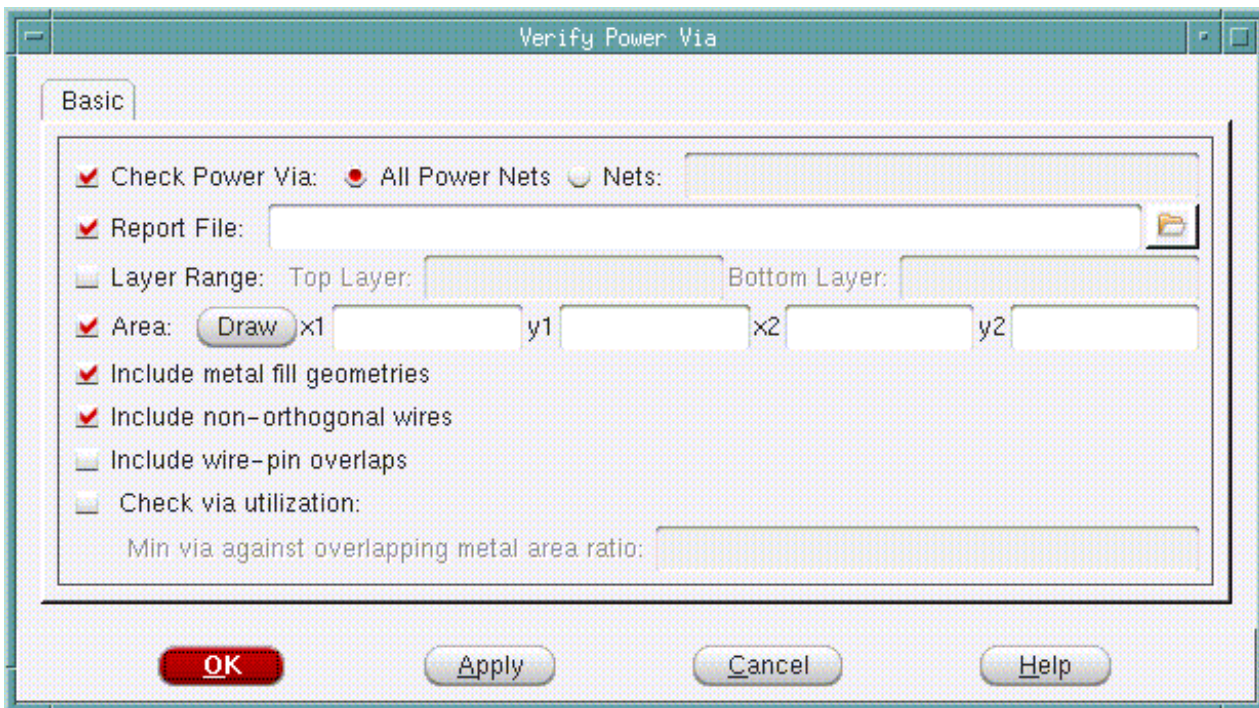
For more information, see "Verify Commands" in the *Voltus Text Command Reference*.

Verify Power Via

Checks for metal geometry overlap on power-grid nets and for missing or open vias in the design.

Note: To verify the design, the power-grid must be routed.

→ *Select Verify - Power Via.*



Fields and Descriptions

<i>Check Power Via</i>	• <i>All Power Nets</i>: Verifies power vias in all the power nets. This is the default option. • <i>Nets</i>: Verifies power vias in the specified net(s).
<i>Report File</i>	Specifies the name of the output file for the report.
<i>Layer Range:</i> <i>Top Layer</i> <i>Bottom Layer</i>	Optional. The layer range checks for missing stacked vias between only bottom and top layers. It also checks for missing vias between all intermediate layer intersections. You can specify standard metal layer names, such as M1, M2, and so on, or metal layer names as defined in the LEF file. <i>Default</i>: All layers. If <i>Bottom Layer</i> is specified but <i>Top Layer</i> is not, it assumes one layer up.
<i>Area:</i> <i>Draw x1 y1 x2 y2</i>	When selected, enables you to draw a box in the Voltus canvas to fill in <i>x1 y1 x2 y2</i> values automatically. Optionally, you can also specify <i>x1 y1 x2 y2</i> coordinates.
<i>Include metal fill geometries</i>	Specifies that metal fill should also be checked. <i>Default</i>: Metal fill will be ignored.
<i>Include non-orthogonal wires</i>	Specifies that non-orthogonal crossing wires should also be checked. <i>Default</i>: Non-orthogonal wires will be ignored.
<i>Include wire-pin overlaps</i>	Specifies that wire-pin overlaps should also be checked. <i>Default</i>: Wire-pin overlaps will be ignored.
<i>Check via utilization: Min via against overlapping metal area ratio</i>	When selected, checks the via utilization. Specifies the minimum via area against overlapping metal geometry ratio. The value must be between 0 and 1.

Tools Menu

- [Design Browser](#)
- [Object Browser](#)
- [Violation Browser](#)
- [Layout Viewer](#)
- [Conformal](#)
- [Constraint](#)
- [Editor](#)
- [Report](#)
- [Write To GIF File](#)
- [Screen Dump](#)
- [Display Screen Dump](#)
- [Create Ruler](#)
- [Clear All Ruler](#)

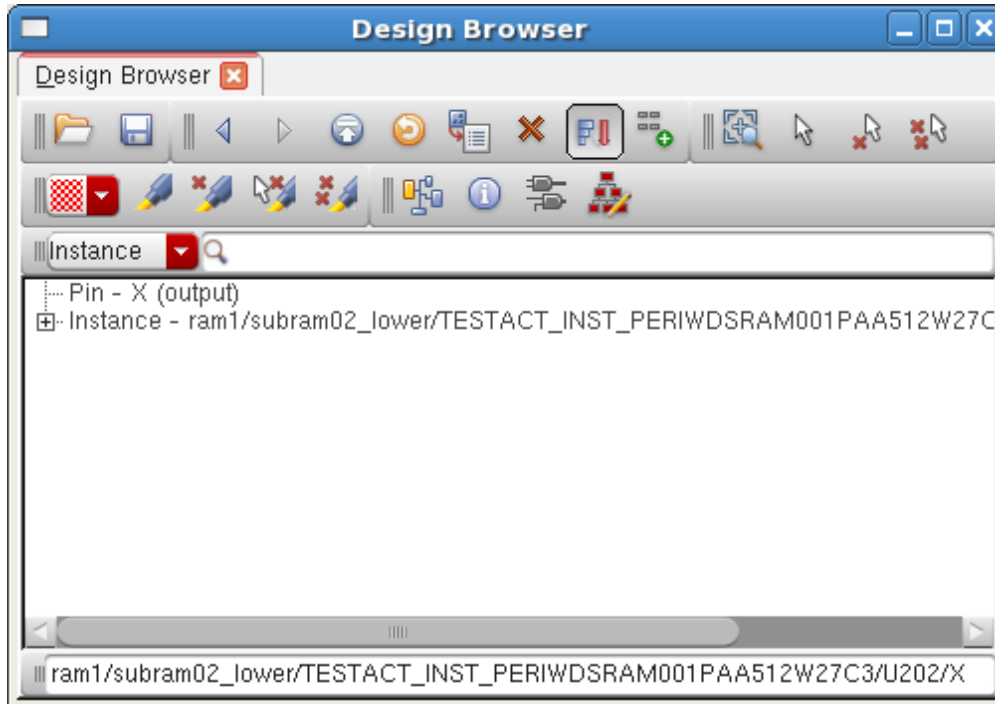
Design Browser

Use the Design Browser to navigate through the chip's design hierarchy. You can use the Design Browser to view the design hierarchy tree at any time after the design is imported. The Design Browser also makes it easier to highlight specific modules, instances, or nets in the design display area.

You can use the widgets in the Design Browser to open forms to navigate through displays, and perform actions. From the Design Browser, you can access the:

- Connectivity Browser to display the number of nets between instances
- Attribute Editor to display an object's type, name, and attributes
- Schematic Viewer to display hierarchical and instance-based schematics

→ Choose *Tools - Design Browser*.



Note: If you select an object in the main window before opening Design Browser, the browser is automatically populated with the details of that object. If multiple objects are selected in the main window, Design Browser automatically displays details of all selected objects.

Design Browser Tool Widgets

You can use the widgets in the Design Browser to navigate through displays, open tools, make selections, and perform actions. The widgets are categorized into groups. You can right-click any empty space on the toolbars and choose which group of widgets to display.

File Widgets	
	<i>Load</i> - Displays the Load Design Browser File form, where you can enter the filename and the format of the placement file.
	<i>Save As</i> - Displays the Save Design Browser File form, where you can specify the filename. The recommended extension is <code>.dbf</code> .

View Widgets	
	<i>Previous Page</i> - Displays viewable data in a reverse direction.
	<i>Next Page</i> - Displays viewable data in a forward direction.
	<i>Top Page</i> - Returns to the top of the design.
	<i>Refresh</i> - Refreshes the Design Browser display.
	<i>Get Selected</i> - Gets the selected object and displays it in the Design Browser.
	<i>Delete Group/Group Member</i> - Deletes an individual instance from a user-defined group or an entire user-defined group.
	<i>Sort By Size</i> - Displays instances in the order of the number of cells in the instances. This toggle button is selected by default. Deselect this button to display instances sorted alphabetically by name.
	<i>Accumulate Results</i> - Traces the hierarchical path for the selected object in the design display area.
Select Widgets	
	<i>Zoom Selected</i> - Zooms in on the object (that you have selected in the Design Browser) in the design display area.
	<i>Select</i> - Selects an object in the design display area.
	<i>Deselect</i> - Deselects an object in the design display area.
	<i>Deselect All</i> - Deselects all objects in the design display area.
Highlight Widgets	
	<i>Edit Highlight Color</i> - Allows you to select a highlight color from the available choices in the drop-down menu.
	<i>Highlight</i> - Highlights an object in the design display area.

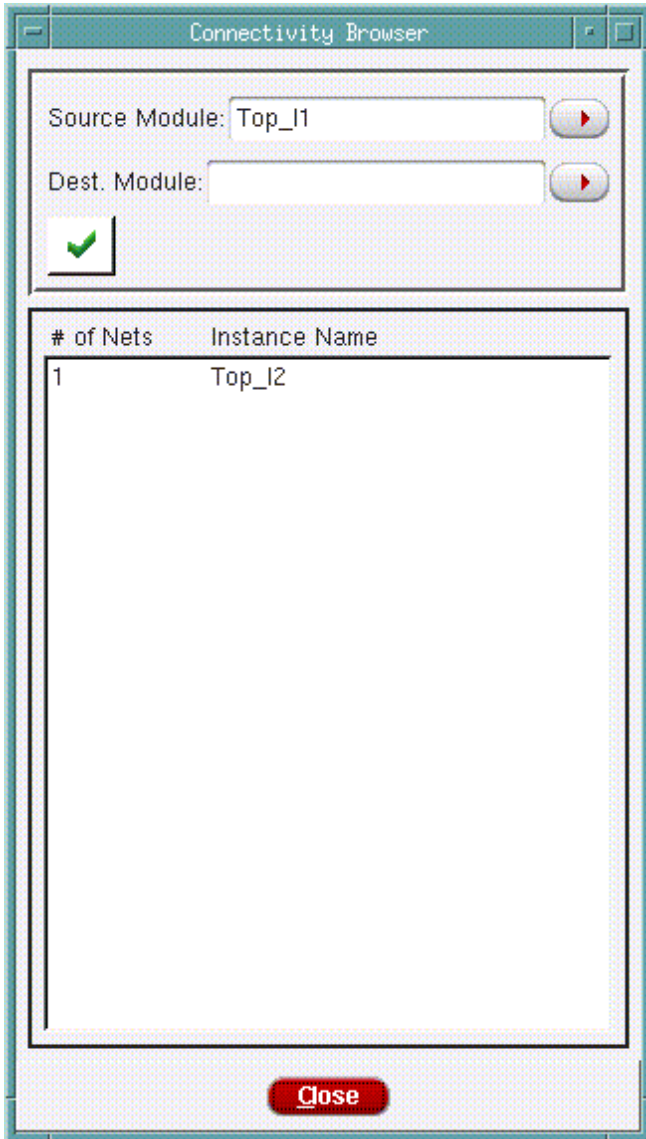
	<i>Dehighlight</i> - Dehighlights an object in the design display area.
	<i>Dehighlight Selected</i> - Dehighlights the selected object.
	<i>Dehighlight All</i> - Dehighlights all objects in the design display area.
Tool Widgets	
	<i>Connectivity Browser</i> - Opens the Connectivity Browser. For more information, see Connectivity Browser .
	<i>Attribute Viewer</i> - Opens the Attribute Viewer for the selected object. For more information, see Attribute Viewer .
	<i>Show Cone Schematic (New)</i> - Opens the Schematic Viewer to display schematic in flattened mode.
	<i>Show Module Schematic (New)</i> - Opens the Schematic Viewer to display schematic in hierarchical mode. For more information, see Schematic Viewer .
Find Bar	

	Enables you to search and display hierarchy information for the object you specify in the adjacent text input box. You can use wildcards to specify object names in the text input box. When you enter text in the text input box, the Clear button (✖) is displayed next to the text input box. Use this button to clear any existing text in the input box. You can view information on the following object types: • <i>Instance</i>: Enables you to search for an instance and display its hierarchy in the Design Browser window. You can traverse to the next hierarchy by double-clicking on the instance name. Along with the instance name, the instance's cell type and number of cells are displayed. • <i>Net</i>: Enables you to search for a net and display its hierarchy in the Design Browser window. Double-clicking the net name displays the net connection. • <i>Group</i>: Enables you to search for an existing instance group and display its hierarchy in the Design Browser window. • <i>Cell</i>: Enables you to search for a cell and display its hierarchy in the Design Browser window.
State Bar	
	Displays the name of selected object on the bar at the bottom of the browser.

Connectivity Browser

Use the Connectivity Browser to display connections from a source module to a destination module, including module guides, blocks, I/O blocks, and standard cells.

→ Choose *Tools- Design Browser* - Highlight a module in the *Design Browser*, and then click the *Connectivity Browser* widget.



Fields and Options

<i>Source Module Instance</i>	Specify the name of the instance
<i>Destination Module</i>	Use this pull-down menu to select the destination module.
<i>Check</i>	Displays the connection information.

Attribute Viewer

Use the Attribute Viewer to display an object's type, name, and attributes. The contents of the Attribute Viewer form change depending on the object selected in the design.

To open the Attribute Viewer from the Layout window, do the following:

→ For a single object, double-left click on an object.

To open the Attribute Viewer from the Design Browser, do the following:

1. Choose *Tools - Design Browser*.
2. In the Design Browser, highlight or left click on an object or a set of objects.
3. Click the *Attribute Viewer* from the Design Browser tool widgets.

Note: The Attribute Viewer displays different attributes, depending on the objects selected.

Attribute Viewer

Object Type: Net

Name	Value	Type
Name	tdigitO_35	String
No. of Terminals	2	Integer
No. of Fanin Terms	1	Integer
No. of FanOut Terms	1	Integer
Total Wire Length	0.0	Double
X-dir Wire Length	0.0	Double
Y-dir Wire Length	0.0	Double
Net Capacitance	0.000000e+00 pf	String
Total Resistance	0.000000	Double
USE	SIGNAL	Enumerate

Close Help

Fields and Options

<i>Object Type</i>	Displays the type of object currently selected.
<i>Close</i>	Closes the Attribute Viewer window.

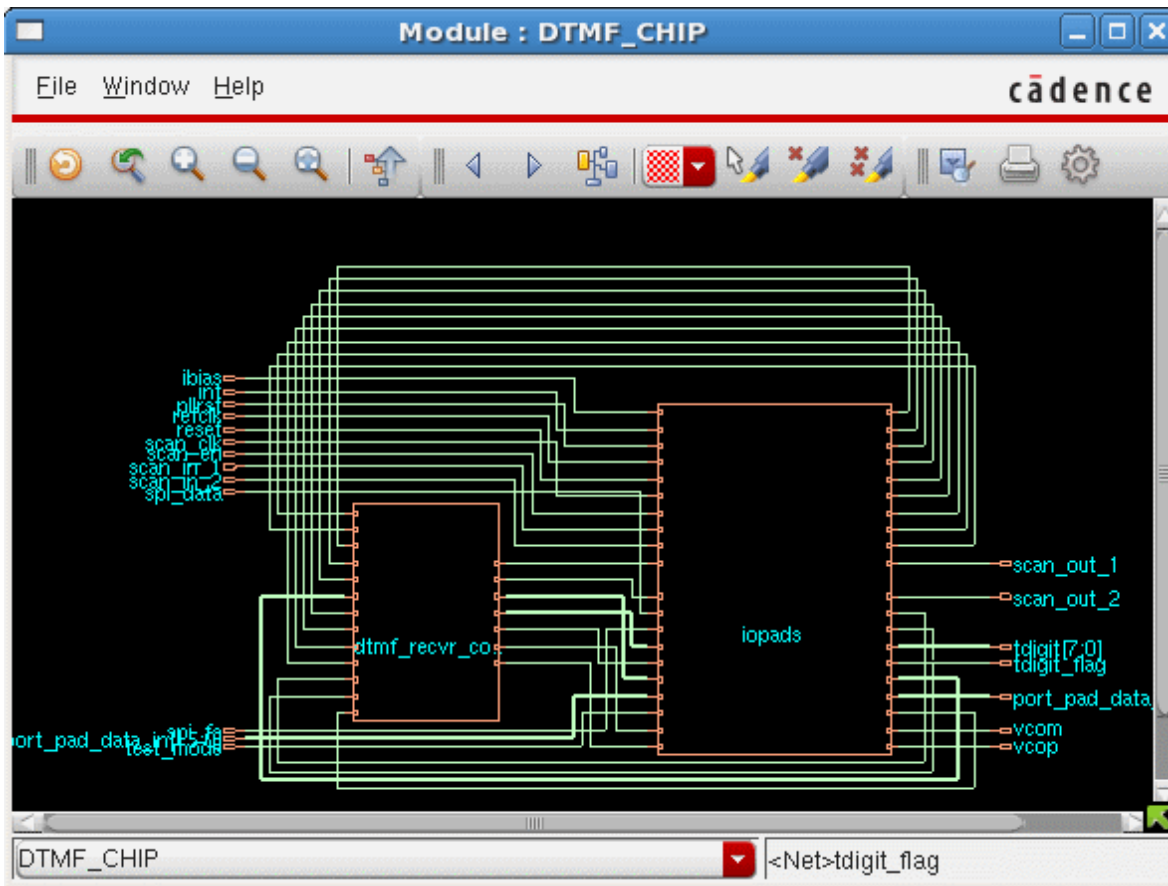
Schematic Viewer

Use the Schematic Viewer with Design Browser to explore a design's connectivity, relationship between modules and instances, and design changes, such as CTS and ECO. You can cross-probe from the Schematic Viewer to the main display to get immediate feedback for debugging and

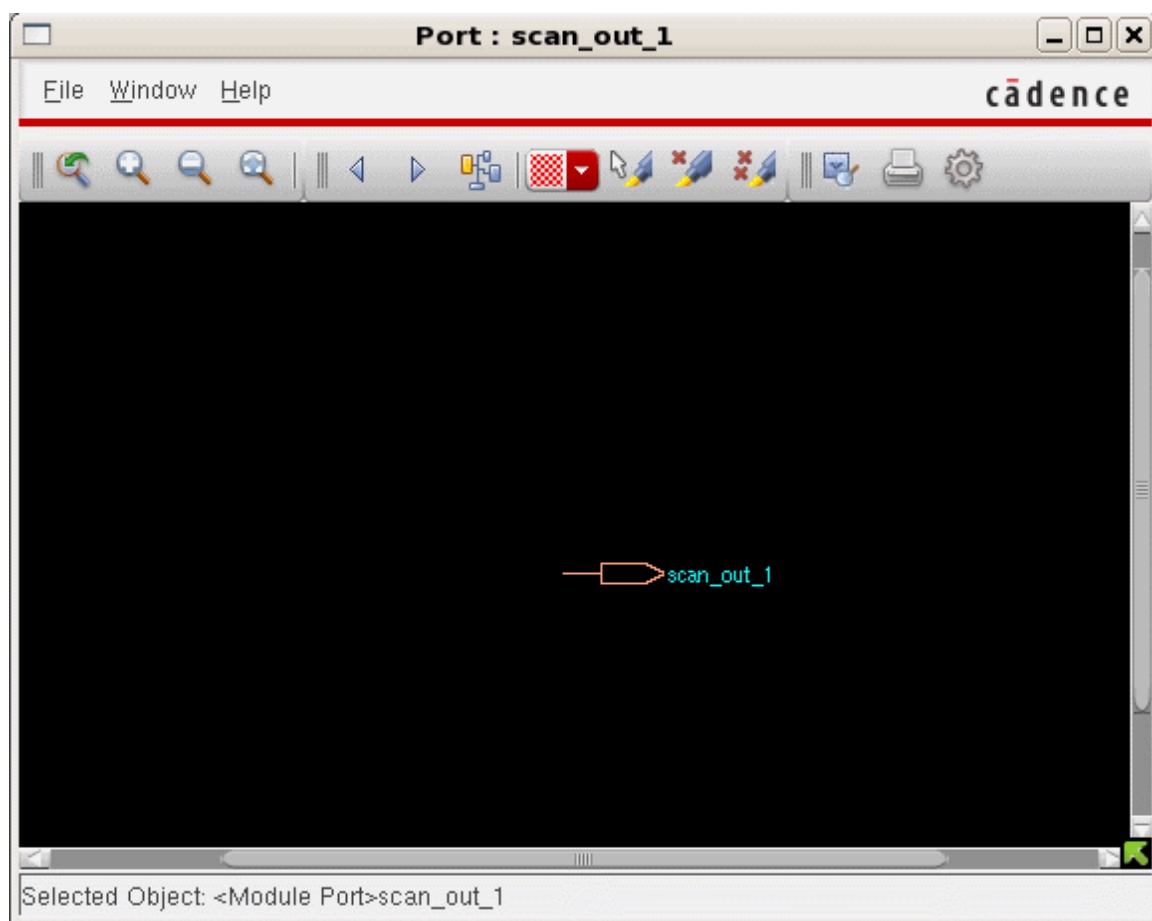
diagnosing your design.

The Schematic Viewer has two main views:

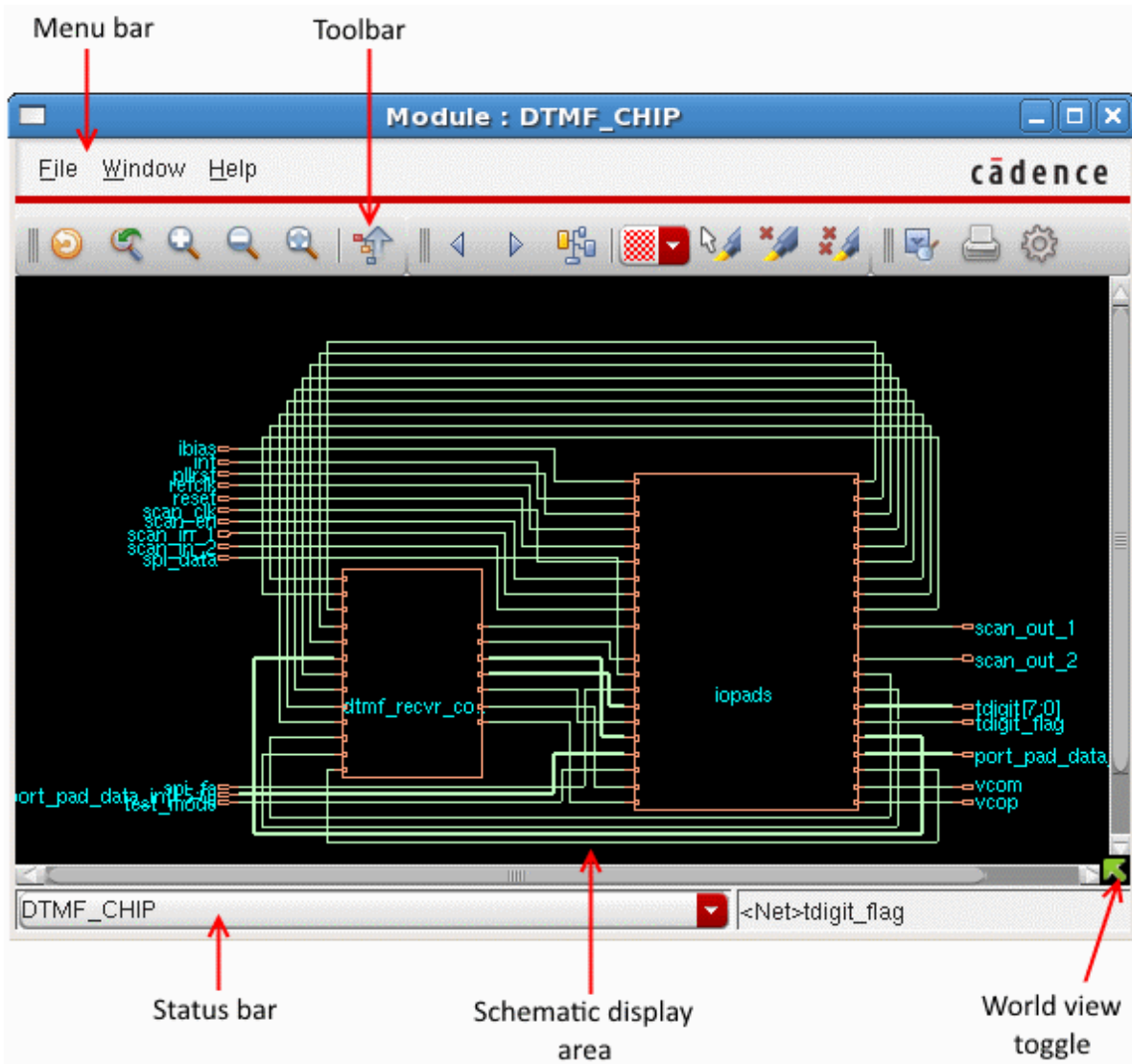
- Hierarchical view - To open Schematic Viewer in Hierarchical view:
 - Select a module, instance, net, or pin in Design Browser and click the *Show Hierarchical Schematic* widget.
 - Select an MSV violation in Violation Browser and click the *Show MSV Violation Schematic* widget.



- Flattened view - To open Schematic Viewer in Flattened view, select an instance, net, or pin in Design Browser and click the *Show Instance Based Schematic* widget.



Schematic Viewer Components



Menu Bar

- File Menu
- Window Menu

File Menu

The File menu enables you to print schematics and close the Schematic Viewer window.

Options	Description
<i>Preferences</i>	Opens the Preferences form where you can set <i>Display</i> , <i>Color</i> , and <i>Font</i> preferences.
<i>Print</i>	Opens the Print Schematic form where you can choose printer options.
<i>Quit</i>	Closes the Schematic Viewer window.

Window Menu

The Windows menu lists all the active Schematic viewer windows for your session. Click on a window name to bring it to the front of your screen.

Choose *Cascade Window* to refresh your desktop and display the main window on top with all other open windows in a cascading view to the left of the main window.

Toolbar

The Schematic Viewer toolbar provides a quick way to access the most commonly used commands. This section describes the toolbar items, which are available for flattened and module schematics, except where noted.

Icon	Action	Description
	Reload View	Reloads or refreshes the current view.
	Last View	Toggles the display between the previous location or zoom level and the current location or zoom level.
	Zoom In	Displays a smaller area of the schematic in greater detail. Each click zooms in one level. The equivalent bindkey is <code>z</code> (<code>Shift-Z</code>).
	Zoom Out	Displays a larger area of the schematic in less detail. Each click zooms out one level. The equivalent bindkey is <code>z</code> .
	Zoom to Full Window	Displays the entire schematic within the display area. The equivalent bindkey is <code>f</code> .

	Push View Up	Displays the parent-level hierarchy. To push the view up to a higher-level schematic, select an object in the schematic display area and click this widget.
	Trace Driver	Traces all possible drivers of the selected object and highlights them in the schematic display area. The default highlight color for drivers is yellow.
	Trace Load	Traces all possible loads of the seobject and highlights them in the schematic display area. The default highlight color for loads is red.
	Trace Connectivity	Traces all possible drivers and loads of the selected object and highlights connectivity from the object's drivers to its loads. The default highlight colors for drivers and loads are yellow and red, respectively.
	Edit Highlight Color	Allows you to select a highlight color from the available choices in the drop-down menu.
	Highlight Selected	Highlights selected object in the schematic display area.
	Dehighlight	Dehighlights an object in the schematic display area.
	Dehighlight All	Dehighlights all objects in the schematic display area.
	Find	Click this widget to open the Find Schematic Object form to find and highlight a specific object in the schematic display area.
	Print	Opens the Print Schematic form, which enables you to print the required schematic display.
	Preferences	Opens the Preferences form where you can set <i>Display</i> , <i>Color</i> , and <i>Font</i> preferences.

Schematic Display Area

The schematic display area constitutes the major portion of the Schematic Viewer window. The schematics of the selected object is displayed here in flattened or hierarchical view. To change the default display properties for objects in the display area, use the *Color* tab on the Preferences form.

World View Window

The world view window identifies the location of the current view in the schematic display area, relative to the entire area. The world view is identified by the pink crossbox. When you display an entire schematic in the area, the crossbox encompasses the world view window. When you zoom and pan through the schematic in the display area, the crossbox identifies where you are relative to the entire schematic.

- To define a new world view, left-click and drag the mouse to create a new crossbox in the world view window.
- To pan left, right, up, or down in the schematic display, right-click and drag the crossbox in the world view window.

Status Bar

The *Status Bar* is located at the bottom of the Schematic Viewer. When you select or hover over an object, this bar displays the object type, name, and ID number.

Context Menu

To access the context menu, right-click an object in the Schematic Viewer window. The options available in context menu varies depending on the:

- Current view of Schematic Viewer - Hierarchical or Flattened
- Object you have selected in the schematic display area

Hierarchical View Context Menu

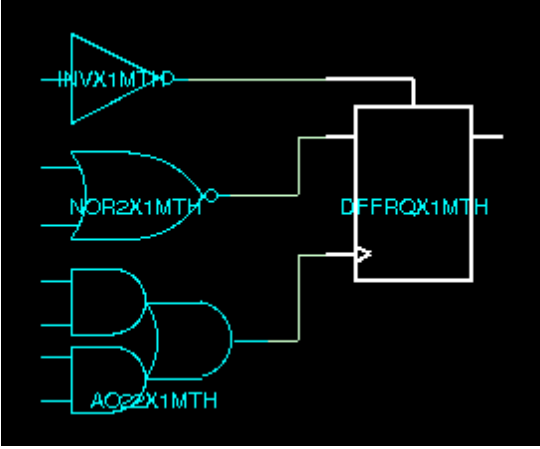
The following table describes the options that are available in the context menu of the Schematic viewer when it is opened in Hierarchical view.

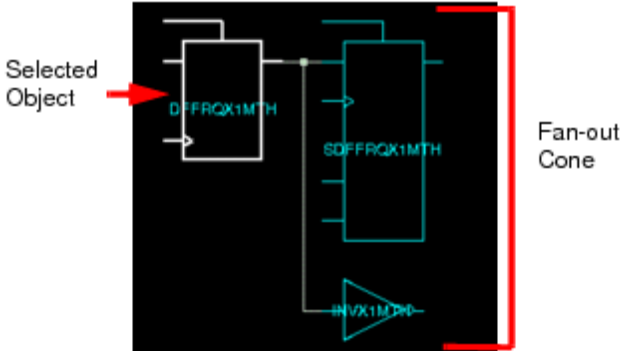
Options	Description
<i>Attribute Editor</i>	Opens the Attribute Editor for the selected object.
<i>Design Browser</i>	Opens the object in Design Browser.
<i>Highlight</i>	Highlights the selected object in the color or pattern you select from the submenu.
<i>Dehighlight</i>	Removes highlight.

<i>Copy Name</i>	Copies the object name.
<i>Open In</i>	Allows you to open selected object in one of the following: • <i>Schematic Viewer (new)</i> • <i>Schematic Viewer (cone)</i> • <i>Schematic Viewer (cone append)</i>
<i>Color Net Segment</i>	Opens the Select Color form from which you can select a different color for the net segment. This option is available only if you right-click a net in the display area.
<i>Open Next Level Instance View</i>	Displays next-level views for instances. This option is available only if you right-click an instance in the display area.
<i>Close Next Level Instance View</i>	Closes a next-level instance view. This option is available only if you right-click an instance in the display area.
<i>Trace Previous Driver</i>	Traces the previous driver of the selected port or instance pin. This option is available only if you right-click a port or instance pin in the display area.
<i>Trace Next Load</i>	Traces the next load of the selected port or instance pin. This option is available only if you right-click a port or instance pin in the display area.
<i>Port to Port Trace</i>	Traces all connections to and from a port and highlights them in the color you select in the Select Color form. This option is available only if you right-click a port in the display area.
<i>Show Critical Path</i>	Displays the critical path.
<i>Change Color</i>	Opens a submenu from which you can change the color of the selected object.
<i>Default Color</i>	Resets the selected object type to the default color.

Flattened View Context Menu

The following table describes the options that are available from the Flattened view context menu of Schematic Viewer.

Options	Description
<i>Attribute Editor</i>	Opens the Attribute Editor for the selected object.
<i>Design Browser</i>	Opens the object in Design Browser.
<i>Highlight</i>	Highlights the selected object in the color or pattern you select from the submenu.
<i>Dehighlight</i>	Removes highlight.
<i>Copy Name</i>	Copies the object name.
<i>Open In</i>	Allows you to open selected object in one of the following: • <i>Schematic Viewer (cone)</i> • <i>Schematic Viewer (cone append)</i>
<i>Open Fan-in Cone</i>	Opens the fan-in cone of the selected object. This option is available in the context menu of instances and block ports. 
<i>Close Fan-in Cone</i>	Closes the fan-in cone of an object.

<i>Open Fan-out Cone</i>	Opens the fan-out cone of the selected object. This option is available in the context menu of instances and block ports.  The diagram shows a circuit schematic on a black background. A red arrow points to a block labeled 'DFFRQX1MTH', which is labeled 'Selected Object'. This block is connected to another block labeled 'SDFFRQX1MTH'. Below 'SDFFRQX1MTH' is a block labeled 'INVX1MTH'. A red rectangular box encloses the 'SDFFRQX1MTH' and 'INVX1MTH' blocks, with a label 'Fan-out Cone' pointing to it.
<i>Close Fan-out Cone</i>	Closes the fan-out cone of an object.
<i>Open Fan-in&out Cone</i>	Opens both the fan-in and fan-out cones of the selected object.
<i>Open Fan-in&out Cone</i>	Closes the fan-in and fan-out cones of an object.

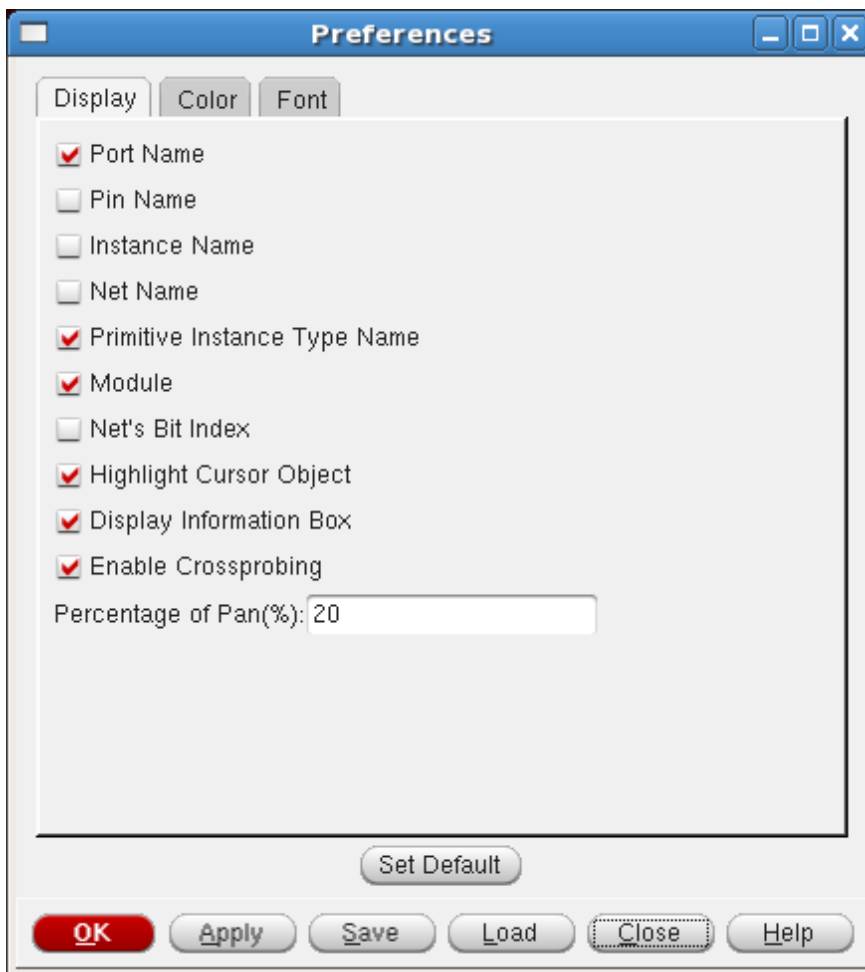
Preferences

The Preferences form enables you to customize the display, colors, and fonts in the schematic display area. It has the following tabs:

- *Display*
- *Color*
- *Font*

Display Tab

Use the *Display* tab to customize the display in the schematic display area.



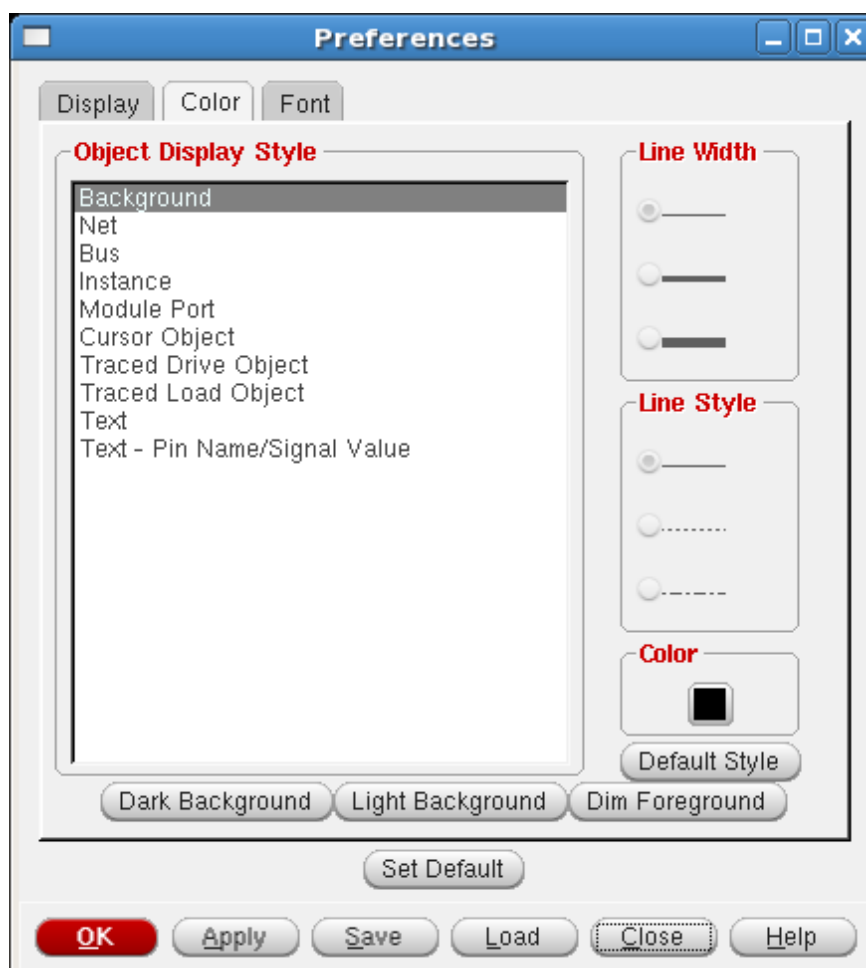
Fields and Options

<i>Port Name</i>	Displays instance/block port names in the schematic display area. Deselect this check box if you do not want to display port names. <i>Default:</i> Selected
<i>Pin Name</i>	Displays I/O pin names in the schematic display area. <i>Default:</i> Unchecked
<i>Instance Name</i>	Displays instance names in the schematic display area. <i>Default:</i> Unchecked

<i>Net Name</i>	Displays net names in the schematic display area. <i>Default:</i> Unchecked
<i>Primitive Instance Type Name</i>	Displays primitive instance type names. <i>Default:</i> Selected
<i>Module</i>	Displays module names in the schematic display area. <i>Default :</i> Selected
<i>Net's Bit Index</i>	Displays bit index for nets in the schematic display area. <i>Default :</i> Unchecked
<i>Highlight Cursor Object</i>	Highlights objects as you move the cursor over them. <i>Default:</i> Selected
<i>Display Information Box</i>	Displays an information box that states the name of the object over which the cursor is currently placed and lists pertinent details about that object. <i>Default:</i> Selected
<i>Enable Crossprobing</i>	Allows cross-probing between the main window and Schematic Viewer. Select or highlight any instance in Schematic Viewer; it will be selected/highlighted in the main window automatically. <i>Default :</i> Selected
<i>Percentage of Pan</i>	Specifies percentage of pan. <i>Default:</i> 20

Color Tab

Use the *Color* tab to change color and line settings for objects in the schematic display area.

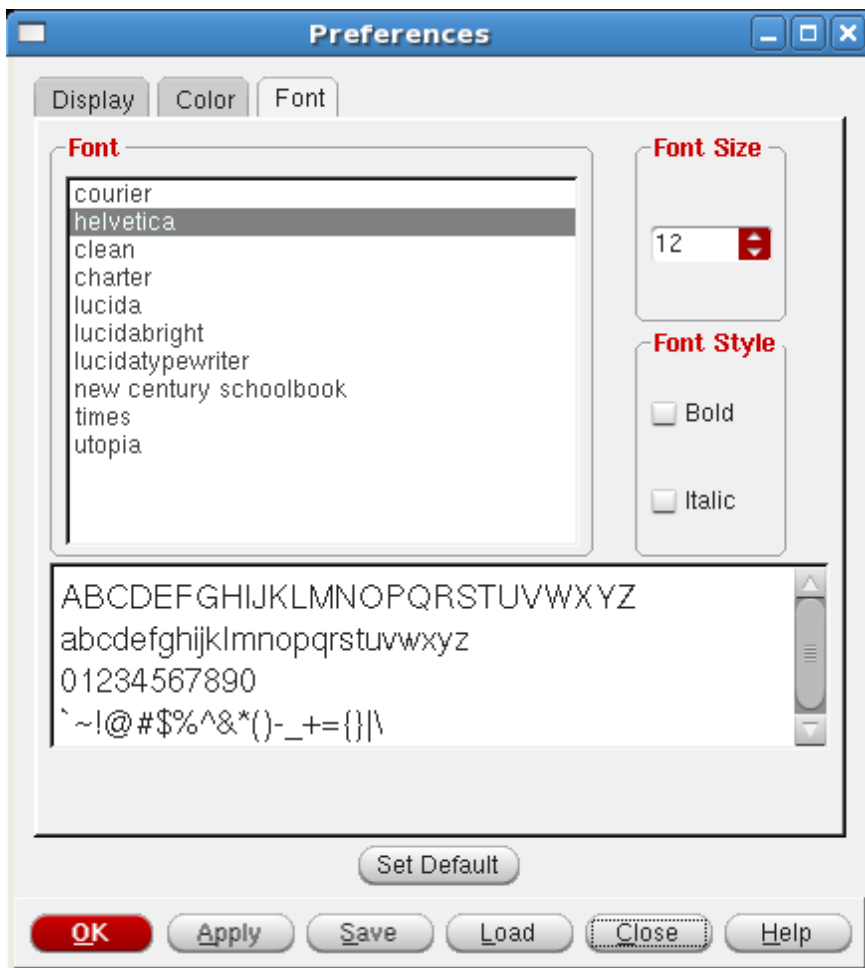


Fields and Options

<i>Object Display Style</i>	Specifies the object type for which you want to change color and line settings. Select any object type from this list and choose appropriate color and line settings, as required.
<i>Line Width</i>	Specifies the line width of the object type selected in <i>Object Display Style</i> list box.
<i>Line Style</i>	Specifies the line style of the object type selected in <i>Object Display Style</i> list box.
<i>Color</i>	Specifies the color of the object selected in <i>Object Display Style</i> list box. Click on the color box to open the Select Color form and choose a new color for the selected object type.
<i>Default Style</i>	Resets the selected object type to the default style.
<i>Dark Background</i> <i>Light Background</i> <i>Dim Background</i>	
	Specifies the background scheme for the <i>Schematic Display Area</i> . The default is <i>Dark Background</i> .
<i>Set Default</i>	Sets the current selection as the default preference.

Font Tab

Use the *Font* tab to change the font size and style settings for the schematic display.

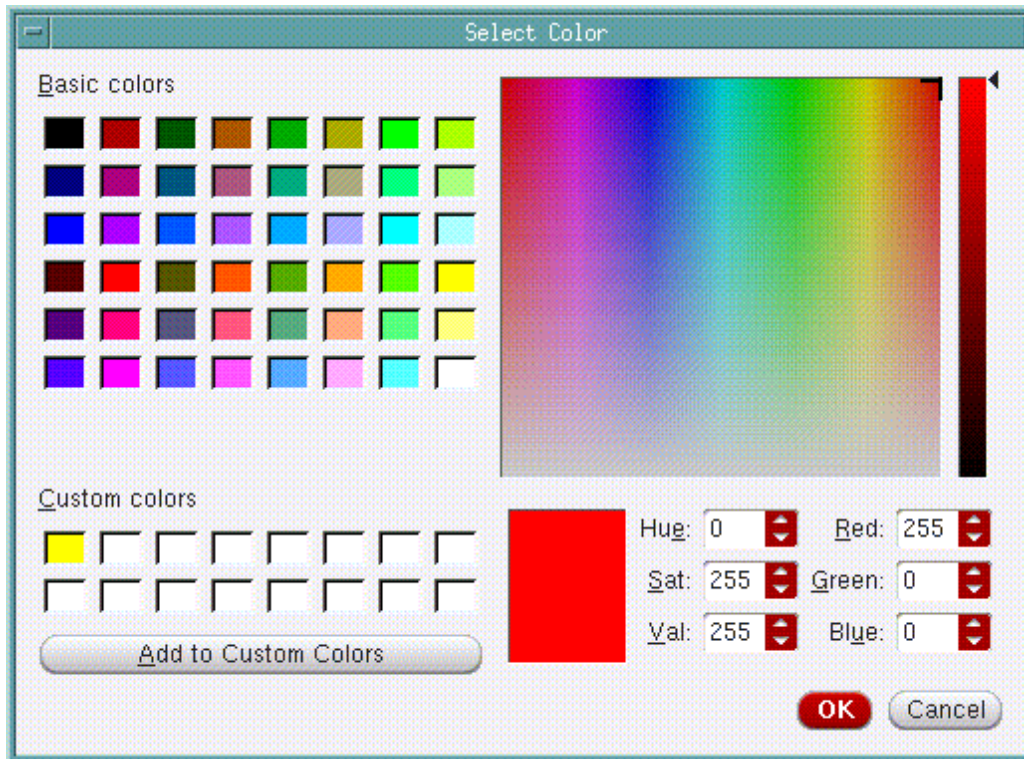


Fields and Options

<i>Font</i>	Specifies the font for the text displayed in the schematic display area.
<i>Font Size</i>	Specifies the size of the schematic display font.
<i>Font Style</i>	Specifies the style to be applied to the schematic display font.
<i>Set Default</i>	Sets the current selection as the default preference.

Select Color

Use the Select Color form to change the color of an object in the schematic viewer. Select the required color from the color palette and click *OK*.

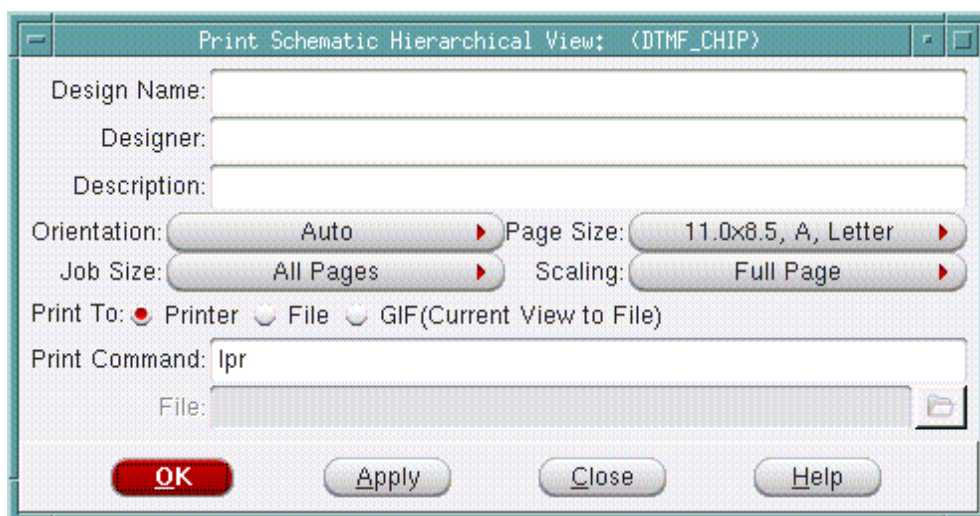


Note: To apply the next available color on the palette, you do not have to re-select the object. Instead, choose *View - Next Color*. You can do this repeatedly.

Print Schematic

Use the Print Schematic form to print the schematic in the display area.

- Choose *File - Print*.



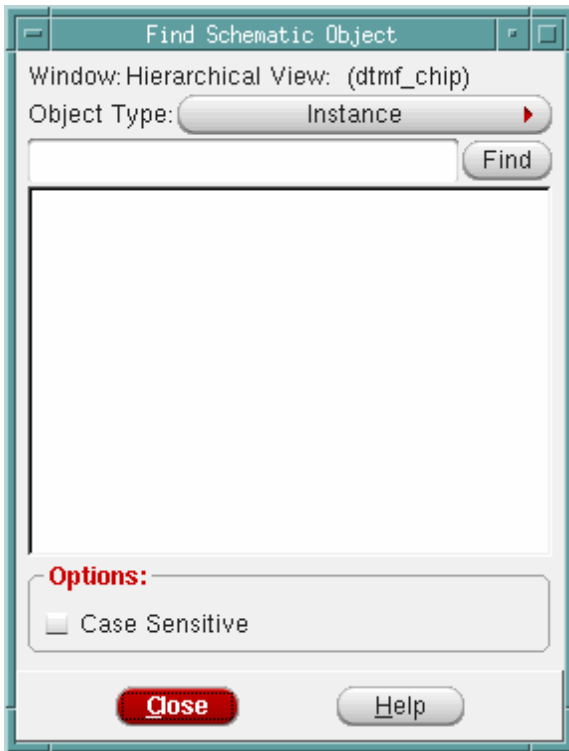
Fields and Options

<i>Design Name</i>	Specifies the name of the design. This information appears in the bottom left corner of the printout.
<i>Designer</i>	Specifies the designer name. This information appears in the bottom left corner of the printout.
<i>Description</i>	Specifies a description. This information appears in the bottom left corner of the printout.
<i>Orientation</i>	Specifies the page orientation. Choose <i>Portrait</i> or <i>Landscape</i> .
<i>Page Size</i>	Specifies the page size. Choose a standard page size of <i>8.5X11</i> or use the pull-down menu to specify a custom size.
<i>Job Size</i>	Specifies the job size. Choose to print all the pages or only the current page.
<i>Scaling</i>	Specifies the scaling. Choose <i>Full Page</i> or <i>Current View</i> .
<i>Print To</i>	Specifies the print output destination. Choose one of the following: • <i>Printer</i> - If you choose this option, type the appropriate print command in the <i>Print Command</i> text box. • <i>File</i> - If you choose this option, the <i>File</i> text box at the bottom is enabled. Type a filename in the <i>File</i> text box or use the adjacent folder button to select the appropriate file. • <i>GIF (Current View to File)</i> - Choose this option to print the current view to the specified file.

Find Schematic Object

Use the Find Schematic Object form to find and highlight a specified object in the schematic display area.

- Click the *Find* widget on the toolbar.



Fields and Options

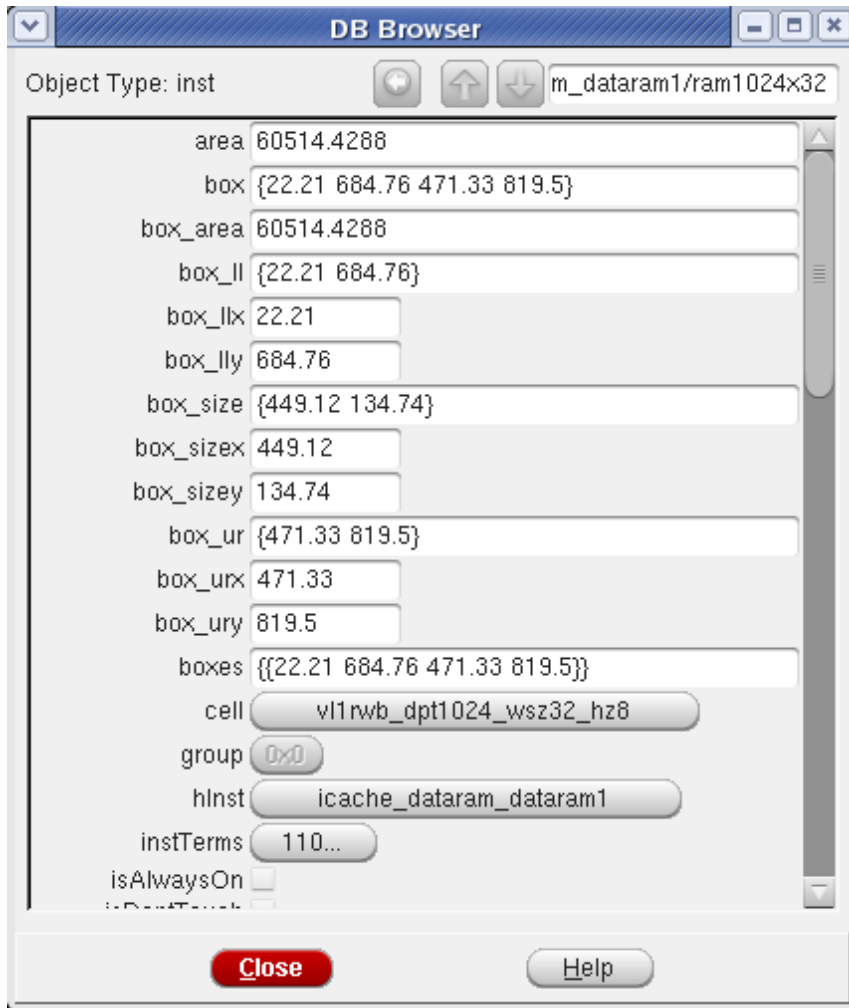
<i>Window</i>	Identifies which schematic window was active when you selected this option.
<i>Object Type</i>	Specifies the object type you want to find.
<i>Find</i>	Specifies the object name or a portion of the name with wildcard characters. Type the object name in the adjacent text box and click <i>Find</i> . The search results are displayed in the list box below this field. Double-click a name to highlight the object in the schematic display area.
<i>Case Sensitive</i>	Allows you to make the search case sensitive.

Object Browser

You can use the object browser (DB Browser) to retrieve the selected object's attributes. The DB Browser displays different attributes, depending on the objects selected in the main window.

→ To launch the DB browser:

1. Select one or more objects in the main window.
2. Select *Tools - DB Browser*.



Fields and Options

<i>Object Type</i>	Specifies the type of object currently selected.
	Click the <i>Show next object in list</i> button to display the attributes of the next object in the sequence. or <i>Prev</i> to view the previous object. Note: This button is available only when multiple objects are selected in the main window.
	Click the <i>Show last object in list</i> button to display the attributes of the previous object in the sequence. Note: These buttons are available only when multiple objects are selected in the main window.
	Click the <i>Back to previous object</i> button to return to the first object in the list. For example, if you select two instances <i>i1</i> and <i>i2</i> . Use the <i>Show next object in list</i> button to move from <i>i1</i> to <i>i2</i> as the active instance, then descend to the <i>i2</i> 's cell. Now if you select <i>Back to previous object</i> button, you return to <i>i1</i> .
<i>Close</i>	Closes the DB Browser.

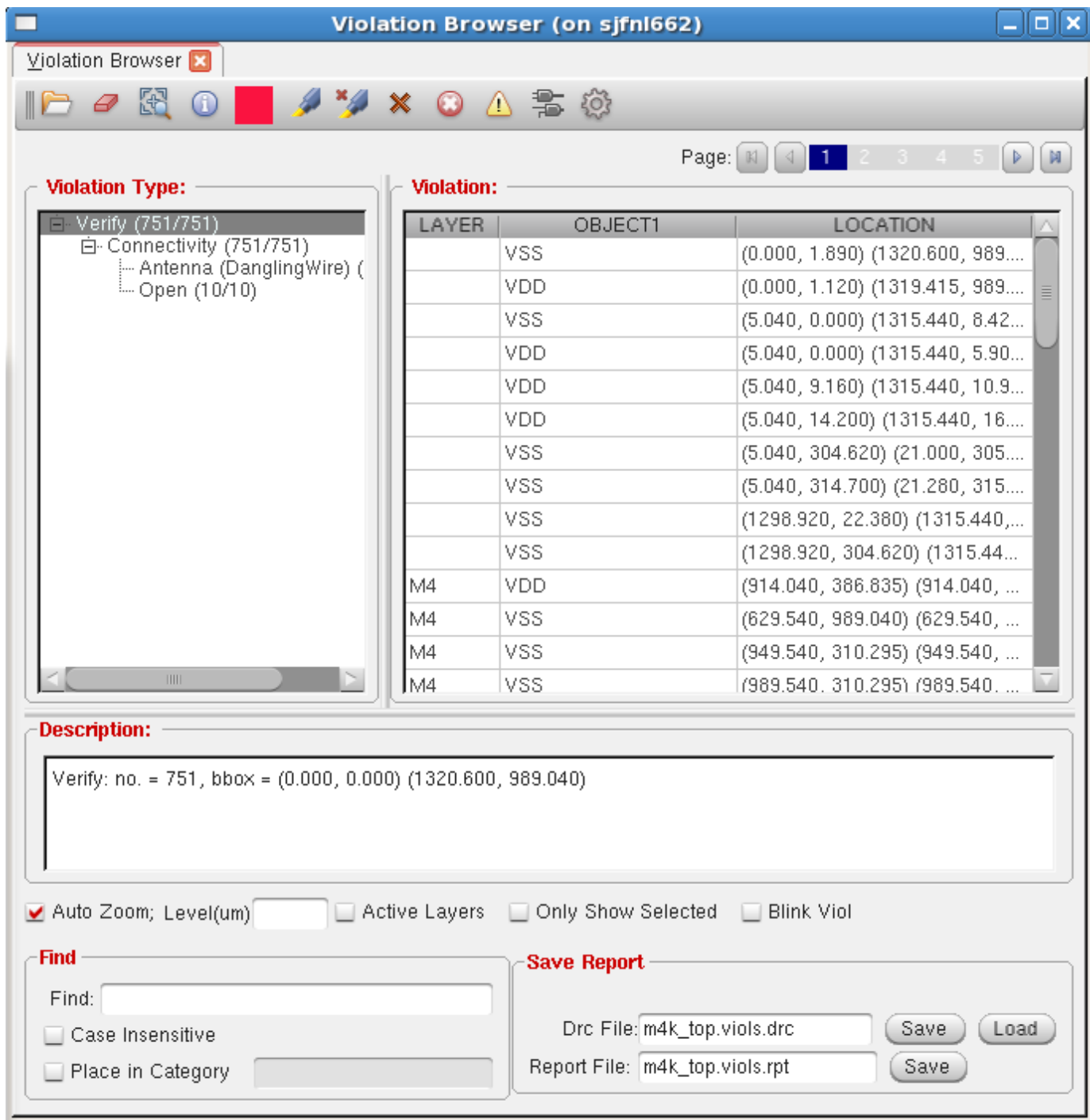
Violation Browser

Use the Violation Browser to display violations and lithography hotspots, mark violation markers as false or true, delete violation markers, and specify a report file.

The Violation Browser also provides access to the following forms:

- Violation Browser Settings
- Load Violation Report

→ Choose *Tools - Violation Browser*.



Fields and Options

	<i>Load Violation Report</i> - Opens the Load Violation Report form. Use this form to load a violation marker file and then view the markers in the Violation Browser.
	<i>Clear Violation</i> - Clears all the violation markers in your design.
	<i>Fit Violation</i> - Displays the violation within the design display area.
	<i>Attribute Editor</i> - Opens the <i>Attribute Editor</i> and populates it with the highlighted violations. Double click on a violation to update fields in the Attribute Editor.
	<i>Highlight Color</i> - Select this button to specify the color for highlighting violations.
	<i>Highlight Violations</i> - Highlights the selected violation(s) in the design display area as well as in the <i>Violation</i> list on the right pane of the Violation Browser. You can select an individual violation in the right pane of the Violation Browser or select violations by type or subtype in the left pane of the browser. To highlight a selected violation or a selected type/subtype of violations, select the highlight color from the color palette and then click the <i>Highlight Violations</i> button.
	<i>Dehighlight Violations</i> - Removes the highlight from the selected violation or selected type/subtype of violations in the design display area.
	<i>Delete Violations</i> - Deletes all selected violations from the design database. Markers are also removed from the design display area and the list of violations.
	<i>Mark Violations as False</i> - Marks the selected violation as a false violation. These violations are stored in the database as false violations. To view the violations that you mark as false, click  in the Violation Browser, select the <i>View False Violations</i> check box in the Violation Browser Settings form, and click <i>OK</i> .
	<i>Mark Violations as True</i> -- Marks the selected violation as a true violation.
	<i>Show MSV Violation Schematic</i> – Displays schematics associated with the highlighted violation in the Schematic Viewer form.
	<i>Setting</i> - Opens the Violation Browser Settings form.

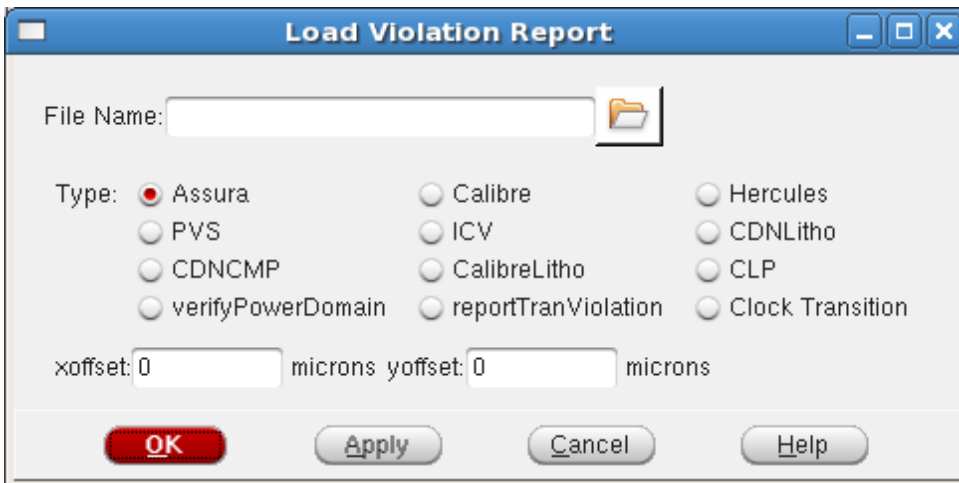
<i>Violation Type</i>	Lists the violation types in a hierarchy window. To show or hide a violation, select it in the <i>Violation</i> or <i>Violation Type</i> window and click the right mouse button. Select <i>Show</i> or <i>Hide</i> from the pop-up window. The numbers in parentheses after each violation type represent the number of violations of that type displayed in the main window and the total number of violations of that type. The count includes true and false violations. For example, if <i>Verify - Connectivity - Open (479/503)</i> means that there are 503 Open violations and 479 of those violations are shown the main window.
<i>Violation</i>	For the highlighted <i>Violation Type</i> , displays the names of the objects with violation markers and their locations.
<i>Description</i>	Displays the layer and bounding box for the violation. Note: Layer is not included for placement violations. You can select text in the <i>Description</i> text box and copy and paste it to another window. You cannot edit text in the <i>Description</i> text box.
<i>Auto Zoom</i>	Automatically zooms in to the selected violations as you browse through the violations. This check box is selected by default.
<i>Zoom Level (um)</i>	Controls the level of zoom for Auto Zoom. When you specify the zoom level, Violation Browser uses the same value for all listed violations so that you do not have to zoom out each time you select a violation.
<i>Active Layers</i>	Keeps only the violation layer(s) visible when you zoom into a violation. If this check box is not selected, the Auto Zoom mode displays layers as per the layer visibility settings on the Layer Control bar.
<i>Only Show Selected</i>	Displays markers for only selected violations. This option is useful if there are many violations in the design. By selecting this option, you can turn off other violation makers in the main window and focus on only the violations you have selected in the browser.
<i>Blink Viol</i>	Makes the marker for the selected violation blink in the main window.
<i>Find</i>	Searches for the specified text and highlights it in the <i>Violation</i> window. This field recognizes pattern matching, so you can type <i>*153*</i> to search for all objects with the string <i>153</i> as part of their <i>LOCATION</i> . Press <i>Enter</i> to find the specified text.
<i>Case Insensitive</i>	Specifies whether the text in the <i>Find</i> text entry box is case sensitive.

<i>Place in Category</i>	Creates a category in the <i>Violation Type</i> window and adds the highlighted object to the category. Categories are not saved.
<i>Drc File - Save</i>	Opens a form that allows you to specify the path and name for saving the DRC file.
<i>Drc File - Load</i>	Opens a form in which you can browse and select the DRC file to be loaded.
<i>Report File - Save</i>	Specifies the report file that contains all the violation information. The browser and the report file list the violations in the same order.

Load Violation Report

Use the Load Violation Report form to load a violation marker file or a hotspot interface format (HIF) file and create markers that the software can interpret. After you load the file, you can view the markers in the Violation Browser.

→ Choose *Tools - Violation Browser - Load Violation Report.* 



Fields and Options

<i>File Name</i>	Specifies the marker file to load.
<i>Type</i>	Specifies the type of marker file. The following file types contain violation markers: • <i>Assura</i> - For Assura violations • <i>Calibre</i> - For Calibre violations • <i>Hercules</i> - For Hercules violations • <i>ICV</i> - For ICValidator violations • <i>PVS</i> - For PVS violations • <i>CDNCMP</i> - For Cadence CMP Predictor (CCP) violations • <i>CLP</i>, <i>verifyPowerDomain</i>, and <i>Clock Transition</i> - For low power violations • <i>reportTranViolation</i> - For DRV transition violations The following types contain hotspot markers, which are used to identify areas susceptible to lithography problems: • <i>CDNLitho</i> (for LPA and NanoRoute litho HIF files) • <i>CalibreLitho</i>
<i>xoffset</i>	Specifies the X offset. When you specify the X and Y offsets, Violation Browser displays the violations at the correct coordinates after accounting for the offsets.
<i>yoffset</i>	Specifies the Y offset, which Violation Browser takes into account to displays violations at the correct coordinates.

Violation Browser Settings

Use the Violations Browser Settings form to specify the settings for displaying violations and lithography hotspots.

→ Choose *Tools - Violation Browser* and click the  button.

Violation Browser Settings

Show Types

☒ All ☐ Geometry ☐ Process Antenna ☐ Overlap
☐ Short ☐ Connectivity ☐ Metal Density ☐ AC Limit

Show Area

☒ Entire area
☐ Specify X1: Y1:
 X2: Y2:

Other Filters

☐ View False Violations
☐ Search Description For Filter String
☒ AND ☐ OR ☐ NOT
Filter Strings:

Delete Violations

☒ Delete Violations By Area
☒ Entire area
☐ Specify X1: Y1:
 X2: Y2:

Fields and Options

<i>Show Types</i>	Lists the types of violations to display or hide in the Violation Browser. Select one or more of the following options:	
	<i>All</i>	Displays all violations.
	<i>Geometry</i>	Displays geometry violations.
	<i>Process Antenna</i>	Displays process antenna violations.
	<i>Overlap</i>	Displays overlap violations.

	<i>Short</i>	Displays short violations.
	<i>Connectivity</i>	Displays connectivity violations.
	<i>Metal Density</i>	Displays metal density violations.
	<i>AC Limit</i>	Displays AC limit violations.
<i>Show Area</i>	Displays violations in the entire area or a specified portion of the area. Specify the portion of the area by using one of the following methods: - Using the <i>Draw</i> button and your mouse to create a rectangular area in the design. The software displays the coordinates in the <i>X1</i>, <i>Y1</i>, <i>X2</i>, and <i>Y2</i> fields. - Manually enter the coordinates in the <i>X1</i>, <i>Y1</i>, <i>X2</i>, and <i>Y2</i> fields.	
<i>Other Filters</i>	Specify additional conditions by selecting one or more of the following options:	
	<i>View False Violations</i>	
		Displays violations that you have marked as false.
	<i>Search Description For Filter String</i>	
		Searches for text in the violation message (in the bottom portion of the Violation Browser window) and the violation description (in the top portion of the Violation Browser window).
	<i>AND OR NOT</i>	
		- The <i>AND</i> and <i>OR</i> conditions work on single strings or multiple strings. - The <i>NOT</i> condition works only on a single string.
	<i>Filter Strings</i>	
		Specifies a text string in the violation message (in the bottom portion of the Violation Browser). When you specify a text string, the Violation Browser searches the message of each violation and displays only the violations whose messages match the search conditions. To specify a list of strings, separate each string with a space. To specify a literal string with space in it, enclose the string in double

quotation marks. For example, you can specify strings such as the following:

- M5
- (600, 500)
- M3 (700, 400)
- "Pin of Net"

You can specify the AND, OR, and NOT conditions with the search strings. Use:

- && between strings to specify the AND condition.
- || between strings to specify the OR condition.
- ! before a string to specify the NOT condition.

The *Filter Strings* field allows you to filter nets based on the AC limit violation type (PEAK/AVERAGE/RMS). For example, you can type the following in the *Filter Strings* field to view the nets with the PEAK violation type in the *Violation Browser* form:

description(*Peak*)

Violation Type:

- Verify (383/383)
 - ACLimit (383/383)
 - ahbmasterin[2] (6/6)
 - ahbmasterin[20] (1/1)
 - ahbmasterin[30] (8/8)
 - ahbmasterin[24] (4/4)
 - ahbmasterin[18] (6/6)
 - ahbmasterin[27] (1/1)
 - ahbmasterin[28] (4/4)
 - ahbmasterin[23] (6/6)
 - ahbmasterin[9] (4/4)
 - ahbmasterin[31] (7/7)
 - ahbmasterin[26] (4/4)
 - ahbmasterin[6] (1/1)

Violation:

LAYER	OBJECT1	LOCATION
M1(1)	ahbmasterin[2]	(36.610, ...
M3(3)	ahbmasterin[2]	(0.050, 1...
M1(1)	ahbmasterin[2]	(2.700, 1...
M2(2)	ahbmasterin[2]	(2.650, 1...
M1(1)	ahbmasterin[2]	(2.790, 1...
M2(2)	ahbmasterin[2]	(36.650, ...

Description:

Pin of Net ahbmasterin[2]
Peak Current: 4.877866 Peak Limit: 4.648000 jN @
bbox = (36.610, 1199.550) (36.700, 1199.650)

☒ Auto Zoom: Level(um) ☐ Active Layers ☐ Blink Viol

Find

Find:

☐ Case Insensitive

☐ Place in Category

Save Report

Drc File: dma_mac.viols.drc

Report File: dma_mac.viols.rpt

<i>Delete Violations</i>	Enables you to delete all violations or delete violations in the specified area only.	
<i>Delete Violations By Area</i>		
	Enables you to delete violations in the area you specify on this form or the area you select with the mouse.	
	<i>Entire area</i>	Deletes the types of violations specified in the entire core design area.
	<i>Specify</i>	Specifies an area from which to delete violations. Only the types of violations specified by this form are deleted.
	<i>Draw ...</i>	Lets you use the mouse to outline an area from which to delete violations.

Selected Clocks	Displays the clock tree driven by the selected clocks.
Add	Adds a clock from the list of crossover, pre-route, or all clocks in the clock list
Delete	Deletes a selected clock from the Selected Clock(s) list.

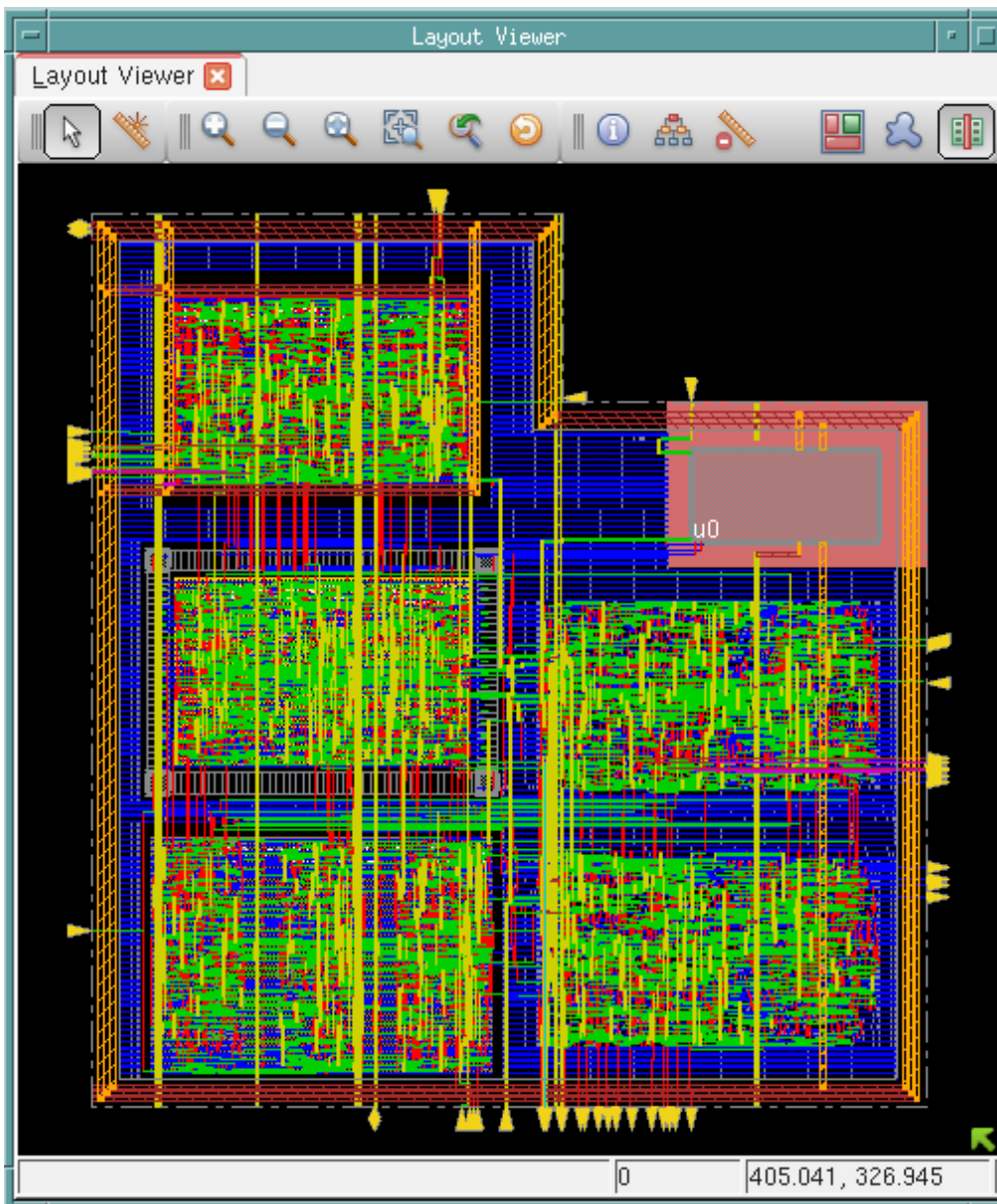
Filter	Specifies a filter pattern for selecting clocks for the clock list.
Crossover Clocks (Clock list)	Lists the cross-over clocks shown in the display area. You can add any of these to the Selected Clock(s) list.
All Clocks (Clock list)	Lists all clocks listed in the CTS spec file. You can then add any of these to the Selected Clock(s) list.
Late	The arrival time of a pin at any given instance has one or more values. Select Late to view the longest time of a path.
Early	The arrival time of a pin at any given instance has one or more values. Select Early to view the shortest time of a path.

Layout Viewer

Use the *Layout Viewer* menu item to open the design in a view-only window. You can launch multiple view-only windows in a single software session. This is useful when debugging as it allows you to view and zoom into different portions of a design simultaneously.

The Layout Viewer window supports all viewing related features, such as *Zoom* and *Pan*. In addition, it supports functions and tools that do not modify the design, such as Attribute Editor and Design Browser.

→ Choose *Tools - Layout Viewer*.



Layout Viewer Tool Widgets

You can use the widgets in Design Viewer to navigate through displays, open tools, and make selections.

Widget	Description
--------	-------------

	<i>Select</i> --Selects an object in the design display area.
	<i>Create Ruler</i> --Click this widget, then left-click in the design display area and drag the mouse to add a ruler (in micrometers), and left-click again to end the ruler. To move the ruler, click and drag it to a new location. Click the ruler again to keep it in the new location.
	<i>In by 2</i> --Click this widget to display a smaller area of the design in greater detail. Each click zooms in two levels.
	<i>Out by 2</i> --Click this widget to display a larger area of the design in less detail. Each click zooms out two levels.
	<i>Fit</i> --Click this widget to display the entire placed design within the design display area.
	<i>Selected</i> --Zooms in on a highlighted selection.
	<i>Previous</i> --Click this widget to toggle the display between the previous location/zoom level and the current location/zoom level.
	<i>Redraw</i> --Refreshes the Design Viewer display.
	<i>Attribute Editor</i> --Opens the Attribute Editor for the highlighted object.
	<i>Design Browser</i> --Opens the Design Browser form.
	<i>Clear All Rulers</i> --Clears all rulers from the display area. Note: To remove a single ruler, place the mouse cursor over the ruler and press <code>Delete</code> .
	<i>Floorplan View</i> --Switches to the Floorplan view of the design.
	<i>Amoeba View</i> --Switches to the Amoeba view of the design.
	<i>Physical View</i> --Switches to the Physical view of the design.

Conformal

The *Conformal* submenu requires Cadence® Tempus™ Timing Signoff Solution license. See the *Tempus Timing Signoff Solution Menu Reference* for details.

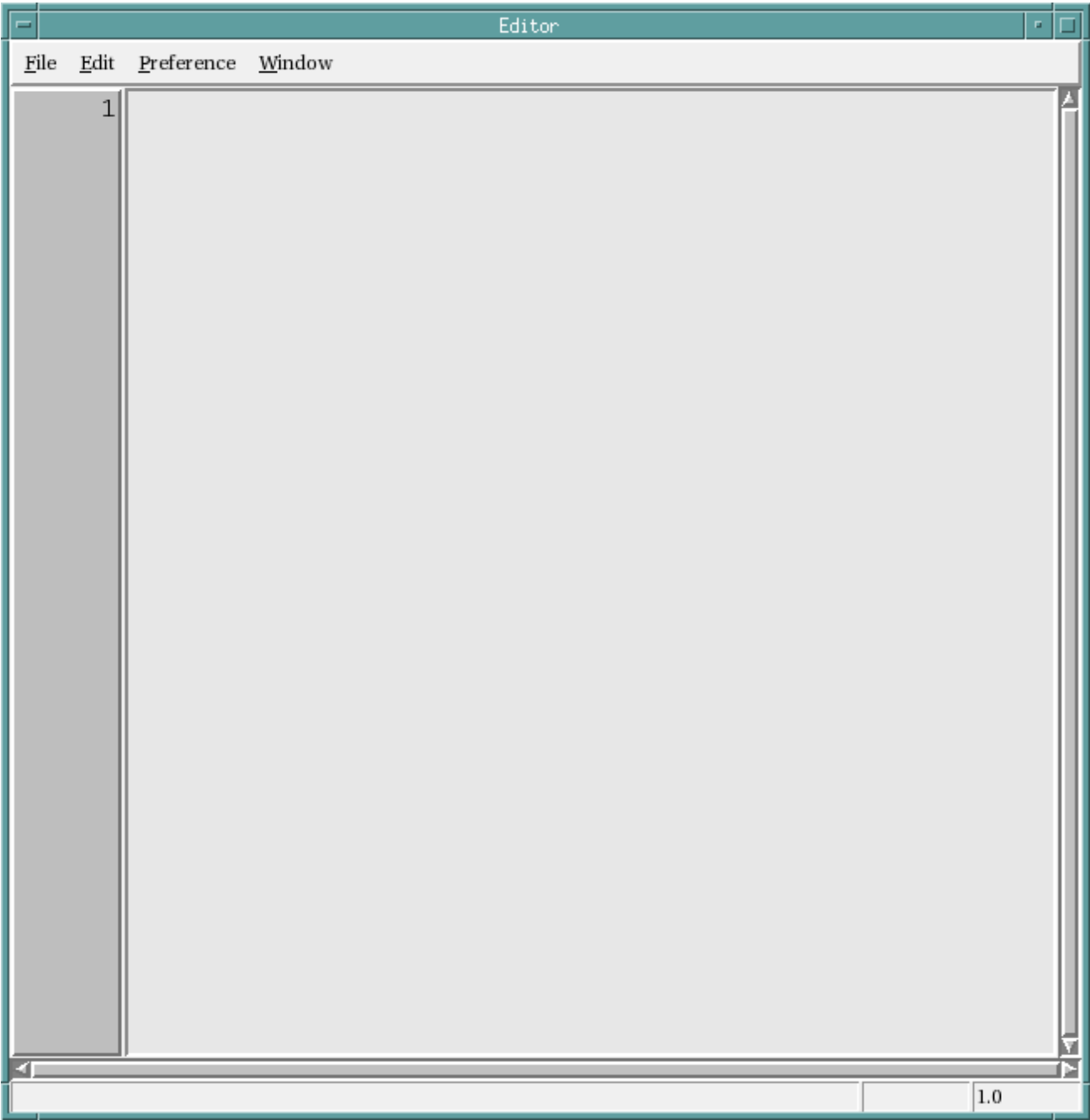
Constraint

The *Constraint* submenu requires Cadence® Tempus™ Timing Signoff Solution license. See the *Tempus Timing Signoff Solution Menu Reference* for details.

Editor

Use the *Editor* submenu to display and edit a text file.

→ Choose *Tools - Editor*.



Fields and Options

Pulldown	Descriptions
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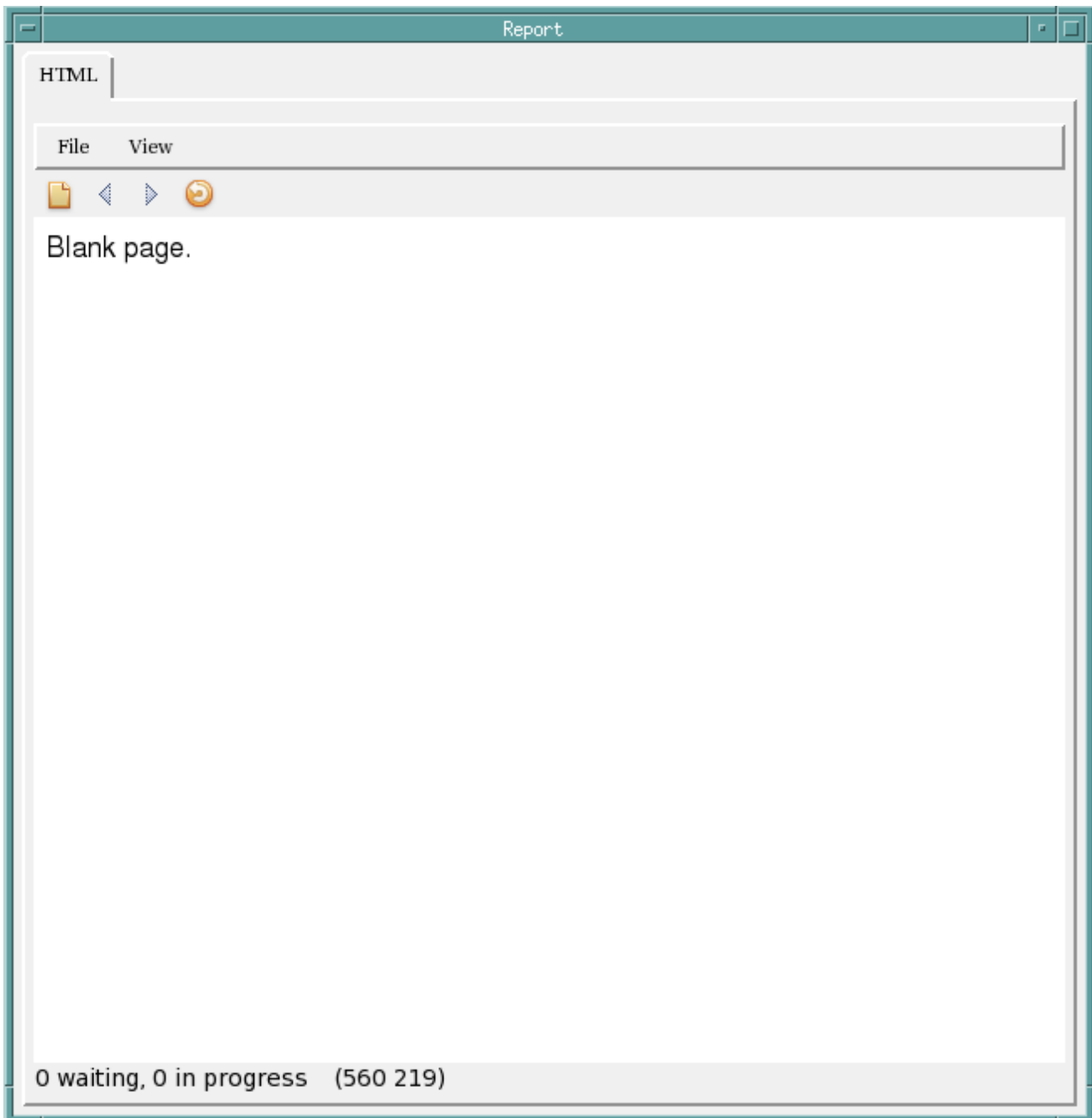
<i>File</i>	Choose from one of the following options:	
	<i>New</i>	Opens a blank file for editing.
	<i>Open</i>	Open an existing file for editing.
	<i>save</i>	Saves a file.
	<i>Save As</i>	Saves a file with a different file name.
	<i>Evaluate</i>	Evaluates the commands in the file.
	<i>Close</i>	Closes the file.
	<i>Exit</i>	Exits the editor window.
<i>Edit</i>	Choose from one of the following options:	
	<i>Undo</i>	Reverses the last operation.
	<i>Cut</i>	Cuts the selected text.
	<i>Copy</i>	Copies the selected text so it can be pasted at another location.
	<i>Paste</i>	Pastes the copied text.
	<i>Find</i>	Opens window to find the specified string in the document.
	<i>Find Next</i>	Finds the next instance of the specified string.
	<i>Find Prev</i>	Finds the previous instance of the specified string.
	<i>Replace</i>	Replaces the specified string with another specified value.
	<i>Goto Line</i>	Go to the specified line number.
	<i>Grep</i>	Searches for all files in the specified directory for the given string.
<i>Preference</i>	Choose form one of the following options:	
	<i>Show Proc Window</i>	
		Displays a right pane with procedure information.
	<i>Show Line Number</i>	
		Shows the line numbers at the left of the screen

	<i>Word Wrap</i>	To wrap words on a line to the next line if they do not fit in window.
	<i>Refresh Highlighting</i>	
		Refreshes highlighting
<i>Window</i>	Provides the ability to switch between any documents that have been opened.	

Report

Use the Report submenu to open any html reports.

→ Choose *Tools - Report*.



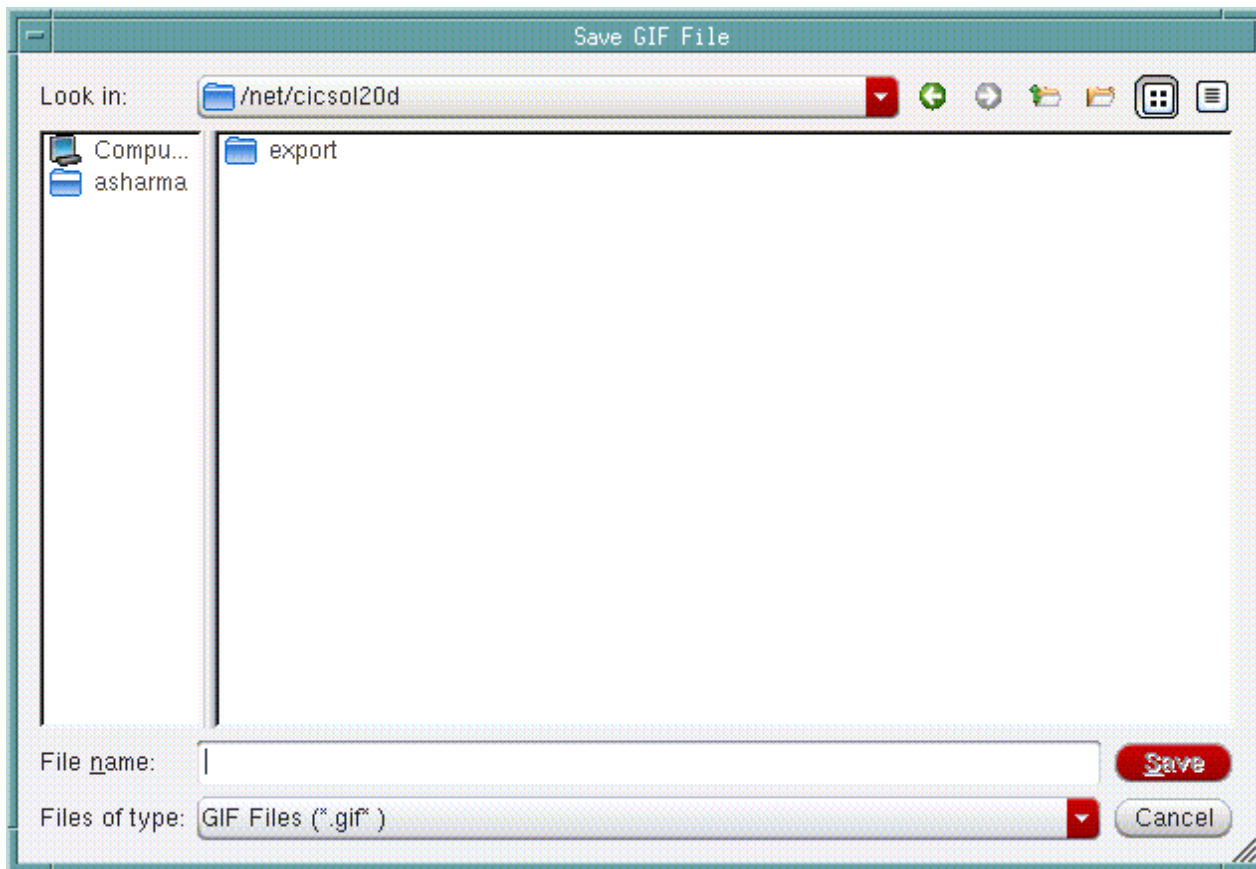
Fields and Options

Menu	Description	
<i>File</i>	Choose from one of the following options:	
	<i>Open File</i>	Opens the specified file.
	<i>Open Tab</i>	Opens a new browser tab.
	<i>Close Tab</i>	Closes the currently selected browser tab.
<i>View</i>	Choose form one of the following options:	
	<i>Gui Font</i>	Specify the GUI font size from a predefined list.
	<i>Browser Zoom</i>	Specify the browser zoom factor from a predefined list.
	<i>Browser Font Scale</i>	
		Specify the browser font size from a predefined list.
	<i>Browser Font Scale Table</i>	
		Specify the browser font size from a predefined list.

Write To GIF File

Use the Write To GIF File menu command to save the snapshot of the current screen to a GIF file in the current directory.

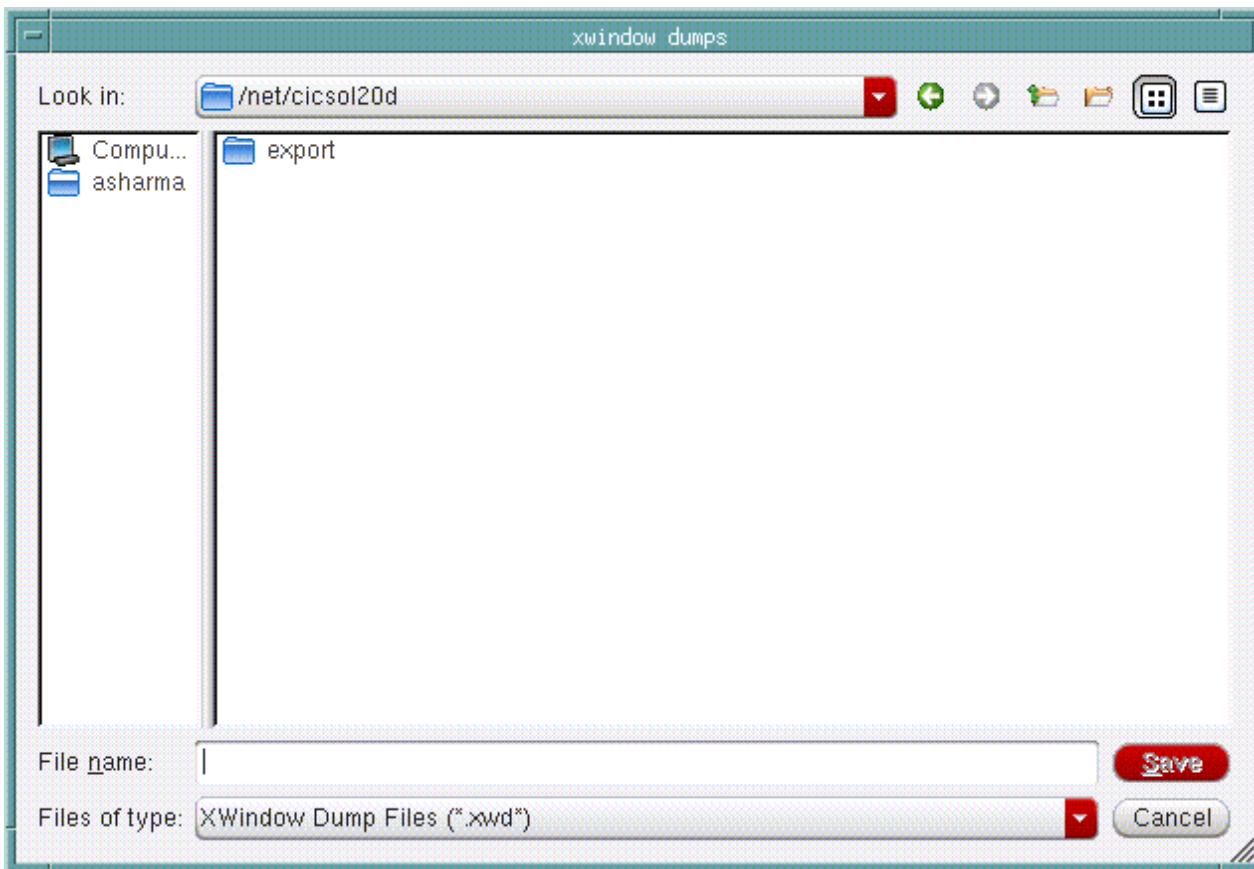
→ Choose *Tools - Screen Capture - Write To GIF File*.



Screen Dump

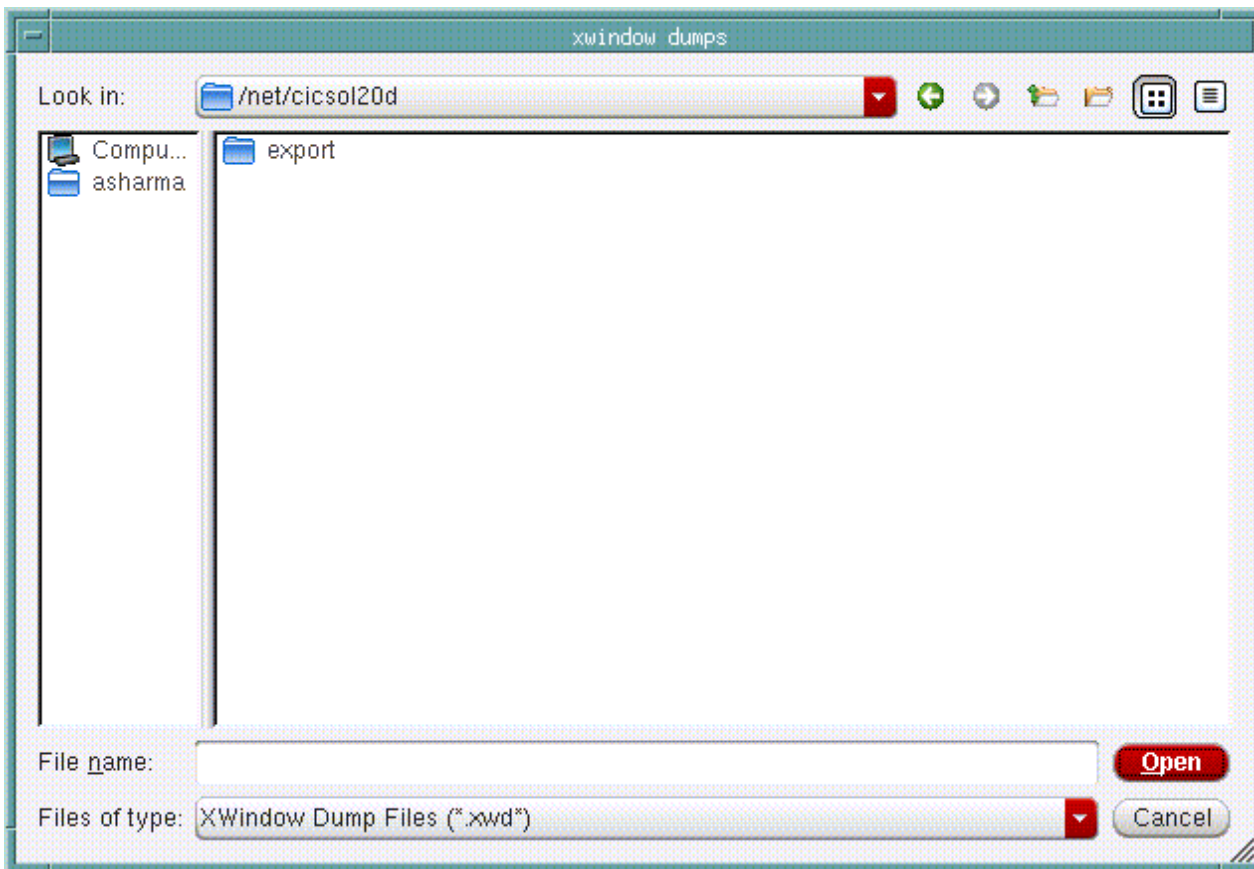
Use the Screen Dump menu command to take a snapshot of the current screen. The file name is in timestamp format that shows when the snapshot was taken. The gzipped file, with the .gz extension, is listed in the text window.

→ Choose *Tools - Screen Capture - Screen Dump*.



Display Screen Dump

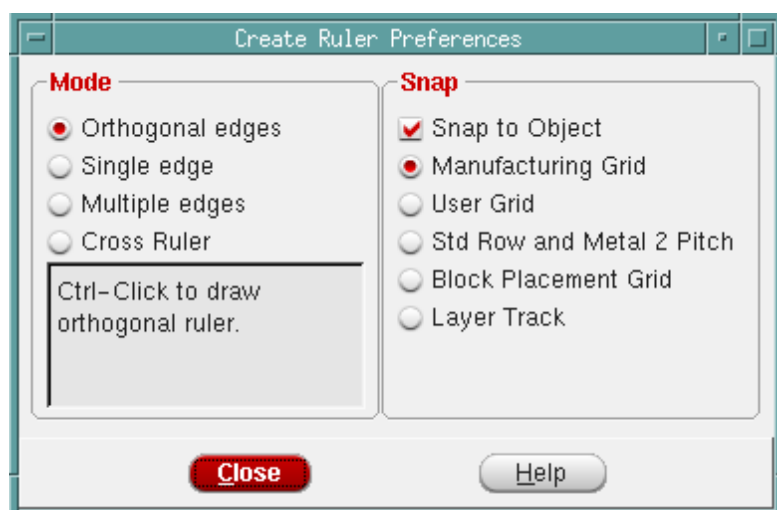
Use the Display Screen Dump menu command to view snapshots that you have taken earlier.
→ Choose *Tools - Screen Capture - Display Screen Dump*.



Create Ruler

Use the Create Ruler Preferences form to set the direction in which ruler lines are drawn.

→ Choose *Tools - Create Ruler*.



Fields and Options

<i>Snap</i>	
Snap to Object	Specifies that the ruler should snap to the nearest floorplan object.
Manufacturing Grid	Specifies that the ruler should snap to the manufacturing grid.
User Grid	Specifies that the ruler should snap to the user specified grid.
Std Row and Metal 2 Pitch	
Block Placement Grid	Specifies that the ruler should snap to the block placement grid.
Layer Track	
<i>Mode</i>	
<i>Orthogonal</i>	Specifies that ruler lines can be drawn in horizontal as well as vertical directions.
<i>Vertical</i>	Specifies that ruler lines will be drawn in the vertical direction.
<i>Horizontal</i>	Specifies that ruler lines will be drawn in the horizontal direction.
<i>X</i>	Specifies that ruler lines can be drawn at a 45 degree angle as well as at a 135 degree angle.
45	Specifies that ruler lines will be drawn at a 45 degree angle.
135	Specifies that ruler lines will be drawn at a 135 degree angle.
Any Angle	Specifies that ruler lines can be drawn in any direction.

Clear All Ruler

Clears all rulers from the display area.

Windows Menu

- [Workspaces](#)
 - [Physical](#)
 - [Schematic](#)
 - [Design Browser + Schematic](#)
 - [Design Browser + Physical](#)
 - [Save Workspace](#)
 - [Delete](#)
 - [Set Default](#)
 - [Menus](#)
 - [Toolbars](#)

Workspaces

Voltus enables you to save and load your customized window settings easily through workspaces. A workspace contains customized main window settings including:

- Size and placement of window
- Specific orientation of toolbars
- Docking status
- Visibility settings

Use the *Workspaces* submenu to manage your workspaces. To access this submenu:

- Choose *Windows - Workspaces*

The *Workspaces* submenu contains the following menu items:

- [Physical](#)
- [Schematic](#)

- [Design Browser + Schematic](#)
- [Design Browser + Physical](#)
- [Save Workspace](#)
- [Delete](#)
- [Set Default](#)

Physical

Use the *Physical* menu item to switch to the Physical view of your current workspace. The Physical view displays the detailed placements of the module's blocks, standard cells, nets, and interconnects. You can move standard cells, blocks, and power and ground objects in this view.

- Choose *Windows - Workspaces - Physical*.

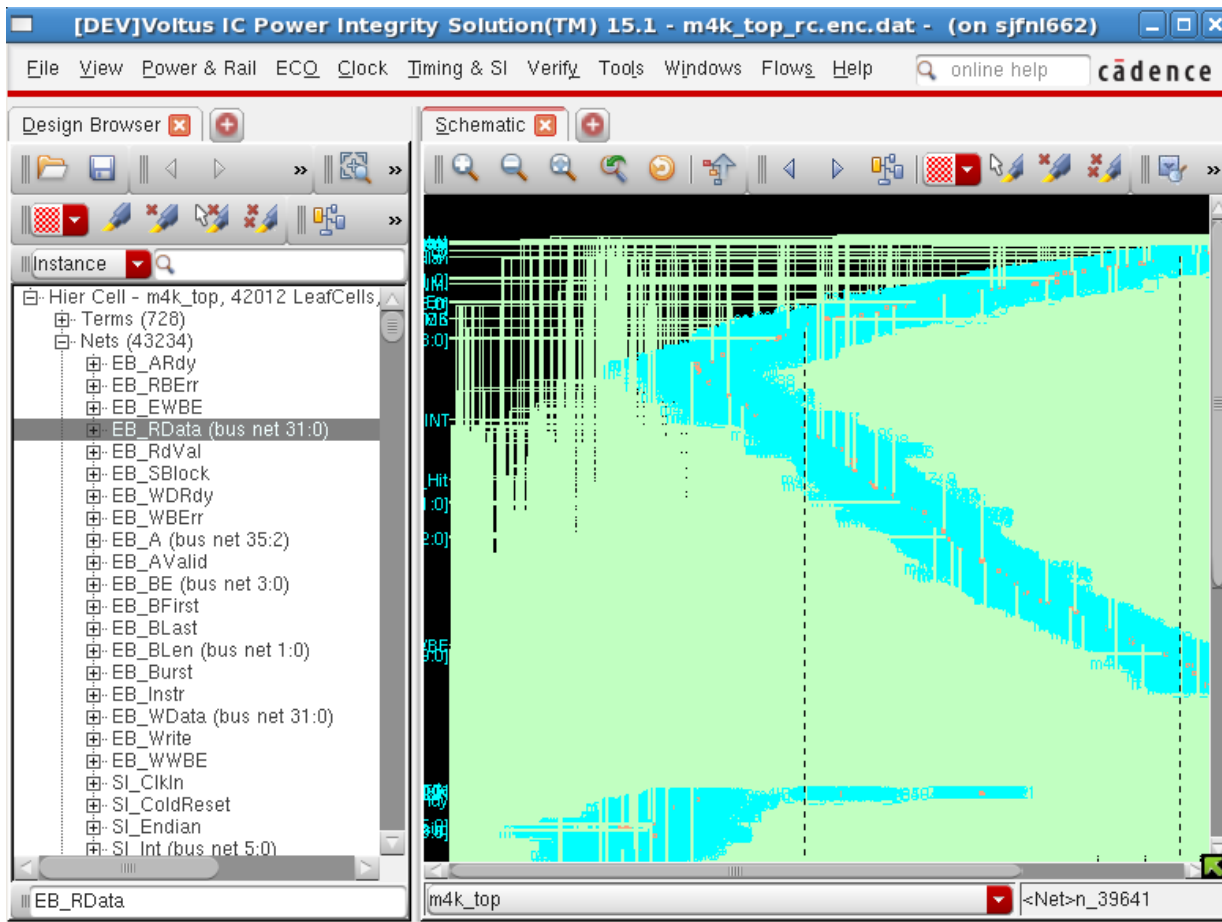
Schematic

Use the *Schematic* menu item to switch to the Schematic design view of your current workspace. The Schematic view is used to explore a design's connectivity, the relationship of modules and instances, and design changes.

- Choose *Windows - Workspaces - Schematic*.

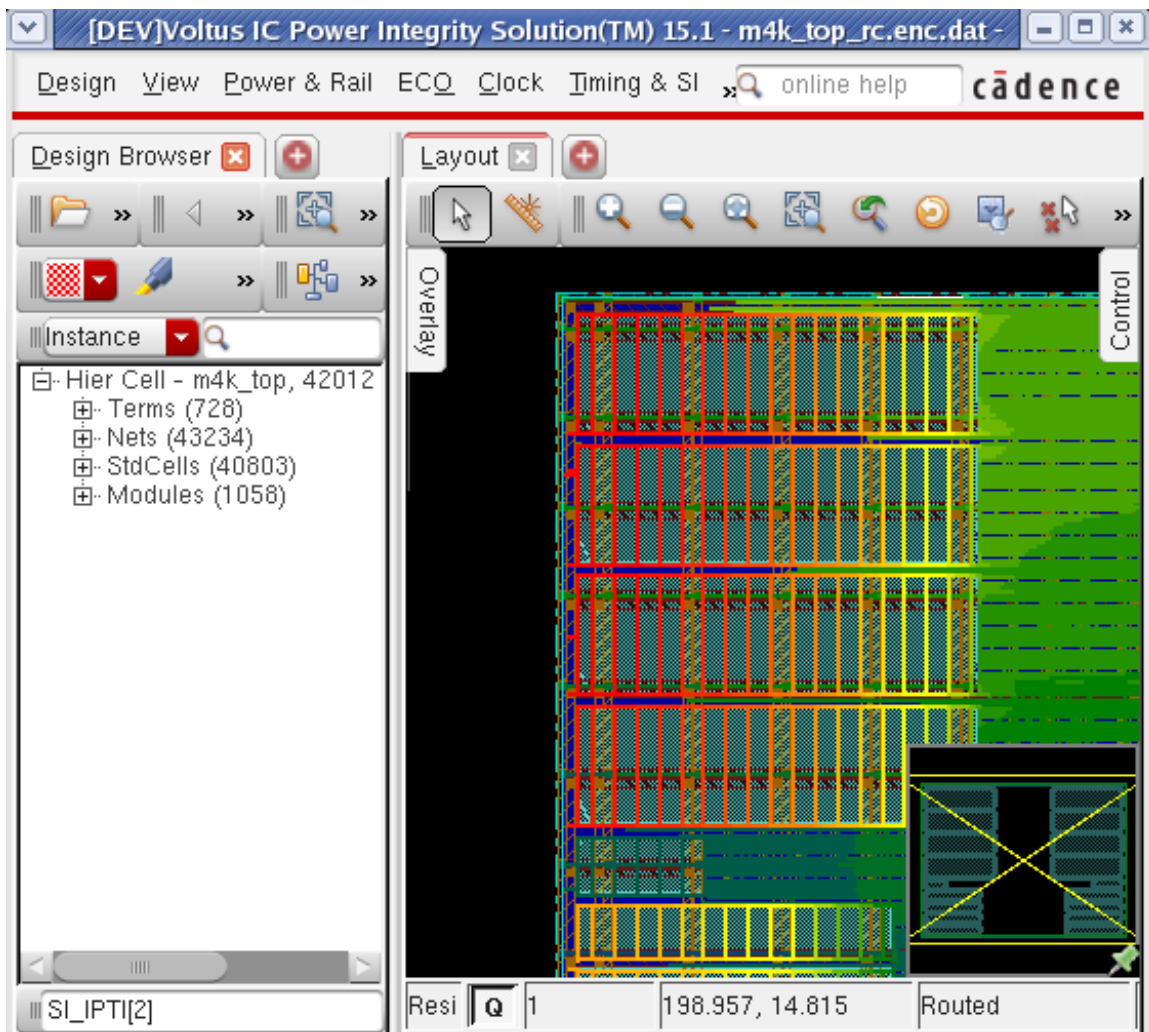
Design Browser + Schematic

Use the *Design Browser + Schematic* menu item to see the Design Browser and the schematic view of the design, side by side.



Design Browser + Physical

Use the *Design Browser + Physical* menu item to see the Design Browser and the physical view of the design, side by side.



Save Workspace

Use the Save Workspace form to save your workspace to a specific directory. You can then load the workspace whenever you want from the specified directory.

- Choose *Windows - Workspaces - Save Workspace*.



Fields and Options

<i>Select workspace name (or enter a new name)</i>	Specifies the name of the workspace you are saving. You can type a new name or, if you have previously saved workspaces, choose an existing name from the drop-down list. The drop-down lists all the workspaces in the selected path.
<i>Select Path</i>	Specifies the location for saving the workspace. You can choose to save the workspace in the current working directory or your home directory. If the workspace is saved to the current directory, it is applied to the current design only. If the workspace is saved to your home directory, it is applicable for all designs. <i>Default:</i> Current working directory

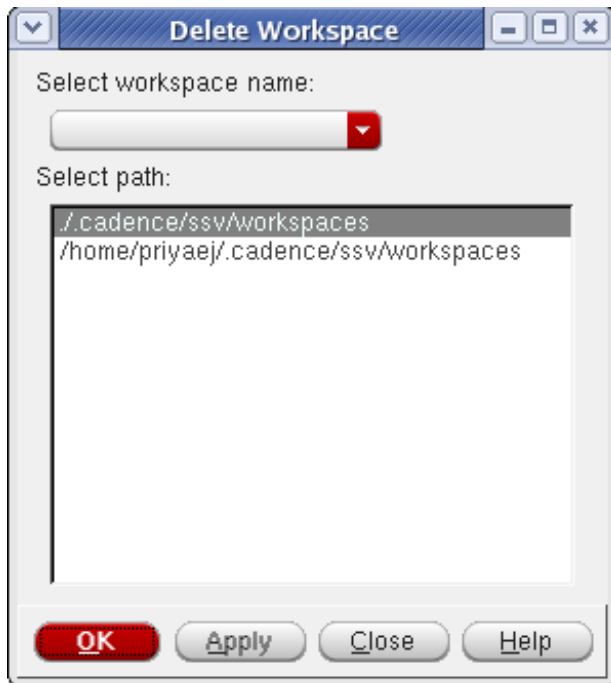
After you have saved a workspace, the saved workspace name appears on the *Workspaces* submenu. Whenever you want to load a workspace, simply click the saved workspace name in the *Workspaces* submenu.

If there are multiple workspaces, the one set as default is loaded automatically. If you have not set any workspace as default, the one that you saved last is loaded automatically. You can then switch to another workspace by selecting the corresponding workspace in the *Workspaces* submenu.

Delete

Use the *Delete Workspace* form to delete the selected workspace.

- Choose *Windows - Workspaces - Delete*.



Fields and Options

<i>Select workspace name</i>	Specifies the name of the workspace you are deleting. The drop-down lists all the workspaces in the selected path.
<i>Select Path</i>	Specifies the location of the workspace. <i>Default:</i> Current working directory

Set Default

Use the Set Default Workspace form to set the selected workspace as default.

- Choose *Windows - Workspaces - Set Default*.

The default workspace is loaded automatically when starting up the software. So, if you want Voltus to load a workspace automatically, you need to first save it as default.



Fields and Options

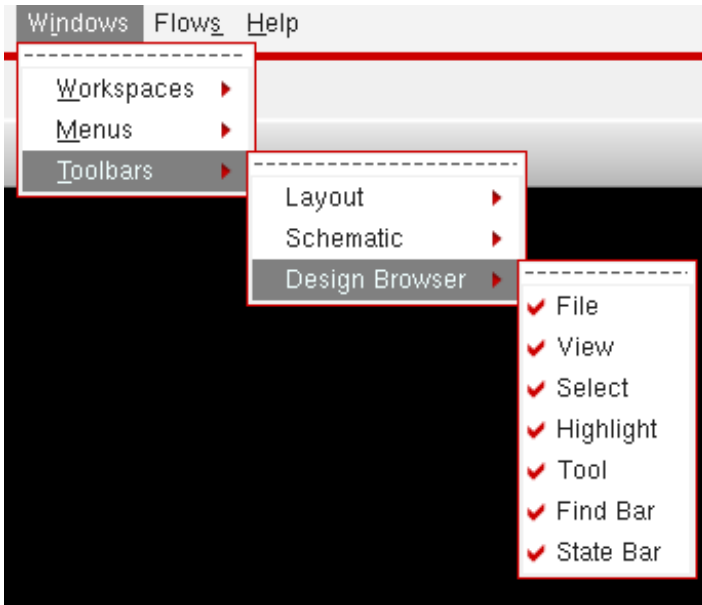
<i>Select workspace name</i>	Specifies the name of the workspace you are setting as default. The drop-down lists all the workspaces in the selected path.
<i>Select Path</i>	Specifies the location of the workspace. <i>Default:</i> Current working directory

Menus

Use the *Menus* submenu to show or hide specific menus, as required.

Toolbars

Use the *Toolbars* submenu to show or hide specific toolbars, as required. For example, you can select the *Highlight* submenu item to remove the Highlight toolbar from the Design Browser form.



Note that menu items under the *Toolbars* submenu appear only when the specific form is open. In the above diagram, the *Layout* tab, *Schematic* tab, and Design Browser form are open.

Flows Menu

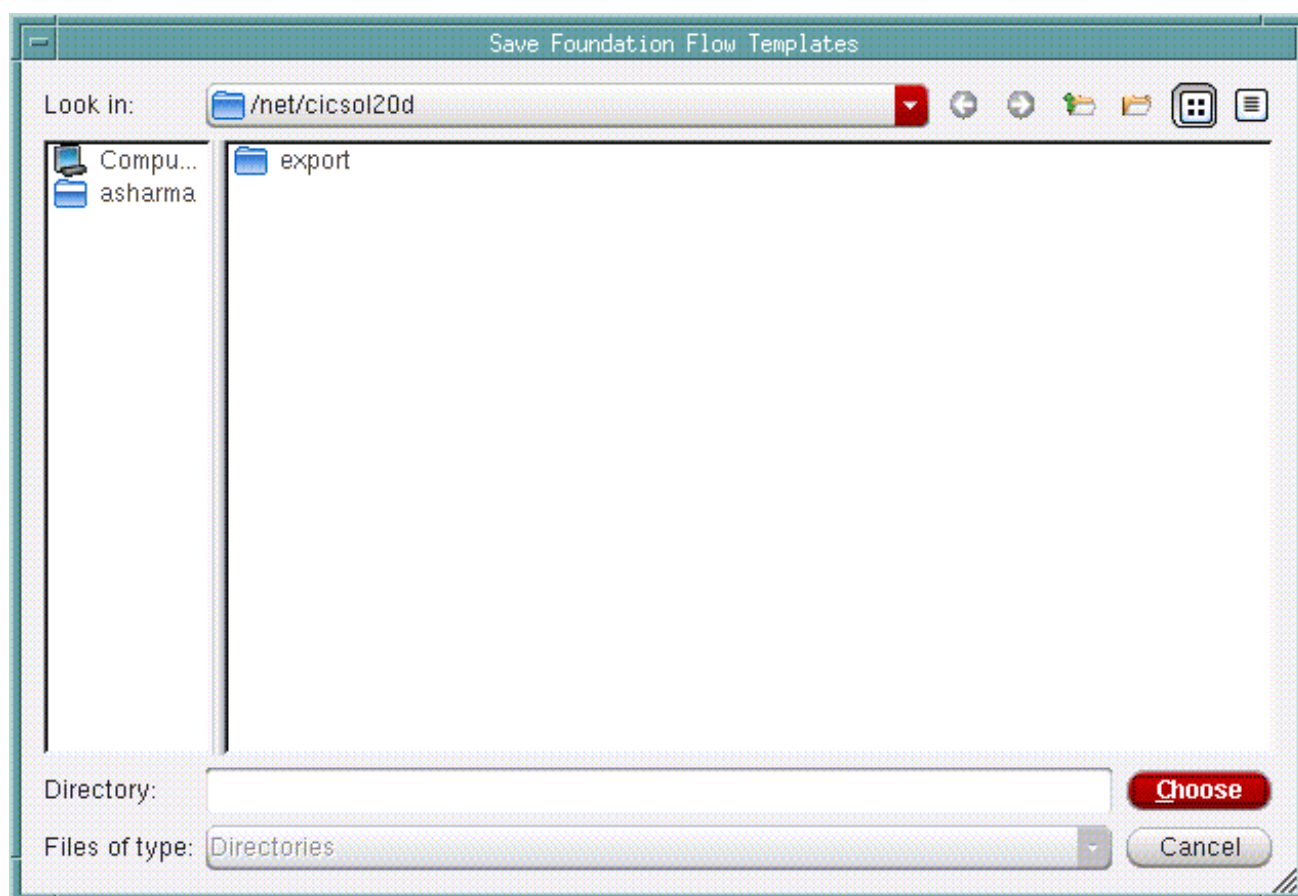
- [Create Template](#)

Create Template

In Voltus, you can enable customized Foundation Flow templates.

→ *Choose Flows - Create Template.*

Use this menu item to save the flow templates to a specified directory. The templates consist of a set of Tcl scripts you can customize for your design and technology needs.

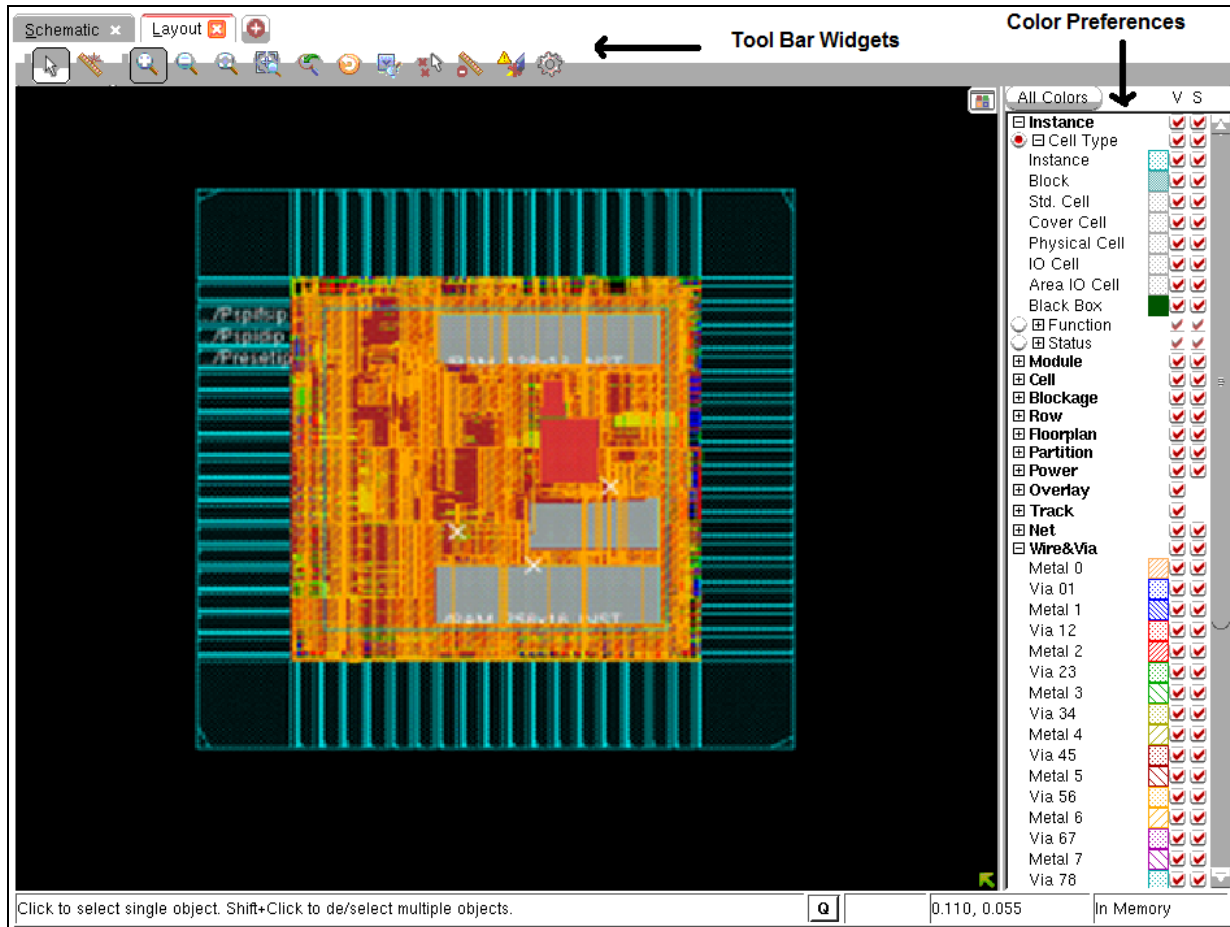


Layout Tab

- [Layout Window Components](#)
- [Toolbar Widgets](#)
- [Find/Select Object](#)
- [Color Preferences](#)
- [Objects Page](#)
- [Bus Guide Color Selection](#)
- [Wire/Via Page](#)
- [View-Only Page](#)
- [Custom Page](#)
- [Cell Layout Page](#)
- [Customizing Layer Color Patterns \(Pattern Window\)](#)
- [Customize Tab](#)
- [Save Color Preference](#)
- [Multicolor Layers](#)
- [Metal Layers and Other Layered Components](#)
- [Creating and Editing Color Groups](#)
- [Color Bar](#)
- [Object Color](#)
- [Create Ruler Preferences](#)
- [Satellite Window](#)
- [Auto Query](#)

Layout Window Components

Use the Layout window to display the physical view of the design in design display area. The components in the Layout window are identified in the following figure, and described in the sections that follow.



Toolbar Widgets

The following row of widgets, located above the design display area, includes shortcuts for some commonly used Voltus commands.



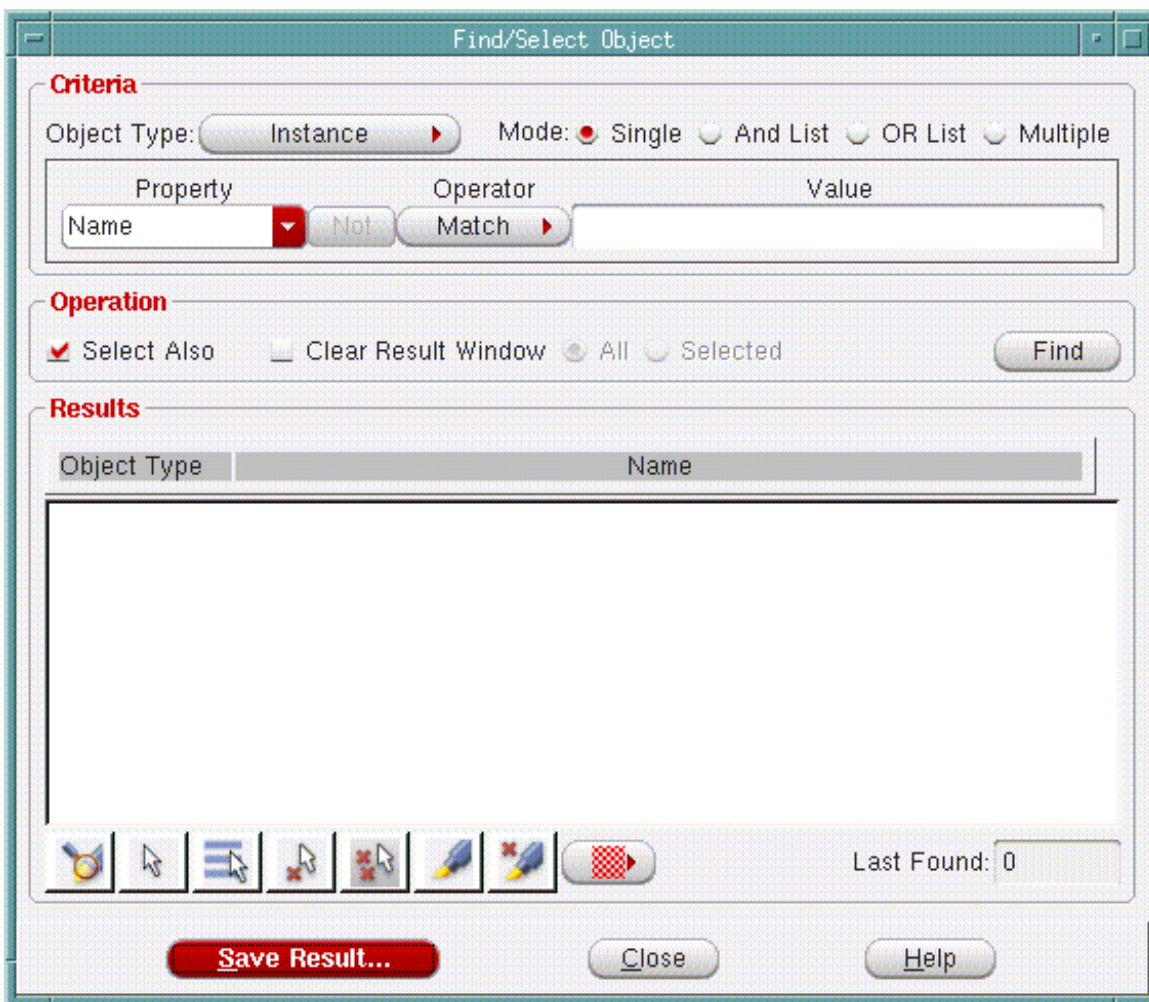
The following table describes the widgets that are located in the Toolbar Widget area:

Widget	Description
	<i>Select</i> --Click this widget, then click a floorplan object to select it. To select multiple objects, use the <i>Shift</i> key or click and drag the mouse.
	<i>Ruler</i>--Click this widget, then left-click in the design display area and drag the mouse to add a ruler (in micrometers), and left-click again to end the ruler. To move the ruler, left-click and drag it to a new location. Left-click the ruler again to keep it in the new location. To remove a ruler, left-click on it and press <i>Esc</i>. To remove all rulers, press <i>⌘</i> (<i>Shift-k</i>). You can set the direction in which rulers can be drawn by clicking the widget, then pressing the <i>F3</i> key to display the Set Ruler Mode form. You can choose to draw orthogonal, vertical, horizontal, or diagonal rulers. You can also snap rulers to floorplan objects.
	<i>Zoom In</i> --Click this widget to display a smaller area of the design in greater detail. Each click zooms in one level. The equivalent bindkey is <i>z</i> .
	<i>Zoom Out</i> --Click this widget to display a larger area of the design in less detail. Each click zooms out one level.
	<i>Fit</i> --Click this widget to display the entire placed design within the design display area. The equivalent bindkey is <i>f</i> .
	<i>Zoom Selected</i> --Select one or more design objects, then click this widget to zoom into the area that contains the selected objects.
	<i>Zoom Previous</i> --Click this widget to toggle the display between the previous location/zoom level and the current location/zoom level. The equivalent bindkey is <i>l</i> (the letter L).
	<i>Redraw</i> --Click this widget to update the display with new colors or stipple patterns that you selected from the More Colors form.
	<i>Clear All Ruler</i> --Clears all rulers from the display area.
	Find/Select Object--Displays the Find/Select Object form.

	Deselect All Objects--Deselects all objects.
	Clear DRC Violations--Clears all DRC violations from design display area.
	Opens the Preferences form, which enables you to customize the schematic display settings, color, and font.

Find/Select Object

Use the Find/Select Object form to find and select specific object types in the design display area.
→ Choose the *Find/Select Object* widget in the Layout Window.



The **Find/Select Object** dialog box is used to search for and select specific objects in the design. It features three main sections: **Criteria**, **Operation**, and **Results**.

- Criteria:** Includes fields for **Object Type** (set to Instance), **Mode** (radio buttons for Single, And List, OR List, Multiple), and a search criteria table with columns for **Property**, **Operator**, and **Value**. The first row shows **Name** with a dropdown arrow, **Not** as the operator, and **Match** as the operator.
- Operation:** Includes checkboxes for **Select Also** (checked), **Clear Result Window**, and radio buttons for **All** and **Selected**. A **Find** button is located to the right.
- Results:** A table with columns **Object Type** and **Name** for displaying search results.

At the bottom, there is a toolbar with icons for various actions (like pan, zoom, select) and a **Last Found: 0** counter. Below the toolbar are buttons for **Save Result...**, **Close**, and **Help**.

Fields and Options

<i>Object Type</i>	Specifies the type of object to find. Choose from: <i>Instance</i>, <i>Pin</i>, <i>Net</i>, <i>Module</i>, <i>Instance Group</i>, or <i>Bump</i>. The <code>Pin</code> object type pin can be used for I/O pins, partition pins, and instance pins. Names of instance pins and/or partition pins should be specified with the full path name. For example, pin A of instance DTMF_CHIP/alu should be specified as: DTMF_CHIP/alu/A You can use the <i>Net</i> option to find/display bus names.
<i>Mode</i>	Specifies the type of find/select operation. Select one of the following: • <i>Single</i>: Performs a simple search for a Property/Value combination. • <i>And List</i>: Performs a find/select operation that matches an AND list of two or more multiple Property/Value combinations. • <i>OR List</i>: Performs a find/select operation that matches an OR list of two or more multiple Property/Value combinations. • <i>Multiple</i>: Performs a find/select operation that matches multiple lists of Property/Value combinations; these lists can be a combination of AND lists and OR lists.
<i>Property</i>	Displays the available properties for the object that you selected in the Object Type field. You can use the pre-defined property <code>Connection</code> to find/select objects by connectivity. The <code>Connection</code> property supports connectivity search for the following objects: • Instance • Pin • Net • Module • Selected objects

<i>NOT</i>	Inverses the value of the operator in the Operator field. For example, If the value in the Operator field is <i><</i> , clicking the NOT button effectively changes the value to NOT <i><</i>
<i>Operator</i>	Specifies the relationship between the property and its value. This field is dynamically populated based on the specified object type.
<i>Value</i>	Specifies the value of the property. This field is dynamically populated based on the specified object type. In some cases, you may have to manually specify a value, for example, the name of a cell.
<i>Select Also</i>	Finds all objects that meet the specified criteria, reports them in the <i>Results</i> window, and selects them in the design display area. <i>Default:</i> The objects are reported in the results window but are not selected in the design display area.
<i>Clear Result Window</i>	Clears objects previously listed in the <i>Results</i> list, then finds the objects that meet the specified criteria and reports them in the Results list. <i>Default:</i> The objects previously listed are retained in the results window along with the results of the current search.
<i>All</i>	Clears all objects previously listed in the Results list.
<i>Selected</i>	Clears those objects previously listed in the Results list that are currently selected in the <i>Results</i> list.
<i>Find</i>	Finds all objects that meet the specified criteria and reports them in the <i>Results</i> window.
Zoom in Highlighted 	Zooms in the highlighted object(s). You can highlight object(s) by selecting them in the results window and clicking the <i>Highlight</i> button
<i>Select</i> 	When you click <i>Select</i> , the objects selected in the Results window are selected in the design display area.
<i>Select All</i> 	When you click <i>Select All</i> , all the objects in the Results window are selected in the design display area.

<i>Deselect</i> 	When you click <i>Deselect</i> , the objects selected in the Results window are deselected in the design display area.
<i>Deselect All</i> 	When you click <i>Deselect All</i> , all the objects in the Results window are deselected in the design display area.
<i>Highlight</i> 	When you click <i>Highlight</i> , the objects selected in the Results window are highlighted in the design display area.
<i>Dehighlight</i> 	When you click <i>Dehighlight</i> , the objects selected in the Results window are dehighlighted in the design display area.

Color Preferences

To set advanced color options, click the *All Colors* button in the Colors panel on the Layout window. This opens the Color Preferences form, which has five pages: *Objects*, *Wire/Via*, *View-Only*, *Custom*, and *Cell Layout*

The visibility settings between the pages and the Layout window are synchronized. That is, as soon you change any setting on any page, the corresponding changes are applied in the Layout window.

After making changes to the All Colors form, you can perform the following actions:

- To apply the changes in visibility, selectability, color, and stipple pattern to objects in the design display area, click *Close*.
- To save a color preference file, click *Save*. For more information, see [Save Color Preference](#).
- To load a preference file, click *Load*.

Objects Page

The screenshot shows the 'Color Preferences' dialog box with the 'Objects' tab selected. The dialog is organized into a table of object types, each with a color swatch, a pattern swatch, and checkboxes for visibility (V) and selectability (S). Below the table is an 'Edit Layer' section with controls for color, pattern, visibility, and selectability, along with an 'Apply' button. At the bottom are buttons for 'Default', 'Save...', 'Load...', 'Customize Tab...', 'Close', and 'Help'.

Object	Color	Pattern	V	S
Area Density	[Swatch]	[Swatch]	☒	☒
Area IO Cell	[Swatch]	[Swatch]	☒	☒
Black Blob	[Swatch]	[Swatch]	☒	☒
Black Box	[Swatch]	[Swatch]	☒	☒
Block	[Swatch]	[Swatch]	☒	☒
Bump(Ground)	[Swatch]	[Swatch]	☒	☒
Bump(Normal)	[Swatch]	[Swatch]	☒	☒
Bump(Other)	[Swatch]	[Swatch]	☒	☒
Bump(Power)	[Swatch]	[Swatch]	☒	☒
Bus Guide	[Swatch]	[Swatch]	☒	☒
Cell Blockage	[Swatch]	[Swatch]	☒	☒
Cover Cell	[Swatch]	[Swatch]	☒	☒
Datapath	[Swatch]	[Swatch]	☒	☒
Fence	[Swatch]	[Swatch]	☒	☒
Guide	[Swatch]	[Swatch]	☒	☒
IO Cell	[Swatch]	[Swatch]	☒	☒
IO Cluster	[Swatch]	[Swatch]	☒	☒
IO Row	[Swatch]	[Swatch]	☒	☒
Instance	[Swatch]	[Swatch]	☒	☒
Metal Fill	[Swatch]	[Swatch]	☒	☒
Module	[Swatch]	[Swatch]	☒	☒
Obstruct	[Swatch]	[Swatch]	☒	☒
Partition Feedthru	[Swatch]	[Swatch]	☒	☒
Partition Pin Block	[Swatch]	[Swatch]	☒	☒
Physical Cell	[Swatch]	[Swatch]	☒	☒
Pin Guide	[Swatch]	[Swatch]	☒	☒
Power Domain	[Swatch]	[Swatch]	☒	☒
Power Graph	[Swatch]	[Swatch]	☒	☒
Region	[Swatch]	[Swatch]	☒	☒
Routing Blockage	[Swatch]	[Swatch]	☒	☒
Standard Row	[Swatch]	[Swatch]	☒	☒
Std. Cell	[Swatch]	[Swatch]	☒	☒
Terminal	[Swatch]	[Swatch]	☒	☒
Violation	[Swatch]	[Swatch]	☒	☒

Edit Layer

Color: [No Change] [Color Buttons]

Pattern: [No Change] [Pattern Buttons]

Visibility: [No Change] ☐ Selectability: [No Change] ☐ [Apply]

[Default] [Save...] [Load...] [Customize Tab...] [Close] [Help]

The Objects page includes the visibility and selectability controls for objects that can be viewed and selected. Each object type has the following features:

- A color button that displays the current color and stipple pattern used to identify the object in the design display area.
 - Click a color button to select it, then click a new color and pattern from the *Edit Layer* section of the form. Click *Apply* to update the Color Preferences form.
 - Double-click the color button for an object to open the Color form.

- A check box for indicating whether the object type is visible in the design display area.
- A check box for making the object type selectable.

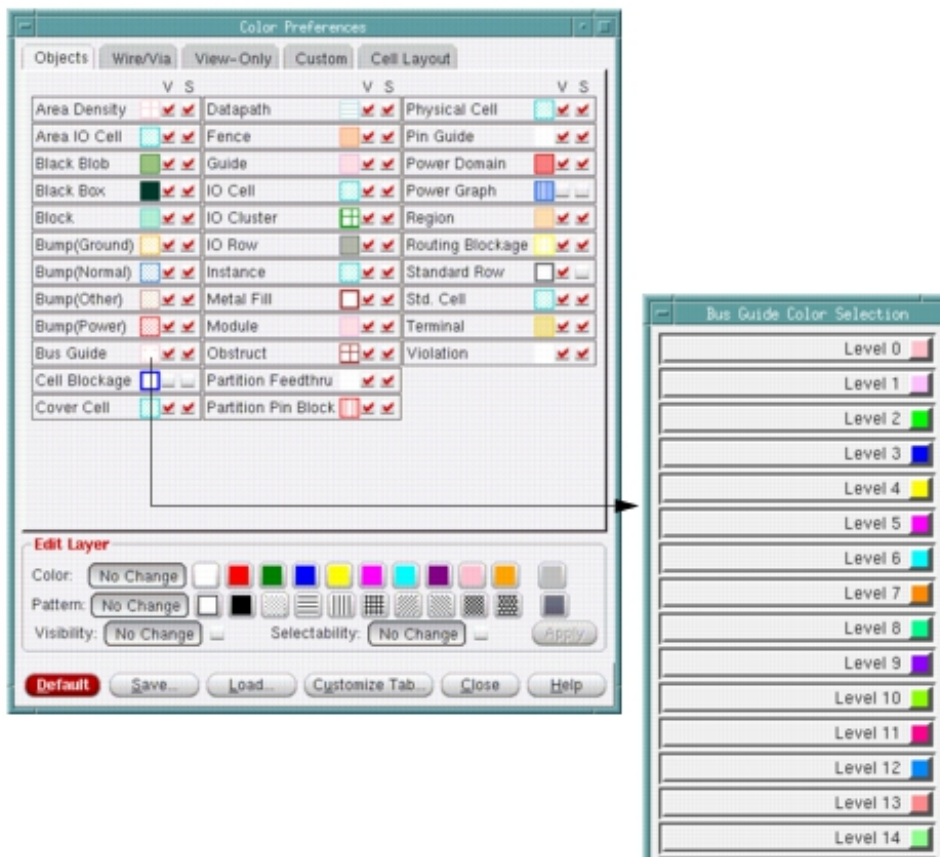
Note: If an object type does not have a Selectability check box, it cannot be selected.

- ✓ When you move the cursor over an object layer name, the equivalent database object name is displayed.

Bus Guide Color Selection

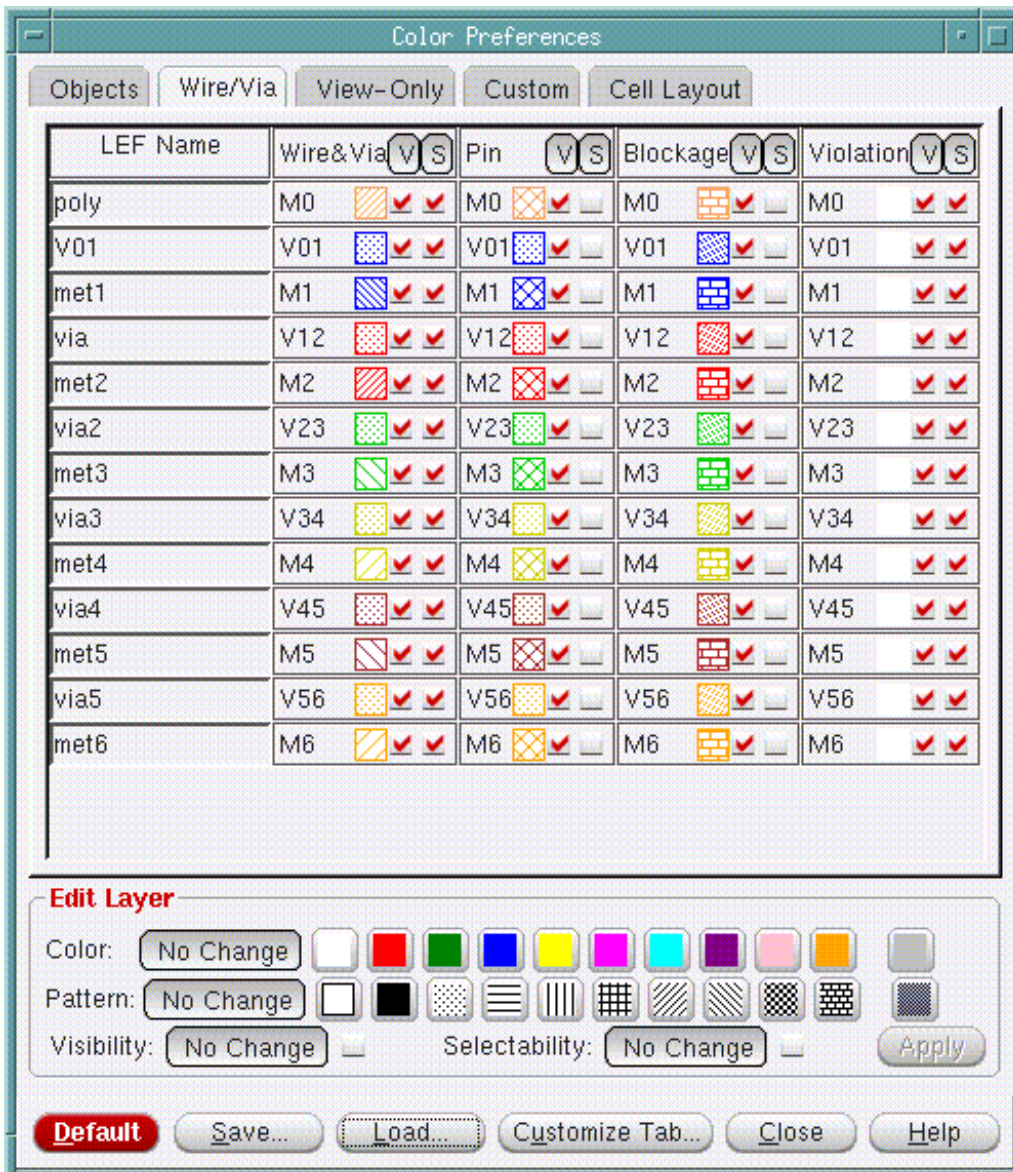
Use the *Bus Guide Color Selection* form to specify multiple colors for bus guide objects.

→ Double-click the *Color Preferences -- Objects -- Bus Guide* object type to display the *Bus Guide Color Selection* form.



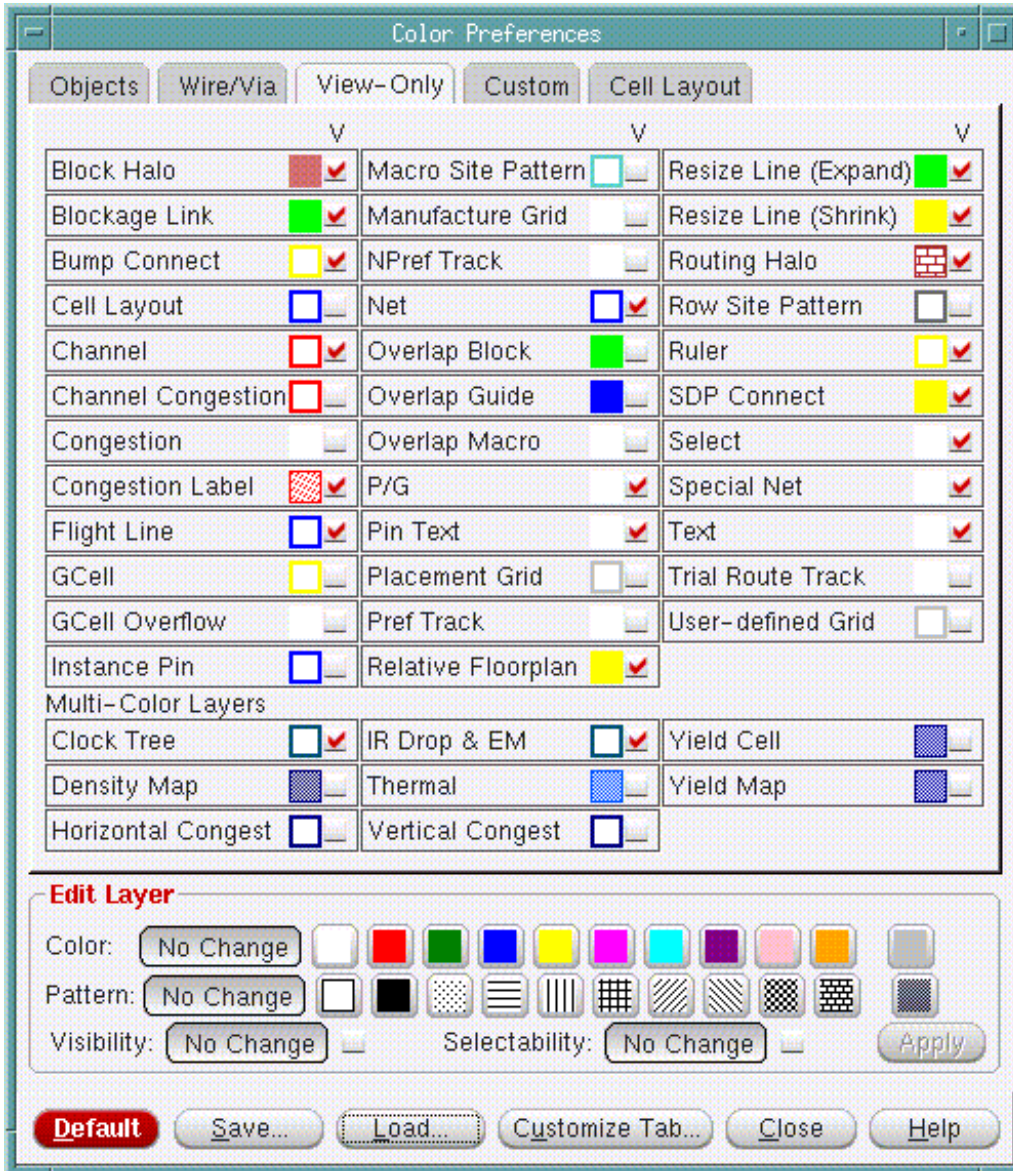
Wire/Via Page

The Wire/Via page includes the visibility and selectability controls for layer-based objects: Wires and Vias, Pins, Blockages, and Violations. You can set global display controls for all objects on the same layer. You can also set controls for individual objects on specific layers.



View-Only Page

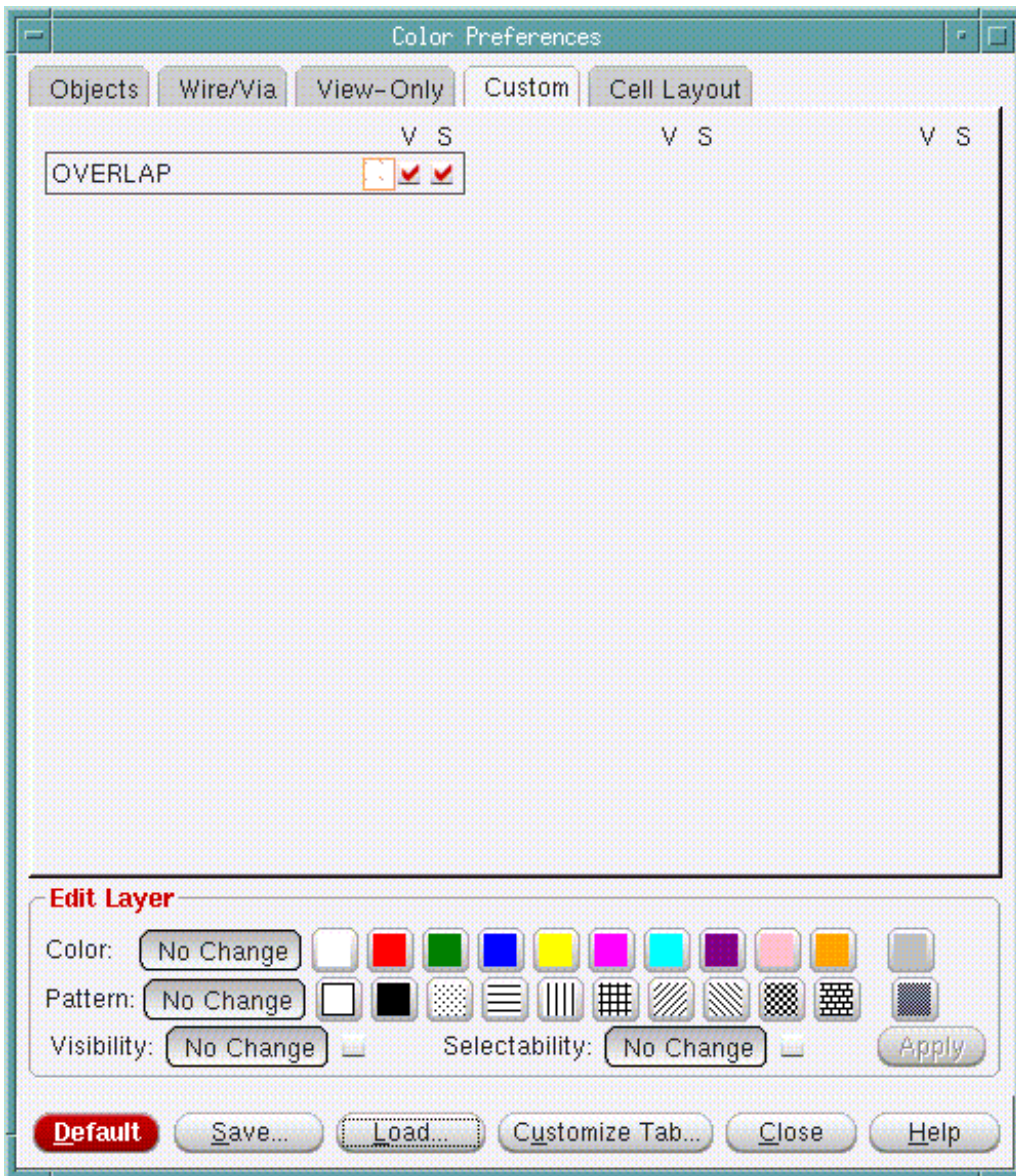
The View-Only page includes the visibility controls for objects that can be viewed but not selected in the Layout window.



Custom Page

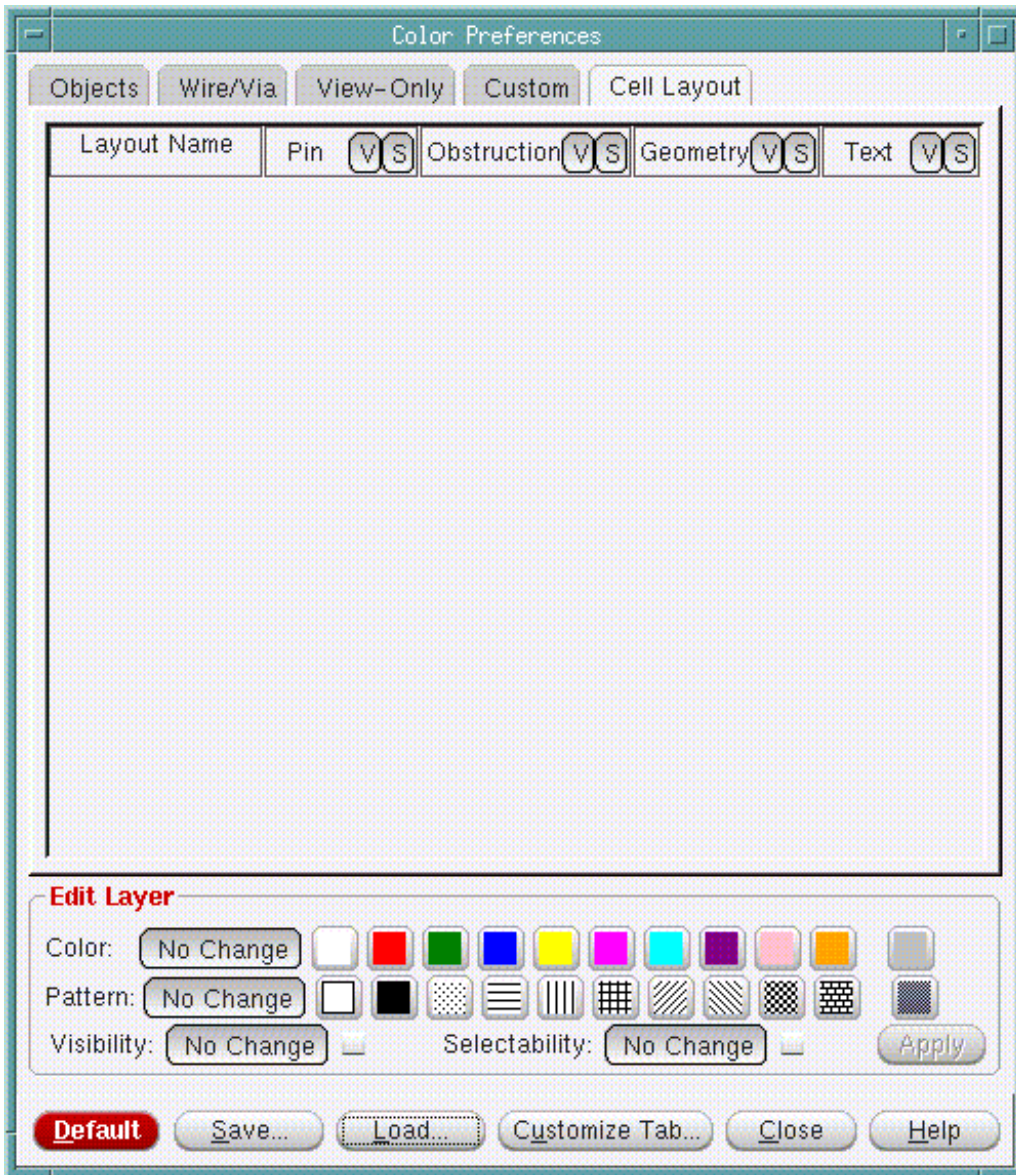
The Custom page contains custom objects.

To select a group of objects that are to have the same color and fill pattern, **Shift**-click the first and last objects in the group, click the new color and the new fill pattern, then click *Close*.



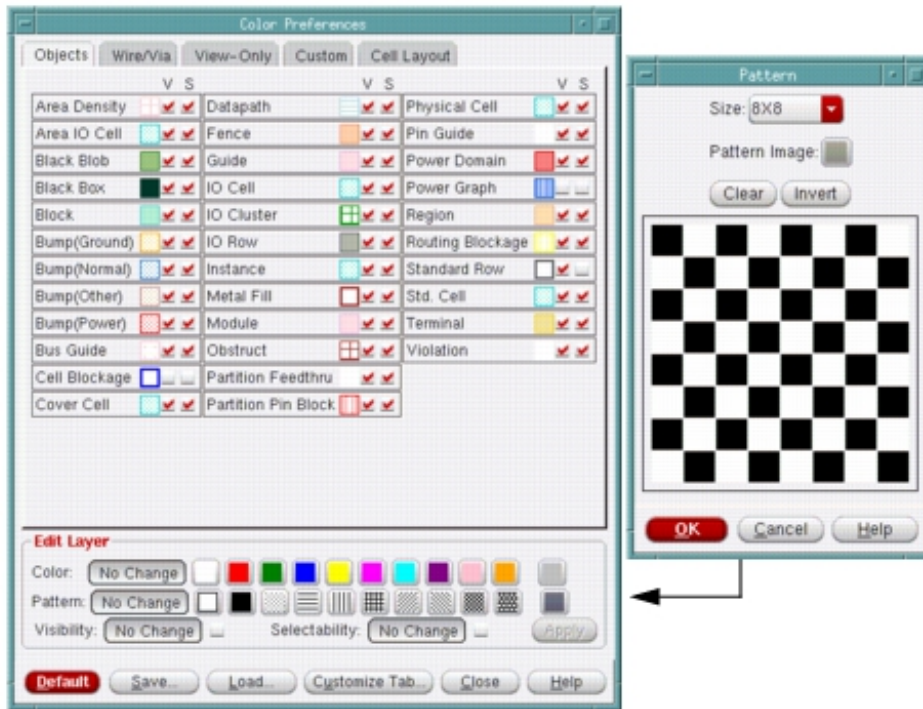
Cell Layout Page

Use the Cell Layout page to set the color preferences for the OA cell layout.

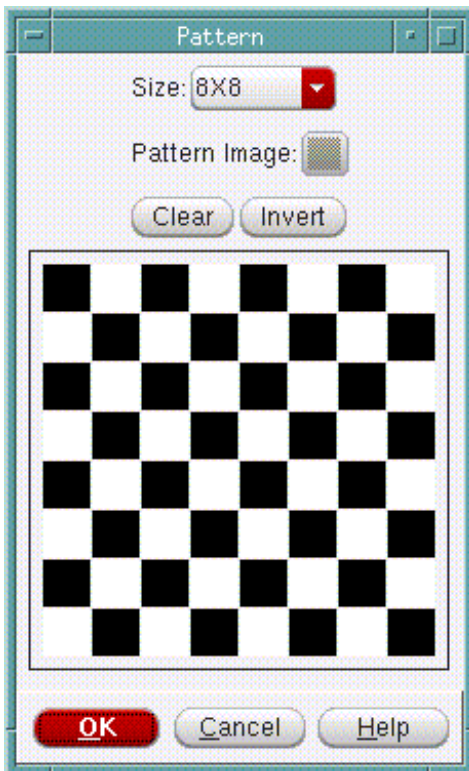


Customizing Layer Color Patterns (Pattern Window)

You can customize the color patterns for the layers by clicking the Custom pattern icon in the Edit Layer group. The Custom pattern icon is the rightmost icon in the Pattern field.



→ Click the Custom icon to display the Pattern form.

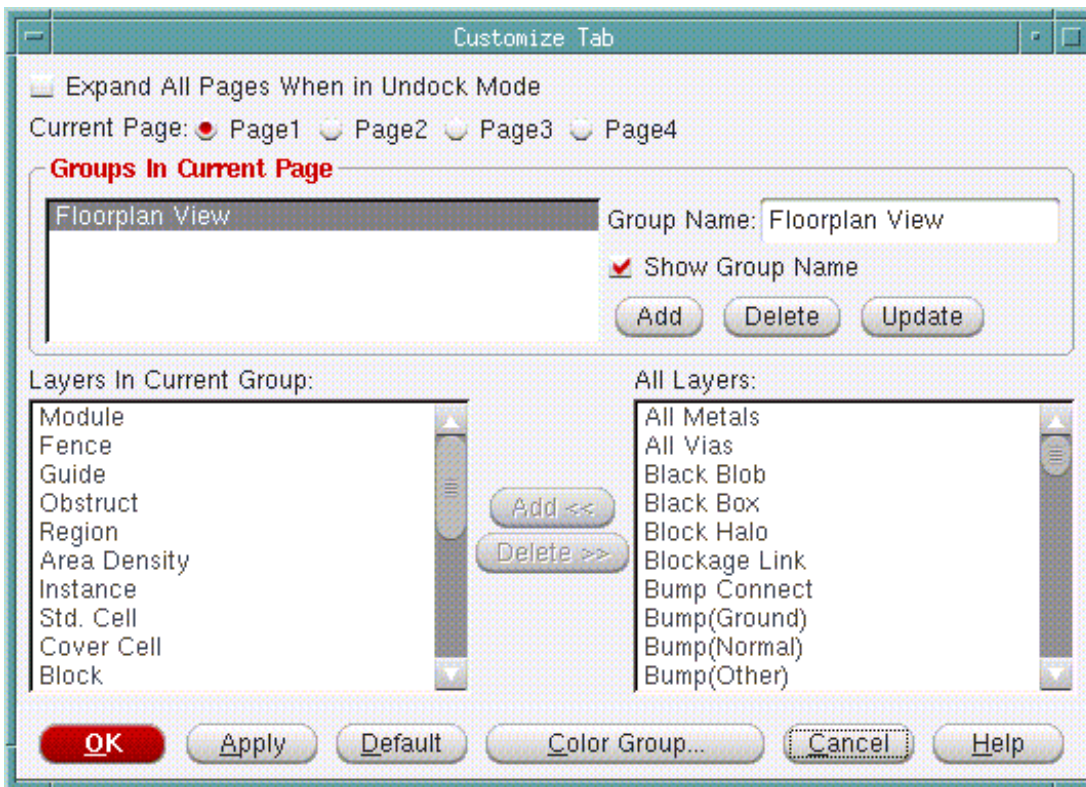


- To clear the pattern, click the *Clear* button.
- To invert the pattern, click the *Invert* button.
- To customize the pattern, do the following:
 - To change a square to black color, click on the square with the left mouse button.
 - To change a square to white color, click on the square with the right mouse button.

Customize Tab

You can use the *Customize Tab* form, accessible by clicking the *Customize Tab* button in any page on the Color Preferences form, to customize or add additional pages to the [Color Bar](#) in the main window.

→ In any of the pages of the [Color Preferences](#) form, click *Customize Tab*.



Fields and Options

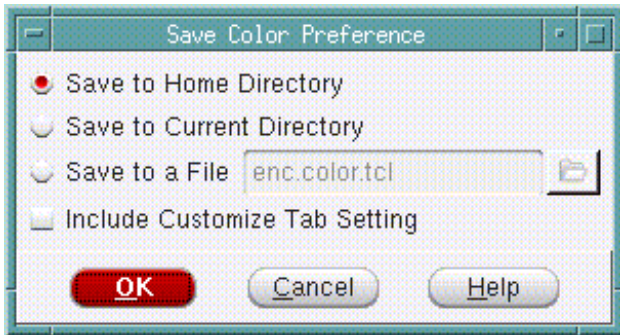
Expand All Pages When in Undock Mode	
	Displays all pages of the Object Color window when you undock the window.
Current Page	Selects the Page in the Color Bar Window that should be customized. By default, the color bar has two pages, but you can create up to four pages.
<i>Groups in Current Page</i>	
	Displays a selectable list of the names of the groups in the current color bar. A group is a collection of objects that are grouped together in the Color Bar. This way you can view the color and visibility controls for related objects together. A page in the color bar can have more than one group. The default first page in the Color Bar displayed in the main window contains the group FPlan View. The default second page contains two groups: Physical View and Layers. You can add, delete, or rename the groups in a Color Bar page.
Group Name	This is an editable field that displays the name of the group currently selected in the <i>Groups in Current Page</i> field.
Show Group Name	Displays the name of the group in the Color Bar page. If this option is not selected, the objects in the group are displayed in the Color Bar page, but the name of the group is not displayed. <i>Default:</i> The default is <i>Selected</i>.
Add	Adds a new group to the page. To add a new group, specify the name of the group in the Group Name field, and then click the Add button. You can then add objects to this group by selecting the objects in the <i>Add Layer</i> field and then clicking the <i>Add</i> button.
Delete	Deletes the group currently selected in the <i>Groups in Current Page</i> field.
Updates	Edits the name of the group currently selected in the <i>Groups in Current Page</i> field. To edit the name of the group, first edit the name in the Group Name field, and then click Update.
All Layers	Displays a list of all objects that you can add to a group.
<i>Layers in Current Group</i>	

	Displays the objects currently part of the group.
<i>Add</i>	Adds the object currently selected in the All Layers field to the Layers in Current Group field.
<i>Delete</i>	Removes the object currently selected in the <i>Layers in Current Group</i> field.

Save Color Preference

Use the Save Color Preference form to save a color preference file. A preference file is a Tcl file that contains color preference information. The recommended filename and extension name is `ets.color.tcl`.

→ In the Color Preference form, click **Save**.



Fields and Options

<i>Save to Home Directory</i>	
	Saves the color preferences file in your home directory.
<i>Save to Current Directory</i>	
	Saves the color preferences file in the current working directory.
<i>Save to a File</i>	Saves the color preferences file in the specified location.
<i>Include Customize Tab Setting</i>	

Along with the color preference information, also saves the settings specified through the <i>Customize Tab</i> form.

Multicolor Layers

During a design session, the following object types do not have a single color, but each object has a color that shows a level of severity and magnitude:

- *Horizontal Congest*
- *Vertical Congest*
- *IR Drop or EM*
- *Clock Tree*

To see the level indicator colors, double-click the color button for the object type. This opens a color preferences form for that object type.

Metal Layers and Other Layered Components

Click the *Wire/Via* tab to see the metal layers available in the design. *M0 Wire* represents the masterslice layer, *M1 Wire* represents the bottom metal layer, and so on. The vias are also shown, with *V01* representing the via between the masterslice layer and the first metal layer, *V12* representing the via between the first and second metal layers, and so on.

You can set preferences for other layered components in addition to wires and vias. For each metal layer, you can set the color, stipple pattern, visibility, and selectability for pins, tracks, blockages, and violations.

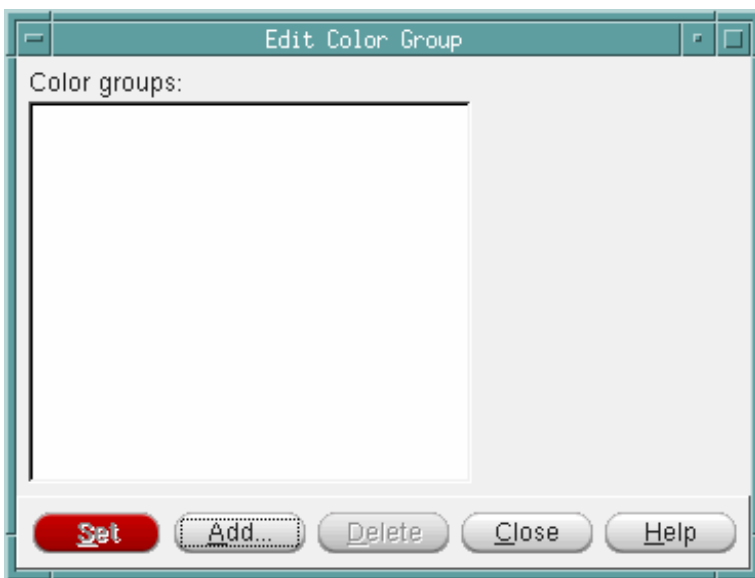
Note: Violations are not selectable.

Creating and Editing Color Groups

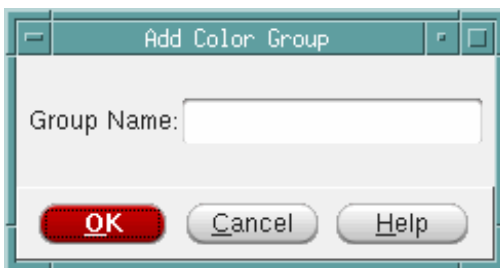
You can create color groups to customize the color settings in Voltus. A color group can contain custom color settings for various objects--for example, you can set the blockage color to red in one color group and to blue in another color group. You can switch between color views during a session. The settings specified in a color group are applied in all views.

To create a Color Group, do the following

- a. In the Layout window, click the *All Colors* button. The *Color Preferences* form appears.
- b. Create the custom color settings that you want for the group. For example, you might want to set the block color, or the visibility of blockages.
- c. Click the *Customize Tab* button. The *Customize Tab* form appears.
- d. Click the *Color Group* button. The *Edit Color Group* dialog box appears.



- a. Click the *Add* button. The Add Color Group form appears.



1. In the Group Name field, enter the name of the new color group, and click OK.

The new color group is created.

To specify or change the color settings for the Color Group, do the following:

1. In the Layout window, click the *All Colors* button. The *Color Preferences* form appears.
2. Create the custom color settings that you want for the group. For example, you might want to change the instance color, or the visibility of blackboxes.
3. Click the *Customize Tab* button. The Customize Tab form appears.
4. Click the *Color Group* button. The Edit Color Group dialog box appears.
5. Select the name of the *Color Group* to which the changed color settings should be applied.
6. Click the *Set* button. The settings are applied to the selected Color Group.

Once you create one or more color groups, the color setting for a particular group can be applied by selecting the group name from a drop-down menu next to the *All Colors* button in the main form. You can thus select the color group from the main window.

Color Bar

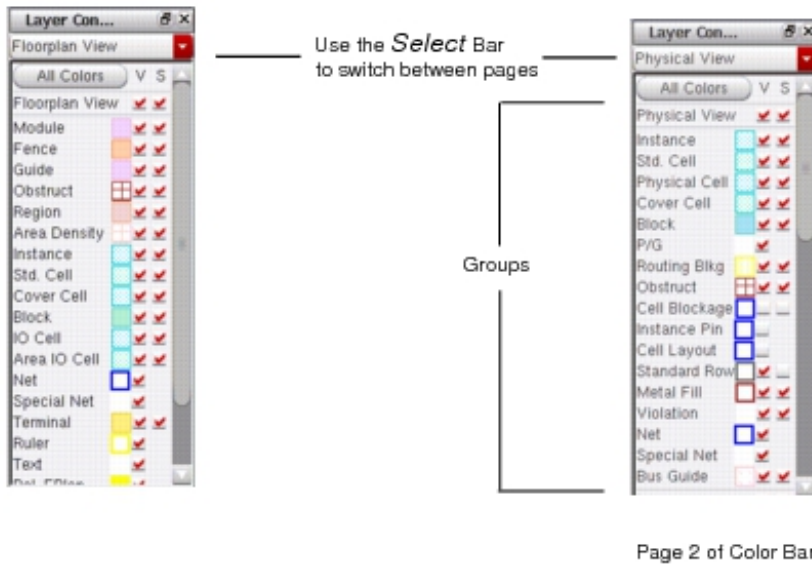
The Layout window includes an Object Color Bar that has color display buttons for objects in the design display area. Each object type has a check box for displaying the object in the design display area and a color button that displays the current color used to identify the object in the design display area.

By default, the Object Color Bar has two pages, but you can create up to four pages. Each page in the color bar can have one or more group. A group is a collection of objects that are grouped together in the Object Color Bar. This way you can view the color and visibility controls for related objects together. The default first page in the Object Color Bar displayed in the Layout window contains the group FPlan View. The default second page contains two groups: Physical View and Layers.

You can use the [Customize Tab](#) form to add pages to the Object Color Bar, add or delete the groups, and to add objects in each group. The Customize Tab form is available on all pages of the [Color Preferences](#) form.

Note: You can right-click on an individual layer to toggle the visibility/selectability of all objects on the layer.

To switch between the pages, click the *Select* bar.



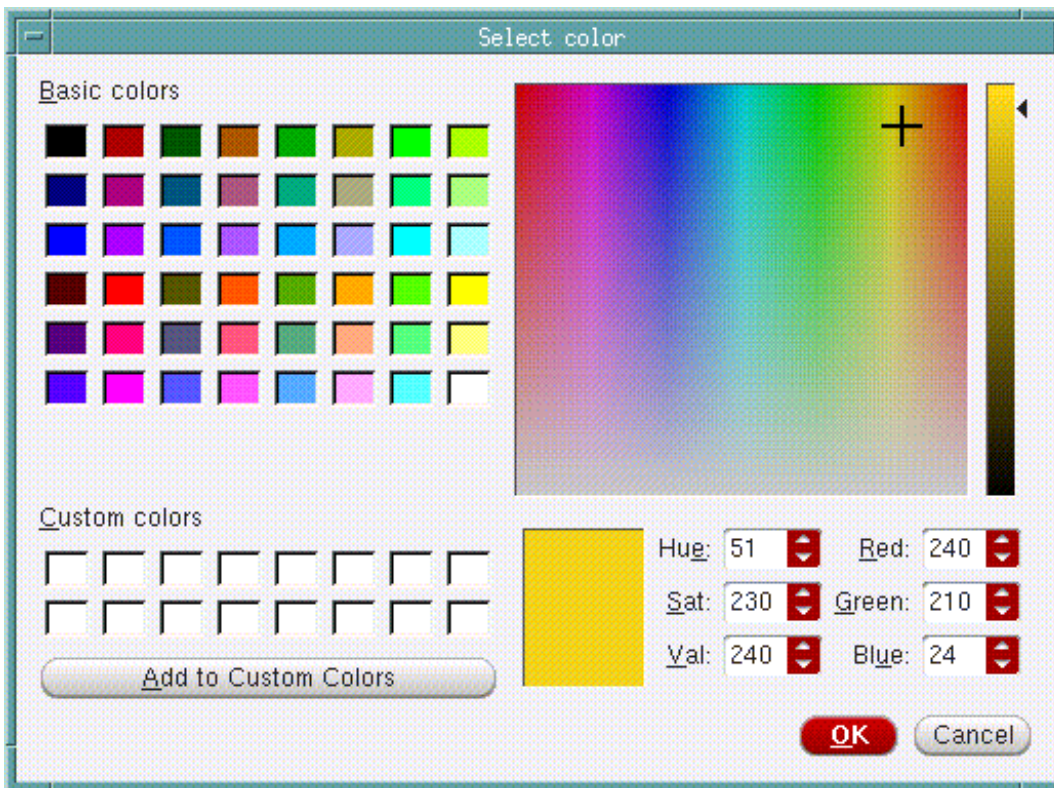
- If you select the *V* (Visibility) check box for an object type, the object type displays in the design display area in the specified color. If you clear the check box, the object type does not display in the design display area.
- If you select the *S* (Selectability) check box for an object type, the object type can be selected in the design display area when you use the *Selection* icon. If you clear the check box, the object type cannot be selected.

Note: Objects that are not visible are not selectable.

Object Color

Use the Select color form to customize the display color of an object.

→ Click on the color indicator of the object type to open the Select color form.



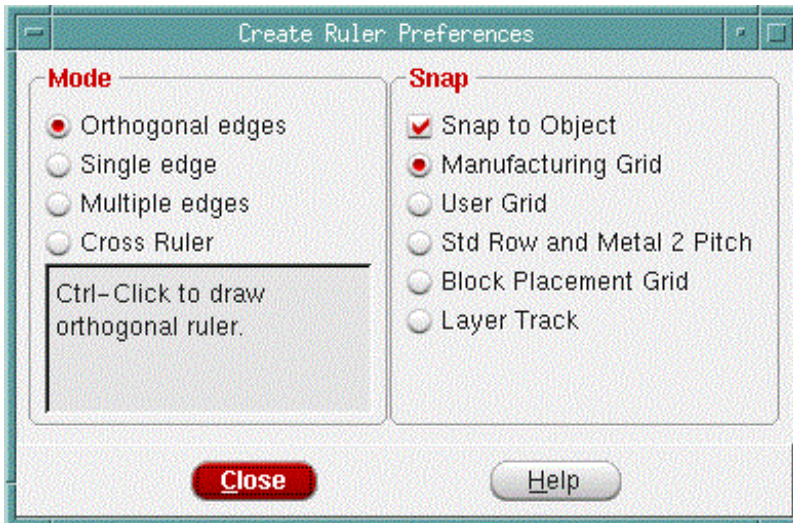
 If you use the text fields, you must press the Return key after your text entry.

Create Ruler Preferences

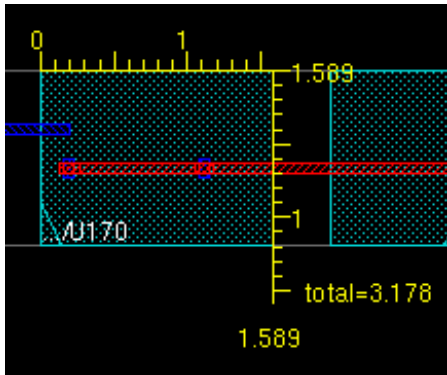
Use the Create Ruler Preferences form to set the direction in which ruler lines are drawn.

To open the Create Ruler Preferences form:

1. Click the Ruler widget 
2. Press the F3 key.



Fields and Options

Mode	Specifies the direction of the ruler.	
	<i>Orthogonal edges</i> Specifies that ruler lines can be drawn in horizontal as well as vertical directions. Use this option if you want to measure orthogonal edges, such as follows:  Note: This is the default ruler mode. Click the <i>Create Ruler</i> widget and start drawing the ruler to measure orthogonal edges. To end the ruler, click or press <code>Enter</code>.	

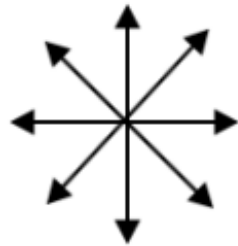
*Single
edge*

Specifies that ruler line can be drawn in a single direction.

Use this option if you want to measure a single edge in **one** of the following directions:

- Vertical
- Horizontal
- Diagonal (45 degrees, 135 degrees)
- Any angle

To draw a vertical, horizontal, or diagonal (45 or 135 degrees) ruler, simply click at the point where you want the ruler to start and move the mouse in the required direction. The ruler line will follow one of the following defined directions automatically.

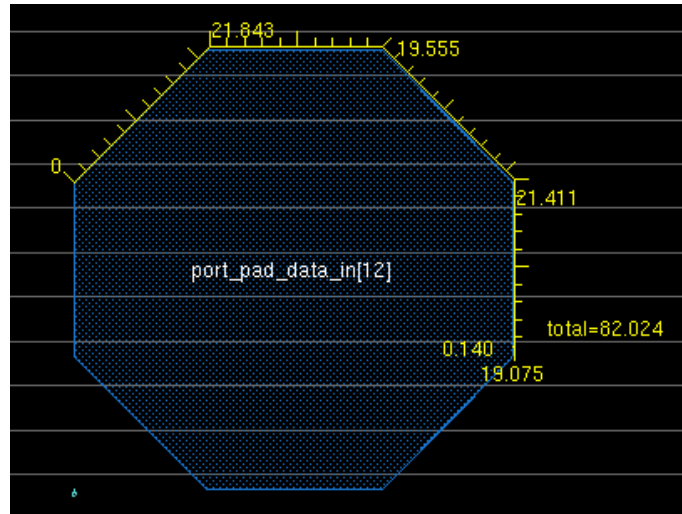


To draw a ruler at an angle other than the above eight directions, keep the `Shift` key pressed and then start drawing the ruler at the required angle.

To end a *Single edge* ruler, either click at the point where you want to end the ruler or press `Enter`.

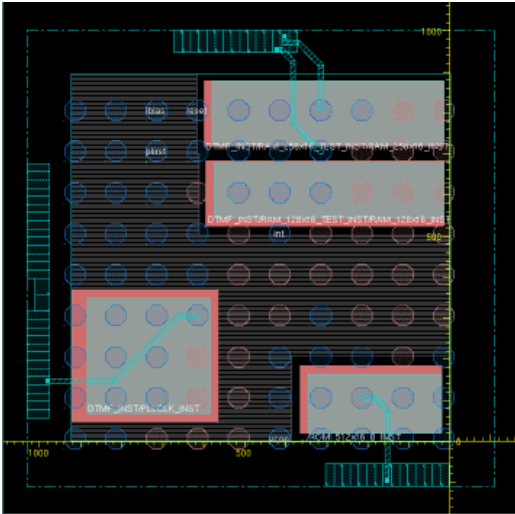
*Multiple
edges*

Specifies that ruler lines can be turned in multiple directions. Use this option if you want to measure multiple segments in a complex pattern.



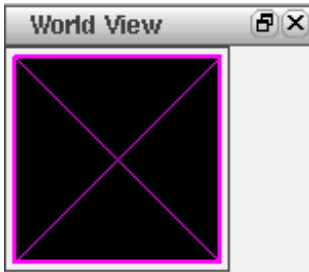
To draw a *Multiple edge* ruler, click at the point where you want the ruler to start. Move the mouse in the required direction, clicking at every point you want the ruler to turn. To end the ruler, press `Enter`.

Note: To draw a ruler in a direction other than horizontal, vertical or diagonal (45 or 135 degrees), keep the `Shift` key pressed and then start drawing the ruler at the required angle.

<i>Cross Ruler</i>	Specifies that ruler lines can be drawn in a + shape. Use this option to measure two edges at the same time, see the overall X and Y length, or to check alignment of macros.
	
To draw a cross ruler, click anywhere in the design. A + ruler is displayed occupying the entire display area in both X and Y directions. The center of the + moves with the cursor. Move the center of the + to the point where you want the origin of the ruler. Press <code>Enter</code> to place and end the ruler.	
<i>Snap</i>	Specifies the snap settings for the ruler.
<i>Snap to Object</i>	Specifies that the ruler should snap to the nearest design object. You can choose which design object to snap to by selecting one of the following options: • <i>Manufacturing Grid</i> (Default) • <i>User Grid</i> • <i>Std Row and Metal 2 Pitch</i> • <i>Block Placement Grid</i> • <i>Layer Track</i>

Satellite Window

The main window includes a satellite window, which identifies the location of the current view in the design display area, relative to the entire design. The chip area is identified by a yellow box, the satellite view is identified by the pink crossbox. When you display an entire chip in the design display area, the satellite crossbox encompasses the chip area yellow box.



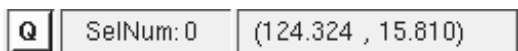
When you zoom and pan through the chip in the design display area, the satellite crossbox identifies where you are relative to the entire chip.

- To move to an area in the design display area, click and drag on the satellite crossbox.
- To select a new area in the design display area, click and drag on the satellite crossbox.
- To resize an area in the satellite window, click with the `Shift` key and drag a corner of the crossbox.
- To define a chip area in the satellite window, right-click and drag on an area.

Auto Query

The Layout window includes a *Q* icon located at the bottom of the design display area that enables and disables the auto query. When you enable auto query, text information displays for any object that is under the cursor. You can use auto query for blockages, hand-drawn or inside cell instances, instance pins, special route wires and vias, and cell instances.

If there are overlapping objects under the cursor, text information displays for the object that is on top. Use the `n` key to get information on the next object, and the `p` key to get information on the previous object. You can also use the F8 binding key to copy auto query text from the message bar in the Layout window to the software console. The text is displayed in the software console with the prefix `Message`.

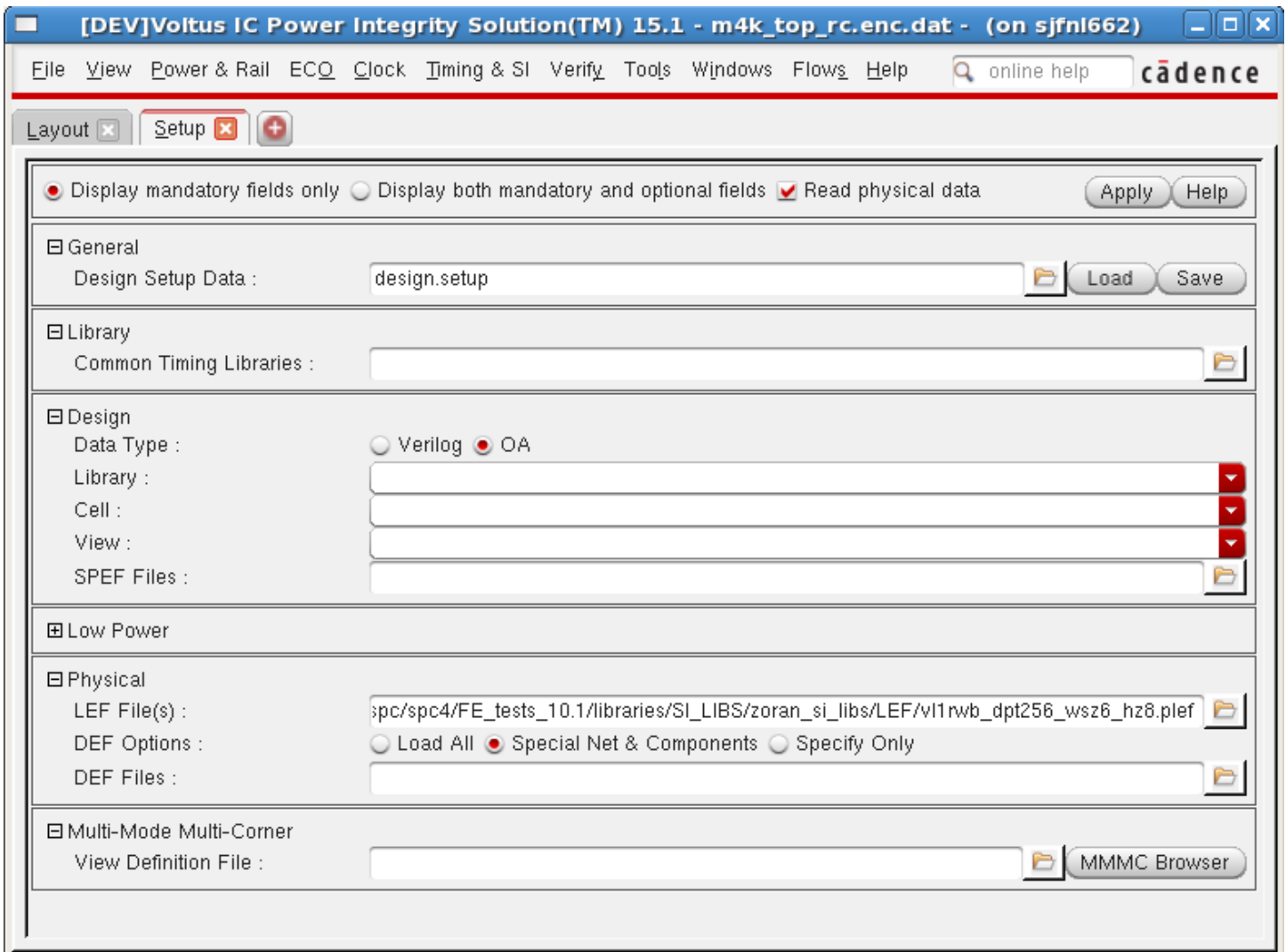


Setup Tab

- [Setup Tab Components](#)

Setup Tab Components

→ Click the *Setup* tab.



Fields and Options

<i>Display mandatory fields only</i>	
	Displays mandatory fields in the form.
<i>Display both mandatory and optional fields</i>	
	Displays both mandatory and optional fields. The optional fields include Low Power, Physical, Multi-Mode Multi-Corner and Statistical Static Timing Analysis.
<i>Read physical data</i>	Reads physical information such as placement, floorplan and routing that you created using Innovus.

<i>Design Setup Data</i>	Loads a saved Innovus database. The database consists of configuration file, and all files listed in the configuration file such as verilog, libraries, SDC. The database also consists of the mode file.
<i>Common Timing Libraries</i>	
	Specifies the name of the libraries common to both the <i>Best Timing Libraries</i> and the <i>Worst Timing Libraries</i> , which are used in both setup and hold analysis.
<i>Worst Timing Libraries</i>	Specifies the name of the timing libraries containing worst-case conditions for setup-time analysis.
<i>Best Timing Libraries</i>	Specifies the name of the timing libraries containing best-case conditions for hold-time analysis.
<i>Worst Noise Libraries</i>	Specifies the noise library (cdB) files used in setup analysis mode. In setup analysis mode, the common cdB files and the max cdB files are used in SI analysis.
<i>Best Noise Libraries</i>	Specifies the noise library (cdB) files used in hold analysis mode. In hold analysis mode, the common cdB files and the min cdB files are used in SI analysis.
<i>Common Noise Libraries</i>	
	Specifies the name of the common noise library (cdB) files used in SI analysis.
<i>Stamp Model Definitions</i>	
	Specifies the Stamp model definition files. Stamp model files contain timing information for a module, such as an instantiated module, block, or partitioned module. You do not need to translate these files before reading them into the software. However, currently you can only read in constant value models. Stamp timing specifications supersede timing information read from the timing library.
<i>Data Type - Verilog</i>	

<i>Verilog Netlist Files</i>	Specifies the gate-level Verilog netlist files to import. If specifying multiple names in the text field, use spaces to separate the names. You can use wildcards (*.<i>extension</i>) or directory names to specify multiple files. You can specify compressed files (.<i>gz</i> extension).
<i>Top Level Design Name</i>	
	Specifies the top module name, and checks for consistency between verilog netlist and timing libraries. Voltus uses the top module for timing analysis.
<i>Timing Constraint File</i>	Specifies the Synopsys Design Constraint (SDC) constraint file to import. You can import one or more constraint files.
<i>Data Type - OA</i>	
<i>Library</i>	Specifies the location of the Open Access database to import. It contains the cell and the view to be loaded.
<i>Cell</i>	Specifies the design cell from the library that is to be loaded.
<i>View</i>	Specifies the view name of the cell that is to be loaded.
<i>SPEF Files</i>	Specifies the name of the SPEF files.
<i>SDF Delay</i>	Specifies the name of the file containing the SDF delays you want to load.
<i>SOCE MSMV File</i>	Specifies the name of the MSMV file. This file uses a format that is similar to the IR-drop file and can be created in Innovus.
<i>Power Net(s)</i>	Specifies the names of the <i>Power Nets</i>. You can enter multiple names by separating the net names with a space, for example: VDD VDD1
<i>Ground Net(s)</i>	Specifies the names of the <i>Ground Nets</i>. You can enter multiple names by separating the net names with a space, for example: GND GND1

<i>LEF Files</i>	Specifies the LEF files to import. The LEF file provides the physical data information. You must specify the technology LEF file first, then specify the standard cell LEF and block LEF in any order.
<i>Floorplan File</i>	Specifies the name of the floorplan file to load for the design session.
<i>Placement File</i>	Specifies the name of the placement file to load for the design session.
<i>Routing File</i>	Specifies the name of the routing file to load for the design session.
<i>DEF Options</i>	Choose from one of the following options: <i>Load All:</i> Loads all DEF files. <i>Special Net & Components:</i> Reads the <code>SPECIALNETS</code> and <code>COMPONENTS</code> section of the DEF file. <i>Specify Only:</i> Loads only the specified files.
<i>DEF Files</i>	Specifies the name of a Design Exchange Format (DEF) file. You can supply hierarchical DEFs to generate a flattened physical database.
<i>View Definition File</i>	Specifies a Tcl command script that contains the analysis view configurations for multi-mode multi-corner timing analysis and optimization. The file contains a list of Tcl commands that create library sets, constraint modes, RC corners, delay corners, and analysis views. Processing the file does not automatically place the software in multi-mode multi-corner analysis mode. If the file also contains a <code>set_analysis_view</code> command, then the software is placed in multi-mode multi-corner analysis mode after the file is processed.
<i>MMMC Browser</i>	Opens the Multi-Mode Multi-Corner Browser (MMMC Browser). The browser displays the multi-mode multi-corner configuration, and allows you to create and edit the various objects and analysis views necessary for multi-mode multi-corner analysis.

<i>SPDF File</i>	Specifies the name of a statistical parameter distribution format (SPDF) file. An SPDF file contains statistics of variation of process parameters and is used for statistical static timing analysis (SSTA). You can specify global, random and spatial variations in this file. You can create this file using the data provided by the foundry.
<i>Statistical SPEF File</i>	Specifies the name of the statistical SPEF (S-SPEF) file. The software requires an S-SPEF file to perform sensitivity-based extraction. Sensitivity-based extraction captures variations in interconnects. In sensitivity based extraction, in addition to nominal values of resistances and capacitances, the software also extracts the sensitivities of R and C to different interconnect process parameters. The S-SPEF file is used for statistical static timing analysis (SSTA).

Related Text Commands

The following text commands provide equivalent or additional functionality:

- `read_design`
- `read_def`
- `read_lib`
- `read_power_domain`
- `read_sdc`
- `read_verilog`
- `set_top_module`

For more information, see the "Setup Commands" in *Voltus Text Command Reference*.

Schematic Tab

- [Schematic Tab Components](#)

Schematic Tab Components

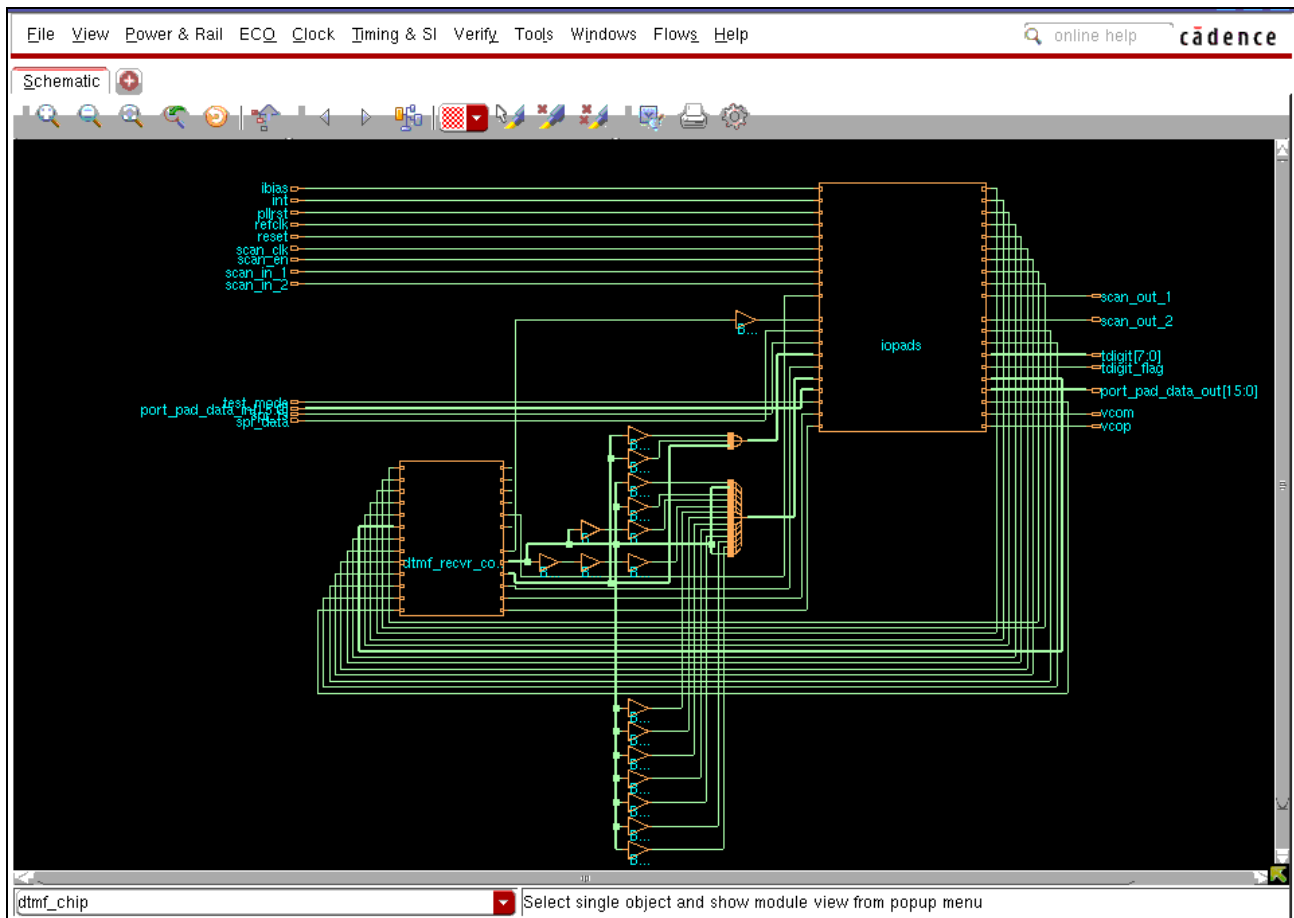
Use the Schematic tab to explore a design's connectivity, the relationship of modules and instances, and design changes, such as CTS, OPT, and ECO. You can cross-probe from the Schematic Viewer to the Layout view to give you flexibility and immediate feedback to debug and diagnose your design.

You can open the Schematic Viewer in the following ways:

- From the main Voltus window:
 - Choose the *Schematic* tab. This opens Schematic Viewer in Hierarchical view.
- From the Design Browser, do one of the following:
 - Select a module and click the *Show Hierarchical Schematic* widget. This opens the Schematic Viewer in Hierarchical view.
 - Select an instance, pin, or net and click the *Show Instance Based Schematic* widget. This opens the Schematic Viewer in Flattened view.
- From the Violation Browser:
 - Select an MSV violation and click the *Show MSV Violation Schematic* widget.

Voltus IC Power Integrity Solution Menu Reference

Schematic Tab



For information on Schematic Viewer, refer to the *Schematic Viewer* section in the [Tools Menu](#) chapter.

Design Browser Tab

For information on Design Browser, refer to the *Design Browser* section in the [Tools Menu](#) chapter.

Layout Viewer Tab

For information on Layout Viewer, refer to the *Layout Viewer* section in the [Tools Menu](#) chapter.

Violation Browser Tab

For information on Violation Browser, refer to the *Violation Browser* section in the [Tools Menu](#) chapter.